

Comparing LLM and human reviews of social science research using data from Unjournal.org

Valentin Klotzbücher David Reinstein Lorenzo Pacchiardi
Tianmai Michael Zhang

2026-02-14

We benchmark how frontier LLMs critique, rate, and prioritize research, comparing AI-generated evaluations to human expert assessments from The Unjournal. Human ratings serve as a reference signal—the core goal is measuring and diagnosing AI reasoning quality: understanding calibration, failure modes, and systematic “taste” differences between AI and human evaluators.

Table of contents

1	Introduction	7
1.1	Research Goals and Questions (overview)	8
1.2	Our work in context	9
1.2.1	AI practical research applications	9
1.2.2	AI peer review, error detection, and research evaluation	9
1.2.3	AI research taste and potential to guide research	11
1.2.4	Our present focus, The Unjournal opportunity	11
1.3	Future work	12
1.4	Acknowledgements	12
2	Data and methods	14
2.1	Unjournal.org evaluations	14
2.2	LLM-based evaluation	14
2.2.1	JSON Schema	15
2.2.2	System Prompt	15
2.2.3	Multi-Model Evaluation	16
2.2.4	GPT-5.2 Pro Focal Run (with Key Issues)	16
2.2.5	Key Issues Comparison with Human Critiques	16
3	Results: Ratings	18
3.1	Evaluation Overview	18
3.1.1	Token Usage and Cost	18
3.2	Human vs LLM Scatter Plot	19
3.3	Model Agreement Statistics	19
3.3.1	Inter-Rater Agreement (Krippendorff's Alpha)	21
3.3.2	Human-Human vs Human-LLM Agreement (Baseline Context)	21
3.3.3	Uncertainty Interval Coverage (LLM Credible Interval Assessment)	22
3.3.4	Statistical Significance of Agreement Measures	22
3.3.5	Information-Theoretic Perspective: Mutual Information	23
3.3.6	Agreement by Evaluation Criterion	24
3.4	Criteria-Level Comparison	24
3.5	Evaluation Priorities: What Predicts Journal Tier Judgments?	24
3.6	Systematic Rating Differences by Paper and Criterion	25
3.7	Papers with Largest Human-LLM Disagreement	25
3.8	Prompt Comparison: Legacy vs Updated GPT-5 Pro	27
4	Results: Critiques & Key Issues	28
4.1	Data Sources	28
4.2	Overview Statistics	28
4.3	Paper-by-Paper Comparison	29

4.4	LLM-Based Assessment	29
4.5	Summary and Observations	31
4.5.1	Aggregate Statistics	31
4.5.2	Observable Structural Differences	32
4.5.3	Questions for Manual Review	32
4.6	Manual Annotation Tool	32
References		34
Appendices		37
A LLM Evaluation Details		37
B Focal Papers (GPT-5.2 Pro with Key Issues)		38
B.1	Acemoglu et al. 2024	38
B.1.1	Ratings Comparison	38
B.2	Adena and Hager 2024	44
B.2.1	Ratings Comparison	44
B.3	Benabou et al. 2023	50
B.3.1	Ratings Comparison	50
B.4	Bilal and Kaenzig 2024	57
B.4.1	Ratings Comparison	57
B.5	Blimpo and Castaneda-Dower 2025	64
B.5.1	Ratings Comparison	64
B.6	Bruers 2021	70
B.6.1	Ratings Comparison	70
B.7	Clancy 2024	76
B.7.1	Ratings Comparison	76
B.8	Dullaghan and Zhang 2022	83
B.8.1	Ratings Comparison	83
B.9	Frech et al. 2023	90
B.9.1	Ratings Comparison	90
B.10	Green et al. 2025	96
B.10.1	Ratings Comparison	96
B.11	McGuire et al. 2024	103
B.11.1	Ratings Comparison	103
B.12	Peterman et al. 2025	109
B.12.1	Ratings Comparison	109
B.13	Weaver et al. 2025	115
B.13.1	Ratings Comparison	115
B.14	Williams et al. 2024	121
B.14.1	Ratings Comparison	121
C Other Papers		129
C.1	Abbink et al. 2025	129
C.1.1	Ratings Comparison	129

C.2	Aghion et al. 2017	133
	C.2.1 Ratings Comparison	133
C.3	Akram et al. 2017	136
	C.3.1 Ratings Comparison	136
C.4	Alatas et al. 2019	140
	C.4.1 Ratings Comparison	140
C.5	Alcott et al. 2024	143
	C.5.1 Ratings Comparison	143
C.6	Angrist et al. 2023	146
	C.6.1 Ratings Comparison	146
C.7	Arora et al. 2023	150
	C.7.1 Ratings Comparison	150
C.8	Bahar et al. 2022	154
	C.8.1 Ratings Comparison	154
C.9	Banerjee et al. 2022	157
	C.9.1 Ratings Comparison	157
C.10	Banerjee et al. 2023	161
	C.10.1 Ratings Comparison	161
C.11	Barberio et al. 2022	164
	C.11.1 Ratings Comparison	164
C.12	Barker et al. 2021	168
	C.12.1 Ratings Comparison	168
C.13	Benjamin et al. 2023	172
	C.13.1 Ratings Comparison	172
C.14	BenYishay et al. 2024	176
	C.14.1 Ratings Comparison	176
C.15	Bergeron et al. 2024	180
	C.15.1 Ratings Comparison	180
C.16	Bettle 2023	183
	C.16.1 Ratings Comparison	183
C.17	Bhat et al. 2022	186
	C.17.1 Ratings Comparison	186
C.18	Bhattacharya and Packalen 2020	190
	C.18.1 Ratings Comparison	190
C.19	Bruehlhart et al. 2021	192
	C.19.1 Ratings Comparison	192
C.20	Buntaine et al. 2023	196
	C.20.1 Ratings Comparison	196
C.21	Carson et al. 2023	199
	C.21.1 Ratings Comparison	199
C.22	Caviola et al. 2022	202
	C.22.1 Ratings Comparison	202
C.23	Chuard et al. 2022	206
	C.23.1 Ratings Comparison	206
C.24	Crawfurd et al. 2023	209
	C.24.1 Ratings Comparison	209
C.25	Denkenberger et al. 2022	213
	C.25.1 Ratings Comparison	213

C.26 Epperson and Gerster 2024	217
C.26.1 Ratings Comparison	217
C.27 Fangwa et al. 2023	220
C.27.1 Ratings Comparison	220
C.28 Hahn et al. 2025	223
C.28.1 Ratings Comparison	223
C.29 Haushofer et al. 2020	227
C.29.1 Ratings Comparison	227
C.30 Hill et al. 2024	231
C.30.1 Ratings Comparison	231
C.31 Jack et al. 2022	235
C.31.1 Ratings Comparison	235
C.32 Jaroszewicz et al. 2024	238
C.32.1 Ratings Comparison	238
C.33 Kirkland et al. 2023	240
C.33.1 Ratings Comparison	240
C.34 Kremer et al. 2020	243
C.34.1 Ratings Comparison	243
C.35 Kremer et al. 2022	247
C.35.1 Ratings Comparison	247
C.36 Kubo et al. 2023	250
C.36.1 Ratings Comparison	250
C.37 Kudymowa et al. 2023	254
C.37.1 Ratings Comparison	254
C.38 Liang et al 2023	257
C.38.1 Ratings Comparison	257
C.39 Liang et al. 2025	261
C.39.1 Ratings Comparison	261
C.40 Naidu et al. 2023	264
C.40.1 Ratings Comparison	264
C.41 Salazar and Restrepo 2024	267
C.41.1 Ratings Comparison	267
C.42 Schoenegger et al. 2025	270
C.42.1 Ratings Comparison	270
C.43 Schuett et al. 2023	274
C.43.1 Ratings Comparison	274
C.44 Trammel and Aschenbrenner 2024	278
C.44.1 Ratings Comparison	278
C.45 Walker et al. 2023	281
C.45.1 Ratings Comparison	281
C.46 Xing et al. 2023	284
C.46.1 Ratings Comparison	284

1 Introduction

Working paper

This is an evolving working paper. Analysis, metrics, and comparisons are under active development.

Is AI good at peer-reviewing? Does it offer useful and valid feedback? Can it predict how human experts will rate research across a range of categories? How can it help academics do this “thankless” task better? Is it particularly good at spotting errors? Are there specific categories, e.g. spotting math errors or judging real-world relevance, where it does surprisingly well or poorly? How does its “research taste” compare to humans?

If AI research-evaluation works it could free up a lot of scientific resources – perhaps \$1.5 billion/year in the US alone ([Aczel, Szaszi, and Holcombe 2021](#)) – and offer more continual and detailed review, helping improve research and address the “[peer-review crisis](#)”. Numerous accounts report a system under strain, with widespread difficulty finding human peer reviewers with appropriate expertise willing to put in careful effort. An influx of AI-generated or AI-human hybrid research seems likely to exacerbate this problem.¹ AI research evaluation could also help characterize methodological strengths and weaknesses across papers, aiding research training and research direction-setting.

Furthermore, “[the leaders of the biggest A.I. labs](#) argue that artificial intelligence will usher in a new era of scientific discovery, which will help us cure diseases and accelerate our ability to address the climate crisis” – e.g., Sam Altman on [cures for cancer](#), etc. Understanding how AI engages with research evaluations may provide a window into its values, abilities, and limitations.²

In this project, we pilot the use of The Unjournal’s evaluations as a benchmark to test the capabilities of current large language models (LLMs) and bespoke agent-based tools, considering how the research evaluations these generate compare to expert human reviews. The Unjournal systematically prioritizes ‘impactful’ research and pays for high-quality human evaluations, structured quantified ratings, claim identification and assessment, and predictions. They further labeled and classified “major consensus critiques” from a subset of these evaluations.³

¹E.g., see [anecdotal reports](#). Proposed solutions include a credit or payment-based model. (Further references to be added.).

²Eger et al. (2025): “When it comes to using AI tools for science, ethics is of overarching importance. This is because the AI tools exhibit various limitations, e.g., they (i) may hallucinate and fabricate content, (ii) exhibit bias, (iii) may have limited reasoning abilities and (iv) sometimes lack suitable evaluation, ... among many other concerns such as risks of fake science, plagiarism, and lack of human authority. Indeed, the European Union has recently released guidelines on the responsible use of AI for science. In it, it points out that, while “[r]esearch is one of the sectors that could be most significantly disrupted by generative AI” where “AI has great potential for accelerating scientific discovery and improving the effectiveness and pace of research and verification processes”

³An individual rater (David Reinstein, Co-director of The Unjournal) curated a pilot version of this [here](#), focusing on critiques identified by multiple evaluators and/or confirmed by the author or evaluation manager. We are currently (Jan 5, 2026) comparing this enumeration to the issues identified by LLM’s (see “Results: Critiques” chapter). Future work will systematically employ multiple enumerators with a pre-defined rubric.

We next prompt several LLMs and agent-based tools⁴ to review the same set of quantitative social science research evaluated by The Unjournal’s human evaluators, with the same set of criteria and guidelines. Each paper is assessed on specific dimensions – for example, the strength of its evidence, rigor of methods, clarity of communication, openness/reproducibility, relevance to global priorities, and overall quality. The LLM will provide quantitative scores (with uncertainty intervals) on these criteria and produce a written evaluation.

Our initial dataset will include the 60 research papers that have existing Unjournal human evaluations.⁵ For each paper, the AI will generate: (1) numeric ratings on the defined criteria, (2) identification of the paper’s key claims, and (3) a detailed review enumerating and discussing the paper’s contributions and weaknesses. We then compare the AI-generated evaluations to the published human evaluations, considering both quantitative metrics (e.g., inter-rater reliability), identification of consensus critiques, and qualitative comparison. The next phase will train and tune models, and measure their performance on research currently in The Unjournal’s pipeline, i.e., where no human evaluation has yet been made public, to enable out-of-time validation, confirmatory hypothesis tests, and less risk of model contamination.

This complements other recent work on AI-evaluation approaches and benchmarks, outlined below. Our work offers in-depth comparisons and oversight in a high-nuance and high-value context: global-priorities-relevant economics and social-science research, and leverages The Unjournal’s ongoing pipeline as a continual experimental and feedback environment.

1.1 Research Goals and Questions (overview)

1. Measure the value of frontier AI research evaluation vs. human peer review

- Is state-of-the-art AI research evaluation reliable and useful enough to replace human peer review in these contexts?
- Reverse question: The Unjournal was asked – “Are *human* evaluations worth it”?
- How well can AI predict what humans will report?
- What are AI’s strengths & weaknesses relative to humans, e.g., in terms of: (1) Predicting journal outcomes, citations, etc. (2) Generating feedback deemed most useful by humans (3) Spotting research errors (4) Diagnosing different types of strengths and limitations (e.g., reasoning transparency, justification of methods and assumptions, communication) (5) Judging content without biases/statistical discrimination based on prestige labels, etc.

2. Compare methods for AI research-evaluation, find the best approaches

- Which approaches come closest to (the desirable features of) human evaluations and in which ways?
- What gets closest to the ‘frontier’, doing better in terms of outcomes like those mentioned above?

3. The nature of frontier AI ‘preferences & tastes’ over research

⁴Dec 2025 focus: OpenAI’s GPT-5.2 Pro model

⁵This is slightly off as of January 2026. We only have 56 papers with official Unjournal evaluations. I suspect this data includes some papers that are either duplicated or in our pipeline for given an independent non-official evaluation

- Which types of papers and approaches do default models tend to ‘like’ more/less relative to humans when given the same guidelines?
- Do we see signs of characteristic preferences that help us understand how AIs are likely to prioritize, conduct, and evaluate research?

4. Hybrid human-AI performance (future trials)

1.2 Our work in context

1.2.1 AI practical research applications⁶

Luo et al. (2025) survey LLM roles from idea generation to peer review, including experiment planning and automated scientific writing. They highlight opportunities (productivity, coverage of long documents) alongside governance needs (provenance, detection of LLM-generated content, standardizing tooling) and call for reliable evaluation frameworks.

Korinek (2025) considers the specific potential for using AI in economic research, offering “working examples and step-by-step code” to help economists use agents for literature reviews, econometric code, fetching and analyzing economic data, and coordinate complex research workflows. He advocates economists using AI tools to boost productivity and “to understand [these tools] capabilities and limitations.” He calls for human researcher oversight “to ensure [these tools] pursue the right objectives and maintain research integrity”. He outlines a range of current limitations and open questions (outlined [here](#)), including hallucinations, mistakes compounding through multi-agent workflows and “brittleness to small variations in prompts”. At the higher level, he argues “the questions that will matter most—what problems deserve investigation, how should we evaluate trade-offs, what constitutes progress—will remain irreducibly human.”

Mitchener, Yiu, et al. (2025) present Kosmos, “an AI scientist for autonomous discovery”. However, evaluations reveal important limitations:

- “Kosmos was found to be only 57% accurate in statements that required interpretation of results, likely due to its propensity to conflate statistically significant results with scientifically valuable ones”
- “although 85% of statements derived from data analyses were accurate, our evaluations do not capture if the analyses Kosmos chose to execute were the ones most likely to yield novel or interesting scientific insights.”... “identifying valuable discoveries Kosmos made is a time-intensive process that relies on human scientists with significant domain expertise”

1.2.2 AI peer review, error detection, and research evaluation

Son et al. (2025) evaluates LLMs on 83 papers paired with 91 confirmed significant scientific errors. This resembles our own approach, although they focus on errata/retraction-worthy errors, while we consider more nuanced methodological limitations. Their strongest (OpenAI o3) model identified only 21.1% of these errors, with only 6.1% precision.⁷

⁶We maintain a NotebookLM of the most relevant and related research [here](#).

⁷They asked domain experts to do qualitative analysis, which found most false positives were hallucinations or student-level misconceptions

Eger et al. (2025) provide a broad review of LLMs in science and a focused discussion of AI-assisted peer review. They argue: (i) peer-review data is scarce and concentrated in CS/OpenReview venues; (ii) targeted assistance that preserves human autonomy is preferable to end-to-end reviewing; and (iii) ethics and governance (bias, provenance, detection of AI-generated text) are first-class constraints.

Dycke, Kuznetsov, and Gurevych (2023) present NLPEER, a “multidomain corpus” of more than 5000 papers and 11k review reports from 2012 to 2022 from five different venues”.

Zhang and Abernethy (2025) propose deploying LLMs as quality checkers to surface critical problems instead of generating full narrative reviews. Using papers from WITHDRARXIV and an automatic evaluation framework that leverages “LLM-as-judge,” they find the best performance from top reasoning models but still recommend human oversight.

Pataranutaporn et al. (2025) asked four nearly state-of-the-art LLM models (GPT-4o mini, Claude 3.5 Haiku, Gemma 3 27B, and LLaMA 3.3 70B) to consider 1220 unique papers “drawn from 110 economics journals excluded from the training data of current LLMs”. They prompted the models to act “in your capacity as a reviewer for [a top-5 economics journal]” and make a publication recommendation using a 6-point scale ranging from “1 = Definite Reject...” to “6. Accept As Is...”. They asked it to evaluate each paper on a 10-point scale for originality, rigor, scope, impact, and whether it was ‘written by AI’. They also (separately) had LLMs rate 330 papers with the authors’ identities removed, or replacing the names with fake male/female names and real elite or non-elite institutions or with prominent male or female economists attached.

They compare the LLMs’ ratings with the RePEC rankings for the journals the papers were published in, finding general alignment. They find mixed results on detecting AI-generated papers. In the names/institutions comparisons, they also find the LLMs show biases towards named high-prestige male authors relative to high-prestige female authors, as well as biases towards elite institutions and US/UK universities.

There have been several other empirical benchmarking projects, including work covered in LLM4SR: A Survey on Large Language Models for Scientific Research and [Transforming Science with Large Language Models: A Survey on AI-assisted Scientific Discovery, Experimentation, Content Generation, and Evaluation](#). (See cited surveys for additional benchmarks.)

Zhang et al. (2025) use AI conference paper data and LLM agents for pairwise manuscript comparisons, finding this approach outperforms traditional rating-based methods for identifying high-impact papers by citation metrics, though with some evidence of biases favoring established research institutions.⁸

We discuss how our approach differs from prior work [below](#).⁹

⁸Key findings from Zhang et al. (2025): (1) Uses AI conference paper data; (2) Employs LLM agents for pairwise manuscript comparisons; (3) Significantly outperforms traditional rating-based methods in identifying high-impact papers by citation metrics; (4) Shows some evidence of biases/statistical discrimination based on characteristics like ‘papers from established research institutions’.

⁹See below. Claude’s brief summary “Our approach differs from prior work by (i) focusing on structured, percentile-based quantitative ratings with credible intervals that map to decision-relevant dimensions used by The Unjournal; (ii) comparing those ratings to published human evaluations rather than using LLM-as-judge; and (iii) curating contamination-aware inputs (paper text extraction with reference-section removal and token caps), with a roadmap to add multi-modal checks when we score figure- or table-dependent criteria”.

1.2.3 AI research taste and potential to guide research

A key motivation for this work stems from broader claims about AI’s potential to accelerate scientific discovery and “revolutionize science.” If AI systems are to contribute meaningfully to research—not just as tools but as evaluators and potential collaborators—we need to understand their “taste”: what they value, what they miss, and whether their priorities align with human expert judgment and real-world research impact.

Understanding LLM preferences and blind spots when evaluating research provides a window into:

- Whether AI can identify truly impactful research vs. stylistic or superficial features
- Potential systematic biases in how AI might direct future research priorities
- The risks and opportunities of AI-assisted or AI-directed research agendas

This connects to broader debates about the returns to science in the presence of technological risk (Clancy 2023), where optimistic forecasts of AI-accelerated science must be weighed against potential downsides and the question of whether AI judgments align with what actually advances knowledge and improves outcomes.

1.2.4 Our present focus, The Unjournal opportunity

Our project distinguishes itself through several unique advantages:

Actual human evaluations: We use *actual* human evaluations of research in economics and adjacent fields, past and *prospective*, including both reports, ratings, and predictions.¹⁰ Other work has relied on collections of research and grant reviews, including NLPEER, SubstanReview, and the Swiss National Science Foundation. That data has a heavy focus on computer-science adjacent fields, and is less representative of mainstream research peer review practices in older, established academic fields. Note that The Unjournal commissions the evaluation of impactful research, often from high-prestige working paper archives like NBER, and makes all evaluations public, even if they are highly critical of the paper.

Rich, structured data: The Unjournal’s 55+ evaluation packages enable us to train and benchmark the models. Their detailed written reports and multi-dimensional percentile ratings with credible intervals allow us to compare the ‘taste’, priorities, and comparative ratings of humans relative to AI models across the different criteria and domains.

Ground truth outcomes: The ‘journal tier prediction’ outcomes provide an external ground-truth¹¹ enabling a human-vs-LLM horse race.

Prospective pipeline: The Unjournal’s pipeline of future evaluations allows for clean out-of-training-data predictions and evaluation, serving as a “live testing lab” for a new judging panel, enabling out-of-time testing and validation from authors and human evaluators.

Hybrid evaluation potential: We are also planning multi-armed trials on these human evaluations (cf. Brodeur et al. 2025) to understand the potential for *hybrid* human-AI evaluation in this context.

¹⁰For additional context on The Unjournal’s approach and evaluation criteria, see our [evaluator guidelines](#).

¹¹These represent ground truths about verifiable publication outcomes, not about the ‘true quality’ of the paper.

The economics, social science, and global-impact policy context provides a unique high-value and nuanced setting to measure and understand AI tendencies and capabilities. According to Korinek (2025) “economists have a special responsibility. We understand incentives, market dynamics, and resource allocation. We grasp the subtleties of human behavior and social coordination. These insights will be crucial for the transition to an AI-augmented research”.

1.3 Future work

Several extensions of this research are natural next steps:

1. **Further statistical analyses:** Information-theoretic and Bayesian measures; multi-level modeling with random and mixed effects for raters, papers, and rating categories
2. **Content-swap bias tests:** Following Pataranutaporn et al. (2025), testing whether LLMs show systematic biases based on author names, institutional affiliations, or other non-content features
3. **Journal-outcome prediction horse-race:** Comparing human vs. LLM predictions against actual publication outcomes and bibliometrics
4. **Systematic model and prompt comparison:** Evaluating how different prompting strategies, models, and agent-based approaches affect evaluation quality
5. **Qualitative evaluation content analysis:** Deeper analysis of LLM-generated written evaluations and reasoning traces
6. **Human enumerator validation:** Employing trained raters to systematically assess whether LLMs identify the same consensus critiques as human experts
7. **Out-of-time validation:** Training predictive models on accumulated evaluations and pre-registering tests on papers entering The Unjournal’s pipeline
8. **Hybrid human-AI trials:** Testing whether human-AI collaboration improves evaluation quality beyond either alone
9. **Decision criteria for deployment:** Developing policy-relevant metrics for when AI evaluation is reliable enough to supplement or replace human review
10. **Implications for evaluation systems:** Drawing lessons for The Unjournal and peer review more broadly

1.4 Acknowledgements

This work is a collaboration with [The Unjournal](#), which has published 55+ detailed evaluation packages building open, quantitative evaluation infrastructure for global-priorities-relevant research in economics, policy, and social science. We thank The Unjournal’s evaluation community for generating the structured human assessments that make this benchmarking possible.

Funding for The Unjournal has been provided by the Survival and Flourishing Fund, the Long Term Future Fund, and EA Funds. Portions of this manuscript were drafted with AI assistance and reviewed by the authors.

2 Data and methods

We draw on two main sources:

- 1) Human evaluations from [The Unjournal’s public evaluation data](#)
- 2) LLM-generated evaluations using a structured JSON-schema prompt resembling The Unjournal’s guidelines, comparing several models including gpt-5-pro-2025-10-06 (knowledge cut-off: 30 September 2024).

2.1 Unjournal.org evaluations

We use The Unjournal’s public data for a baseline comparison. At The Unjournal each paper is typically evaluated (aka ‘reviewed’) by two expert evaluators¹ who provide quantitative ratings on a 0–100 percentile scale for each of seven criteria (with 90% credible intervals),² two “journal tier” ratings on a 0.0 - 5.0 scale,³ a written evaluation (resembling a referee report for a journal), and identification and assessment of the paper’s “main claim”. For our initial analysis, we extracted these human ratings and aggregated them, taking the average score per criterion across evaluators (and noting the range of individual scores).

All papers have completed The Unjournal’s evaluation process (meaning the authors received a full evaluation on the Unjournal platform, which has been publicly posted at unjournal.pubpub.org). The sample includes papers spanning 2017–2025 working papers in development economics, growth, health policy, environmental economics, and related fields that The Unjournal identified as high-impact. Each of these papers has quantitative scores from at least one human evaluator, and many have multiple (2-3) human ratings.

2.2 LLM-based evaluation

Following The Unjournal’s [standard guidelines for evaluators](#) and their [academic evaluation form](#), evaluators are asked to consider each paper along the following dimensions: **claims & evidence**, **methods**, **logic & communication**, **open science**, **global relevance**, and an **overall** assessment. Ratings are interpreted as percentiles relative to serious recent work in the same area. For each metric, evaluators are asked for the midpoint of their beliefs and their 90% credible interval,

¹Occasionally they use 1 or 3 evaluators.

²See their guidelines [here](#); these criteria include “Overall assessment”, “Claims, strength and characterization of evidence”, “Methods: Justification, reasonableness, validity, robustness”, “Advancing knowledge and practice”, “Logic and communication”, “Open, collaborative, replicable science”, and “Relevance to global priorities, usefulness for practitioners”

³“a normative judgment about ‘how well the research should publish’ ” and “a prediction about where the research will be published”

to communicate their uncertainty. For the journal rankings measure, we ask both “what journal ranking tier should this work be published in? (0.0-5.0)” and “what journal ranking tier will this work be published in? (0.0-5.0)”, with some further explanation. The full prompt can be seen in the code below – essentially copied from the Unjournal’s guidelines page.

We captured the versions of each paper that was evaluated by The Unjournal’s human evaluators, downloading from the links provided in The Unjournal’s Coda database.

We evaluate each paper by passing the PDF directly to the model and requiring a strict, machine-readable JSON output. This keeps the assessment tied to the document the authors wrote. Direct ingestion preserves tables, figures, equations, and sectioning, which ad-hoc text scraping can mangle. It also avoids silent trimming or segmentation choices that would bias what the model sees.

2.2.1 JSON Schema

We enforce a strict JSON Schema for the output. The model must return one object for each metric, including a midpoint rating and a 90% credible interval. This guarantees that every paper is scored on the same fields with the same types and bounds. We request credible intervals (as we do for human evaluators) to allow the model to communicate its uncertainty rather than suggest false precision.

2.2.2 System Prompt

The system prompt is built from modular sections, each addressing a specific aspect of the evaluation task. We present each section below with brief explanations.

We instruct the model to act as an expert evaluator while explicitly preventing it from using author reputation, publication venue, or any prior knowledge of the paper’s reception as evidence of quality.

Before scoring, we ask for a written assessment identifying key issues in the manuscript. This “think first, score second” approach encourages the model to ground its ratings in specific observations.

We define what percentile rankings mean and establish the reference group: “serious research in the same area encountered in the last three years.” This anchors the model’s judgments to a consistent baseline.

We define each of the seven percentile metrics, closely following The Unjournal’s [guidelines for evaluators](#). Note the emphasis on global priorities and practical relevance over pure academic novelty.

We explain what credible intervals mean and how to construct them. This guidance helps ensure the model produces well-calibrated uncertainty estimates rather than artificially narrow intervals.

In addition to percentile metrics, we ask for journal tier predictions on a 0-5 scale. These provide an external benchmark and, for papers that eventually publish, allow us to assess predictive accuracy.

Finally, we include instructions for self-consistency checks and the required JSON output format. The validation section encourages the model to ensure its scores align with its written assessment.

The sections above are concatenated to form the complete system prompt:

The `evaluate_paper` function uploads a PDF to the API and submits a background job for evaluation:

Relying on GPT-5 Pro, we use a single-step call with a reasoning model that supports file input. One step avoids hand-offs and summary loss from a separate “ingestion” stage. The model reads the whole PDF and produces the JSON defined above. We do not retrieve external sources or cross-paper material for these scores; the evaluation is anchored in the manuscript itself.

The Python pipeline uploads each PDF once and caches the returned file id keyed by path, size, and modification time. We submit one background job per PDF to the OpenAI Responses API with “high” reasoning effort and server-side JSON-Schema enforcement. Submissions record the response id, model id, file id, status, and timestamps.

We then poll job status and, for each completed job, retrieve the raw JSON object, and write the responses to disk.

2.2.3 Multi-Model Evaluation

To compare model performance, we collect evaluations from multiple providers. This allows us to assess whether different models exhibit systematic biases or calibration differences.

For OpenAI models without reasoning (e.g., GPT-4o), we use a simpler synchronous call:

Anthropic’s API differs from OpenAI: PDFs are passed as base64-encoded content, and calls are synchronous (no background jobs). We use Claude’s native PDF support.

Google’s Gemini API supports PDF via file upload:

A unified runner collects evaluations from all providers:

2.2.4 GPT-5.2 Pro Focal Run (with Key Issues)

This section evaluates 14 focal papers using GPT-5.2 Pro with an extended schema that outputs a numbered list of key issues alongside the standard metrics.

2.2.5 Key Issues Comparison with Human Critiques

To validate how well the LLM identifies substantive issues, we compare its `key_issues` output against human expert critiques from The Unjournal’s Coda database. These human critiques were written by domain experts during the standard evaluation process, providing a ground-truth benchmark for issue identification.

The comparison proceeds in two stages: first, we extract and align the data sources; then, optionally, we use an LLM to systematically assess the degree of alignment between machine-generated and human-identified issues.

The comparison pipeline uses a manually curated input and produces structured outputs:

Input: - `results/key_issues_comparison.md`: A manually curated markdown document pairing GPT-identified issues with human expert critiques for each paper.

Outputs: - `results/key_issues_comparison.json`: Parsed data in JSON format for programmatic use. - `results/key_issues_comparison_results.json`: LLM-assessed alignment metrics including coverage, precision, missed issues, and overall ratings.

The coverage metric estimates what fraction of human-identified issues the LLM captured; precision estimates what fraction of LLM issues are genuinely relevant. Together with the qualitative ratings, these provide a structured assessment of how well the LLM’s issue identification aligns with expert judgment.

3 Results: Ratings

We compare LLM-generated research evaluations against human expert reviews from [The Unjournal](#). This analysis covers 60 papers evaluated by 5 frontier LLMs (GPT-5 Pro, GPT-5.2 Pro, GPT-4o-mini, Claude Sonnet 4, Gemini 2.0 Flash), with 45 papers having both human and LLM evaluations for direct comparison. Human evaluations come from 49 expert reviewers across 49 papers in The Unjournal’s database.

3.1 Evaluation Overview

The previous [Methods](#) chapter explains our process and pipeline.¹ This generally mirrors The Unjournal’s rubric and [guidelines](#), requesting a range of percentile metrics and normative and predictive ‘journal tiers’ metrics.²

Each numeric field carries a midpoint (model’s 50% belief) and a 90% credible interval; wide intervals indicate acknowledged uncertainty.

Future Models

We are working to extend this comparison to newer frontier models as they become available, as well as adapting these to bespoke agent-based tools such as Refine.ink.

3.1.1 Token Usage and Cost

Token costs are computed from API-reported input/output tokens (GPT-5 Pro runs with `reasoning = high`).

¹We submit each PDF through a single, schema-enforced Responses call. The system prompt mirrors the Unjournal rubric, blocks extrinsic priors (authors, venue, web context), and requires strict JSON: a ~1k-word diagnostic summary plus numeric midpoints and 90% credible intervals for every metric. We enforce well-formed intervals (lower < midpoint < upper for 0–100 metrics; ci_lower < score < ci_upper for 0–5 tiers). GPT-5 Pro runs with reasoning effort = “high”, so traces are available for audit.

²0–100 percentiles relative to a reference group of ~‘all serious work you have read in this area in the last three years’, overall (global quality/impact), claims_evidence (claim clarity/support), methods (design/identification/robustness), advancing_knowledge (contribution to field/practice), logic_communication (internal consistency and exposition), open_science (reproducibility, data/code availability), abd global_relevance (decision and priority usefulness). Journal tiers: (0–5, continuous) capture where the work should publish vs. will publish.

Table 3.1: Token usage and estimated cost per model

Model	Papers	Avg input	Avg output	Avg reasoning	Total cost	Cost/paper
GPT-5 Pro	60	38,516	6,400	4,717	\$57.71	\$0.962
GPT-5.2 Pro	14	45,143	2,889	707	\$11.91	\$0.850
Claude Sonnet 4	15	116,386	731	—	\$5.40	\$0.360
GPT-4o-mini	60	71,581	470	—	\$0.66	\$0.011
Gemini 2.0 Flash	59	17,821	1,022	—	\$0.10	\$0.002

Total cost across all 208 LLM evaluations: **\$75.77**.

3.2 Human vs LLM Scatter Plot

Each point represents a paper, with human mean rating on the x-axis and LLM rating on the y-axis. Points on the diagonal indicate perfect agreement.

Note: human means carry their own variance; correlations here are bounded by human inter-rater noise.

3.3 Model Agreement Statistics

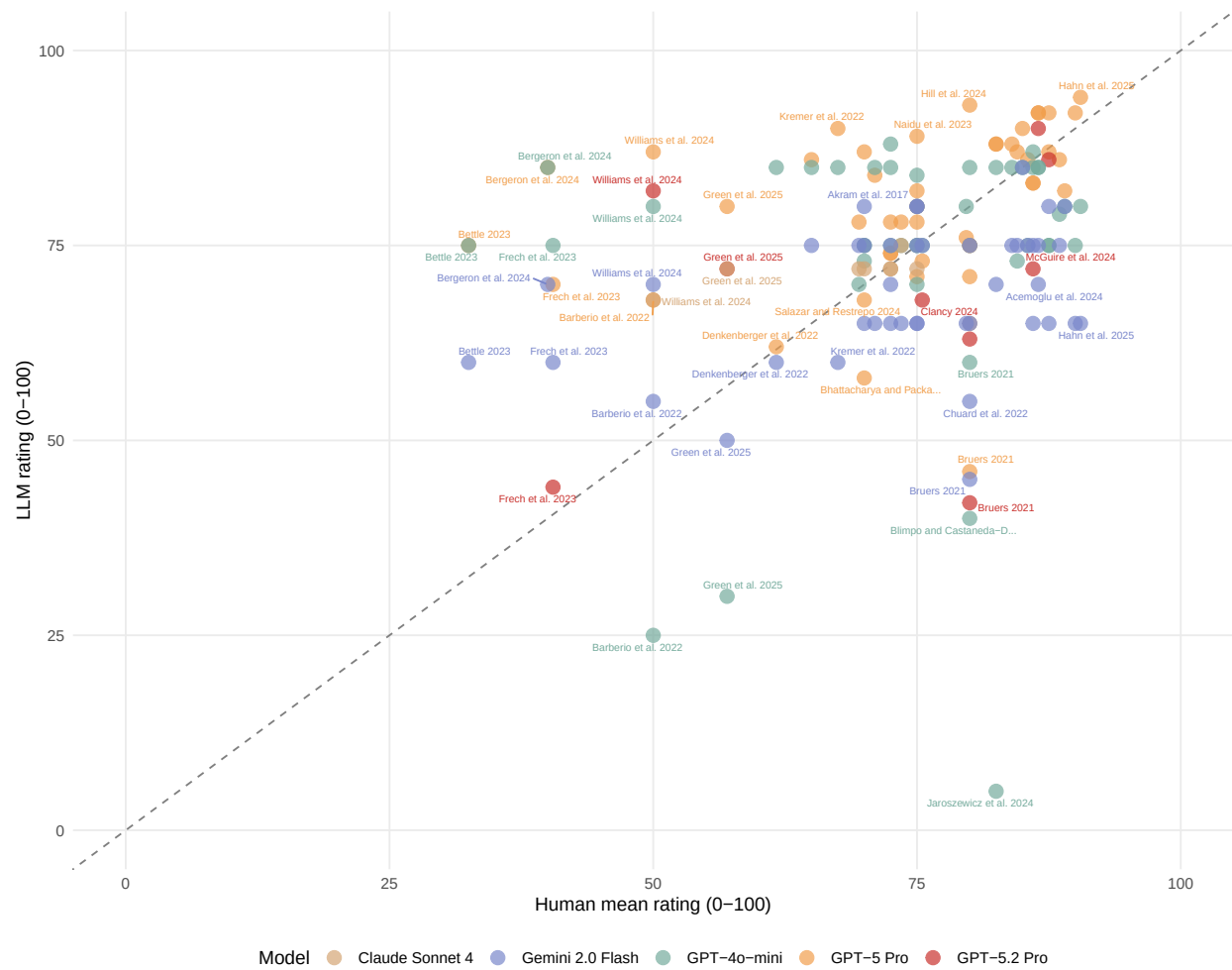
We assess agreement between LLM and human evaluators using multiple metrics:

- **Pearson r**: Linear correlation between ratings
- **Spearman** : Rank-based correlation (robust to outliers)
- **Mean bias**: Average difference (LLM – Human); positive = LLM rates higher
- **RMSE/MAE**: Error magnitude measures

Table 3.2: Agreement between each LLM and human mean ratings (overall criterion)

Model	N	Pearson r	Spearman	Mean bias	RMSE	MAE
Gemini 2.0 Flash	44	0.351	0.321	-5.0	14.4	11.6
GPT-5 Pro	45	0.342	0.517	+6.4	15.4	10.3
GPT-5.2 Pro	9	0.333	0.471	-2.7	19.0	14.7
Claude Sonnet 4	10	0.197	0.272	+1.5	9.1	6.3
GPT-4o-mini	45	0.135	0.116	+0.0	20.2	13.1

Figure 3.1: Human vs LLM overall ratings



3.3.1 Inter-Rater Agreement (Krippendorff’s Alpha)

Krippendorff’s alpha measures agreement among multiple raters, accounting for chance agreement. Values above 0.8 indicate good agreement; 0.67–0.8 is acceptable for exploratory research. Ceiling effects apply because human evaluators are not perfectly aligned.

Table 3.3: Inter-rater agreement: Krippendorff’s alpha by criterion

Criterion	Krippendorff’s	Interpretation
Overall	0.221	Poor
Claims	0.335	Poor
Methods	0.192	Poor
Adv. Knowledge	0.144	Poor
Logic & Comms	0.021	Poor
Open Science	0.093	Poor
Global Relevance	0.189	Poor

3.3.2 Human-Human vs Human-LLM Agreement (Baseline Context)³

To interpret human-LLM agreement, we must compare against **human-human agreement** as an upper bound. If human evaluators only agree with each other at $\alpha = 0.5$, then expecting LLMs to achieve $\alpha = 0.8$ with humans would be unrealistic. The table below shows both:

- **HH**: Krippendorff’s alpha among human evaluators (treating each evaluator as a separate rater)
- **HL**: Krippendorff’s alpha between human mean and each LLM model

Table 3.4: Human-human vs Human-LLM agreement by criterion (Krippendorff’s α)

Criterion	α_{HH}	α_{HL} (GPT-5 Pro)	α_{HL} (GPT-5.2 Pro)	α_{HL} (GPT-4o-mini)	α_{HL} (Claude Sonnet 4)	α_{HL} (Gemini 2.0 Flash)
Overall	0.55	0.25	0.36	0.14	0.14	0.26
Claims	0.42	0.20	0.26	0.19	0.21	0.18
Methods	0.54	0.29	0.07	0.03	-0.22	0.16
Adv. Knowledge	0.20	0.18	0.20	0.13	0.08	0.07
Logic & Comms	0.33	-0.04	-0.19	-0.00	0.25	-0.05
Open Science	0.03	-0.06	-0.61	-0.16	0.30	0.01
Global Relevance	0.33	0.25	0.51	0.04	-0.20	0.18

³This section was added with AI assistance to address feedback about contextualizing agreement metrics.

Interpretation: If r_{HL} approaches or exceeds r_{HH} , the LLM is performing at or above the level of agreement humans achieve with each other—a meaningful benchmark for “human-level” evaluation consistency.

3.3.3 Uncertainty Interval Coverage (LLM Credible Interval Assessment)⁴

LLMs report 90% credible intervals alongside their point estimates. If these intervals are **well-specified**, approximately 90% of human mean ratings should fall within them. Lower coverage suggests the LLM is overconfident; higher coverage suggests unnecessary uncertainty.

Table 3.5: Share of human mean ratings within LLM 90% credible intervals

Criterion	GPT-5 Pro	GPT-5.2 Pro	GPT-4o-mini	Claude Sonnet 4	Gemini 2.0 Flash
Overall	64%	44%	49%	70%	66%
Claims	39%	44%	44%	43%	56%
Methods	59%	22%	55%	40%	60%
Adv. Knowledge	55%	44%	52%	60%	65%
Logic & Comms	36%	44%	62%	50%	68%
Open Science	33%	33%	40%	60%	48%
Global Relevance	49%	89%	44%	60%	59%

Interpretation: Coverage near 90% indicates well-specified uncertainty. Coverage below 70% suggests systematic overconfidence; coverage above 95% may indicate overly wide intervals that provide little information.

3.3.4 Statistical Significance of Agreement Measures⁵

To assess whether observed correlations are statistically meaningful and to compare models, we compute **bootstrap 95% confidence intervals** for Pearson correlations. This nonparametric approach makes no distributional assumptions and accounts for the small sample sizes typical in evaluation studies.

Table 3.6: Bootstrap 95% CIs for human-LLM correlation (overall ratings)

Model	n	r	95% CI
Gemini 2.0 Flash	44	0.35	[0.10, 0.56]
GPT-5 Pro	45	0.34	[0.09, 0.59]
GPT-5.2 Pro	9	0.33	[-0.47, 0.88]
Claude Sonnet 4	10	0.20	[-0.66, 0.94]
GPT-4o-mini	45	0.13	[-0.15, 0.46]

⁴This analysis addresses feedback about LLM uncertainty quantification.

⁵Bootstrap inference was added with AI assistance to quantify uncertainty in agreement metrics.

Interpretation: Non-overlapping confidence intervals suggest statistically distinguishable performance. Wide intervals reflect uncertainty due to limited sample size—a common constraint in evaluation research where obtaining more human judgments is costly.

3.3.5 Information-Theoretic Perspective: Mutual Information⁶

Beyond correlation, we can ask: **how much information do LLM ratings convey about human ratings?** We use *mutual information* (MI), a fundamental concept from information theory (Shannon 1948; Cover and Thomas 2006).

What is mutual information? MI quantifies the *reduction in uncertainty* about one variable when you learn the value of another. Formally, $MI(X;Y) = H(X) - H(X|Y)$, where H is entropy (average surprise/uncertainty). In plain terms:

- Before seeing the LLM rating, you have some uncertainty about what the human rating will be
- After seeing the LLM rating, how much does your uncertainty decrease?
- MI measures this decrease, in *bits* (the same unit used in data compression and communication)

Why use MI instead of (or alongside) correlation?

Property	Correlation (r)	Mutual Information
Captures linear relationships	Yes	Yes
Captures nonlinear relationships	No	Yes
Detects threshold effects	No	Yes
Scale	-1 to +1	0 to ∞ bits
Sensitive to outliers	Yes	Less so

A practical example: suppose the LLM predicts human ratings perfectly when the paper is methodology-focused, but randomly when it’s theory-focused. Correlation might show $r = 0.5$ (moderate), but MI would reveal this is *inconsistent* performance rather than *uniformly moderate* performance. This distinction matters for deciding when to trust LLM evaluations.

How to interpret MI values (for 5-bin discretization):

- **0 bits:** Independence—LLM ratings tell you nothing about human ratings
- **~0.5 bits:** Weak association—knowing LLM rating slightly reduces uncertainty
- **~1.0 bits:** Moderate association—knowing LLM rating roughly halves your uncertainty
- **~1.5 bits:** Strong association—substantial predictive value
- **2.32 bits (maximum):** Perfect prediction—the LLM quintile perfectly determines the human quintile

We discretize ratings into quintiles (5 bins) because MI is formally defined for discrete distributions. This preserves ordinal structure while avoiding overfitting to noise in continuous ratings.

⁶Information-theoretic interpretation added with AI assistance.

Table 3.8: Mutual information between human and LLM overall ratings

Model	n	MI (bits)	Normalized
GPT-5 Pro	45	0.60	26%
GPT-5.2 Pro	9	1.61	70%
GPT-4o-mini	45	0.30	13%
Claude Sonnet 4	10	0.77	33%
Gemini 2.0 Flash	44	0.14	6%

Interpretation: Normalized MI above 50% indicates substantial predictive value. If MI is low despite moderate correlation, investigate which paper characteristics the LLM struggles with—this points toward targeted improvements or hybrid human-AI workflows where humans handle cases the LLM finds difficult.

3.3.6 Agreement by Evaluation Criterion

Table 3.9: Pearson and Spearman correlations by evaluation criterion

Criterion	Claude Sonnet 4	GPT-4o-mini	GPT-5 Pro	GPT-5.2 Pro	Gemini 2.0 Flash
Overall	0.20 / 0.27	0.13 / 0.12	0.34 / 0.52	0.33 / 0.47	0.35 / 0.32
Claims	0.46 / 0.28	0.18 / 0.28	0.21 / 0.39	0.25 / 0.32	0.21 / 0.05
Methods	-0.47 / -0.61	0.03 / 0.12	0.34 / 0.54	0.02 / 0.05	0.19 / 0.19
Adv.	0.17 / 0.23	0.12 / 0.04	0.30 / 0.46	0.16 / 0.15	0.10 / 0.10
Knowledge					
Logic & Comms	0.52 / 0.56	-0.01 / -0.02	0.19 / 0.28	-0.25 / -0.27	0.07 / 0.08
Open Science	0.29 / 0.44	-0.13 / -0.13	0.17 / 0.12	-0.60 / -0.51	-0.00 / -0.09
Global	-0.25 / -0.21	0.03 / 0.09	0.35 / 0.43	0.50 / 0.36	0.23 / 0.20
Relevance					

3.4 Criteria-Level Comparison

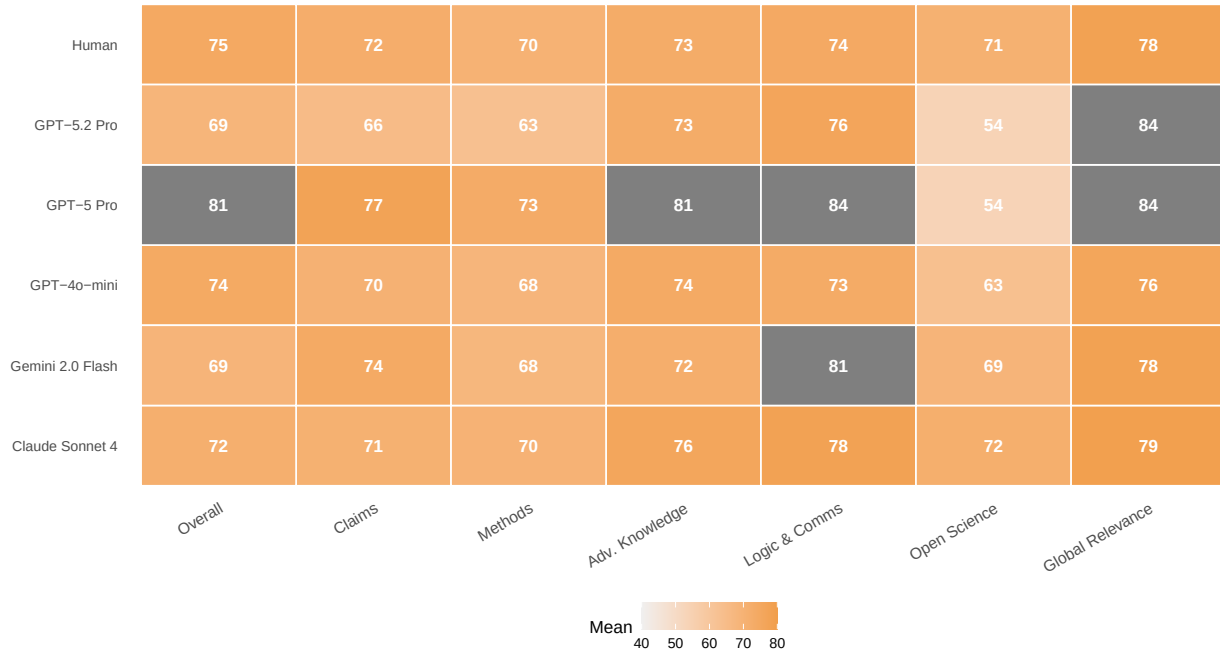
3.5 Evaluation Priorities: What Predicts Journal Tier Judgments?⁷

How do evaluators translate quality assessments into journal tier predictions? By correlating each criterion with “where should this publish?” judgments, we can see which factors each evaluator type weighs most heavily. Divergence here reveals systematic differences in evaluation priorities—what we might call “taste.”

Tier prediction column not found in human data.

⁷This analysis was added with AI assistance to reveal differences in what humans vs. LLMs emphasize when predicting journal outcomes.

Figure 3.2: Mean rating by model and criterion



Interpretation: Large divergences indicate different evaluation philosophies. For example, if humans weight “open science” heavily but LLMs don’t, this suggests the LLM may undervalue reproducibility when forming overall judgments.

3.6 Systematic Rating Differences by Paper and Criterion⁸

The heatmap below shows **Human** — **LLM** differences for each paper and criterion. Green indicates humans rated higher than the LLM; orange indicates the LLM rated higher. Papers are ordered by their overall rating difference.

This visualization helps identify: (1) papers with systematic disagreement across all criteria, (2) criteria where humans and LLMs consistently diverge, and (3) potential patterns related to paper characteristics.

3.7 Papers with Largest Human-LLM Disagreement⁹

Understanding *why* specific papers generate disagreement can inform both LLM limitations and potential human biases. The table below shows papers where humans and LLMs diverged most.

⁸This visualization was added with AI assistance to identify patterns in human-LLM disagreement.

⁹This table was added with AI assistance to facilitate qualitative investigation of outliers.

Figure 3.3: Human – LLM rating differences by paper and criterion

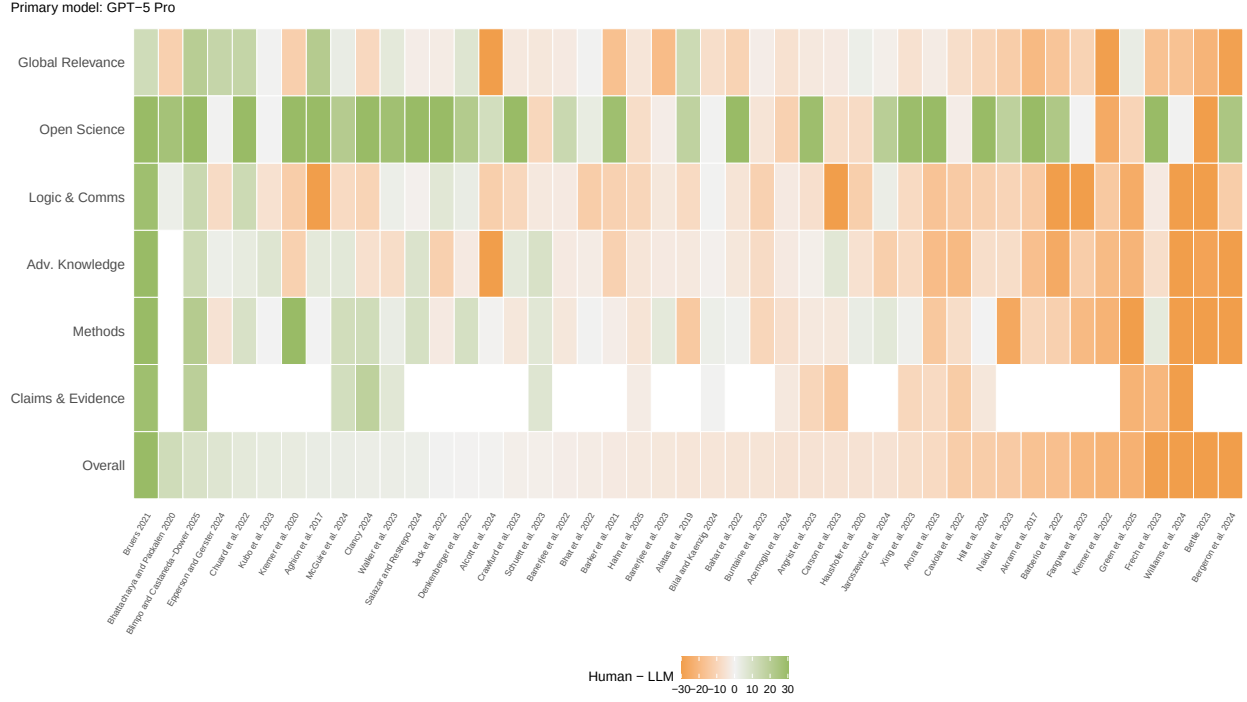


Table 3.10: Papers with largest human vs. LLM rating differences

Paper	Human Rating	LLM Rating	Difference	Direction
Bruers 2021	80	46	+34	Human > LLM
Bhattacharya and Packalen 2020	70	58	+12	Human > LLM
Blimpo and Castaneda-Dower 2025	80	71	+9	Human > LLM
Epperson and Gerster 2024	89	82	+7	Human > LLM
Chuard et al. 2022	80	75	+5	Human > LLM
Bergeron et al. 2024	40	85	-45	LLM > Human
Bettle 2023	32	75	-42	LLM > Human
Williams et al. 2024	50	87	-37	LLM > Human
Frech et al. 2023	40	70	-30	LLM > Human
Green et al. 2025	57	80	-23	LLM > Human

These outliers merit qualitative investigation: examining the paper characteristics, human evaluator comments, and LLM rationales may reveal systematic patterns in what drives disagreement.

Note: GPT-5 Pro evaluations include extended reasoning traces. See the [Appendix](#) for full assessment summaries and reasoning traces per paper.

3.8 Prompt Comparison: Legacy vs Updated GPT-5 Pro

We compare GPT-5 Pro evaluations between two prompt versions:

- **Legacy (Oct 2024):** Original combined schema without assessment summary
- **Updated (Jan 2026):** Modular prompt with explicit diagnostic assessment summary

We matched 57 papers across both runs.

Table 3.11: Summary statistics: Legacy vs Updated prompt

Criterion	Legacy Mean	Updated Mean	Mean Diff	SD Diff
Adv. Knowledge	84.5	80.4	-4.1	4.8
Claims	83.2	77.5	-5.7	4.8
Global Relevance	85.8	82.6	-3.2	6.3
Logic & Comms	86.4	83.6	-2.8	4.8
Methods	81.3	73.6	-7.7	4.6
Open Science	67.9	54.8	-13.1	8.9
Overall	86.0	80.4	-5.6	4.4

The updated prompt with explicit diagnostic assessment appears to produce systematically different ratings. The summary table above quantifies the shift; scatter and boxplot visualizations of the prompt comparison are available in the full online version.

4 Results: Critiques & Key Issues

This chapter compares *qualitative critiques*: the key methodological and interpretive issues identified by the LLM model — against human expert critiques documented by Unjournal evaluation managers in Coda.

We first assess alignment using LLM-based comparison: GPT-5.2 Pro evaluates coverage (what proportion of human concerns GPT identified) and precision (whether GPT issues are substantive). Human-checks on this alignment are underway. See [LLM-Based Assessment](#) for aggregate metrics.

4.1 Data Sources

GPT-5.2 Pro Key Issues: Structured output from the focal evaluation run (January 2026), where the model was prompted to identify 5-15 key issues per paper, ordered from most to least important.

Human Expert Critiques: Curated content from the “Key critiques & issues with paper” column in The Unjournal’s internal tracking database (Coda), written by evaluation managers synthesizing evaluator feedback. These use severity labels (“Necessary”, “Optional but important”, “Unsure”) and cite specific evaluator comments.

We matched **14 papers** with both GPT-5.2 Pro key issues and human expert critiques.

4.2 Overview Statistics

Table 4.1: Summary of key issues comparison data

Metric	Value
Papers Compared	14.0
Avg GPT Issues per Paper	11.6
Min GPT Issues	10.0
Max GPT Issues	12.0
Avg Coda Critique Length (chars)	2311.0
Min Coda Length	1197.0
Max Coda Length	4998.0

4.3 Paper-by-Paper Comparison

Each paper section shows:

- **Assessment badge:** Coverage and precision metrics with overall rating
- **LLM Assessment Summary:** GPT-5.2 Pro’s justification of the comparison
- **Missed Issues:** Human expert issues that GPT failed to capture
- **Extra Issues:** GPT issues not present in human critique
- **Full texts:** Expandable sections with complete human critique and GPT issues

Paper-by-paper comparisons with interactive collapsible panels are available in the [online HTML version](#).

4.4 LLM-Based Assessment

The comparison between GPT key issues and human critiques was assessed using GPT-5.2 Pro, which evaluated coverage (what proportion of human concerns GPT identified) and precision (whether GPT issues are substantive).

- **Coverage:** Proportion of *consensus* human issues that have any LLM match (match quality 30%)
- **Weighted Coverage:** Mean match quality across all human issues (treating unmatched issues as 0%)
- **Precision:** Proportion of LLM issues that match a human concern

Note: Any interpretation or narrative commentary in this section is written by an LLM (Codex). The values themselves come from the GPT-5.2 Pro comparison outputs.

i Caveat on Precision

“Precision” only reflects whether LLM issues match the *consensus* human issues curated in Coda—those prioritized by evaluation managers as the most important concerns. Many LLM-identified issues may have been noted by one or more individual human evaluators but were not included in this curated set. A low precision score does not necessarily mean the LLM raised irrelevant concerns; it may simply reflect issues that weren’t prioritized in the consensus summary.

Table 4.2: LLM assessment of GPT vs human critique alignment

Metric	Value
Papers Assessed	14
Mean Coverage (%)	74.9
Mean Precision (%)	46.9
Coverage Range	40-100
Precision Range	17-83

Figure 4.1: Coverage vs Precision across papers (LLM-assessed)

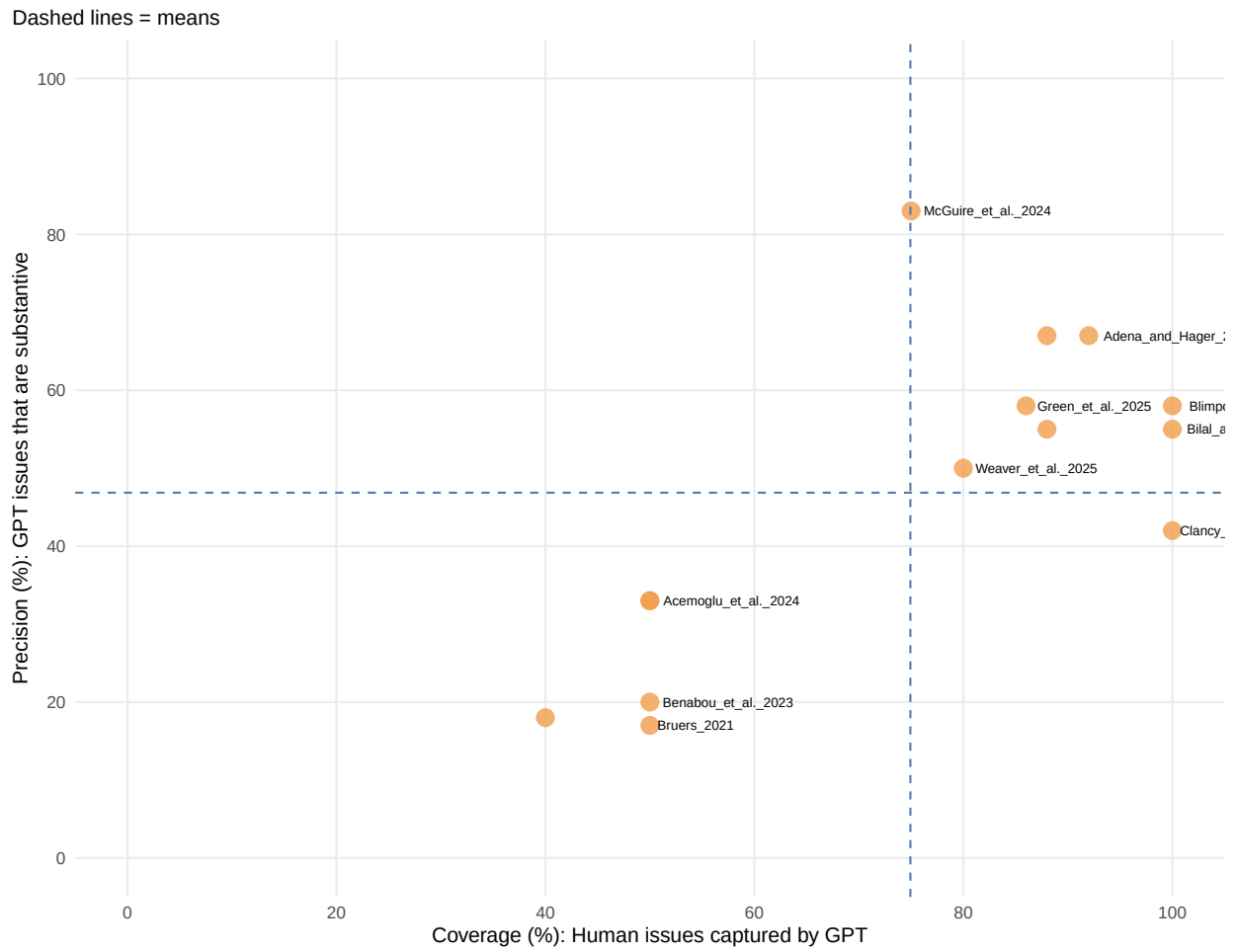


Table 4.3: Per-paper LLM assessment results

Paper	Coverage (%)	Precision (%)	Rating
Bilal_and_Kaenzig_2024	100	55	Good
Blimpo_and_Castaneda-Dower_2025	100	58	Good
Clancy_2024	100	42	Good
Adena_and_Hager_2024	92	67	Good
Dullaghan_and_Zhang_2022	88	55	Good
Williams_et_al._2024	88	67	Good
Green_et_al._2025	86	58	Good
Weaver_et_al._2025	80	50	Moderate
McGuire_et_al._2024	75	83	Good
Acemoglu_et_al._2024	50	33	Moderate
Benabou_et_al._2023	50	20	Poor
Bruers_2021	50	17	Moderate
Frech_et_al._2023	50	33	Moderate
Peterman_et_al._2025	40	18	Poor

Interpretation Guide

- **Coverage:** Percentage of human-identified issues that GPT also captured (in some form). Higher = GPT missed fewer human concerns.
- **Precision:** Percentage of GPT issues that are substantive rather than generic or spurious. Higher = GPT’s critiques are more targeted.
- **Overall Rating:** Qualitative assessment (Excellent/Good/Moderate/Poor) based on both coverage and precision metrics.

4.5 Summary and Observations

4.5.1 Aggregate Statistics

Table 4.4: Aggregate statistics across all papers

Metric	Value
Papers Compared	14.0
Total Human Issues Parsed	53.0
- Necessary	7.0
- Optional	37.0
- Unsure/Other	9.0
Total LLM Issues	163.0
Avg Human Issues per Paper	3.8
Avg LLM Issues per Paper	11.6

4.5.2 Observable Structural Differences

The following are **verifiable structural differences** between the two data sources (not assessments of quality or coverage):

Aspect	GPT-5.2 Pro	Human Expert (Coda)
Format	Numbered bullet points (array of strings)	Free-form prose
Structure	Ordered list (prompted: “most to least important”)	Often uses severity labels (“Necessary”, “Optional but important”)
Source attribution	None (single model output)	Often cites specific evaluators (E1, E2, names)
Length	Constrained by prompt (~10-12 issues)	Unconstrained, highly variable

4.5.3 Questions for Manual Review

The side-by-side comparisons above are provided for **manual expert assessment**. Key questions to investigate:

1. **Coverage:** What proportion of human-identified issues appear in the GPT output (in some form, to some extent)? (This is the focal question).
2. **Precision:** Are GPT issues substantive or does the model identify spurious/generic concerns?
3. **Severity alignment:** Does GPT’s importance ordering correlate with human severity labels?
4. **Missed issues:** Are there critical human concerns that GPT systematically misses?
5. **Novel issues:** Does GPT surface valid concerns that humans overlooked?

💡 LLM Assessment Complete

The comparisons above have been assessed using GPT-5.2 Pro. See the [LLM-Based Assessment](#) section for coverage and precision metrics.

4.6 Manual Annotation Tool

To systematically assess concordance between human and LLM critiques, use the [Issue Annotation Tool](#).

ℹ For developers: Regenerating annotation data

To update the annotation data after changes to human critiques or LLM responses:

```
# Generate annotation data (parses human critiques into individual issues)
python3 tools/build_issue_annotation_data.py
```

Annotation workflow:

For each human-identified issue:

1. **Match score (0-1):** How well do LLM issues capture this concern?
 - 0 = Not addressed at all
 - 0.5 = Partially captured or tangentially related
 - 1 = Fully captured by one or more LLM issues
2. **Confidence (0-1):** How certain are you of this assessment?
3. **Context flag:** Check this box if the human critique references information the LLM didn't have (e.g., appendix or preregistration materials not shared with the LLM at the point this evaluation was done)
4. **Link to LLM issues:** Select which LLM issues (L1, L2, ...) correspond to this human issue
5. **Discussion:** Explain your reasoning, note partial matches, or flag ambiguities

Export annotations as JSON/CSV for analysis. Annotations are auto-saved to browser localStorage.

Annotation Data Schema

The tool parses human critiques using heuristics:

- **Severity labels** normalized to: **necessary**, **optional**, **unsure**
- **Evaluator attributions** (E1, E2, DR) preserved in issue text
- **Issue boundaries** detected via enumeration, sentence breaks, and section headers

Review parsed issues in the UI and edit/add/remove as needed before annotating.

References

WARNING: One or more problems were discovered while enumerating dependencies.

```
# /Users/valik/repos/llm-uj-research-eval/results_ratings.qmd -----
Error: <text>:9:1: unexpected '/'
8: scales::dollar(total_cost, accuracy = 0.01)
9: /
  ^
```

```
# /Users/valik/repos/llm-uj-research-eval/slides_vk/index.qmd -----
Error: Parser error: while parsing a block mapping at line 8, column 7 did not find expected k
```

Please see `?renv::dependencies` for more information.

We used R v. 4.5.2 (R Core Team 2025) and the following R packages: DT v. 0.34.0 (Xie et al. 2025), ggalluvial v. 0.12.5 (Brunson 2020; Brunson and Read 2023), ggbreak v. 0.1.6 (Shuangbin Xu et al. 2021), ggforce v. 0.5.0 (Pedersen 2025a), ggrepel v. 0.9.6 (Slowikowski 2024), glue v. 1.8.0 (Hester and Bryan 2024), here v. 1.0.2 (Müller 2025), irr v. 0.84.1 (Gamer, Lemon, and <puspendra.pusp22@gmail.com> 2019), janitor v. 2.2.1 (Firke 2024), kableExtra v. 1.4.0 (Zhu 2024), knitr v. 1.51 (Xie 2014, 2015, 2025), patchwork v. 1.3.2 (Pedersen 2025b), plotly v. 4.11.0 (Sievert 2020), renv v. 1.0.0 (Ushey and Wickham 2023), reticulate v. 1.44.1 (Ushey, Allaire, and Tang 2025), rmarkdown v. 2.30 (Xie, Allaire, and Golemund 2018; Xie, Dervieux, and Riederer 2020; Allaire et al. 2025), scales v. 1.4.0 (Wickham, Pedersen, and Seidel 2025), stringi v. 1.8.7 (Gagolewski 2022), tidyverse v. 2.0.0 (Wickham et al. 2019), viridis v. 0.6.5 (Garnier et al. 2024).

Aczel, Balazs, Barnabas Szaszi, and Alex O Holcombe, “A billion-dollar donation: Estimating the cost of researchers’ time spent on peer review,” *Research integrity and peer review*, 6 (2021), 1–8 (Springer).

Allaire, JJ, Yihui Xie, Christophe Dervieux, Jonathan McPherson, Javier Luraschi, Kevin Ushey, Aron Atkins, Hadley Wickham, Joe Cheng, Winston Chang, and Richard Iannone, “[rmarkdown: Dynamic documents for r](#),” (2025).

Brodeur, Abel, David Valenta, Alexandru Marcoci, Juan P Aparicio, Derek Mikola, Bruno Barbarioli, Rohan Alexander, Lachlan Deer, Tom Stafford, Lars Vilhuber, and others, “[Comparing human-only, AI-assisted, and AI-led teams on assessing research reproducibility in quantitative social science](#),” *Institute for Replication (I4R) Discussion Paper*, 195 (2025).

Brunson, Jason Cory, “[ggalluvial: Layered grammar for alluvial plots](#),” *Journal of Open Source Software*, 5 (2020), 2017.

Brunson, Jason Cory, and Quentin D. Read, “[ggalluvial: Alluvial plots in ‘ggplot2’](#),” 2023.

Clancy, Matt, “[The returns to science in the presence of technological risk](#),” *arXiv preprint arXiv:2312.14289*, (2023).

- Cover, Thomas M, and Joy A Thomas, “Elements of information theory,” (Wiley-Interscience, 2006).
- Dycke, Nils, Ilia Kuznetsov, and Iryna Gurevych, “[NLPeer: A unified resource for the computational study of peer review](#),” in *Proceedings of the 61st annual meeting of the association for computational linguistics (volume 1: Long papers)*, Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, eds., 2023.
- Eger, Steffen, Yong Cao, Jennifer D’Souza, Andreas Geiger, Christian Greisinger, Stephanie Gross, Yufang Hou, Brigitte Krenn, Anne Lauscher, Yizhi Li, Chenghua Lin, Nafise Sadat Moosavi, Wei Zhao, and Tristan Miller, “[Transforming science with large language models: A survey on AI-assisted scientific discovery, experimentation, content generation, and evaluation](#),” *arXiv preprint arXiv:2505.05151*, (2025).
- Firke, Sam, “[janitor: Simple tools for examining and cleaning dirty data](#),” (2024).
- Gagolewski, Marek, “[stringi: Fast and portable character string processing in R](#),” *Journal of Statistical Software*, 103 (2022), 1–59.
- Gamer, Matthias, Jim Lemon, and Ian Fellows Puspendra Singh <puspendra.pusp22@gmail.com>, “[irr: Various coefficients of interrater reliability and agreement](#),” (2019).
- Garnier, Simon, Ross, Noam, Rudis, Robert, Camargo, Antônio Pedro, Sciaini, Marco, Scherer, and Cédric, “[viridis\(Lite\) - colorblind-friendly color maps for r](#),” (2024).
- Hester, Jim, and Jennifer Bryan, “[glue: Interpreted string literals](#),” (2024).
- Korinek, Anton, “[AI agents for economic research](#),” *SSRN Working Paper*, (2025).
- Luo, Ziming, Zonglin Yang, Zexin Xu, Wei Yang, and Xinya Du, “[LLM4SR: A survey on large language models for scientific research](#),” *arXiv preprint arXiv:2501.04306*, (2025).
- Mitchener, Ludovico, Ethan Yiu, and others, “[Kosmos: An AI scientist for autonomous discovery](#),” *arXiv preprint arXiv:2511.02824*, (2025).
- Müller, Kirill, “[here: A simpler way to find your files](#),” (2025).
- Pataranutaporn, Pat, Nattavudh Powdthavee, Chayapatr Achiwaranguprok, and Pattie Maes, “[Can AI solve the peer review crisis? A large scale cross model experiment of LLMs’ performance and biases in evaluating over 1000 economics papers](#),” 2025.
- Pedersen, Thomas Lin, “[ggforce: Accelerating ‘ggplot2’](#),” (2025a).
- , “[patchwork: The composer of plots](#),” (2025b).
- R Core Team, “[R: A language and environment for statistical computing](#),” (Vienna, Austria, R Foundation for Statistical Computing, 2025).
- Shannon, Claude E, “A mathematical theory of communication,” *The Bell System Technical Journal*, 27 (1948), 379–423.
- Shuangbin Xu, Meijun Chen, Tingze Feng, Li Zhan, Lang Zhou, and Guangchuang Yu, “[Usgbreak to effectively utilize plotting space to deal with large datasets and outliers](#),” *Frontiers in Genetics*, 12 (2021), 774846.
- Sievert, Carson, “[Interactive web-based data visualization with r, plotly, and shiny](#),” (Chapman; Hall/CRC, 2020).
- Slowikowski, Kamil, “[ggrepel: Automatically position non-overlapping text labels with ‘ggplot2’](#),” (2024).
- Son, Jaeho, and others, “[FLAWS: A benchmark for error identification and localization in scientific papers](#),” *arXiv preprint arXiv:2511.21843*, (2025).
- Ushey, Kevin, JJ Allaire, and Yuan Tang, “[reticulate: Interface to ‘Python’](#),” (2025).
- Ushey, Kevin, and Hadley Wickham, “[renv: Project environments](#),” (2023).
- Wickham, Hadley, Mara Averick, Jennifer Bryan, Winston Chang, Lucy D’Agostino McGowan, Romain François, Garrett Grolemund, Alex Hayes, Lionel Henry, Jim Hester, Max Kuhn, Thomas Lin Pedersen, Evan Miller, Stephan Milton Bache, Kirill Müller, Jeroen Ooms, David Robinson,

- Dana Paige Seidel, Vitalie Spinu, Kohske Takahashi, Davis Vaughan, Claus Wilke, Kara Woo, and Hiroaki Yutani, “[Welcome to the tidyverse](#),” *Journal of Open Source Software*, 4 (2019), 1686.
- Wickham, Hadley, Thomas Lin Pedersen, and Dana Seidel, “[scales: Scale functions for visualization](#),” (2025).
- Xie, Yihui, “knitr: A comprehensive tool for reproducible research in R,” in *Implementing reproducible computational research*, Victoria Stodden, Friedrich Leisch, and Roger D. Peng, eds. (Chapman; Hall/CRC, 2014).
- , “[Dynamic documents with R and knitr](#),” (Boca Raton, Florida, Chapman; Hall/CRC, 2015).
- , “[knitr: A general-purpose package for dynamic report generation in R](#),” (2025).
- Xie, Yihui, J. J. Allaire, and Garrett Golemund, “[R markdown: The definitive guide](#),” (Boca Raton, Florida, Chapman; Hall/CRC, 2018).
- Xie, Yihui, Joe Cheng, Xianying Tan, and Garrick Aden-Buie, “[DT: A wrapper of the JavaScript library ‘DataTables’](#),” (2025).
- Xie, Yihui, Christophe Dervieux, and Emily Riederer, “[R markdown cookbook](#),” (Boca Raton, Florida, Chapman; Hall/CRC, 2020).
- Zhang, Tianmai M, and Neil F Abernethy, “[Reviewing scientific papers for critical problems with reasoning LLMs: Baseline approaches and automatic evaluation](#),” *arXiv preprint arXiv:2505.23824*, (2025).
- Zhang, Yaohui, Haijing Zhang, Wenlong Ji, Tianyu Hua, Nick Haber, Hancheng Cao, and Weixin Liang, “[From replication to redesign: Exploring pairwise comparisons for LLM-based peer review](#),” *arXiv preprint arXiv:2506.11343*, (2025).
- Zhu, Hao, “[kableExtra: Construct complex table with ‘kable’ and pipe syntax](#),” (2024).

A LLM Evaluation Details

This appendix presents detailed LLM evaluations for each paper, including:

- **Numerical ratings** from all available models and human evaluators
- **Assessment summaries** (collapsible) — the model’s written evaluation
- **Key issues** (GPT-5.2 Pro only) — numbered list of methodological and evidential concerns
- **Reasoning traces** (collapsible) — the model’s internal reasoning process (GPT-5 Pro only)

The appendix is organized into two sections:

1. **Focal Papers** — 14 papers evaluated with GPT-5.2 Pro using an extended prompt that produces structured key issues
2. **Other Papers** — remaining papers with standard evaluations

Click on the callout headers to expand/collapse each section.

Found 60 papers with LLM evaluations

B Focal Papers (GPT-5.2 Pro with Key Issues)

These 14 papers were evaluated with GPT-5.2 Pro using an extended prompt that identifies key methodological and evidential issues.

B.1 Acemoglu et al. 2024

B.1.1 Ratings Comparison

Metric	GPT-5 Pro	GPT-5.2 Pro	Gemini 2.0 Flash	Human
Overall	92 [87–97]	90 [82–96]	70 [55–80]	86 [75–95] (n=2)
Claims	90 [84–95]	86 [76–94]	75 [65–85]	86 [78–92] (n=2)
Methods	88 [82–93]	91 [82–97]	65 [50–75]	82 [70–92] (n=2)
Adv. Knowledge	92 [86–96]	92 [83–98]	75 [60–85]	89 [78–95] (n=2)
Logic & Comms	92 [87–96]	86 [77–93]	80 [70–90]	89 [80–95] (n=2)
Open Science	68 [55–78]	78 [62–90]	70 [60–80]	56 [45–70] (n=2)
Global Relevance	95 [90–99]	88 [76–96]	80 [70–90]	89 [80–95] (n=2)
NA	—	—	—	4 [3–5] (n=2)
NA	—	—	—	4 [3–5] (n=2)
NA	—	—	—	89 [80–95] (n=2)

GPT-5 Pro Assessment

This paper investigates whether support for an entrenched authoritarian incumbent is sustained in part by voter misperceptions about democratic erosion and the benefits of democratic institutions, using a combined online survey experiment and a very large field experiment around Türkiye’s May 2023 elections. Strengths are substantial: both studies are preregistered/IRB-approved; the online experiment includes carefully designed placebo texts to diagnose experimenter demand; the field experiment randomizes at the neighborhood level across more than half a million voters, uses administrative ballot-box outcomes (presidential first and second rounds, parliamentary election, and one-year-later municipal election), and instruments treatment intensity (share of completed canvassing interactions) with randomized assignment. Estimated impacts are consistent across modalities: information correcting misperceptions about media independence/democracy and their real-world implications increases beliefs aligned with research and V-Dem trends, raises opposition voting intentions online by ~3.7 pp (with near-zero placebo effects), and increases opposition vote shares in the field by ~2.4 pp (no turnout effects). Heterogeneity patterns also align with the core mechanism: larger effects among individuals/neighborhoods ex ante more misinformed or leaning toward

the incumbent, and higher effects when canvassers are non-affiliated rather than opposition party affiliates, supporting a credibility channel. The analysis is clearly written, with transparent estimation, strong first stages, balance checks, consistency across alternative control sets and alternative endogenous treatment measures (completed vs initiated conversations or pamphlet drop-offs), and reduced-form estimates reported. Several issues merit careful consideration. (1) Identification and interpretation. The field experiment necessarily lacks an individual-level post-treatment survey (the authors attempted and failed to collect follow-up), so mechanisms linking belief updating to ballot outcomes cannot be established within the same subjects; the mechanism is inferred from the online experiment and neighborhood-level heterogeneity in the field. This triangulation is reasonable but should be framed as “consistent with,” not definitive. Moreover, the field “basic” message contains slogans and a call to action, and the CHP/İYİ pamphlets contain partisan branding, creating a bundle of information plus mobilizing/partisan cues. The “high-credibility” variant is closer to non-partisan facts, but still includes, in party-affiliated arms, party imagery that could weaken the claim of neutral information. The authors mitigate this with non-affiliated canvasser arms and by showing no turnout effects, yet some persuasion could still operate via motivational priming rather than pure belief correction. The authors appropriately present results for high-credibility and basic treatments separately, but the main headline effects are often presented in a bundled fashion; more emphasis on the unbundled contrasts and on non-affiliated effects would isolate the information channel further. (2) External validity and timing. Izmir is an opposition-leaning province; while effects concentrate in neighborhoods that were previously less favorable to the opposition, generalizing to other provinces or countries with different repression/propaganda equilibria is uncertain. The salience of the catastrophic 2023 earthquake and corruption concerns plausibly amplify treatment potency; this context-dependence is a feature but limits generalization. (3) SUTVA/spillovers. Neighborhood-level randomization with door-to-door canvassing risks contamination if treated messages diffuse into adjacent control neighborhoods through social networks or canvasser routing. The protocols geocode and route canvassers, but the paper does not present explicit spatial spillover checks (e.g., outcomes by distance to treated neighborhoods or exclusion buffers). If spillovers exist, ITT estimates could be attenuated; however, in principle they could also bias results if spillovers are asymmetric. (4) Treatment measurement and fidelity. The endogenous regressor is the share of completed conversations from canvasser logs; self-reported logs may have reporting error. The authors address this via instrumental variables, present strong first stages, and show robustness using initiated conversations and pamphlet drops. Still, fidelity audits (spot checks, audio capture, or independent observers) would further bolster confidence that treatment delivery was uniform across canvasser types and treatments. (5) Estimation presentation. The field results emphasize 2SLS LATE/TOT magnitudes (~ 2.4 pp); these are well complemented by reduced form (~ 0.8 pp), but readers may conflate the two. The figures clearly note 2SLS; still, consistently pairing TOT with RF in the main text would avoid misinterpretation. (6) Outcome multiplicity and adjustments. The paper examines multiple outcomes (beliefs, intentions, three 2023 races, turnout, 2024 municipal election) and heterogeneity dimensions. The strongest results have very small p-values and replicate across settings, mitigating multiple-testing concerns, but no formal adjustment is reported. (7) Foundations of factual claims. The paper treats V-Dem’s indices as ground truth for democracy/media decline, and the informational prompts rely on research summarizing relationships between democracy/media and outcomes (disaster mortality, corruption). These are reasonable choices and widely used,

but there is measurement and causal identification debate in the literatures. The authors present these as “according to research” and cite canonical studies; providing sensitivity to alternative indices (e.g., FH, Polity) in the online prompt or additional figures in the appendix would defuse claims that the information is contestable. (8) Placebos and demand effects in the field. Because outcomes are administrative vote shares, experimenter demand is unlikely; still, the lack of a field placebo arm leaves open whether any pro-opposition contact could drive similar effects. The lack of turnout changes and the credibility gradient by canvasser affiliation partially mitigate this, but a cluster-randomized placebo with non-political content would be a valuable addition. (9) Long-run effects. The one-year municipal effect is marginally significant and small; given intervening events, it should be interpreted cautiously (though randomization helps), and power calculations for long-run effects are not presented. (10) Sampling and exclusion. The field sample excludes remote neighborhoods requiring more time to canvass; this is operationally reasonable but limits representativeness within Izmir. For the online experiment, Facebook-recruited respondents are not representative, and attrition is non-trivial (though balanced). The authors are transparent about both points and use the online component primarily for mechanisms and belief updating. (11) Ethics and risk. The paper reports dual IRB approvals and registry, and indicates a non-partisan invitation to all major parties. The non-partisan arm is a strength, though the governing parties’ refusal to participate constrains balance across affiliations. Overall, the design, execution, and triangulation between online and field evidence are strong. The central claims are well supported within the caveats above: voters systematically misperceive democratic erosion and the value of democracy/media; credible, accurate information reduces misperceptions and shifts vote choices toward opposition; effects are concentrated among the misinformed and incumbent-leaning; and credibility/non-partisanship matters. The paper meaningfully advances empirical understanding of persuasion under authoritarianism and offers practical lessons for civic communication strategies. Replication is feasible in principle (administrative outcomes, clear estimators, and randomization), but would benefit from full code release, canvasser logs, and treatment materials in a public repository alongside the registry.

i GPT-5.2 Pro Assessment

This manuscript studies whether support for an increasingly authoritarian incumbent can partly reflect misperceptions about (i) the extent of democratic and media erosion and (ii) the practical value of democracy/media freedom for salient outcomes, rather than a genuine preference for authoritarianism. It combines (a) an online survey experiment recruited via Facebook ads (with placebo “demand” messages) and (b) a large neighborhood-randomized door-to-door canvassing field experiment in one major city, linked to administrative ballot-box election returns. The core empirical results—information shifts beliefs and increases opposition vote intention/actual opposition vote share by roughly 2–4 pp—are internally coherent across the two experiments and are strengthened by the field study’s use of administrative outcomes (reducing concerns about experimenter demand or social desirability in the main endpoint). The paper is also unusually transparent about design choices (registration, IRB, stratification, rerandomization, canvasser identity cross-randomization) and conducts a set of sensible robustness checks (controls, alternative “contact” measures, reduced form vs IV, heterogeneous effects patterns that align with their mechanism story). Nevertheless, several issues in the manuscript materially affect interpretability and generalization. First, the core

construct “misperception” is defined relative to V-Dem indices treated as ground truth; while V-Dem is a high-quality expert-coded dataset, it is not direct “official data” and is not an unambiguous truth benchmark for citizens’ beliefs. The treatment scripts and pamphlets sometimes label these indicators as “official data,” which risks overstating epistemic certainty and could itself affect credibility and persuasion; this matters because the paper’s conceptual claim hinges on “accurate information correcting misperceptions.” Relatedly, the causal claims embedded in the informational scripts (“as democracy strengthens, deaths/building damage from disasters decrease”; “as media independence worsens, corruption increases”) are presented in a strongly causal way, but the cited evidence is largely observational/cross-country (and context dependent). Even if the literature supports an average association, presenting it as a simple law may be contestable; if respondents update because they perceive a broad normative/valence message (or because they view the intervention as an opposition argument packaged as science), the interpretation “beliefs corrected by accuracy” becomes less clean.

Second, the field experiment’s treatment content and delivery raise identification and interpretation questions about what exactly is being estimated. The “high-credibility” pamphlet is closer to factual claims, but it is still an opposition-aligned valence message in a highly polarized campaign environment; the “basic” pamphlet is explicitly mobilizational and, in versions shown, includes candidate/party branding. The paper emphasizes non-partisanship in approach, but the realized intervention was implemented with opposition parties and included overt opposition branding in at least some arms. This complicates the interpretation that the effect is driven primarily by informational correction rather than standard partisan persuasion/mobilization channels (even if turnout does not change). In addition, the paper’s preferred field estimates are 2SLS effects of “completed conversations” instrumented by neighborhood assignment. This is reasonable as a compliance-adjusted estimand, but the exclusion restriction is not fully innocuous: assignment affects not only conversation completion but also pamphlet exposure, canvasser presence, perceived campaign intensity, and possibly social spillovers within neighborhoods. The authors do show reduced forms and alternative endogenous variables (initiated conversations, pamphlets left), which helps, but the narrative sometimes slides from an IV-LATE per conversation to a broader claim about “information” effects. Interference/spillovers are plausible (adjacent neighborhoods, within-building networks, media/social discussion) and could bias reduced-form effects toward zero or alter interpretation; the paper does not deeply model or bound spillovers.

Third, external validity is limited in ways that deserve more emphasis. The field experiment occurs in one province/city with distinctive political composition (often described as an opposition stronghold), and neighborhood sampling excludes remote/time-costly areas. The mechanism claims about persuading incumbent supporters are supported by stronger effects in “below-median opposition share” neighborhoods, but this is a within-city comparison and may not transport to national contexts, rural settings, or places with different media ecosystems/repression. Likewise, the online experiment is recruited via Facebook ads, with a sizable drop-off from clickers to completers and a lottery-like incentive; even with balance/attrition checks, representativeness is weak, and the degree to which online respondents resemble marginal persuadables in the electorate is uncertain. The paper generally treats the online experiment as mechanism-support (belief updating, placebo check) and the field experiment as the main outcome test, which is appropriate, but readers may still overgeneralize.

Fourth, some measurement/integrity details merit clarification. In the appendix summary statistics, turnout in administrative data exceeds 1 in some ballot boxes (e.g., max 1.16), which

is not impossible in some administrative systems (e.g., list updates, ballot box reassignment, denominator mismatch), but it requires explicit explanation because it can signal mismatched denominators or merging issues. In the online study, there is a potential inconsistency in the definition of the “Turnout” variable (appendix definition appears inverted relative to how it is used in figures/means), which could be a documentation error but should be resolved for replicability. Additionally, the “placebo” scripts still contain substantive propositions (“some people think media independence worsened and corruption increased”), which may not be a pure demand-only placebo; if anything, this makes the placebo a weak-information condition, complicating the interpretation that placebo effects isolate demand.

Finally, the heterogeneity and longer-term persistence results are intriguing but should be treated cautiously. Many subgroup analyses are median splits and may not all be pre-registered; multiple testing and researcher degrees of freedom could inflate apparent structure. The municipal-election effect a year later is potentially important, but persistence claims need more discussion of competing explanations (differential subsequent campaigning, migration, compositional changes of ballot boxes, or correlated neighborhood trends) and of how stable ballot-box mapping is across elections (the paper notes ballot boxes cannot be tracked over time in some contexts). Overall, the manuscript provides strong experimental evidence that credible-sounding institutional-performance information can move beliefs and election outcomes at the margin under competitive authoritarian conditions, but the conceptual leap from “information treatment” to “correction of misperceptions about true institutional state/causal impacts” is somewhat overstated, and the field implementation’s partisan entanglement and possible interference limit how cleanly one can attribute the effect to neutral information per se.

Gemini 2.0 Flash Assessment

This paper investigates the impact of providing information about the state of democracy and media freedom on voter behavior in Turkey. The authors conducted both an online experiment and a field experiment in Izmir during the lead-up to the May 2023 elections. The online experiment involved exposing participants to information about the decline of democracy and media freedom, as well as their importance in addressing issues like natural disasters and corruption. The field experiment involved door-to-door canvassing in select neighborhoods of Izmir, where residents received similar information. The key finding is that providing information increased support for the opposition, particularly among voters who were initially more supportive of the incumbent government and held misperceptions about the state of democratic institutions. This suggests that at least some support for authoritarian regimes comes from misperceptions rather than inherent preferences. The authors conclude that providing accurate information can bridge ideological divides and sway voters. They also find that the source of information matters, with non-affiliated canvassers being more effective than those affiliated with opposition parties.

Several issues arise upon review. First, there is the question of external validity. The study focuses on Turkey, a country with specific political and social dynamics. It is unclear how well these findings would generalize to other countries with different contexts. The authors acknowledge this to some extent but do not fully explore the limitations. Second, there are potential issues with experimenter demand effects in the online experiment. While the authors include placebo treatments, it’s still possible that participants altered their responses

to align with what they believed the researchers wanted to see. The authors acknowledge this limitation and argue that the field experiment, which relies on actual voting behavior, helps to mitigate this concern.

Third, there are questions about the robustness of the findings to different specifications and control variables. The authors address this to some extent by presenting results with different sets of controls, but it's still possible that other factors could be driving the results. Fourth, there may be limitations in the field experiment due to the non-random assignment of treatments at the individual level. Instead, treatment was assigned at the neighborhood level, potentially leading to spillover effects and other forms of interference. The authors acknowledge that this may lead to varying proportions of treated households. Fifth, the authors state that the governing coalition parties did not accept to take part in their study and they were therefore unable to assess how the information campaign impacted their supporters. It may be that the information shared had limited reach to these voters or was ignored because of its source.

Sixth, it is not clear how the information being shared was regarded by citizens in light of propaganda and misinformation. It could be that any information shared by any party or group would be disregarded completely as being untrustworthy. Seventh, the door-to-door information campaign was implemented in Izmir, Turkey's third-largest province. The extent to which this can be generalized to the rest of Turkey, including more rural areas is not clear. Eighth, the basic informational treatment used in the field experiment contained less impartial and more partisan language which calls into question whether citizens viewed the experiment as unbiased. Overall, these issues somewhat temper the positive findings of the research.

⚠️ GPT-5.2 Pro: Key Issues Identified

1. 1) The paper operationalizes “misperceptions” by comparing respondents’ beliefs to V-Dem indices treated as ground truth; V-Dem is expert-coded (not citizens’ lived experience) and not “official data,” so the epistemic basis for labeling beliefs as misperceived needs more nuance.
2. 2) Treatment scripts/pamphlets present causal claims (democracy → fewer disaster deaths; media independence → less corruption) as near-laws, but the cited evidence is largely observational/context-dependent; this risks conflating persuasion by normative/valence messaging with “accurate information correction.”
3. 3) Field implementation is intertwined with opposition parties and includes explicitly partisan branding in some arms (especially the basic treatment), complicating interpretation that effects reflect non-partisan information rather than standard partisan persuasion.
4. 4) The main field estimand is a 2SLS effect of “completed conversations” instrumented by assignment; exclusion restrictions are not fully defended because assignment also changes pamphlet exposure, canvasser presence/intensity, and possibly other neighborhood-level channels.
5. 5) Potential interference/spillovers across neighborhoods (social discussion, adjacency, shared spaces) are plausible in a city setting; the design and analysis largely assume SUTVA without bounding or testing spillovers.
6. 6) External validity is limited: the field experiment is in a single city/province with selective neighborhood inclusion, and the online sample is Facebook-ad recruited

with non-representative composition; generalization to national electorates or different authoritarian contexts is uncertain.

7. 7) Administrative turnout values exceeding 1 in some ballot boxes (max >1) and documentation inconsistencies about the online “turnout” variable require explicit reconciliation to ensure data integrity and interpretability.
8. 8) The placebo messages still contain substantive propositions (“some people think ... has worsened”), so the placebo may not cleanly isolate experimenter demand; this could attenuate differences and complicate interpretation.
9. 9) Heterogeneity analyses rely heavily on median splits and multiple subgroup comparisons; if not fully pre-registered, multiple-testing adjustments or a clearer separation of confirmatory vs exploratory analyses are needed.
10. 10) Persistence claims using the 2024 municipal election are suggestive but need stronger discussion of alternative explanations (subsequent campaigns, compositional/administrative changes, ballot-box mapping stability across elections).
11. 11) The paper sometimes slides from compliance-adjusted per-contact effects to broad claims about the impact of “accurate information campaigns”; clearer separation of reduced-form assignment effects, LATE, and mechanism interpretation would improve causal communication.
12. 12) The “non-partisan approach” narrative is somewhat inconsistent with the realized partisan delivery; clarifying what was attempted vs what was implemented would improve transparency.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

B.2 Adena and Hager 2024

B.2.1 Ratings Comparison

Metric	GPT-5 Pro	GPT-5.2 Pro	Gemini 2.0 Flash
Overall	85 [77–92]	78 [66–88]	75 [65–85]
Claims	82 [72–90]	80 [67–90]	80 [70–90]
Methods	86 [78–92]	82 [70–92]	75 [65–85]
Adv. Knowledge	84 [72–91]	76 [60–88]	70 [60–80]
Logic & Comms	90 [84–96]	86 [76–94]	85 [75–95]
Open Science	54 [40–68]	62 [45–78]	80 [70–90]
Global Relevance	72 [62–85]	72 [55–86]	60 [50–70]

i GPT-5 Pro Assessment

This manuscript reports a large-scale, geo-randomized natural field experiment testing whether Facebook video fundraising increases charitable giving. Roughly 7,686 German postal codes

(after excluding extreme Facebook-reach PLZs) were assigned in blocks to either a 14-day Save the Children video campaign or to pure control, with a 2×2 factorial variation in video content (empathy vs. effectiveness) and impression allocation strategy (Facebook free allocation vs. fixed postal-code budgets proportional to estimated potential $\times$ reach). Outcomes are daily donation revenue and counts to Save the Children at the PLZ-day level, aggregated across all donation channels (online and offline), over 31 pre-treatment days, 14 treatment days, and 38 post-treatment days (with some checks extending into early January). The study additionally leverages two external datasets of donations to other charities (a 23-charity alliance and betterplace.org projects) to assess competitive spillovers, and presents several robustness checks, including difference-in-differences, randomization inference, and sensitivity to post-campaign window length. The central findings are: (i) significant increases in donation frequency both during and (especially) over the combined during+post period; (ii) a positive effect on revenue that is marginally significant in the long-term window, with point estimates implying immediate profitability (€1.45 per €1 spent) and higher long-run returns under a lifetime value assumption; (iii) evidence of crowding out of other children-related or similar charities/projects; (iv) little long-run difference between empathy and effectiveness content despite stronger short-run attention and clicks for the empathy video; (v) limited performance differences between free and fixed impression allocation (with some short-run advantages for free allocation); and (vi) local spillovers to neighboring PLZs suggesting interference (thus main effects are lower bounds on total impact).

Most important issues and limitations: - Identification and pre-trends: For Save the Children outcomes, pretreatment differences across treatment/control PLZs are small and statistically insignificant, and the preferred specifications include the lagged dependent variable and block fixed effects. For the alliance data, pretreatment differences are significant; the paper appropriately relies on controls and difference-in-differences estimators. While randomization should ensure unbiasedness, the presence of pretreatment imbalances in some auxiliary datasets raises residual concerns about time-varying shocks or regression-to-the-mean effects. The paper provides randomization inference and robustness checks which help, but an explicit pre-registered primary analysis for competition outcomes and a systematic treatment of multiplicity would strengthen inference. - Interference and spillovers: The authors document positive spillovers from treated nearby PLZs (30 km), especially from urban neighbors and affecting rural areas more, implying violations of SUTVA and that ITT estimates are lower bounds on total effects. Their post-hoc adjustment using the share of treated neighbors offers a useful diagnostic but is necessarily model-based and not pre-registered; the magnitude and interpretation of “total effects” remain somewhat sensitive to specification. A design explicitly accommodating network/geo spillovers or two-stage randomization would sharpen these estimates. - Exposure and dosage: In the “free allocation” arm, the platform endogenously varied impressions across treated PLZs, creating meaningful heterogeneity in treatment intensity. Analyses are reported at the ITT level; the paper does not instrument for dosage or present dose-response analyses at PLZ level (impressions per capita), which could clarify mechanisms and effect heterogeneity. The factorial variation (free vs. fixed allocation) partly addresses this, but a complier- or intensity-based analysis would be valuable. - Profitability and long-run impact: The ROI calculations extrapolate from the long-run point estimate with wide confidence intervals and employ an assumed donor lifetime value ($LTV = 1.75$) and persistence figures drawn from prior work/industry norms, not estimated directly on ad-acquired donors. This is reasonable for a managerial back-of-the-envelope but introduces uncertainty; retention and average

gift sizes could differ for ad-acquired donors versus incumbents. The paper correctly flags the uncertainty but could further explore plausible ranges and scenario analyses. - Competition and net sector effects: The paper provides valuable evidence of crowding out using two non-overlapping-but-not-disjoint data sources (an alliance of 23 charities and betterplace.org). However, both are online-only (whereas Save the Children outcomes include all channels), making it hard to infer full general equilibrium effects across the sector. Overlap between the data sources, missing offline donations for competitors, and significant pretreatment differences in the alliance data complicate any precise estimate of net sectoral welfare. The direction (crowding out) is reasonably supported; the magnitude remains uncertain. - Measurement and aggregation: Outcomes are aggregated at the PLZ-day level and winsorized at €1,000 on PLZ-day observations (68 affected). The authors report that results are robust without winsorization but with less precision; still, winsorization at the PLZ-day level, rather than donor-level, is a pragmatic compromise that could slightly affect distributional inferences. Normalizing per-million-inhabitants-per-day and then scaling to treated population is clear; reporting both per-capita ATE and population-weighted totals is appropriate. The lack of an individual-level click-to-donation linkage (especially under free allocation where click data are semi-aggregated by group) limits mechanistic interpretation when contrasting intermediate metrics with eventual donations. - Pre-registration and analytic flexibility: The design was preregistered, but the blocking variable set was expanded (to include Facebook reach) and the impression allocation strategy arm was added post registration. The core main effects analysis aligns with the experimental design, but multiple secondary analyses (heterogeneity, spillovers, competition, and intermediate metrics) are extensive. A formal multiplicity adjustment is not presented; readers should interpret marginal p-values with caution. - External validity: The setting is Germany in November–December (peak giving season) with Save the Children, a large, well-known charity, and a largely untargeted campaign. This maximizes policy relevance for general audiences and non-targeted campaigns but may not generalize to less salient charities, non-seasonal periods, different geographies, or more targeted/different ad channels. The authors’ heterogeneity results suggest that targeting higher-potential PLZs would likely improve ROI materially. - Open science and replicability: The study leverages proprietary donation data and platform metrics; data are aggregated and anonymized, and not public. The experiment was preregistered and the online appendix is detailed, but no analysis code or replication package appears to be provided. Reproducibility by third parties is therefore limited without data access, although the methodological descriptions are sufficiently clear to permit re-implementation given similar data.

Overall, the design is strong: large-scale randomization, outcomes covering all donation channels for the focal charity, a meaningful post-campaign horizon, attention to spillovers, and careful discussion of substitution versus expansion. The principal limitations concern interference (and the resulting interpretation of ITT as a lower bound), incomplete sector coverage for competition outcomes (and channel mismatch), reliance on assumed LTV for profitability, and limited replicability due to proprietary data. The claims are generally well-calibrated to the evidence: the increase in donation frequency is robust; the revenue effect is positive but only marginally significant; profitability is plausible but uncertain; and click-based intermediate metrics can be misleading relative to long-run donations. The paper provides practically useful guidance to fundraisers and a valuable contribution to the economics of charitable giving and digital advertising, with particularly useful insights on general equilibrium effects and on the (limited) inferential value of engagement metrics for ultimate fundraising outcomes.

This manuscript studies whether Facebook video fundraising increases charitable giving using a large geo-randomized field experiment across 7,686 German postal codes, linking treatment assignment to the full universe of the focal charity’s donations at the postal-code-by-day level and (partially) to donations to other charities/projects. The design is unusually ambitious and well-motivated: by randomizing at geography and measuring total donations rather than click-through conversions, it directly targets common advertising-evaluation problems (channel substitution, untracked delayed/offline conversions, and low power from rare individual-level donation events). The main identification argument—treated and control postal codes are comparable due to block randomization, so differences in donation outcomes during/after the campaign can be attributed to the ads—is plausible, supported by balance checks, and strengthened by randomization-inference robustness. However, several internal threats and interpretive risks remain. First, interference/spillovers are likely in geo experiments with platform targeting (misclassification of location, commuting, and cross-area social exposure); the paper finds positive “treated neighbors” associations and interprets them as spillovers implying lower-bound direct effects, but this spillover analysis is not itself experimentally manipulated, and the “share treated within 30km” is mechanically correlated with many spatial features and with block structure; thus it is suggestive but not definitive evidence about interference magnitude or direction. Given potential interference, conventional SUTVA-based estimands are ambiguous: the reported treatment effect is closer to an “assignment-to-treatment intensity” effect under partial interference rather than a clean individual exposure effect, and the paper’s conversion of coefficients into campaign-level profit implicitly assumes a particular mapping from assignment to exposure that may not hold if spillovers are material. Second, the mapping from postal-code assignment to actual impression delivery is only partially observed and is mediated by Facebook’s delivery algorithm; while the paper reports aggregate reach/impressions and a factorial manipulation of allocation strategy, there is still substantial scope for differential delivery within treated postal codes (e.g., toward cheaper-to-reach users) that could create heterogeneous treatment intensity unrelated to charitable propensity, complicating interpretation as a per-capita causal effect. Third, statistical strength differs across outcomes: donation frequency effects are consistently significant, but donation revenue effects are weaker (long-term revenue significant only at $p < 0.1$ in the preferred specification), making the profitability conclusion sensitive to functional-form assumptions and tail behavior. The paper addresses outliers via winsorization at €1,000 at the PLZ-day level; while it reports similar point estimates without winsorization, inference and ROI can still be sensitive because the revenue effect is borderline and charitable gifts are heavy-tailed. Also, the main outcome construction aggregates into three periods and scales “per million inhabitants per day,” which is sensible for comparability but obscures dynamic treatment effects and makes the choice of post period (ending Dec 31) consequential; the justification is that effects should fade after the fiscal-year giving season, yet this choice intertwines treatment persistence with strong seasonality, and the “sensitivity to days included” figure helps but does not fully resolve whether the late-December revenue uptick is treatment-driven or a differential seasonal pattern that remains by chance in treated areas. Fourth, competition/crowding-out results are valuable but less clean: for the alliance of 23 charities the authors find pretreatment imbalance and therefore rely on lagged-outcome controls and DiD-style changes. These approaches are reasonable, but the outcomes differ in scope (only online giving for competitors vs all channels for the focal charity), overlap between datasets is acknowledged but unresolved, and

the analysis cannot measure total sector-wide giving; therefore the welfare/sector conclusion (“redistribution/lost to the sector”) remains speculative. For the donation platform data, category tagging and the project-level aggregation introduce classification and compositional issues (projects vs organizations; multiple tags) that can attenuate or misattribute substitution. Fifth, the 2×2 factorial “mechanism” claims are cautiously worded but still at risk of over-interpretation: video-type differences in donations are mostly not statistically distinguishable, while intermediate metrics (views/clicks) differ strongly; this supports the managerial point that clicks are poor proxies for donations, but the paper’s broader inference (“power of asking” is main mechanism) is not directly tested against alternative mechanisms (mere salience, trust signals, or contemporaneous external events). Sixth, the heterogeneity exercise reports many interactions and suggests targeting improvements; these are likely exploratory (and appear not fully preregistered), raising multiplicity and overfitting concerns, especially because the blocking procedure itself used a prediction model based on past donations and covariates, which can interact with subsequent heterogeneity patterns. Finally, open-science and reproducibility are mixed: preregistration is a major strength and the appendix is extensive, but key inputs (donation microdata, competitor data, platform delivery logs) are proprietary/partially aggregated, and it is unclear whether replication code and complete analysis scripts are available; this limits independent verification of data processing decisions (winsorization, aggregation, population scaling, neighbor calculations) and of exact specification search. Overall, the paper’s central empirical contribution—a large-scale, plausibly causal estimate of incremental donations from Facebook fundraising with evidence of some substitution across charities—is credible and useful, but several conclusions (especially profitability magnitude, spillover-adjusted “total effects,” and sector-level implications) should be treated as more uncertain and assumption-dependent than the main narrative sometimes suggests.

i Gemini 2.0 Flash Assessment

This paper investigates the effect of online fundraising on charitable giving using a geo-randomized field experiment on Facebook. The authors partnered with Save the Children and randomly assigned almost 8,000 German postal codes to either receive fundraising videos or serve as a control group. The study examines changes in donation revenue and frequency, addressing challenges like substitution and measurement issues inherent in studies based on click-through data. The authors find that video fundraising increased donation revenue and frequency for Save the Children during and after the campaign, that the campaign was profitable, and that the effects were similar regardless of video content or impression assignment strategy. They also find evidence of crowding out of donations to other similar charities and projects, and demonstrate that click data may be an inappropriate proxy for donations. The study’s strengths lie in its large-scale geo-randomized design, which circumvents common issues in online advertising research and enhances external validity. The authors preregistered their design and provide robustness checks, additional analyses, and a discussion of profitability. However, the study has some limitations. First, the conclusions are limited to the specific implementation of the campaign. Second, there is a potential for spillovers across postal codes, which may bias the main results downwards. Third, the estimated effects on competing charities are based on data from a subset of charities, and there may be unobserved heterogeneity in the types of projects funded. Fourth, some of the variables are semi-aggregated, which limits the ability to reliably test treatment differences. Finally, while the authors winsorize

the data to reduce the influence of outliers, they are unable to do so at the individual level. Specific issues to address:

1. **Potential Spillovers:** The authors acknowledge the potential for spillovers across postal codes, which could bias their main results. While they attempt to address this issue by including a variable indicating the share of treated postal codes within 30 kilometers, this may not fully capture all spillover effects.
2. **Limited Data on Competing Charities:** The authors rely on data from a subset of charities to examine the effects of the Save the Children campaign on other organizations. This raises concerns about the representativeness of the data and the potential for unobserved heterogeneity in the types of projects funded.
3. **Semi-Aggregated Variables:** The authors acknowledge that some of their variables are semi-aggregated, which limits their ability to reliably test treatment differences. This is a potential limitation of the study's design.
4. **Winsorizing Data:** While the authors winsorize the data to reduce the influence of outliers, they are unable to do so at the individual level. This may not fully address the issue of extreme values in the data.
5. **External Validity:** The conclusions are necessarily limited to the specific implementation of the campaign, and it is unclear how the results would generalize to other contexts.

⚠️ GPT-5.2 Pro: Key Issues Identified

1. 1) Interference/SUTVA: Geo-randomized Facebook targeting plausibly induces cross-postal-code spillovers and location misclassification; the paper's neighbor-share regression is suggestive but not a randomized test, so the magnitude/direction of interference remains uncertain and complicates causal estimands.
2. 2) Treatment intensity vs assignment: Postal-code assignment does not ensure uniform exposure; Facebook's delivery algorithm can create endogenous within-area intensity and composition, making per-capita effects harder to interpret as "advertising causes X euros per million residents."
3. 3) Revenue effects are borderline: Donation revenue results are weaker than frequency (e.g., long-term $p < 0.1$), so ROI/profitability conclusions are sensitive to specification, tail behavior, and aggregation choices.
4. 4) Winsorization and heavy tails: PLZ-day winsorization at €1,000 may affect inference and ROI in a heavy-tailed donation distribution; robustness is mentioned but should be more fully quantified (e.g., alternative cutoffs, quantile/Poisson models).
5. 5) Post-period choice and seasonality: Ending outcomes at Dec 31 is substantively motivated but couples treatment persistence to extreme seasonal dynamics; more explicit dynamic/event-study estimates would better separate persistence from seasonal noise.
6. 6) Competitor/crowding-out measurement mismatch: Competitor datasets are incomplete (often online-only), overlap across sources is unresolved, and sector-wide giving is unobserved, limiting the strength of conclusions about total charitable-sector

- effects.
7. 7) Pretreatment imbalance for the 23-charity alliance outcomes: The need for DiD/lagged-outcome adjustment is appropriate, but residual confounding and differing seasonal patterns across areas remain possible; inference is also relatively weak (often $p < 0.1$).
 8. 8) Mechanism claims exceed direct tests: The conclusion that effects primarily reflect the “power of asking” is plausible but not directly isolated from alternative mechanisms (salience, trust, concurrent campaigns/news shocks).
 9. 9) Factorial design interpretation: Differences across video type and impression allocation are mostly not statistically distinguishable; emphasizing “no matter the content, online fundraising works” should be framed as limited power to detect meaningful differences rather than equivalence.
 10. 10) Heterogeneity/multiplicity risk: Many subgroup interactions are reported with strong managerial recommendations, but these appear exploratory and risk false positives without multiplicity control or holdout validation.
 11. 11) Profitability depends on assumptions: Long-run ROI uses an assumed lifetime-value multiplier and implicitly assumes competitor crowd-out does not offset social impact; uncertainty propagation around ROI should be more explicit.
 12. 12) Replication constraints: Although preregistered and well-documented, proprietary and aggregated data plus unclear code availability limit independent reproducibility of key transformations (aggregation, neighbor calculations, data merges).

GPT-5 Pro reasoning trace available in the [online HTML version](#).

B.3 Benabou et al. 2023

B.3.1 Ratings Comparison

Metric	GPT-5 Pro	GPT-5.2 Pro	Gemini 2.0 Flash
Overall	84 [76–91]	80 [70–88]	65 [45–80]
Claims	82 [74–89]	82 [72–90]	75 [60–85]
Methods	78 [70–86]	85 [75–92]	65 [50–75]
Adv. Knowledge	83 [75–90]	76 [62–86]	70 [50–85]
Logic & Comms	86 [80–92]	88 [80–94]	80 [70–90]
Open Science	64 [50–78]	72 [55–85]	70 [50–80]
Global Relevance	58 [45–70]	66 [50–80]	60 [40–75]

i GPT-5 Pro Assessment

This paper develops and implements a broad, preregistered experimental paradigm to study conflicts between consequences and means across six ends-versus-means (EVM) dilemmas,

alongside six standard self-versus-other (SVO) tasks. The design is carefully documented (extensive instructions, comprehension and attention checks, aligned-variant robustness study) and the central empirical findings are clear: (i) sizeable shares of participants choose nonconsequentialist options in each EVM task (about 20–44%); (ii) a real-stakes trolley implementation (Saving a Life paradigm) yields similar choices to hypothetical versions; (iii) however, EVM choices show little individual-level consistency across tasks, whereas SVO choices are relatively consistent; (iv) EVM choices are essentially uncorrelated with prosociality. The study is well-executed, but several methodological and interpretive issues merit attention.

Construct validity of the EVM dimension: The six EVM tasks plausibly tap distinct subcomponents of deontological constraints—harm–benefit tradeoffs (trolley), lying aversion (lying game), corruption avoidance (bribe game), expressive adherence to a morally preferred but non-pivotal act (group donation), avoidance of a repugnant statement (statement choice), and rule-following absent instrumental value (rule task). This breadth is a strength, but also complicates aggregation: lack of cross-task consistency may reflect multidimensionality (distinct deontological subdomains) rather than the absence of stable underlying preferences per se. The paper’s conclusion that there are no stable deontological types across contexts may thus be too strong; an alternative is that there exist multiple latent moral constraints with limited overlap. While the principal component analysis supports low unidimensionality in EVM, a more formal latent-variable analysis (e.g., confirmatory factor models testing multi-factor structures) could separate a global factor from task-specific factors and quantify reliability. The study partially addresses measurement concerns (binary vs continuous metrics, biserial correlations), yet the remaining possibility of low reliability and multidimensionality jointly dampening correlations is not fully ruled out.

Mapping of “means” and “ends” across tasks: The classification of options as deontological vs consequential is mostly persuasive but not always clean. In the lying game, the consequential link hinges on receiver compliance to the message, conveyed to senders as >90% based on a pilot; realized compliance rates in the main receiver pool are not reported, so the perceived vs actual probability of achieving ends could differ. In the bribe game, the strategy-method mapping is transparent and the pairing ensures the de facto bribe tradeoff, but the moral content (avoiding bribery) may be qualitatively different from other tasks. In the group-donation task, Option A’s deontological reading (expressing a principle though inconsequential) is coherent but depends on participants recognizing the donation is already foregone and interpreting Option A as a purely expressive, duty-aligned act—a point the paper backs with recall checks. The rule-following task introduces a self-interest component (breaking a non-instrumental rule raises the participant’s own payoff), unlike the other five EVM tasks that aim to leave the decision-maker’s material/social payoffs unaffected; this asymmetry may draw in a distinct preference (intrinsic norm compliance or self-control) and limit comparability within the EVM block. The statement-choice task measures willingness to endorse a repugnant statement to achieve a good outcome; however, the moral domain here (environment) differs from the charity target (children’s cancer) and the sample is highly pro-environment, likely inflating deontological choices in this task. These heterogeneities underscore the likely multidimensional nature of EVM decisions and caution against reading the absence of pairwise correlations as the absence of stable moral motives more generally.

Real-stakes trolley implementation: The Saving a Life paradigm is carefully built, with transparency about cost-effectiveness calculations (expected lives saved via Operation ASHA). Still, the real-consequences treatment probability (10%) is modest, and while the authors show in

a separate self-versus-other robustness study that similarly probabilistic implementation impacts behavior strongly, domain differences might matter (sacrificial dilemmas vs prosocial tradeoffs). Furthermore, the trolley translation replaces direct physical harm with redirection of limited treatment resources—more akin to allocating medical resources or letting die vs killing distinctions. The authors mitigate this by comparing classical and SAL versions (finding similar rates) and by careful framing, but some readers may still view the deontological status as not fully isomorphic to classic sacrificial cases.

Exclusions and sample: Excluding the top 1% fastest respondents and those preferring €2 to another participant over a €15 charity donation in the baseline arguably improves interpretability for the EVM tasks (uniform moral ranking of ends). The authors replicate findings in the full sample, which helps. Still, the sample is mostly students from one country and notably pro-environment; external validity is limited, especially for the statement-choice task. Order randomization of tasks reduces, but does not eliminate, concerns about carryover, moral balancing, or learning effects; the paper does not report pre-registered tests of order effects. **Statistical inference and multiple testing:** The paper presents many pairwise correlations; main conclusions are driven by the near-zero correlations across EVM tasks and between EVM and SVO blocks, which are robust to alternative codings and sample restrictions. Where isolated correlations are positive (e.g., SAL trolley with lying game ~ 0.16) they are small and would not survive conservative multiple-testing adjustments. The power analysis for the SAL real vs hypothetical contrast targets a 10 percentage-point effect; smaller true effects would not be detected, although the observed differences are indeed very small.

Alternative interpretations and design extensions: Because each EVM task fixes a single tradeoff magnitude (e.g., €15 donation vs €2 transfer; one vs three expected lives), the design identifies prevalence but not willingness-to-pay to violate or respect a deontological constraint. Staircasing the stakes within tasks would permit structural estimation of individual-specific thresholds and enable tests of cross-task parameter correlations, which may reveal consistency at the level of tradeoff intensity rather than binary choices. Further, presenting harmonized moral domains (e.g., all tasks involving the same beneficiary type) would reduce domain confounds. Finally, reporting realized receiver compliance in the lying game and additional checks on perceived vs actual mappings would strengthen the consequentialist characterization.

Open science and replicability: The study is preregistered (registry number provided), has ethics approval, and includes very detailed instructions and comprehension checks. However, based on the PDF alone, it is unclear whether analysis code and de-identified data are publicly available; replication of the SAL trolley requires institutional partnerships and specific cost-effectiveness assumptions. The aligned-version robustness experiment, while very helpful for attention/comprehension validation, uses a separate sample; within-subject counterbalancing or crossover designs could further strengthen claims about comprehension vs moral motives. Overall, the paper offers a valuable methodological contribution and strong, carefully triangulated evidence on the prevalence and context-dependence of ends-versus-means decisions, and on the limited predictive value of trolley judgments for other EVM settings. The main interpretive caution is to avoid overgeneralizing the lack of cross-task consistency as absence of stable moral structure; the evidence is equally consistent with multiple, relatively orthogonal deontological subcomponents. The applied relevance is moderate—policy designers should measure behavior in the specific context of interest rather than extrapolate from trolley-style dilemmas—yet the direct policy quantification is limited. Communication is clear, and robustness checks are extensive; further latent-structure modeling and parametric elicitation would

sharpen identification of underlying moral preferences.

i GPT-5.2 Pro Assessment

This manuscript reports a large, preregistered online experiment (two sessions one week apart; ~550 usable subjects after preregistered exclusions) designed to measure how often people choose “ends” (consequences) over “means” (rule-like or duty-like constraints), and whether such choices reflect stable individual types. The study’s strengths are its ambitious within-subject design across multiple incentivized tasks, careful comprehension checks, and a thoughtful attempt to move beyond purely hypothetical trolley vignettes by implementing a “saving-a-life” (SAL) paradigm tied to real donations that translate into statistical lives saved. The authors also include “aligned” versions of the ends-versus-means (EVM) games in a separate robustness sample to show that subjects mostly choose the option that is simultaneously favored by both consequentialist and deontological reasoning, suggesting attention/understanding is not the primary driver of the main results.

The core empirical finding is that substantial minorities (roughly 20–44% depending on the task) choose the “deontological” option when it conflicts with higher social surplus, but that these choices correlate weakly across EVM tasks, including very low predictive power of trolley choices for other EVM dilemmas. The paper contrasts this with sizable cross-task consistency within a standard “self-versus-other” (SVO) prosociality battery, and reports that EVM behavior is largely uncorrelated with prosociality measures.

The biggest threat to interpretation is construct validity: whether the binary “deontological vs consequentialist” coding is measuring a single underlying moral principle, rather than a bundle of domain-specific norms and aversions (e.g., aversion to lying, aversion to bribery/corruption, purity/sanctity reactions to a “repugnant” statement, preference for rule-following, expressive choice under non-pivotality, omission/commission effects in the SAL trolley). These are plausibly distinct moral (and psychological) constructs, and the observed low cross-task correlations could reflect multi-dimensionality rather than an absence of stable moral traits. The manuscript leans toward the stronger conclusion—“no stable preference types”—but the data, as presented, are also consistent with (i) multiple stable but partially independent deontological dimensions; (ii) stable traits that manifest only in subsets of contexts; or (iii) a trait plus strong situation-specific costs (e.g., moral “prices” varying across tasks). The principal component analysis showing no dominant first component in EVM tasks is informative, but it does not by itself rule out structured multidimensional latent traits; a more direct latent-variable/IRT/factor-mixture approach (explicitly modeling measurement error and task-specific intercepts) would better support the “no types” claim.

Relatedly, the evidentiary basis for “lack of consistency” relies heavily on raw Pearson correlations among dichotomous indicators. Binary measurement attenuates correlations, and heterogeneity in base rates across tasks can mechanically limit achievable correlations. The authors partially address this by dichotomizing SVO measures in robustness tables (showing SVO consistency remains), but they do not provide reliability-adjusted estimates or latent-trait modeling for the EVM block. The distribution-of-counts (“inverse-U”) is suggestive, yet it could also be generated by a mixture of several stable sub-traits that apply to different “means” (honesty, anti-corruption, purity, rule-following) combined with variation in how salient or costly each “means” feels. In short, the paper convincingly shows low *unidimensional* consistency across its chosen EVM tasks, but it is less conclusive that individuals lack

stable moral structure altogether.

The SAL trolley implementation is a substantial methodological contribution, but it also introduces interpretation complexity. The act is framed as redirecting a preset donation between two locations with different multipliers, saving statistical lives via a charity rather than directly killing or saving identifiable persons. Deontological reluctance to “pull the lever” could reflect omission bias, default effects, fairness concerns across regions, skepticism about the multiplier, or discomfort with being responsible for reallocating aid, not only “never use persons as means.” The paper reports extensive information provision and comprehension checks, and finds no significant difference between hypothetical and 10%-implemented “real” versions of the SAL trolley. However, the “realness” manipulation is relatively weak (10% probability), and while the authors provide a separate robustness demonstration that a similar probabilistic implementation affects SVO decisions, it remains possible that (a) moral responsibility is perceived categorically rather than probabilistically; (b) subjects treat “statistical lives via donation” differently from personal payoff tradeoffs; or (c) the SAL setting changes which moral heuristics are triggered relative to classic trolley variants (e.g., “do not withdraw promised help” vs “do not kill”). These issues do not negate the main empirical pattern but suggest caution in mapping the SAL result back to standard trolley debates and especially to policy analogies.

A further limitation is external validity and sample composition: the pool is mostly students from a specific lab population, with online administration. The study is internally strong, yet generalization to broader populations (and to the applied settings invoked in the introduction, such as autonomous vehicles, triage, bribery in real institutions) is uncertain. Additionally, many EVM tasks share a common “ends” structure (15€ donation to a child-cancer charity vs 2€ to another subject), which standardizes consequences but may also impose a particular moral framing (charity salience, deservingness, perceived legitimacy of charities vs direct transfers). Some participants may have stable preferences about “charity vs direct giving,” or about “obeying experimenter-supplied rules,” that interact with the “means” manipulations. On balance, the manuscript is careful, transparent, and methodologically strong for experimental work: preregistration, comprehension checks, randomized order of tasks, a between-subject manipulation of real vs hypothetical stakes in the SAL trolley, and a thoughtfully designed aligned-versions robustness study. The main descriptive claims—nontrivial prevalence of non-consequential choices and weak cross-task correlations among these EVM indicators—are well supported. The main inferential leap that deserves tempering is the interpretation that there are “no stable types” rather than “no single stable type across heterogeneous moral domains,” and the paper would be stronger with explicit modeling of multidimensionality and measurement error, plus additional tasks that isolate omission/default, promise-keeping, and purity/sanctity mechanisms from “means-versus-ends” per se. Replicability appears plausible given the detailed appendices and preregistration, though the evaluation would be strengthened by clearly stated public links to code/data/materials (not fully verifiable from the manuscript text alone).

Gemini 2.0 Flash Assessment

This paper investigates the prevalence and consistency of consequentialist and deontological decision-making. The authors conducted a series of online experiments with 593 subjects, presenting them with six main games that involve ends-versus-means tradeoffs, supplemented by

six classical self-versus-other choice tasks. The ends-versus-means games include a real-stakes version of the trolley dilemma, and novel games involving lying, bribery, making repugnant statements, and rule-following. The self-versus-other tasks measure prosociality using dictator, trust, public goods, and moral-luck games. The authors preregistered their procedures and hypotheses.

The paper's key findings are:

1. A significant fraction of subjects (20-44%) make non-consequentialist choices, indicating that standard economic models that assume consequentialism may be inaccurate.
2. The trolley dilemma, even with real stakes, does not predict behavior in other ends-versus-means scenarios.
3. There is little individual-level consistency in deontological vs. consequentialist decision-making across the six ends-versus-means games, suggesting that moral choices are context-dependent.
4. Ends-versus-means decision-making is unrelated to general prosocial preferences measured in the self-versus-other games.

Several potential weaknesses and limitations should be noted:

- **Identification threats:** The study relies on incentivized experiments, which may not perfectly capture real-world moral decision-making. The artificiality of the experimental settings could influence subjects' choices and limit the generalizability of the findings.
- **Sample Limitations:** The sample consists primarily of students, which may not be representative of the broader population. This could limit the external validity of the results.
- **Measurement Issues:** The paper acknowledges some limitations regarding the measures used. There may be an inherent difficulty in cleanly separating consequentialist vs. deontological motivations; the games used may inadvertently tap into other decision-making dimensions or psychological constructs. Some may see the various scenarios as involving altruism or self-interest.
- **Missing Robustness Checks:** While the authors conducted robustness checks, including an aligned robustness experiment and analyses excluding confused subjects, it would be helpful to provide correlations after conducting the PCA, rather than simply claiming 'these robustness checks do not affect our conclusions'.
- **Interpretation of Lack of Correlation:** The paper interprets the lack of correlation between trolley dilemma choices and other EVM games as evidence against a stable individual preference type. However, one could also argue that this simply reflects the unique features of the trolley dilemma or that the other games lack sufficient power. It would be helpful to explore more nuanced interpretations of the data patterns.
- **Limited Exploration of Heterogeneity:** While the authors tested for heterogeneous treatment effects, they could have explored other dimensions of heterogeneity in moral decision-making, such as individual differences in cognitive style, emotional regulation, or moral reasoning abilities. These factors could potentially moderate the relationship between consequentialism, deontology, and prosociality.

Overall, this research makes a valuable contribution to understanding the complexities of moral decision-making. The findings challenge the assumptions of standard economic models

and highlight the importance of context in shaping moral choices. However, the limitations discussed above should be considered when interpreting the results and drawing conclusions about the nature of moral preferences.

⚠ GPT-5.2 Pro: Key Issues Identified

1. Construct validity: the “deontological” option differs across games (harm/omission in trolley, honesty in lying, anti-corruption in bribe, purity in repugnant statement, rule-following, expressive/non-pivotal voting), so weak cross-task correlations may reflect multidimensional moral domains rather than absence of stable traits.
2. The headline conclusion of “no stable individual preference types” is stronger than what low pairwise correlations among binary indicators can establish; latent-variable models (multidimensional factor/IRT, mixture models) could test for structured heterogeneity more directly.
3. Reliance on Pearson correlations of dichotomous choices risks attenuation from measurement error and base-rate differences; reliability-adjusted associations or alternative estimators would better quantify “consistency.”
4. SAL trolley interpretation is confounded by default effects, omission/commission, reallocation of promised aid, fairness across regions, and the indirect nature of “statistical lives via donations,” which may not cleanly map to standard trolley ‘instrumental harm’ constructs.
5. The “real stakes” manipulation in the trolley is probabilistic (10% implementation); while a robustness exercise shows similar probabilities matter in SVO tradeoffs, moral responsibility may respond differently, so null effects should be interpreted cautiously.
6. Many EVM tasks share a common consequence frame (15€ charity donation vs 2€ to another subject); stable preferences about charity vs direct transfers, or about ‘experimenter-intended morality,’ could interact with means manipulations.
7. External validity is limited by an online, mostly-student lab pool; the extent to which observed context-dependence generalizes to high-stakes institutional settings (triage, bribery, AI policy) is uncertain.
8. Some games (e.g., group donation with non-pivotality, rule-following task) may tap compliance/norm sensitivity toward the experiment itself rather than moral philosophy distinctions, complicating interpretation as deontology vs consequentialism.
9. Multiple-hypothesis considerations are noted for some correlations, but the paper’s narrative may still overweight small significant links (e.g., trolley–lying $r = 0.16$) without a unified error-rate control for exploratory patterns.
10. Open-science/replication usability would be clearer if the manuscript explicitly provided permanent links to anonymized data, code, and experimental materials beyond the preregistration reference.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

B.4 Bilal and Kaenzig 2024

B.4.1 Ratings Comparison

Metric	GPT-5 Pro	GPT-5.2 Pro	Gemini 2.0 Flash	Human
Overall	92 [85–97]	86 [76–92]	65 [45–80]	88 [80–95] (n=2)
Claims	88 [80–95]	82 [70–90]	70 [55–85]	88 [80–98] (n=2)
Methods	82 [74–90]	79 [66–88]	60 [40–75]	84 [72–95] (n=2)
Adv. Knowledge	95 [88–99]	91 [83–96]	75 [60–90]	94 [92–96] (n=2)
Logic & Comms	92 [85–97]	87 [78–93]	80 [70–90]	92 [88–98] (n=2)
Open Science	62 [45–78]	58 [40–75]	60 [40–75]	62 [55–70] (n=2)
Global Relevance	97 [92–99]	96 [90–99]	85 [70–95]	90 [85–95] (n=2)
NA	—	—	—	5 [4–5] (n=2)
NA	—	—	—	5 [4–5] (n=2)
NA	—	—	—	90 [85–95] (n=2)

i GPT-5 Pro Assessment

This manuscript makes an ambitious and consequential claim: macroeconomic damages from climate change are much larger than the mainstream panel literature has inferred, because global mean temperature shocks better capture the full climate system’s damaging manifestations (especially extremes) than country-level temperature variation. The empirical core uses time-series local projections on a newly assembled climate–economy dataset (global aggregates and a large country panel, 1960–2019, with extensions back to 1900 for a subset) and defines temperature “shocks” as multi-step forecast errors (Hamilton-type filter) to purge common trends. The headline result is that a 1°C global temperature shock lowers world real GDP per capita by about 12% at peak, with persistent effects beyond a decade; the analogous local-temperature panel effect is ~1% and often imprecise. The paper argues that global temperature shocks are more tightly linked to extreme heat, drought, extreme precipitation, and extreme winds, and that these channels quantitatively explain a substantial share (~two thirds) of the estimated global GDP response. The paper then structurally maps the reduced-form impulse response into a neoclassical growth model by identifying a productivity damage function that reproduces the output IRF and validates non-targeted capital dynamics. Under a moderate warming path (to ~3°C above pre-industrial by 2100), the model implies ~25% present welfare losses and an SCC around \$1,367/ton (95% CI roughly \$469–\$2,264), robust across discount rates and climate sensitivity ranges; the domestic cost of carbon for a large country is argued to exceed typical abatement cost estimates, implying unilateral decarbonization is privately cost-effective. Strengths include: (i) careful construction of globally relevant climate variation and rigorous detrending consistent with climate physics (external and internal variability), (ii) transparent local projection estimation with attention to serial correlation and lag structure, (iii) numerous robustness checks (alternative shock constructions, VAR/different detrending, ENSO/volcano controls, alternative samples back to 1900, panel estimates with country fixed effects and region-specific trends, jackknife influence checks, and alternative shock persistence deconvolution), (iv) mechanism evidence linking

global temperature shocks to increases in multiple extreme event exposures, (v) thoughtful tests for general-equilibrium spillovers using trade- and distance-weighted external shocks, and (vi) a structural mapping that is internally consistent and identifies damage persistence with closed-form Jacobians, including a useful validation on capital responses. That said, several important issues limit full certainty. Identification: the core aggregate time-series variation remains data-sparse (T 60) and unavoidably vulnerable to residual confounding by co-moving global shocks, even with rich global controls, recession dummies, and ENSO/volcano adjustments. Because the global shock is common, time fixed effects cannot be used; instead the authors rely on macro controls whose adequacy cannot be fully verified *ex post*. The authors mitigate concerns through multiple controls, pre-trend tests, Granger non-causality of shocks from macro/financial variables, sample stability, and country-panel specifications, but some risk of omitted common factors (e.g., world financial conditions, geopolitical shocks imperfectly captured by controls) remains. Scaling and linearity: the 12% “per 1°C” figure is derived by linearly scaling observed shocks that are typically ~0.1–0.3°C; moreover, the stated “1°C shock” produces a persistent temperature path (remaining ~0.5°C above baseline for years), so the IRF partly reflects accumulated elevated temperatures rather than a one-off perturbation. The authors are transparent here (reporting a ~5% peak effect under a counterfactual purely transitory shock), but the headline phrasing could be misinterpreted as a steady-state elasticity. Within-sample tests find little nonlinearity among observed shocks, but out-of-sample linear scaling up to +1°C or more could be inaccurate if climate–economy damages are convex or exhibit thresholds; conversely, if large-scale nonlinearities (tipping points) exist, linear extrapolation could understate damages. Reverse causality: the paper corrects for emissions-to-temperature feedback using published impulse responses for CO₂, CH₄, and SO₂ and standard emissions–GDP elasticities, finding negligible adjustments. This is reassuring and plausibly correct given the small short-run temperature response to annual emission fluctuations, but it relies on calibrated climate responses and constant elasticities. Mechanism decomposition: the extreme-event evidence is compelling and consistent with climate science (ocean-driven humidity and circulation alter precipitation and winds globally), yet the quantitative “share explained” combines separate IRFs for each extreme exposure multiplied by their dynamic responses to global temperature shocks and assumes linear additivity. Because extremes co-occur and load on common circulation modes, some double counting or omitted interaction terms are possible; the authors frame this as a lower-bound bottom-up reconstruction, but exact decomposition is inherently challenging. Measurement of extremes uses reanalysis data with fixed 1950–1980 thresholds; though robustness checks (alternative thresholds/weights and smoothing for noisy variables) are provided, reanalysis biases and cross-dataset consistency could affect magnitudes. External validity over longer horizons: while samples back to 1900 for advanced economies strengthen stability claims and suggest limited historical adaptation, identifying persistent effects beyond 10 years remains difficult. The model mapping: estimating the structural damage kernel $(s)=A(e^{\{-Bs\}}-e^{\{-Cs\}})$ from the output IRF is elegant and parsimonious, but imposes a specific functional form and that damages enter TFP levels (not growth). This is conservative relative to “growth damage” models, but still a strong restriction; the identification uses only the first 10 years, and longer-term damage persistence is imposed rather than learned from the data. The growth model is a representative-agent neoclassical benchmark without explicit sectoral heterogeneity, migration, or endogenous adaptation/abatement investments; hence the SCC and welfare results reflect that abstraction. The SCC construction uses a conservative temperature re-

sponse to a CO₂ pulse (75% of multi-model mean), but the broader SCC comparison to integrated assessments depends on normative discounting and damage channels; the paper's sensitivity analysis is informative and shows large SCC even under higher discount rates, but readers should note these normative choices. Aggregation and weighting: the use of PWT world aggregates (PPP-based) may overweight damages in poorer countries; the welfare construct is global and not distributionally explicit. The domestic cost-of-carbon calculation for a large country is only summarized; more detail on how domestic damages are derived from global damages (e.g., shares, heterogeneity, trade linkages) would aid policy interpretation. Statistical inference: the use of lag-augmentation (and Driscoll–Kraay in panels) is appropriate; bootstrapping that incorporates shock-estimation uncertainty shows similar coverage. Still, multiple horizons raise concerns about joint inference; the authors present significance windows and uncertainty bands but do not emphasize joint confidence regions. Replicability and open science: data sources are public and the appendix is detailed; however, the paper does not clearly provide a replication package (code and processed data). Given the policy salience, a full replication archive (including the shock-construction scripts, local projection code, panel aggregations, and model solution/estimation scripts) would enhance credibility and reuse. Constructive suggestions: (i) add placebo tests (e.g., projecting GDP on future temperature shocks; or using pseudo-shocks from bootstrapped or permuted temperature to assess false positives), (ii) explore alternative climate indices as instruments (e.g., stratospheric aerosol optical depth, AMO/PDO) to isolate specific exogenous drivers, (iii) report results using monthly world industrial production or other high-frequency global proxies to increase degrees of freedom, (iv) expand sensitivity to alternative reanalysis datasets for extremes, (v) present market-exchange-rate aggregation alongside PPP, and (vi) release a complete replication package. Overall, the empirical approach is thoughtfully executed, the mechanisms are plausibly grounded in climate dynamics, and the structural mapping is coherent; while the strong claims rest on necessarily limited time-series variation, the breadth of robustness checks and consistency across methods lend the results considerable credibility. The work is highly relevant to policy and is likely to materially shift debates on macro-climate damages and optimal carbon pricing.

i GPT-5.2 Pro Assessment

This manuscript makes an ambitious and consequential claim—global warming damages world GDP far more than prior panel estimates—by shifting identification from local (country) temperature variation to time-series variation in global mean temperature. The empirical core is a set of time-series and panel local-projection estimates of world/country GDP responses to “global temperature shocks,” defined as multi-step forecast errors from a Hamilton (2018) detrending/innovation procedure. The headline response is large (12% peak GDP loss per +1°C global shock at ~6 years) and persistent, and the paper then maps these reduced-form responses into a structural productivity damage kernel in a neoclassical growth model to compute large welfare losses and an SCC above \$1,000/tCO₂.

The most important technical concern is identification in a short global time series. Even with many robustness checks, the design fundamentally relies on the assumption that innovations to global temperature (after removing predictable components and adding macro controls) are as-good-as-random with respect to other global shocks that affect GDP at multi-year horizons. This is much harder to guarantee than in subnational/country panel designs. The

time-series sample for the headline estimate is ~60 annual observations (1960–2019). With local projections out to 10 years, inference and finite-sample robustness are delicate: overlapping dependent variables (cumulative growth), persistent regressors (their shock has autocorrelation by construction), and multiple controls/dummies consume degrees of freedom. The authors use lag augmentation and various controls; however, small-sample distortions, sensitivity of long-horizon LPs, and implicit researcher degrees of freedom (horizon choice, lags, controls, recession dummies, filtering choices) remain potential threats. The paper provides many alternative specifications and finds similar shapes, which is reassuring, but the “global shock” setting remains more vulnerable to unobserved common factors than local-weather panel work.

A second concern is conceptual: whether “natural variability” shocks to global mean temperature are externally valid for damages from anthropogenic warming. The paper argues that global mean temperature is the relevant state variable and that global shocks better predict extremes than idiosyncratic local shocks; still, natural variability episodes (ENSO, volcanic forcing, solar) differ from sustained radiative forcing in spatial pattern, seasonality, ocean/land distribution, and the joint behavior of other climate variables. Even if the authors show robustness to controlling for ENSO and volcano years, the remaining variation is still a mix of mechanisms. The mapping from these historical deviations to a smooth $+2^{\circ}\text{C}$ path by 2100 assumes (i) local-linear damages in temperature over a range much larger than realized annual global shocks ($\sim 0.3^{\circ}\text{C}$ max), and (ii) stability of the joint relationship between global temperature and damaging extremes under different forcing regimes. The manuscript discusses nonlinearity briefly and finds little evidence within-sample, but the data are not informative about tails and tipping behavior; equally, they are not very informative about large-scale adaptation under slow, anticipated warming.

Third, the interpretation of the 12% “per 1°C shock” is subtle because their shock is persistent in temperature levels. The GDP impulse response partly reflects accumulated exposure to elevated temperatures following the shock, not just the contemporaneous 1°C increase. They provide a decomposition that yields a smaller peak effect ($\sim 5\%$) for a purely transitory temperature increase. For welfare/SCC, the crucial object is the damage kernel (the mapping from a temperature path to productivity), and here the identification hinges on correctly estimating both (a) the temperature persistence response and (b) the GDP response. Any bias in persistence (from filtering choices) can mechanically amplify the inferred long-run damages.

Fourth, the “extreme events” mechanism is suggestive but not fully pinned down causally. The paper shows global temperature shocks predict increases in several reanalysis-based extreme indices, whereas local shocks do not, and a bottom-up aggregation explains $\sim$ two-thirds of the GDP response. However, measurement choices (percentile thresholds, baseline 1950–1980, smoothing of wind/precipitation, reanalysis quality in early decades, cross-country aggregation) and potential simultaneous global shocks affecting both extremes reporting/measurement and economic activity could influence these results. The aggregation exercise mixes dynamic responses and requires assumptions about persistence and separability across channels; it is informative, but not a definitive mediation analysis.

Fifth, the structural translation to welfare and SCC, while transparent, is model-dependent. The neoclassical growth model is intentionally simple (representative agent, exogenous labor, no climate adaptation investment, no endogenous mitigation, no sectoral/region heterogeneity). The estimated “damage function” is effectively a reduced-form productivity kernel that absorbs all channels; that is consistent with a top-down approach, but it also means the SCC

is sensitive to (i) assuming damages operate only through TFP (not depreciation, mortality, amenities, risk, inequality), (ii) the functional-form restriction used to extrapolate beyond the 10-year empirical horizon, (iii) the assumption of persistent level effects rather than growth effects, and (iv) the assumed future temperature path and discounting. The authors do a fair amount of sensitivity analysis for discounting and warming levels, but uncertainty propagation remains partial: e.g., uncertainty in climate sensitivity/impulse response, in the representativeness of historical variability, and in long-horizon extrapolation of the damage kernel is not fully integrated.

Finally, replicability and transparency look decent in terms of data provenance (public datasets, detailed appendices, many robustness figures), but the paper does not clearly state code/data release, versioning, and exact pipelines for constructing shocks and extreme indices; given the complexity (multiple datasets, weighting schemes, filters, and smoothing), full computational reproducibility would depend on sharing code and intermediate artifacts. Overall, this is a high-quality, carefully executed contribution with extensive robustness work and a clear argument for why global temperature may capture damages that local-temperature panel designs difference out. The key uncertainty is whether the time-series identification and external validity are strong enough to support the magnitude of the headline effect and the resulting SCC; even moderate attenuation from residual confounding or non-representativeness could materially change welfare numbers. The paper's strengths—novel perspective, comprehensive datasets, and transparent mapping to welfare—are substantial, but readers should treat the quantitative magnitudes as more uncertain than the within-paper confidence intervals imply, because several deep uncertainties are not (and arguably cannot be) fully identified from the historical time series used here.

Gemini 2.0 Flash Assessment

This paper investigates the macroeconomic impacts of climate change by comparing the effects of global versus local temperature changes on GDP. The authors find that global temperature changes have a significantly larger impact on GDP than local temperature changes, estimating that a 1°C increase in global temperature reduces world GDP by 12%. They attribute this difference to global temperature's stronger correlation with extreme climatic events. The authors then estimate damage functions in a neoclassical growth model, projecting a 25% present welfare loss and a Social Cost of Carbon (SCC) of \$1,367 per ton. They argue that these findings suggest unilateral decarbonization policies are cost-effective for large countries. The research uses a time-series local projection approach and constructs a new climate-economy dataset spanning the last 120 years. They address identification concerns such as omitted variable bias and reverse causality through various robustness exercises. The core identification strategy involves using temperature shocks derived from the Hamilton (2018) method. The paper then explores the margins of GDP reduction and regional impacts, finding stronger negative effects in warmer and lower-income countries. It also examines the role of economic spillovers and reconciles the impacts of global and local temperature. Finally, it embeds these reduced-form estimates into a neoclassical growth model in order to arrive at welfare cost estimates and the social cost of carbon.

Important issues to consider are:

1. **Identification Threats:** The identification strategy relies heavily on the assumption that the constructed temperature shocks are exogenous. While the authors address

potential omitted variable bias and reverse causality, there might be unobserved confounders that are not fully accounted for. The exogeneity of temperature shocks derived from the Hamilton (2018) filter needs careful scrutiny.

2. **Data Limitations:** The study uses a combination of datasets, including Penn World Tables and Berkeley Earth temperature data. While these are standard resources, they have inherent limitations in terms of data quality, coverage, and consistency across time and regions. The extrapolation of these data, especially over a 120-year period, could introduce measurement errors and biases.
3. **Model Assumptions:** The neoclassical growth model relies on several simplifying assumptions, such as perfect foresight, a representative agent, and specific functional forms (e.g., Cobb-Douglas production function). These assumptions may not fully capture the complexities of the real world, particularly in the context of climate change and its heterogeneous impacts.
4. **Extrapolation and Linearity:** The estimation of damage functions involves extrapolating from observed temperature shocks to larger, hypothetical warming scenarios. The assumption of linearity in the damage function might not hold, especially as climate change reaches tipping points or exhibits non-linear effects.
5. **Aggregation Issues:** The aggregation of country-level data into global aggregates could mask important heterogeneities and distributional effects. The impacts of climate change are likely to vary significantly across regions and populations, and these differences may not be fully captured in the aggregate analysis.
6. **Replication Barriers:** While the authors describe their methods and data sources, the availability of code and data would be crucial for independent replication and verification of the results. Clear documentation and open access to these resources would enhance the credibility and transparency of the research.
7. **External Validity:** The study's findings are based on historical data and may not fully reflect future adaptation efforts, technological changes, or policy interventions. The extent to which the estimated relationships hold under different socio-economic and environmental conditions remains uncertain.
8. **Justification of choices:** The choice of the horizon h in the shock creation and the lag length are not justified beyond reference to the climate science literature – it is not clear that the relevant horizon for climate's *economic* effect is being captured. The use of population weights vs area weights when aggregating the data is not motivated, particularly considering the results' end use – in welfare/utility calculations – where one might expect population weighting to be more relevant. The damage functions are heavily constrained – and these constraints are not strongly motivated.

GPT-5.2 Pro: Key Issues Identified

1. 1) Time-series identification risk: global temperature innovations may still correlate with unobserved global shocks (policy, finance, technology, geopolitics) that affect

- GDP at multi-year horizons, and the small annual sample (60) makes this hard to rule out conclusively.
2. 2) External validity: natural variability shocks (ENSO/volcano/other internal variability) may not be representative of damages from sustained anthropogenic warming, due to different spatial/seasonal patterns and joint dynamics of climate variables.
 3. 3) Magnitude interpretation: the 12% “per 1°C shock” effect partly reflects the post-shock persistence of temperature; the welfare mapping depends critically on how persistence is estimated and translated into a damage kernel.
 4. 4) Limited information about nonlinearities and tails: the historical global shocks are small; extrapolating linearly to multi-degree warming and to potential tipping points/adaptation regimes is weakly identified.
 5. 5) Long-horizon LP and finite-sample inference: overlapping cumulative-growth outcomes, persistent regressors, multiple controls/dummies, and 10-year horizons can generate biased standard errors and sensitivity to specification choices despite robustness checks.
 6. 6) Temperature-shock construction: the Hamilton (2018) multi-step forecast-error approach (choice of h and p) is defensible but not uniquely grounded; shock definitions can embed assumptions about persistence that influence implied damages.
 7. 7) Mediation via extreme events is suggestive but not causal: extreme indices rely on reanalysis and threshold/smoothing choices; the bottom-up aggregation assumes separability and correct dynamic persistence and may omit other correlated channels.
 8. 8) Panel-country evidence still lacks time fixed effects for global shocks; while global controls and alternative designs are explored, any residual common-factor confounding remains a concern.
 9. 9) Structural welfare/SCC translation is model-dependent: representative-agent neo-classical model with damages only through TFP omits adaptation, sectoral/region heterogeneity, risk/uncertainty, and endogenous mitigation; the functional-form restriction for the damage kernel extrapolates beyond observed horizons.
 10. 10) Uncertainty propagation: reported confidence intervals mainly reflect reduced-form estimation uncertainty; deeper uncertainties (climate impulse response, external validity, long-horizon extrapolation, measurement error in extremes) are not fully integrated.
 11. 11) Replicability: while data sources are public and appendices are detailed, full computational reproducibility likely requires releasing code and exact data-processing pipelines (shocks, weighting, extreme event construction, smoothing choices, and versioning).

GPT-5 Pro reasoning trace available in the [online HTML version](#).

B.5 Blimpo and Castaneda-Dower 2025

B.5.1 Ratings Comparison

Metric	Claude Sonnet 4	GPT-5 Pro	GPT-5.2 Pro	Gemini 2.0 Flash	Human
Overall	65 [55–75]	71 [62–79]	63 [50–76]	65 [45–80]	80 [75–95] (n=2)
Claims	70 [60–80]	66 [55–75]	60 [45–74]	70 [55–85]	85 [75–90] (n=1)
Methods	62 [52–72]	64 [53–75]	62 [46–76]	65 [50–75]	85 [75–90] (n=1)
Adv. Knowledge	68 [58–78]	72 [60–82]	70 [55–82]	75 [60–85]	85 [75–90] (n=1)
Logic & Comms	75 [70–80]	76 [66–86]	73 [60–84]	80 [70–90]	90 [80–100] (n=1)
Open Science	55 [45–65]	28 [15–45]	38 [20–55]	70 [50–80]	85 [75–95] (n=1)
Global Relevance	72 [62–82]	70 [58–80]	80 [65–90]	75 [60–85]	90 [80–100] (n=1)
NA	—	—	—	—	4 [4–5] (n=1)
NA	—	—	—	—	4 [4–5] (n=2)
NA	—	—	—	—	90 [80–100] (n=1)

GPT-5 Pro Assessment

This randomized field experiment studies whether providing micro and informal enterprises (MIEs) in Lomé (Togo) with training on the tax code and the purpose of taxation (the “fiscal contract”) affects tax participation and economic activity. The design is straightforward: 424 firms were sampled from an earlier representative listing (after dropping ambulant vendors and rare firm types), randomized into treatment and control, and 383 were re-contacted ~10 months later to run the intervention and endline (191 treated, 192 control). The paper documents large and durable gains in tax knowledge among treated firms and finds a sizable reduction in the extensive margin of tax payment (around 9–14 percentage points), alongside increased economic activity (measured by “open orders” at survey time). It also reports a positive effect on average tax paid per firm (self-reported, receipt-verified where available), interpreting this as a shift toward higher-revenue firms paying. A simple model posits that greater knowledge raises firms’ bargaining power vis-à-vis tax collectors, moving enforcement away from low-revenue firms toward higher-revenue firms. The paper is clearly written and situates the contribution in the literature on taxation and information/transparency, with a welcome focus on the extensive margin in the informal sector. That said, several internal validity, measurement, and reporting issues merit attention. First, the measurement of tax payments at the intensive margin is fragile. There is substantial missingness in the “amount paid” variable, and the analysis imputes all missing values and non-payments as zero. This conflates non-response with non-payment and risks differential misclassification across arms if treatment affects record-keeping, awareness of receipts, or willingness/ability to produce receipts. The claim in the abstract that tax revenue collected increased is too strong given this imputation strategy and the limited number of validated receipts (27 control vs. 33 treatment); it would be better labeled as suggestive and accompanied by robustness checks: (i) report effects only among those with verified receipts; (ii) bound the effect using worst/best-case assumptions; (iii) reweight using inverse-probability weights for receipt availability; and (iv) include “receipt availability” as an outcome to check differential documentation. Second,

although balance at baseline among the 383 endline firms appears acceptable, the paper does not present a formal differential attrition analysis from the 424 randomized firms. Even if attrition is similar in counts across arms, the composition may differ. A CONSORT-style flowchart, attrition by arm, and tests of whether treatment predicts endline observation (with baseline covariates) would strengthen credibility; if imbalance exists, use IPW or Lee bounds for key outcomes. Third, standard errors are reported as heteroskedasticity-robust, but the sampling and implementation plausibly induce within-area or within-enumerator correlation (32 enumeration areas, and field teams trained to deliver a standardized intervention). Not clustering by enumeration area (and/or enumerator) and not using a wild bootstrap may understate uncertainty, especially for binary outcomes near 40%. Re-estimating main outcomes with area-level (and enumerator) clustering and reporting wild cluster bootstrap p-values would be prudent. Fourth, multiple testing and outcome specification: knowledge outcomes are numerous, and there are several main and secondary outcomes (participation, open orders, revenue, amount paid). There is no mention of a pre-analysis plan or registration, and no adjustment for multiple comparisons. Reporting a pre-specified primary outcome set (or constructing standardized indices) and applying BH/FDR or FWER corrections would clarify which results are robust to multiplicity. Fifth, measurement of economic activity via “open orders” is intuitive for service-oriented MIEs but could be sensitive to day-of-week, time-of-day, or seasonality. The paper does not report whether survey timing was balanced across arms, nor whether time/day fixed effects were included. Adding those controls, or showing balance in timing, would bolster the interpretation that the treatment increased activity. The alternative “revenue last month” measure is noisy; the paper partially mitigates this using an asset-based proxy but could provide more detail on the predictive model (fit, cross-validation) and consistency across measures. Sixth, mechanism evidence is suggestive but not determinative. The “empowerment” channel is plausible (heterogeneity by baseline perceived unpredictability; differential responses conditional on tax-collector visits; knowledge of statutory determinants), yet the conditional-on-visit comparisons may be endogenous if visits themselves are affected by treatment. An IV strategy for “visit” (e.g., using area-level baseline intensity or collector coverage) or pre-specified heterogeneity by exogenous proxies for bargaining power would be stronger. The LATE analyses instrument “knowledge” with treatment, but the composite “Accountability” component is defined as “not saying ‘I don’t know’” about the use of taxes; this may reflect attitude or confidence rather than knowledge per se. Reporting 2SLS results using alternative, tighter knowledge constructs (e.g., statutory determinants only; principal components of knowledge items) and overidentification tests would help. Seventh, spillovers/contamination are discussed qualitatively but not tested. Because firms are sampled with spatial proximity, simple checks—effects by distance to treated firms, or including block/EA fixed effects and proportion-treated—could bound concerns. Eighth, the sample excludes ambulant vendors and very rare firm types; the inference is therefore for relatively fixed-location MIEs, heavily tailors and barbers. This circumscribes external validity. Moreover, Togo’s tax regime has since changed; the paper rightly notes this but could discuss how predictions might change under lump-sum/bracketed forfait regimes where bargaining margins differ. Ninth, the ethical dimension deserves brief discussion. The intervention plausibly reduced the extensive margin of tax payment among low-revenue firms by improving knowledge; while authors frame this as bringing practices closer to legal intent, Readers would benefit from a short ethics/IRB statement and discussion of potential unintended consequences. Finally, transparency/open science practices are limited in the manuscript. There is no mention of pre-registration, nor

provision of data and code. Given the policy importance and the nuanced findings, making de-identified data, survey instruments, and replication code available would substantially raise credibility and reusability. Overall, the core experimental result that civic information increased knowledge and reduced extensive-margin tax payment is convincing and important; the uptick in economic activity is promising but would benefit from timing controls; and the inference that total revenue per firm increased is, at present, suggestive rather than conclusive due to measurement/missingness. Strengthening inference with robustness to clustering, attrition, missingness, and multiplicity—and sharpening mechanism tests—would elevate the paper’s contribution further.

i GPT-5.2 Pro Assessment

This manuscript studies whether closing “civic information” gaps about the fiscal contract changes tax participation among informal microenterprises in Lomé, Togo. The core design is a firm-level RCT (N 383) where treated firms receive training and follow-up consultations on the tax code and the purposes/uses of taxation. The headline results are internally coherent but methodologically fragile in several ways that affect how confidently one can interpret (i) the counterintuitive decline in extensive-margin tax payment, and (ii) the claimed increase in economic activity and total tax revenue. The strongest part of the evidence is that the intervention substantially increased measured tax knowledge and related beliefs at endline (large intent-to-treat effects on correctly identifying taxes, determinants of liability, and having an opinion about whether taxes are used well). However, several threats arise from outcome measurement, missingness, and interpretation of mechanisms.

First, key outcomes rely heavily on self-reports with selective verification. “Pays taxes” is a binary indicator of paying “any taxes,” but the paper’s institutional description suggests multiple taxes (municipal, national, TPU), varied collection practices, and the possibility of partial payments, bribes, or in-kind seizures. Without clearer operationalization (what counts as “paid,” how recall is anchored, whether any receipt was required for the binary, and how inconsistencies were handled), misclassification is plausible—especially if the training changes how respondents label or understand what they paid (e.g., treated firms may reclassify previous payments as non-tax fees or bribes, or be more precise about TPU vs other levies). The finding in Table 6 (many control payers do not know whether they paid TPU) suggests considerable baseline misunderstanding that could translate into differential reporting after treatment.

Second, the paper’s tax revenue amount outcome is extremely underpowered and subject to strong selection: only 60 firms show receipts (27 control, 33 treatment). The authors impute all missing values (including those missing because no receipt was shown) as zero. This conflates true nonpayment with non-verification and makes the estimate sensitive to differential willingness/ability to produce receipts, which plausibly depends on treatment (treated firms may be more likely to request/keep receipts, or more willing to show them to enumerators). The direction of bias is ambiguous, but the interpretation “tax revenue increased” is not secure without missing-data diagnostics, bounds (e.g., Lee bounds), or models that separate “paid” from “verified amount.” Relatedly, revenue and orders are measured at endline with only limited discussion of seasonality, day-to-day volatility, and the extent to which enumerator presence or repeated contact (treatment firms had more interactions) could generate Hawthorne/attention effects.

Third, causal interpretation of the mechanism (“empowerment” and bargaining) rests mainly

on heterogeneity patterns and descriptive splits (e.g., conditional on tax collector visit; revenue quartiles; baseline revenue-to-assets ratio). These patterns are suggestive but not strongly identified. “Visited by a tax official” is plausibly post-treatment and could itself be affected by treatment through changes in visibility or behavior, making conditioning on it a collider and complicating inference. Similarly, subgroup analyses are not accompanied by a pre-specified analysis plan, multiple-testing adjustments, or a transparent accounting of how many interactions were explored, which raises risks of over-interpreting exploratory heterogeneity.

Fourth, the 2SLS/LATE exercise is not fully persuasive as evidence that “knowledge causes reduced tax participation.” The “take-up” variable is constructed from endline responses (knowledge of obligation and accountability opinion) and then instrumented with assignment. While this is a standard “encouragement design” framing, here the instrument likely affects tax payment through multiple channels beyond those two survey items (e.g., salience, perceived scrutiny, trust, norms, changes in record-keeping), violating the exclusion restriction for interpreting the IV coefficient as the causal effect of “knowledge.” In addition, the product-of-indicators construction is ad hoc and may mechanically amplify measurement error and nonlinearity; the large IV magnitudes (-0.55 to -0.78) are difficult to reconcile with realistic behavioral margins without more careful interpretation.

Fifth, the empirical specifications mostly use robust (heteroskedasticity-robust) standard errors, but the sampling approach clusters firms within enumeration areas and neighborhoods. If outcomes are correlated within these areas (tax enforcement intensity, local collector behavior, local shocks), not clustering could understate uncertainty. The paper also reports the treatment effect becoming more significant with covariates; while ANCOVA can increase precision, the fact that significance moves from 10% to 1% depending on specification heightens sensitivity concerns, especially given modest sample size and a minimum detectable effect close to the estimated effect.

Sixth, there is some ambiguity about timelines and what constitutes “baseline” given the initial sampling-frame survey and the later experiment start; clearer separation of pre-randomization measures from post-randomization interactions would help assess risks of differential attrition or baseline measurement drift. The authors report high tracking and no refusals conditional on location, which supports internal validity, but the initial loss from 424 selected to 383 located (10%) could still induce imbalance if correlated with treatment assignment or with firm characteristics in ways not fully observed.

Despite these limitations, the paper tackles an important and under-studied margin (extensive participation among informal firms) and documents a plausible, policy-relevant possibility: civic-information interventions can change bargaining dynamics and reallocate compliance toward higher-revenue firms, potentially increasing efficiency even if participation falls. The institutional detail is valuable, and the primary experimental contrast is credible. But the strongest conclusions should be narrowed to: (i) the intervention robustly increased measured tax knowledge and fiscal-contract salience; (ii) it likely reduced reported extensive-margin payment in the short run; and (iii) evidence for increased total revenue and increased economic activity is suggestive but not definitive given missingness, verification selection, and attention confounds. Improvements in measurement, pre-analysis transparency, clustering/robustness, and missing-data handling would materially strengthen the contribution.

Claude Sonnet 4 Assessment

This manuscript presents a randomized controlled trial examining how providing civic information about the tax code and fiscal contract affects tax participation among informal microenterprises in Lomé, Togo. The intervention trained 191 firms about their tax obligations and the purpose of taxation, with 192 firms serving as controls. The main finding is counter-intuitive: treated firms were 9-14 percentage points less likely to pay taxes, but had increased economic activity (58% more open orders) and higher total tax revenues collected (\$4.49 increase). The authors rationalize this through a theoretical model suggesting empowerment effects - low-revenue firms feel empowered to challenge unfair assessments while high-revenue firms are more likely to self-declare. While the experimental design is sound with successful randomization and high tracking rates (90%), several methodological concerns limit the robustness of findings. The tax revenue measure suffers from substantial missing values since verification through receipts was required, with only 60 of 383 firms providing receipts. The theoretical rationalization, while plausible, relies heavily on exploratory heterogeneity analysis rather than pre-specified tests. The mechanism relies on constructed measures of 'take-up' based on post-treatment knowledge, which may suffer from endogeneity issues. The paper explores alternative explanations in the appendix but cannot definitively rule out competing mechanisms like decreased government satisfaction or increased awareness of non-compliance costs. The qualitative follow-up interviews provide supportive evidence but are conducted on a small, non-representative sample. The external validity is questionable given the specific institutional context of Togo's tax system and the particular intervention design. Missing robustness checks include alternative specifications for the main outcomes, sensitivity to different definitions of tax participation, and formal mediation analysis.

Gemini 2.0 Flash Assessment

This paper investigates the impact of providing information to informal firms in Togo about their tax obligations and the benefits of taxation. The authors conducted a randomized controlled trial (RCT) in Lomé, Togo, where they provided training to a randomly selected group of informal firms on the tax code and the purpose of taxation. The study's primary goal was to examine how improving these firms' understanding of the fiscal contract affects their tax participation. The researchers hypothesized that better-informed firms would be more likely to comply with tax regulations, leading to increased tax revenue. The study, however, reveals a nuanced and counterintuitive set of findings. While the intervention improved firms' knowledge of the tax code, it paradoxically led to a decrease in the number of firms actively paying taxes. This reduction in tax participation was statistically significant. The authors found that the treatment effect ranged from -9.2 to -13.7 percentage points, which translates to a 22%-33% reduction relative to the control mean. Despite the decline in tax participation, the study found that the intervention increased economic activity among the participating firms, as measured by the number of open orders. Furthermore, the total tax revenue collected actually increased, suggesting a shift in the distribution of taxpayers toward higher-revenue firms. The authors propose a theoretical model to rationalize these findings, suggesting that the information intervention empowered low-revenue firms to challenge tax officials' assessments, making them less likely to pay taxes. Conversely, high-revenue firms, encouraged to self-declare rather than hide income, became more likely to pay taxes. The validity of the study

hinges on the assumption that the treatment and control groups are statistically similar in terms of both observed and unobserved variables. The authors conduct a covariate balance check to verify the randomization process, finding that the observable characteristics of firms and firm owners are balanced. However, the study acknowledges that unobserved variables could still potentially influence the results. The study's findings suggest that simply providing information about tax obligations may not be sufficient to increase tax compliance among informal firms. The authors call for further research to investigate the complex interplay between information, bargaining power, and tax behavior in developing countries. A potential limitation of the study is the reliance on self-reported data, particularly for tax revenue. To address this, the authors required verification via a tax receipt, but this resulted in a significant number of missing values. Furthermore, the external validity of the study may be limited by the specific context of Togo, including its tax institutions and practices. The authors acknowledge these limitations and call for further research to explore the generalizability of their findings to other countries and contexts. The authors' theoretical model suggests that the intervention increased the audit cost for tax collectors, leading them to target larger revenue firms. This, in turn, induced greater participation among high-revenue firms, resulting in a change in the composition of taxpayers. Overall, this is a well-executed field experiment that sheds light on the complexities of tax compliance in developing countries. The study's findings challenge the conventional wisdom that simply providing information is sufficient to increase tax revenue. The study has a few weaknesses. The number of firms that showed payment receipts was low. While the paper attempts to address this with data imputation, this clearly leads to a lower bound estimate of this effect, as the paper later acknowledges. Also, one could also argue that the ex-post quantitative surveys were not adequately randomized, as firms were selected based on whether they changed their tax behavior after treatment, introducing the possibility of selection bias.

⚠️ GPT-5.2 Pro: Key Issues Identified

1. 1) Tax revenue amounts are based on a small subset with receipts (60/383) and the analysis imputes all missing/unverified amounts as zero, conflating nonpayment with non-verification and making the “revenue increased” claim highly sensitive to missing-data assumptions.
2. 2) The binary “pays taxes” outcome may be differentially misreported after training because the intervention changes respondents’ understanding/classification of payments (TPU vs other taxes/fees/bribes), threatening comparability across arms.
3. 3) Treatment firms received substantially more contact (training + follow-up consultations) than controls, so effects on economic activity and even tax behavior may partly reflect attention/Hawthorne effects rather than information about the fiscal contract per se.
4. 4) Conditioning on (or splitting by) post-treatment variables such as “visited by a tax official” risks post-treatment selection/collider bias; such results should be framed as descriptive rather than evidence of bargaining-power mechanisms.
5. 5) The 2SLS “knowledge take-up” LATE relies on a constructed endline measure (including an ‘accountability opinion’ proxy) and likely violates exclusion (assignment affects outcomes through channels other than the measured knowledge items), so IV coefficients should not be interpreted straightforwardly as causal effects of ‘knowl-

- edge.’
6. 6) Standard errors are not clearly clustered by enumeration area/neighborhood despite clustered sampling and likely correlated enforcement environments; uncertainty may be understated.
 7. 7) Main treatment effect significance varies notably by specification (from 10% without controls to 1% with controls), suggesting sensitivity and the need for robustness checks (e.g., randomization inference, clustered SEs, alternative specifications, or pre-specified primary model).
 8. 8) Economic activity measures (open orders; estimated monthly revenue) are potentially noisy and seasonally variable, and the paper does not fully rule out survey-timing or reporting effects tied to differential enumerator engagement across arms.
 9. 9) Heterogeneity analyses (revenue quartiles, revenue-to-assets ratio, tax unpredictability) appear exploratory without multiple-testing adjustments or a clear count of attempted interactions, raising the risk of selective emphasis.
 10. 10) The mechanism story (‘empowerment’ reallocates audits from low- to high-revenue firms) is plausible but not directly tested with administrative enforcement data; evidence is indirect and partly based on self-reports and descriptive patterns.
 11. 11) Timeline/baseline clarity could be improved (sampling-frame survey vs experimental baseline), to confirm all baseline covariates and outcomes used in ANCOVA are strictly pre-treatment and comparable across arms.
 12. 12) Replicability/open-science elements (data/code availability, pre-registration, analysis reproducibility) are not evident in the manuscript, limiting external verification and reuse.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

B.6 Bruers 2021

B.6.1 Ratings Comparison

Metric	GPT-5 Pro	GPT-5.2 Pro	Gemini 2.0 Flash	Human
Overall	46 [33–58]	42 [25–60]	45 [25–65]	80 [70–90] (n=2)
Claims	42 [30–55]	38 [20–58]	55 [35–70]	70 [65–75] (n=2)
Methods	35 [20–48]	30 [15–50]	40 [20–60]	65 [60–70] (n=2)
Adv. Knowledge	57 [45–70]	47 [30–65]	60 [40–75]	90 [85–95] (n=2)
Logic & Comms	62 [50–72]	55 [40–70]	70 [55–80]	90 [85–95] (n=2)
Open Science	20 [10–35]	22 [10–40]	60 [40–75]	80 [75–85] (n=2)
Global Relevance	68 [55–80]	72 [55–85]	70 [50–85]	80 [70–90] (n=2)
NA	—	—	—	80 [70–90] (n=2)

This paper tackles an important and underexplored measurement problem: monetizing external animal welfare costs of meat by eliciting a “representative agent” willingness to pay (WTP) and willingness to accept (WTA) to avoid/experience the life of a farm animal via a highly hypothetical “colored pill” scenario. The study is transparent about severe limitations, and the data indeed show substantial fragility. Key identification and validity concerns arise at several levels. First, the core construct—mapping respondents’ stated WTP/WTA for a lifelike but memory-less, present-time experience of living as an animal onto the animal’s utility contribution to a non-anthropocentric social welfare function—rests on strong, contestable assumptions: additive separability of “extended utility,” equal animal-human welfare weights ($\alpha=1$), an existence baseline interpreted via a “neutral animal” or via critical-level utilitarianism, and respondents’ ability to reliably imagine the entirety of an animal’s life. The scenario is epistemically transformative, so hypothetical bias is likely large; the paper acknowledges this. Second, the survey data quality and treatment raise reliability concerns. Of 500 starters, only 301 completed and passed a single comprehension check. A further 61 were dropped for “inconsistencies” (e.g., preferring to avoid the happiest animal more than a conventional farm animal, or preferring a “better life” chicken less), which could reflect misunderstanding but may also remove complex preferences, compounding selection. For median estimates, all respondents below the sample’s median confidence level (~66%) were also excluded (N 120), leaving very small effective N for WTA (N=57) and WTP (N=66). For means, the authors remove extreme WTA outliers with a bespoke threshold (10x adjacent value; practical cutoff ~€1,000,000) and separately winsorize at 90% in sensitivity analysis. Distributions are extremely skewed with orders-of-magnitude differences between mean and median; results are highly sensitive to outlier handling and the confidence-based exclusions (Table 2 vs Table 5), such that central quantitative conclusions vary considerably under defensible alternative choices. Third, external validity is limited: the sample is a Flemish-speaking Belgian panel (18–65) selected to match gender and age but not necessarily broader socio-cultural characteristics; applying these valuations to US price elasticities and climate costs in later simulations mixes contexts without calibration. Fourth, measurement and interpretation of “relative WTP/WTA” (subtracting the neutral animal response) are sensible given identity-loss aversion and “perspective-taking aversion,” yet introduce additional uncertainty: the neutral animal shows non-zero means (with zero medians), and the critical-level reinterpretation alters the welfare baseline. These steps may over- or under-correct individual biases in unknown ways. Fifth, statistical analysis is constrained: non-parametric rank tests find expected ordering (happiest animal < “better life” chicken < conventional animals), but power is low and subgroup comparisons are mostly null. No regression or pre-registered analysis plan is provided; sequencing/order effects and comprehension heterogeneity cannot be credibly assessed. The absence of correlations between the representative-agent valuations and consumer marginal WTP measures is plausible given noise, but it also highlights construct misalignment and low power. Sixth, the policy analysis combining these valuations with a partial equilibrium tax simulation is illustrative but carries further weaknesses: (i) large uncertainty in animal welfare cost inputs (median vs mean WTA differ by orders of magnitude); (ii) cross-country parameter mixing (Belgian preferences with US prices/elasticities); (iii) high sensitivity to cross-price elasticities and substitution patterns; (iv) lack of general-equilibrium or supply-side adjustments; and (v) no explicit consideration of heterogeneity in welfare weights. The conclusion that a carbon tax “could possibly” increase total external costs

(via beef-to-chicken substitution) is directionally reasonable under some parameterizations, but the numerical magnitudes span from modest decreases (when using median WTA) to extremely large increases (when using mean WTA), rendering strong policy prescriptions premature. Seventh, several normative steps (e.g., $\alpha=1$; treating median WTA as a preferred estimator; recommending a flat meat tax) are not derived from a formal welfare-optimization with multiple externalities; a principled solution would internalize marginal external damages by species (climate plus animal welfare), not necessarily a flat tax. The flat-tax suggestion seems driven by the desire to avoid substitution toward high animal-welfare-cost species under a climate-only tax, but it is a second-best rule of thumb rather than an optimal policy result. Eighth, the abstract occasionally overgeneralizes relative to the body: for example, the claim that animal welfare costs are “likely much larger” than the consumer utility or environmental costs holds under mean WTA—especially for chicken—but is not supported under the study’s preferred conservative median WTA for pork and beef, and is borderline for chicken when compared to the higher consumer utility estimates. Finally, replicability and transparency are limited: the instrument text is summarized, but there is no link to the full questionnaire, dataset, or code; no pre-registration; and only aggregated results are presented. The authors’ discussion is candid about hypothetical bias, distributional skew, selection and outlier sensitivity, and the limits of statistical power. The main qualitative insight—that many respondents judge conventional farm animal lives as “not worth living” and that per-kg animal welfare costs are likely far higher for chicken than beef—seems robust to direction but not to magnitude. As an exploratory, hypothesis-generating effort, the paper is informative and thought-provoking, but the quantitative estimates are too fragile for decision-caliber policy inference without substantial additional work (larger samples, pre-registration, improved vignettes, incentive-compatible designs, careful outlier protocols, open materials/data, and a formal multi-externality optimal tax model).

i GPT-5.2 Pro Assessment

This manuscript proposes a contingent-valuation-style survey to monetise “external animal welfare costs” of meat production by asking Belgian respondents to imagine a pill that makes them live (without memory or later recall) the full life of a farm animal, and then eliciting WTP/WTa to avoid or accept that experience. The paper’s central quantitative claims (e.g., median welfare cost of chicken meat €10/kg; mean many orders of magnitude larger; welfare costs likely exceed consumer surplus and climate costs; carbon taxes could backfire by shifting demand from beef to chicken; flat meat tax preferable) rest on a long chain of conceptual and empirical steps, several of which are fragile or weakly supported by the presented evidence. Conceptually, the mapping from the non-anthropocentric social welfare function to a “representative agent” WTP/WTa requires strong assumptions (additive separability between the respondent’s human utility and the imagined animal utility, a defensible zero point for “non-existence”/neutral welfare, and the comparability/weighting $\alpha=1$ across species and between animals and humans). The survey scenario attempts to justify separability via “no memories” and “lifelike present experience,” but it does not demonstrate that respondents actually interpret the vignette in a way that isolates animal welfare from other disutilities (loss of agency, identity/continuity, fear of bodily harm, aversion to being non-human, claustrophobia of confinement, disgust, moral distress, etc.). The author recognises this and introduces a “welfare-neutral animal” plus relative measures (subtracting neutral-animal WTP/WTa), but

the data themselves show that many respondents assign positive values even to the neutral animal (large mean WTA/WTP), which suggests the neutral reference does not successfully anchor “zero welfare” and that the relative measure may still conflate animal welfare with non-welfare-related aversion to the pill scenario.

Empirically, the effective sample for the core estimates collapses dramatically: 500 starters → 301 completers who pass a comprehension check, then further exclusions for any inconsistency (N=61 removed), then median estimates exclude all below-median confidence respondents (N=120 removed), leaving N 57 (WTA) and N 66 (WTP) for medians. This creates substantial selection risk: high-confidence respondents may systematically differ (in numeracy, ethical views, extremity of preferences, susceptibility to framing), and excluding low-confidence (who tend to report lower WTA and many zeros, as the paper notes) mechanically increases estimated costs. The outlier handling for mean WTA is ad hoc (a “10× jump” rule in rank-ordered absolute WTA, effectively cutting at €1,000,000), and the sensitivity analysis shows estimates can shift by orders of magnitude depending on truncation/winsorisation and confidence weighting. Because the headline policy implications depend on whether one uses mean vs median, WTA vs WTP, and absolute vs relative measures, the paper’s own results imply very high uncertainty in the core quantity being estimated; yet the narrative sometimes treats particular choices (e.g., median relative WTA) as “conservative” and policy-relevant without a principled decision-theoretic justification tied to a social objective or loss function.

A further major concern is interpretive: the elicited WTP/WTA is, at best, a measure of respondents’ moralised beliefs and affective reactions to a thought experiment about experiencing animal lives, not a direct estimate of animals’ experienced welfare or of a welfare-relevant cardinal utility scale. The paper at times acknowledges this (measuring “expectations by humans”), but then uses the numbers as if they are external costs in the standard Pigouvian sense, comparable to climate damages in €/kg and suitable to plug into tax formulas. This requires that the elicited valuations behave like stable, coherent welfare tradeoffs; the observed large WTA–WTP gap, extreme skew/outliers (including astronomically large WTAs), and lack of correlation between “how bad animals’ lives are” and respondents’ MWTP for cruelty-free meat all point to substantial instability and construct validity problems.

The statistical analysis is limited and not well matched to the inferential goals. Nonparametric tests (Friedman, Mann–Whitney) are used, but with small and selected subsamples, low power, and multiple comparisons; null subgroup results are unsurprising. There is no multivariate modelling to adjust for beliefs/knowledge, diet, engagement, or comprehension beyond a single check, and the paper’s own claim that larger samples are needed leaves many interpretations unresolved (e.g., whether high valuations reflect informed beliefs about suffering, general empathy, or misunderstanding). Conversions to €/kg depend on assumed edible meat yields per animal; while plausible, this conversion is a key driver of the “chicken is far worse per kg” conclusion and is not stress-tested with alternative yield assumptions or production systems.

The policy analysis extends beyond what the data support. The carbon-tax “backfire” calculation combines Belgian survey-derived welfare costs with US prices, US consumption, US demand elasticities, and climate cost estimates from other contexts; it assumes linear demand and applies substitution patterns in a stylised way. Given the enormous uncertainty in welfare-cost estimates (and in cross-price elasticities, which the paper notes can be near zero or negative), the paper’s quantitative claim that a carbon tax raises animal welfare costs by \$32/year (or, under mean WTA, by \$150,000/year) reads more like a sensitivity illustration

than a credible forecast; the manuscript should present it as such and avoid strong prescriptive conclusions (“governments could implement a flat tax on meat”) without a clearer welfare framework, calibration, and uncertainty propagation.

Strengths include: a clearly motivated attempt to integrate animal welfare into welfare economics; an innovative elicitation vignette aimed at approximating impartiality and direct experience; transparency about severe limitations; and the useful juxtaposition between “consumer MWTP for welfare improvements” and the per-kg welfare implications of different species. However, the core measurement lacks validated construct mapping, results are extremely sensitive to exclusion and outlier decisions, and downstream policy claims overreach relative to the evidential base.

i Gemini 2.0 Flash Assessment

This paper explores the external animal welfare costs of meat consumption using a survey of Belgian consumers. The authors employ a contingent valuation method with hypothetical scenarios, asking respondents about their willingness to pay (WTP) to avoid the experiences of farm animals and their willingness to accept (WTA) compensation for undergoing those experiences. The core idea is to elicit the ‘representative agent’ WTP and WTA, quantifying the trade-off between human and animal utilities. The paper finds that many participants believe farm animals have lives not worth living. The median estimate for the animal welfare cost of chicken meat is 10 euro/kg, while the mean is much higher. The paper also discusses how a carbon tax on meat could inadvertently increase animal welfare costs due to shifts in consumption patterns (e.g., from beef to chicken). The analysis relies on a survey administered to 500 Flemish-speaking Belgian citizens. The survey includes questions on WTP/WTA, certainty scales, knowledge about farm animal living conditions, and demographics. Several exclusion criteria are applied to filter out inconsistent responses and outliers. The results suggest that external animal welfare costs can be substantial, potentially exceeding the consumer benefit of meat consumption. However, the authors acknowledge significant limitations, including hypothetical bias, sensitivity to exclusion criteria, and the difficulty of accurately measuring complex welfare trade-offs. The paper uses non-parametric tests and sensitivity analysis to address some of these issues. The paper offers a thoughtful and well-structured exploration of a difficult topic, but the reliance on hypothetical scenarios and the sensitivity of the results to data processing choices raise concerns about the robustness and generalizability of the findings. The conclusions are appropriately caveated, emphasizing the exploratory nature of the research and the need for further investigation.

Several issues impact the credibility and generalizability of the findings:

1. **Hypothetical Bias:** The contingent valuation method is inherently susceptible to hypothetical bias, as respondents are not making real purchasing decisions. The authors acknowledge this limitation, but the extent to which it affects the results is difficult to quantify. The ‘cheap talk’ disclaimer may not fully mitigate this bias.
2. **Survey Design:** The hypothetical scenario involving colored pills and experiencing the life of an animal is complex and may be difficult for respondents to fully grasp. This complexity could introduce noise and distort the results. The choice of colors (yellow and blue) may also inadvertently introduce bias, even if the authors intended them to be neutral.

3. **Exclusion Criteria and Outlier Treatment:** The authors apply several exclusion criteria to remove inconsistent responses and outliers. While these steps are intended to improve data quality, they also introduce the risk of selectively filtering the data and biasing the results. The sensitivity analysis reveals that the WTA estimates are particularly sensitive to these choices.
4. **Sample Representativeness:** While the survey aims for a representative sample of the Flemish population, the achieved sample (n=500, reduced to smaller sizes after exclusions) may not fully reflect the diversity of Belgian consumers. The reliance on an online panel could also introduce selection bias.
5. **WTA-WTP Discrepancy:** The large difference between WTA and WTP estimates is a common issue in contingent valuation studies, but it raises questions about which measure is more appropriate for estimating animal welfare costs. The authors discuss several explanations for this gap, but the choice of measure remains somewhat arbitrary.
6. **Scope Insensitivity:** The authors note a potential issue of scope insensitivity, where respondents do not adequately differentiate between the suffering of different animals based on the amount of meat they produce. This could lead to inaccurate estimates of the relative welfare costs of different types of meat.
7. **Statistical Power:** The small sample size, particularly after exclusions, limits the statistical power of the analysis. This makes it difficult to detect significant differences between groups and to control for confounding factors in regression analyses.
8. **Generalizability to Other Countries:** The survey was conducted in Belgium, and the results may not be generalizable to other countries with different cultural values, dietary habits, and animal welfare regulations.
9. **Lack of Robustness Checks:** The paper could benefit from more extensive robustness checks, such as exploring alternative weighting schemes or using different statistical methods to analyze the data.
10. **Interpretation of WTP/WTA:** The interpretation of WTP/WTA as a direct measure of animal welfare costs may be too simplistic. These measures reflect human preferences and perceptions, which may not perfectly align with the actual welfare of animals.

GPT-5.2 Pro: Key Issues Identified

1. 1. The identification of “external animal welfare costs” with respondents’ pill-scenario WTP/WTA relies on strong, unvalidated assumptions (additive separability, correct interpretation of the vignette, $\alpha=1$ interpersonal/species comparability, and a meaningful zero point tied to non-existence/neutral welfare).
2. 2. The welfare-neutral animal is intended to anchor the zero point, but many respondents still exhibit positive valuations for the neutral animal (especially in means), indicating the relative measure may not isolate animal welfare from general aversion to the scenario (identity loss, being non-human, fear/disgust).
3. 3. Extreme attrition and selection: core median estimates use very small subsam-

- ples after exclusions (e.g., N 57/N 66), and excluding below-median confidence responses likely biases estimates upward (the paper notes low-confidence respondents often report zero/low WTA).
4. 4. Outlier handling for mean WTA is ad hoc (rank “10× jump” rule; effective €1,000,000 cutoff) and changes headline results by orders of magnitude; no pre-registered or theory-grounded robust estimation strategy is provided.
 5. 5. The WTA–WTP gap is enormous and central; interpreting WTA as the “right” welfare measure is not justified within a clear rights baseline or welfare-theoretic loss function, and the paper alternates between mean/median and WTA/WTP to motivate conclusions.
 6. 6. Construct validity: responses may reflect moral attitudes, social desirability, or misunderstanding rather than welfare judgments; the lack of correlation between representative-agent valuations and MWTP for cruelty-free meat suggests the measures are capturing different constructs or are noisy/unstable.
 7. 7. Statistical inference is limited: small effective samples, multiple tests, low power, and no multivariate modelling; key heterogeneity questions (knowledge, diet, engagement, comprehension) are left largely unresolved.
 8. 8. The per-kg conversion (animal valuation divided by assumed edible meat yield) is a key driver of the “chicken » beef” conclusion but is not stress-tested for alternative yield assumptions, production systems, or uncertainty in yields.
 9. 9. External validity is narrow (Flemish-speaking Belgian online panel, 18–65); nevertheless, results are used to motivate broad global/practitioner conclusions and compared numerically to other countries’ parameters.
 10. 10. The carbon-tax backfire calculation mixes contexts (Belgian welfare costs with US prices/elasticities/consumption) and relies on strong functional-form assumptions; uncertainty is not propagated, yet the policy recommendation (flat meat tax) is stated relatively strongly.
 11. 11. Reporting/replicability gaps: the manuscript does not (in the provided text) specify full questionnaire wording in an appendix, detailed data cleaning scripts, or provide data/code, limiting reproducibility and auditability.
 12. 12. Interpretation sometimes overstates what the data show: given sensitivity analyses and acknowledged unreliability, claims about welfare costs “likely” exceeding consumer surplus and climate costs should be framed as highly contingent scenarios rather than robust estimates.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

B.7 Clancy 2024

B.7.1 Ratings Comparison

Metric	GPT-5 Pro	GPT-5.2 Pro	Gemini 2.0 Flash	Human
--------	-----------	-------------	------------------	-------

Overall	73 [65–82]	68 [55–80]	75 [60–85]	76 [44–88] (n=2)
Claims	64 [55–73]	65 [50–78]	80 [70–90]	82 [43–90] (n=2)
Methods	58 [48–68]	60 [42–74]	70 [55–80]	70 [66–64] (n=3)
Adv. Knowledge	80 [72–88]	72 [55–85]	85 [70–95]	74 [50–94] (n=3)
Logic & Comms	85 [78–92]	78 [65–88]	80 [70–90]	74 [55–85] (n=3)
Open Science	43 [30–56]	50 [30–70]	60 [40–75]	80 [64–93] (n=3)
Global Relevance	84 [76–92]	86 [72–94]	90 [80–99]	75 [62–86] (n=3)
NA	—	—	—	2 [0–4] (n=1)
NA	—	—	—	4 [3–5] (n=1)
NA	—	—	—	70 [44–87] (n=3)

i GPT-5 Pro Assessment

This report develops a quantitative welfare model to assess the social returns to accelerating science in the presence of biotechnology-related catastrophic risks. Its strongest features are clear problem framing, transparent algebraic derivations, and an earnest attempt to parameterize key quantities with external evidence (notably from the Existential Risk Persuasion Tournament, XPT). However, several methodological choices, data transformations, and normative assumptions substantially drive the results and are only partially stress-tested. Core identification and modeling issues include: (1) Heavy reliance on a piecewise-constant “time of perils” hazard specification (zero additional hazard until a sudden discrete onset) with a single annual excess mortality rate d , and later a separate extinction hazard dx , conditional on remaining in the current epistemic regime. This discretization simplifies exposition but likely misrepresents the dynamics (e.g., gradual increases in capability and countermeasures, correlated shocks, feedbacks from AI progress outside academic science). (2) The conversion from XPT cumulative five-year event probabilities (for $>1\%$ mortality pandemics by 2030/2050/2100) to annualized hazards assumes year-to-year independence and stationarity within each interval, and then layers a further adjustment to “conditional on staying in the current epistemic regime.” This multi-step inversion is sensitive to assumptions: the annual probability of “exiting the regime” p is itself assembled from heterogeneous sources (AI-induced transformative growth probabilities, total extinction forecasts, and one macro TFP trend-break), summed and treated as a constant per-year hazard. This mixture is plausible but fragile; no formal uncertainty propagation is presented. (3) The paper adopts ad hoc severity distributions to translate event probabilities into expected mortality (e.g., conditional means of 2% for 1–10% events, 20% for 10–99.9%, extinction after perils only), and fixes the time of perils around 2037–2038; these choices are defensible as illustrative but not empirically grounded. Alternatives (e.g., heavier tails for severity; allowing pre-perils extinction probability >0 ; continuous hazard growth) are not explored beyond high-level discussion. (4) The “epistemic discount” p is used to weigh current-regime welfare versus an unmodeled future regime; in most of the paper p is held constant and assumed policy-invariant, then in Section 8 allowed to interact with extinction risk. The calibration $p=0.98$ (2%/year exit probability) is justified as conservative via a hand-built combination of TAI, non-bio extinction, and macro break risks. Given p heavily affects long-run benefits of science, this deserves probabilistic sensitivity analysis; currently the break-even thresholds and ROI estimates lack uncertainty bands. (5) The utility framework inherits Open Philanthropy’s normative assumptions: log utility in income plus a fixed 2 utils per healthy year. This yields large weight on health gains relative to income in

the simplified model, then substantially smaller in the “more realistic health” model. While the author acknowledges the moral/philosophical stakes, there is limited exploration of alternative welfare weights (e.g., different value-of-life-year, concavity differences, risk aversion), which could materially shift the ROI magnitudes and breakeven d .

(6) *Health modeling and demography*: the baseline model assumes a constant net population growth rate s , with pausing science inducing a permanent decrease $s \rightarrow s'$ via slower mortality declines; this compounds into very large long-run population differences, which dominate results. The author appropriately develops a more realistic health/population model (logistic survival improvement using US SSA cohort life tables mapped to global, constant births $\sim 130M$, diminishing marginal health returns, and transitory benefits from delay), but this relies on US data as a proxy for global survival curves and assumes equal exposure to the perils hazard across ages. The switch in models demonstrates sensitivity, yet parameter uncertainties (e.g., the 56% share of mortality reduction attributable to “science,” the 7-month set-back from a one-year pause, the age-invariant “unnatural” hazard) are large and not propagated.

(7) *Economic growth channel*: the mapping from pausing science to one-year slowdown in frontier growth and downstream diffusion uses $G=1\%$ per-capita frontier growth from science and $g=0.75\%$ (a 25% reduction) based on a stylized partition of technology into science-dependent classes (30% high-dependence slowed by 50%; 50% indirectly dependent slowed by 20%). This is consequential and only lightly justified; alternative shares or elasticities could materially change the income channel. The chosen lag $T=74$ years (20 years science $\rightarrow$ technology plus 54 years for global diffusion) is defensible from literature, but again treated as a point mass; the timing of benefits is crucial and would benefit from distributional sensitivity analyses.

(8) *Transformation of XPT forecasts*: the paper conditions XPT cumulative probabilities on “no regime exit” by 2100 using group-level medians and a constant p ; the XPT medians themselves hide wide dispersion (especially among domain experts) and small biosecurity expert sample size (~ 14). The paper qualitatively notes this and offers robustness via a “too late” case and immediate-onset case, but does not present stochastic uncertainty (e.g., Monte Carlo drawing from forecast distributions and severity priors).

(9) *Treatment heterogeneity* is largely absent: analysis focuses on “average science,” though risks plausibly depend nonlinearly on domain (e.g., some wet-lab or LLM-enabled bio-design work may raise d and dx more than metascience or physics). The paper discusses differential tech development and “racing to safety” qualitatively but quantitative scenarios where acceleration focuses on defense-relevant capabilities versus offense-prone areas are sketched only minimally (Section 9 reduces d by $\sim 0.26\%$ proportionally; effects are small mainly because d is small).

(10) The approach to extinction risk outside the current regime is structurally elegant (deriving an additional penalty term proportional to dx and the value of the next regime), but empirically unconstrained: (the value of the next epistemic regime in “population-years at today’s welfare”) is treated as a free normative parameter and then mapped to implied discount rates; the core policy conclusion—results hinge on whether one weights the far future at $\sim 10^2$ vs $\sim 10^4$ population-years—remains sensitive to debatable ethical priors.

(11) *Claims versus evidence*: the main quantitative claims are the ROI-to-cash multiples (e.g., $\sim 331\times$ in the simplified model without perils; $\sim 60\text{--}70\times$ in the more realistic health model; still positive under superforecaster- and domain-expert-implied d ; break-even $d \sim 0.13\text{--}0.15\%$), and qualitative policy guidance (“accelerate science but treat risk reduction as a separate objective”). These claims follow from the model but lack formal uncertainty intervals; they appear to be reasonably internally consistent but depend on stacked assumptions.

(12) *Replicability and openness*: the appendices provide clear formulas and parameter

values, but code for the Python model and the referenced spreadsheet for A3 calibration are not provided here; readers cannot reproduce the numerical outputs straightforwardly or test parameter sensitivity programmatically. To strengthen the work, the author could: provide the full code and data workbook; perform probabilistic sensitivity/uncertainty analysis over key parameters (p , T , G/g , s , science-attributable share of health, severity distribution, t_1 , and the conditionalization step); explore continuous hazard models and correlation structures; analyze domain-specific acceleration (defense-heavy vs offense-heavy fields); test alternative welfare weights and risk aversion; validate the health model by reproducing global life table patterns rather than US proxies; and add empirical checks on the assumed shares of science-dependent innovation flows and their slowdown under a pause. Overall, the paper is thoughtful, careful in exposition, and policy-relevant, but methodologically still exploratory with many calibrated moving parts and limited robustness/replication support. These issues motivate moderate ratings on methods and claims/evidence, higher on logic/communication and global relevance, and mid-low on open science.

i GPT-5.2 Pro Assessment

This report offers an explicit, quantitatively worked-through attempt to value “a year (or dollar) of science” while accounting for downside technological risks, focusing on advanced biotechnology. The manuscript’s core contribution is a transparent welfare accounting framework that decomposes benefits (income, health) and costs (a brought-forward “time of perils” with higher mortality; and, in an extension, extinction/civilization-collapse risk). The main vulnerabilities are not arithmetic mistakes but the extent to which results are driven by stacked structural assumptions, parameterizations that are only loosely tied to data, and strong functional-form choices that often dominate the sign and magnitude of conclusions. The report appropriately emphasizes sensitivity and produces multiple model variants, but many uncertainties are not well represented by the limited scenario analysis and point calibrations, so readers could easily over-interpret point ROI multiples (e.g., 60x, 300x) as decision-ready. A primary identification/interpretation issue is that the object being valued is ambiguous: the report alternates between “the average dollar of science today,” “a one-year pause in all science,” and “accelerating science via metascience,” and then implicitly treats these as comparable. However, marginal philanthropic acceleration is unlikely to resemble (i) a global one-year stop of science or (ii) the average productivity of baseline public+private basic research. Returns to inframarginal global science spending are not the same as returns to incremental metascience improvements, especially under diminishing returns, selection effects in research portfolios, and general equilibrium responses (crowding out/in, reallocation across fields, substitution toward private R&D, induced regulation, etc.). The report acknowledges some of this but still presents ratios like “70x” in a way that can be read as an actionable marginal ROI.

The welfare model relies heavily on a representative-agent approach, with global utility in a year defined as a fixed “2 utils” for a healthy life-year plus log income. This embeds a specific moral and empirical stance: (a) log income implies globally comparable marginal utility of consumption and strong curvature, (b) the “2 log points per DALY” normalization makes health very weighty relative to income, and (c) the model values future cohorts through an aggregate summation that is sensitive to population ethics. The author recognizes the “never born vs die” issue and introduces a variant with fixed births; that is a useful robustness check,

but it also undercuts comparability between model versions and leaves unresolved what moral stance the headline ROI is intended to represent.

Discounting and “epistemic regimes” are central and somewhat fragile. The paper motivates discounting primarily by the probability that historical trends cease to apply (transformative AI, catastrophe, etc.), setting $p = 0.98$ as a “conservative” annual survival of the current regime. But this treatment conflates at least three things: pure time preference, extinction/catastrophe risk, and model uncertainty/forecast invalidation. Collapsing them into a single constant hazard is analytically convenient but can misstate the relative weight on long-run consequences versus near-term hazards; importantly, the paper’s break-even thresholds and the extinction-risk extension are extremely sensitive to how the regime-change hazard interacts with science policy (which is often assumed away by construction, then later partially reintroduced). The “epistemic regime” construct also creates interpretive opacity: if the next regime’s value is assumed independent of science policy, the analysis essentially bakes in that science cannot materially affect transformative AI timing, alignment, or other macro transitions—yet these are exactly the channels many readers might believe dominate the long-run stakes.

The modeling of technological risk uses a step-function “time of perils” that increases annual mortality by d and may introduce an annual extinction probability d_x . This is a tractable abstraction, but it is a strong structural prior. Real risk likely changes gradually, is endogenous to institutions, and is path-dependent (information hazards, regulation, countermeasure build-up, and strategic interactions). The report later explores “immediate onset” and “too late” bounds, which is helpful, but the main results still rest on the assumption that a one-year change in science shifts the perils’ onset by exactly one year (and not, e.g., by weeks, or by zero if other drivers dominate). Also, the costs of “accelerating science” are treated largely as earlier arrival of bio-capability, while benefits have long lags (74 years baseline); this asymmetry is plausible but the chosen lag structure is itself a major driver and is only indirectly supported by evidence (patent citations, diffusion proxies). The diffusion calibration (frontier-to-global) via GDP and life expectancy gaps is inventive, but it is a coarse proxy: convergence in outcomes reflects institutions, trade, governance, and capital deepening in addition to science diffusion, and the mapping from these lags to marginal effects of additional science is uncertain.

A key evidential dependence is the conversion of Existential Risk Persuasion Tournament (XPT) forecasts into annualized mortality and extinction hazards. The pipeline involves multiple transformations: adjusting for “epistemic regime exit” using forecasts of extreme GDP growth and extinction; imposing a specific hazard shape (q_0, q_1, q_2 with a discrete t_1); annualizing 5-year event probabilities; and mapping “>1% killed” to assumed average severities (2%, 20%, 100%). Each step is plausible as a modeling choice, but together they create substantial model risk, and the final d and d_x should be viewed as highly distribution-sensitive. Moreover, the report treats XPT medians as “gold standard” defaults, but does not fully propagate uncertainty (within-forecaster uncertainty, between-group disagreement, dependence across questions, selection of biosecurity experts, calibration of superforecasters on this domain). The manuscript later argues for preferring superforecasters over domain experts using intersubjective accuracy and correlated pessimism; these arguments are interesting but not decisive and may themselves be confounded (e.g., “intersubjective accuracy” could correlate with social cognition rather than truth; cross-domain correlation could reflect a latent ‘risk realism’ factor rather than bias).

The “more realistic health” model is a meaningful improvement over constant population-growth compounding, but it introduces new concerns: it uses US Social Security cohort sur-

vival curves, maps global life expectancy to an earlier US cohort, and fits a logistic function with $\log(\text{birthyear}-1800)$ to impose diminishing returns. This is internally coherent, yet it hardcodes a long-run ceiling (age 120) and assumes medical progress trends resemble SSA projections for the US; global health progress and catastrophic disease dynamics may not follow this. It also focuses on mortality rather than morbidity (DALYs), despite earlier reliance on DALY-like valuations; this mismatch can bias the relative weight of health gains and the break-even d.

Finally, the report's "ROI relative to giving \$1 to a \$50k earner" is a clear communication device but a weak policy comparator. It is not the relevant alternative for most marginal funders (who face choices among targeted global health, biosecurity, AI safety, climate, etc.). The conversion also depends on the estimate of global basic research spending (~\$357B) and assumes linear scaling from a one-year global pause to per-dollar marginal spending effects, which is unlikely. Overall, the work is a thoughtful and unusually transparent applied modeling exercise with valuable decomposition and scenario logic; its main limitation is that headline quantitative conclusions are contingent on many contestable structural assumptions and parameter mappings, and uncertainty is understated relative to the real decision uncertainty about marginal interventions to "accelerate science."

i Gemini 2.0 Flash Assessment

The paper addresses the crucial and timely question of whether accelerating scientific progress is always beneficial, especially considering the potential for misuse of advanced technologies like synthetic biology. The author develops a quantitative economic model to compare the historical benefits of science against the forecasted costs of potential catastrophes, like civilization-ending pandemics. The analysis incorporates the long-run impacts of science on per capita income and health, integrating it into a utility framework inspired by Open Philanthropy's work. A key aspect of the model is the consideration of an 'epistemic regime', representing the predictability of the future based on historical trends, and the potential for exiting this regime due to transformative events. The paper leverages forecasts from the Existential Risk Persuasion Tournament (XPT) to estimate the probabilities of catastrophic events. The author compares superforecaster and domain expert estimates, noting significant discrepancies between them. A central finding is that the desirability of accelerating science depends heavily on the expected value of the long-run future and the assumed probabilities of extinction-level events. The paper also explores the potential for 'better science' to mitigate risks, finding that such improvements could significantly alter the calculus. The author ultimately leans towards preferring superforecaster estimates and advocates for a metascience policy that treats accelerating science and reducing risks as separate, but complementary, objectives.

However, several potential issues are worth noting. The model's reliance on specific utility functions and parameter values introduces sensitivity to these choices. The extrapolation of historical trends into the distant future, especially in the context of potential paradigm shifts, is inherently uncertain. The use of forecasts from the XPT, while representing some of the best available data, is subject to the limitations of expert elicitation and potential biases. The model's treatment of extinction risk, while sophisticated, involves assigning values to scenarios that are inherently difficult to quantify. The paper's analysis of 'better science' relies on assumptions about the effectiveness of metascience interventions in reducing specific risks. The model simplifies the complex dynamics of scientific progress, technological diffusion, and

societal responses to emerging threats. Replicability might be challenging due to the complexity of the model and the reliance on specific datasets. Overall, while the paper provides a valuable framework for analyzing the trade-offs associated with accelerating scientific progress, the conclusions are contingent on a number of assumptions and parameter choices, and may not fully capture the complexities of the issue.

⚠️ GPT-5.2 Pro: Key Issues Identified

1. 1. The analysis conflates (i) the average return to global science spending, (ii) the effect of a hypothetical one-year global pause in science, and (iii) the marginal impact of metascience/philanthropic acceleration; these are not equivalent under diminishing returns, portfolio composition, and general equilibrium responses.
2. 2. The “epistemic regime” discounting (constant annual p) combines forecast invalidation, catastrophic risk, and structural change into one hazard and assumes policy has no effect on regime transition in the baseline; conclusions about long-run benefits vs perils are highly sensitive to this construct.
3. 3. The “time of perils” is modeled as a discrete step increase with constant hazards (d , d_x) and a one-year shift from pausing/accelerating science; real risk likely evolves endogenously and may not be meaningfully shifted by marginal changes in science pace, especially if AI/engineering dominate capabilities.
4. 4. Mapping XPT forecasts into annual mortality and extinction hazards involves many strong transformations (conditioning on regime exit via GDP/extinction questions; imposing $q_0/q_1/q_2$ step form; annualizing 5-year probabilities; severity assumptions for $>1\%$ and $>10\%$ events); uncertainty from this pipeline is not propagated into results.
5. 5. The welfare function embeds strong normative commitments (log income, 2 log-points per healthy year, aggregation over persons and time) that materially drive results; the treatment of population ethics shifts across model versions (constant growth vs fixed births), affecting comparability and interpretation of the headline ROI.
6. 6. The baseline model’s health/population mechanism (a one-year pause permanently reduces population growth) likely overstates long-run effects; the “more realistic health” correction helps but introduces other hardcoded assumptions (US SSA survival curves, ceiling at age 120, diminishing returns via $\log(\text{birthyear}-1800)$).
7. 7. The choice and calibration of long lags (20 years science→invention plus ~54 years diffusion) rely on indirect proxies (patent citation patterns, GDP/life expectancy convergence) that conflate science with other determinants; results depend on these lags because costs are assumed near-immediate while benefits are delayed.
8. 8. The comparison metric (ROI vs giving \$1 to a \$50k earner) is not a policy-relevant counterfactual for most funders, and the conversion from a one-year global pause to per-dollar marginal effects assumes near-linearity and a correct denominator for “science spending.”
9. 9. The claim that non-extinction mortality risks from advanced biotech are too small to offset historical benefits hinges on assuming historical benefit trends persist and are attributable to ‘science’ at the stated shares (e.g., 56% of life expectancy gains); these attribution fractions are only weakly identified.

10. 10. The extinction-risk extension uses a reduced-form term for the value of the next regime (W) without a consistent treatment of discounting, population, and income dynamics across regimes; the resulting “break-even ” numbers can be misleadingly precise.
11. 11. Arguments for preferring superforecasters over domain experts (intersubjective accuracy, correlated pessimism, limited track-record evidence) are suggestive but not robustly validated; conclusions about whether to accelerate science under extinction risk remain epistemically fragile.
12. 12. Replicability is limited: the report describes a Python simulation and spreadsheet steps but does not clearly provide code/data artifacts, and many parameter choices involve manual judgment calls that would benefit from packaged reproducible workflows and probabilistic sensitivity analysis.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

B.8 Dullaghan and Zhang 2022

B.8.1 Ratings Comparison

Metric	Claude Sonnet 4	GPT-5 Pro	GPT-5.2 Pro	Gemini 2.0 Flash
Overall	35 [25–45]	72 [65–80]	55 [40–68]	55 [35–75]
Claims	40 [30–50]	68 [58–77]	52 [35–67]	65 [45–80]
Methods	30 [20–40]	58 [47–68]	42 [28–58]	50 [30–70]
Adv. Knowledge	45 [35–55]	70 [60–80]	63 [45–76]	60 [40–80]
Logic & Comms	60 [50–70]	77 [68–85]	75 [62–85]	75 [60–90]
Open Science	55 [45–65]	78 [70–88]	70 [50–83]	75 [60–90]
Global Relevance	65 [55–75]	80 [70–90]	82 [68–92]	80 [60–95]

i GPT-5 Pro Assessment

This report sets out to elicit and aggregate probabilistic forecasts about cultured meat production volumes through 2051, with the goal of informing philanthropic strategy in farmed animal welfare. The design—29 clearly specified questions spanning volume thresholds, price/volume pairs, species restrictions, and potential signposts (funding, researchers, input costs, retail presence, public opinion)—is helpful for operationalizing otherwise vague claims. The authors transparently characterize the work as a first-pass mapping of uncertainty rather than a definitive forecast; they disclose important limitations and errors, share anonymized forecasts and code, and discuss cruxes in a way that makes their reasoning auditable. The headline result (e.g., a 54% probability of never exceeding 100,000 metric tons annually by 2051) is likely to be influential for decision-makers; thus, the internal validity and robustness of the forecasting procedure merit close scrutiny. The most salient limitations concern sample size, forecaster

composition, anchoring, and aggregation choices. The paid panel comprises six individuals (five experienced generalist forecasters and one cultured meat scientist), each working up to ~15 hours, without observed interactions or iterative updating beyond a single round; while independence reduces herding, the very small N and limited domain expertise beyond one specialist create substantial risks of idiosyncratic bias and under-diversified model coverage. Selection procedures for the panel were documented, but there is no weighting by historical forecasting skill or domain calibration, despite skill heterogeneity being known to matter in probability aggregation. The chosen aggregation method (geometric mean of odds) is reasonable and widely used, yet there is no sensitivity analysis to alternative pooling rules (e.g., arithmetic mean of probabilities, extremized pools, performance-weighted pooling), no back-testing (not feasible here), and no assessment of internal coherence across nested questions beyond face validity. Several measurement and definitional issues could affect forecasts and downstream interpretation. First, the key >51% “true cultured meat” criterion was intended to be “by weight,” but this was not specified to forecasters; inconsistent interpretations (by weight vs. volume or composition) could have shifted some probabilities. Second, the “no transformative AI” condition was imposed but not operationalized; without a common definition, forecasters may have applied this conditional inconsistently. Third, an acknowledged error in the amino-acid cost question wording (including “recombinant proteins” and pointing to transferrin and insulin in the reference materials) plausibly biased some forecasters toward higher assumed media costs, which could propagate to lower production-volume forecasts; the report notes this but does not quantify the direction or magnitude of its impact on the aggregated production probabilities. Fourth, two “Manhattan project” questions lacked full results due to an error. Resolution criteria for some signposts (e.g., “impartial effective animal advocacy evaluators,” “credible industry reports” for retail offerings) may be contestable and could introduce subjectivity at resolution time; pre-registered, more objective criteria would improve replicability. The correlation analysis of beliefs between signposts and outcomes is underpowered and at risk of spurious inference: with only six forecasters, many pairwise correlations were tested, conventional p-values were used, and no multiple-testing adjustments were applied; the authors appropriately downplay substantive findings, but visual emphasis on “statistically significant” correlations may still mislead readers given the high false-positive risk in such a small-N, many-comparisons setting. Evidence-synthesis choices emphasize TEAs (especially Humbird 2020) and critical appraisal of consultancy projections; this is appropriate given the topic, and the report even identifies and documents an error in CE Delft (2021). However, given that most panelists are not domain experts, heavy anchoring on one TEA’s pessimistic inputs (e.g., sterility requirements, growth factor and amino-acid costs) without a structured expert-elicitation protocol to reconcile competing techno-economic models increases the chance of systematic bias; the qualitative section on cruxes is valuable but could be supplemented with structured parameter elicitations and explicit uncertainty propagation (e.g., Monte Carlo over media cost and cell-density assumptions) to map how key techno-economic uncertainties translate into volume probabilities. The question set wisely focuses on realized production volumes rather than claimed prices or nameplate capacity, mitigating subsidy and PR confounds; the inclusion of paired price/volume thresholds still provides a useful bridge to cost-competitiveness. Yet the time horizons (2031/2036/2051) and thresholds (100k, 1M, 10M, 50M tons) are somewhat arbitrary; a brief justification is given (philanthropic planning over decades vs. century-scale), but scenario coverage could be improved by including intermediate milestones (e.g., 250k–500k tons) and explicit hybrid-product trajectories (<51% cell

content), which the authors note will be addressed in a forthcoming tournament. The work is admirably transparent about sources, limitations, and potential updates that would change beliefs, and it provides concrete, action-relevant implications (e.g., prioritize long-term enabling infrastructure and research, consider alternatives to cultured meat in near-term animal welfare strategies). Still, key robustness checks are missing: no exploration of how results change under alternative aggregators; no re-aggregation excluding the amino-acid question error; no domain-expert-only vs. generalist-only comparisons; and no post-hoc monotonicity/coherence audits of aggregated probabilities across nested thresholds and years (though the presented tables appear monotone). Reproducibility of the analysis is reasonably supported by shared spreadsheets and scripts; however, full replication would require access to the same forecasters, which is not feasible. Overall, this is a thoughtfully scoped, well-communicated initial elicitation with strong transparency practices and clear utility for practitioners, but its inferential strength is constrained by small-N panel forecasting, some avoidable measurement/wording errors, limited sensitivity analysis, and an underpowered correlation exercise. Future iterations would benefit from a larger, mixed panel of domain and forecasting experts, performance-weighted or hierarchical aggregation, explicit pre-registration of resolution criteria, careful parameterized elicitations tied to TEA uncertainties, robustness checks to pooling assumptions, and a live tournament to incorporate news and score calibration over time.

i GPT-5.2 Pro Assessment

This manuscript reports an elicitation exercise: a small panel of forecasters (five highly ranked generalist forecasters plus one cultured-meat scientist) provided probabilities for 29 medium/long-horizon questions about cultured meat, especially whether annual global sales will exceed specified production-volume thresholds by 2031/2036/2051. The paper’s central empirical output is a set of aggregated probabilities (via geometric mean of odds) that generally imply low likelihood of very large production volumes by mid-century. The paper is valuable as a transparent, decision-oriented snapshot of uncertainty and as a corrective to consultancy-style projections that often lack public calculations. However, the evidential status of the reported probabilities is weaker than the narrative may suggest, mainly because the forecasts are not anchored in a formal quantitative model, the panel is extremely small and not demonstrably representative of relevant expertise, and several design/measurement issues introduce avoidable noise and potential bias.

The most important limitation is the elicitation design and sampling. With only six forecasters, the aggregate is highly sensitive to idiosyncratic priors, reference classes, and interpretation of question wording. The selection strategy (top Metaculus performers plus one domain scientist) plausibly improves calibration versus a convenience sample, but the manuscript does not present any calibration evidence specific to these forecasters on analogous long-horizon technology-scaling questions, nor does it justify that “highly ranked” on Metaculus translates to this domain and horizon. There is also no attempt to quantify between-forecaster uncertainty beyond reporting dispersion (e.g., hierarchical aggregation, uncertainty bands for the aggregate, robustness to leaving-one-out, or alternative pooling rules). As a result, readers cannot easily distinguish “panel disagreement” from “model uncertainty,” and the point probabilities risk being over-interpreted.

Question specification introduces additional ambiguity. The definition of cultured meat relies on “>51% produced directly from animal cells,” but the instructions did not specify whether

this is by weight, volume, calories, or revenue share; the authors note this post hoc, implying the resolved outcomes could be evaluated differently than some forecasters intended. Several questions are complex and contain multiple clauses (e.g., production volume “sold at any price within a continuous 12-month span before the end of year X”), which is precise but may be difficult to operationalize reliably given the likely opacity of global production/sales data. The manuscript partly mitigates this by referencing “credible sources,” but it does not pre-commit to a concrete resolution protocol or list of acceptable data sources and dispute procedures for the key production-volume endpoints, which are exactly the endpoints most likely to be noisy, proprietary, and politically contested.

A substantive measurement error is acknowledged in the amino-acids question: the inclusion of “recombinant proteins” and references to transferrin/insulin (not amino acids) could have biased forecasts upward. This is important because the manuscript interprets inputs (amino acids, growth factors, sterility requirements) as cruxes and potential levers; an error in a “crux variable” threatens the interpretability of correlations and qualitative synthesis. More broadly, the reference packet given to forecasters may have induced anchoring on particular techno-economic analyses (notably Humbird 2020) or on highly uncertain company claims about costs, without a structured debiasing or “consider the opposite” protocol.

The aggregation method—geometric mean of odds—is reasonable as one pooling rule for probabilities, but the manuscript does not justify why this is preferable here (e.g., versus linear opinion pools, performance-weighted pools, extremizing, or hierarchical Bayesian aggregation), nor does it show sensitivity of results to the pooling rule. Given the tiny N and heavy-tailed individual beliefs, this choice can materially affect the headline numbers. Similarly, the manuscript presents correlation and “statistical significance” claims across forecast questions, but with only six forecasters the effective sample size for cross-person correlation is extremely small, and p-values from conventional tests are not meaningful without strong assumptions. Additionally, many pairwise correlations are examined; without multiple-comparisons correction, the reported “significant” correlations are likely to include false positives. This weakens the evidential basis for claims about which “signposts” co-move with production expectations. Interpretively, the paper is careful in places (“describing uncertainty and models, not justifying any specific probability”), but it still drifts into strategic conclusions (e.g., recommending long-term support vs near-term nudges) that are only indirectly supported by the elicitation. The elicitation measures beliefs about outcomes, not the marginal effect of interventions by EA funders. Without a causal model linking philanthropic actions to the probability of crossing production thresholds, it is hard to move from “likely low volumes” to “there may be more benefit in ensuring long-term support” versus “shift to other alternatives,” because both recommendations depend on leverage, tractability, and counterfactuality rather than only on baseline probabilities. The manuscript also notes (but does not analyze) a crucial welfare assumption: that cultured meat displaces conventional production rather than adding supply, reshaping demand, or affecting animal numbers via indirect market responses.

Despite these issues, the manuscript has notable strengths: it clearly defines decision-relevant thresholds, distinguishes volume from price as an indicator, explicitly compares against conventional meat scale, elicits conditional questions (e.g., consumer willingness conditional on some production), and transparently lists forecasts and (apparently) provides data/scripts for replication. The discussion of “cruxes” (innovation efficiency, fundamental constraints, reference classes, reliance on TEAs) is useful for structuring disagreement and future forecasting tournaments. Overall, this is a thoughtful applied forecasting report with good transparency

and framing for practitioners, but its quantitative outputs should be treated as a preliminary, high-uncertainty elicitation rather than robust estimates; the smallest set of improvements that would substantially raise credibility would be (i) tighter resolution criteria and unambiguous definitions, (ii) a larger and more systematically sampled panel (including multiple domain experts), (iii) pre-registered aggregation and analysis (including sensitivity checks), and (iv) either a formal quantitative scaling model or explicit baselines/reference-class growth-rate calculations integrated into the forecasts.

i Claude Sonnet 4 Assessment

This paper examines cultured meat production forecasts through 2050 using expert elicitation from a panel of forecasters. The work addresses an important policy-relevant question about alternative protein development and animal welfare impacts. However, several methodological limitations undermine the reliability of the findings. The forecasting panel is small (6 people) with limited domain expertise - only one cultured meat scientist among mostly Metaculus forecasters. The aggregation method (geometric mean of odds) lacks justification and may not be optimal for this context. Key technical assumptions show inconsistencies, such as incorrectly including recombinant proteins in amino acid cost questions, which the authors acknowledge may have biased estimates upward. The paper's heavy reliance on the Humbird (2020) techno-economic analysis, while understandable given limited alternatives, creates potential for systematic bias if that analysis contains errors. The 'cruxes' analysis relies primarily on qualitative interpretation rather than rigorous quantitative decomposition of disagreement sources. Reference class comparisons (solar, biotech, etc.) are mentioned but not systematically analyzed with proper causal frameworks. The work lacks robustness checks, sensitivity analysis, or validation against external benchmarks. While the extensive documentation of forecaster reasoning and the comprehensive question set are valuable, the methodological weaknesses limit confidence in the specific probability estimates presented. The paper is more useful as an exploratory mapping of uncertainty than as definitive forecasts for decision-making.

i Gemini 2.0 Flash Assessment

This report focuses on forecasting the production volume of cultured meat through 2050, utilizing a panel of paid forecasters and analyzing various factors influencing its development. The central question revolves around the potential for cultured meat to significantly impact the food supply and whether the Effective Altruism (EA) community should invest in accelerating its growth. The authors develop forecasting questions around cultured meat production volume targets and hypothesized signposts of progress, considering factors like funding, research, input costs, and public support. They aggregate probabilities from the forecasting panel and analyze correlations between different inputs and production volumes.

Strengths:

- **Clear Focus:** The report clearly defines its research question and objectives, focusing on the specific issue of cultured meat production volume.
- **Comprehensive Approach:** The authors consider a wide range of factors influencing cultured meat production, including technological, economic, and social aspects.

- **Expert Consultation:** The report refers to TEAs, commentary, and discussion with scientists. This suggests an engagement with existing literature as well as primary research. In particular, it seems they are able to identify methodological problems in existing work.
- **Transparent Methodology:** The authors clearly describe their methodology, including the selection of forecasters, the formulation of questions, and the aggregation of probabilities.
- **Nuanced Analysis:** The report acknowledges the uncertainties and complexities involved in forecasting technological advancements and provides a nuanced analysis of the factors driving the forecasts.
- **Practical Implications:** The report discusses the implications of the forecasts for the EA community and suggests potential strategies for maximizing the impact of investments in cultured meat.
- **Open Science:** The report makes anonymized forecasts available, which increase the accessibility of the research.

Weaknesses and Limitations:

- **Forecasting Panel Limitations:** The size and composition of the forecasting panel may limit the generalizability of the results. As well, it seems that the forecasters were not, on average, domain experts. This raises concern about the validity of the probability aggregation.
- **Reliance on Expert Opinion:** The reliance on expert opinion and techno-economic analyses (TEAs) introduces subjectivity and potential biases into the forecasts. Furthermore, the TEAs themselves may be biased or inaccurate. I am concerned because these analyses are highly uncertain, but may drive the forecasts.
- **Uncertainty Quantification:** The report acknowledges uncertainty, but the methods used to quantify it (e.g., geometric mean of odds) may not fully capture the range of possible outcomes. The use of geometric mean of odds is not standard, and is not well explained. It is unclear that this method is a reasonable approach to aggregating expert forecasts.
- **Limited Scope:** The report primarily focuses on production volume and does not delve deeply into other important aspects of cultured meat, such as consumer acceptance, regulatory hurdles, and environmental impact.
- **Correlation vs. Causation:** The report identifies correlations between different inputs and production volumes, but it does not establish causal relationships. As the author states, “Analyzing the correlation of production volume predictions and predictions on the other inputs (media prices, public funding, biotech funding, and researchers) does not reveal much.” I am not sure this analysis adds much value.
- **Temporal Scope:** The report relies on data up until 2022, which means some more recent developments are necessarily excluded. The long-term forecasting nature of the study means this is more likely to introduce error.

Potential Improvements:

- **Expand the Forecasting Panel:** Increasing the size and diversity of the forecasting panel could improve the robustness of the results.

- **Incorporate Sensitivity Analysis:** Conducting sensitivity analysis on key assumptions and parameters could help to better understand the uncertainty surrounding the forecasts.
- **Develop Causal Models:** Developing causal models to explain the relationships between different inputs and production volumes could provide more actionable insights.
- **Address Scope Limitations:** Broadening the scope of the analysis to include other important aspects of cultured meat could provide a more comprehensive picture of its potential impact.

Overall Assessment:

Overall, this report presents a valuable contribution to the understanding of cultured meat production and its potential impact. The authors employ a rigorous methodology and provide a nuanced analysis of the factors influencing the forecasts. However, the limitations of the forecasting panel, the reliance on expert opinion, and the limited scope of the analysis should be considered when interpreting the results. The authors acknowledge the limitations appropriately and provide a balanced view on the opportunities and challenges for cultured meat production. I believe the authors should have focused less on identifying correlations between variables, as the analysis did not provide meaningful insights. I also think that the method of aggregating the probability estimates is not well justified.

The paper might be improved by incorporating other possible viewpoints such as more recent developments in the cultured meat industry. More importantly, the authors should justify or omit the use of geometric means of probabilities for aggregation.

⚠️ GPT-5.2 Pro: Key Issues Identified

- 1) Very small and potentially unrepresentative panel (n=6) for long-horizon technology-scaling forecasts; no demonstrated calibration for this domain/horizon, and no quantification of aggregate uncertainty or sensitivity to individual forecasters.
- 2) Ambiguous key construct: “>51% from animal cells” was not specified as weight/volume/calories/value; this can change both forecasters’ interpretations and eventual resolution, especially for hybrid products.
- 3) Resolution and measurement risk for global production/sales volumes is high (proprietary data, inconsistent reporting); the paper does not fully pre-commit to a detailed resolution protocol and authoritative data sources for the main endpoints.
- 4) Acknowledged wording/reference error in the amino-acids input-cost question (mixing “recombinant proteins” and non-amino-acid examples) plausibly biases an important ‘crux’ variable and complicates interpretation of correlations and narratives.
- 5) Aggregation choice (geometric mean of odds) is not justified or stress-tested against alternative pooling rules (linear pool, extremizing, performance weighting, leave-one-out), despite the tiny panel and potentially influential outliers.
- 6) Correlation and “statistical significance (p<0.05)” claims are not credible with n=6 and many tested pairs; multiple-comparisons and model assumptions are not addressed, risking spurious inferences about which signposts matter.
- 7) The report mixes descriptive forecasts with strategic recommendations, but does

not model counterfactual philanthropic leverage; probabilities of outcomes do not directly imply which interventions are best.

8. 8) Heavy reliance on a few techno-economic analyses as anchors without a structured approach to combining first-principles engineering constraints with reference-class growth; this may over-weight specific assumptions (e.g., sterility standards, media costs).
9. 9) Some questions are compound and cognitively demanding (multiple thresholds, time windows, price conditions, species restrictions), increasing noise and interpretation variance across forecasters.
10. 10) Key welfare mechanism (degree of displacement of conventional meat, rebound effects, market expansion) is acknowledged but not analyzed, limiting direct usefulness for animal-welfare impact estimation.
11. 11) The “conditional on no transformative AI” assumption is underspecified and may complicate interpretation; the paper does not discuss how sensitive conclusions are to broader macro/tech scenarios (policy shocks, pandemics, geopolitical shifts).

GPT-5 Pro reasoning trace available in the [online HTML version](#).

B.9 Frech et al. 2023

B.9.1 Ratings Comparison

Metric	GPT-5 Pro	GPT-5.2 Pro	Gemini 2.0 Flash	Human
Overall	70 [60–80]	44 [30–58]	60 [40–75]	40 [22–58] (n=2)
Claims	66 [55–76]	38 [24–55]	65 [50–75]	45 [28–65] (n=2)
Methods	58 [45–68]	35 [22–52]	60 [40–70]	62 [48–82] (n=2)
Adv. Knowledge	62 [50–72]	50 [35–65]	70 [55–80]	55 [42–68] (n=2)
Logic & Comms	78 [68–88]	62 [48–74]	75 [65–85]	75 [62–88] (n=2)
Open Science	45 [35–60]	33 [18–52]	40 [25–55]	100 [100–100] (n=2)
Global Relevance	82 [70–90]	73 [60–85]	80 [70–90]	65 [54–85] (n=2)
NA	—	—	—	2 [1–3] (n=1)
NA	—	—	—	2 [1–3] (n=2)
NA	—	—	—	65 [54–85] (n=2)

i GPT-5 Pro Assessment

This paper frames pharmaceutical innovation as a global public good and operationalizes each country’s “contribution” as the quasi-rents from brand-name drug sales: revenues above marginal production and distribution costs. The authors synthesize public-goods and alliance theories (Samuelson; Olson–Zeckhauser) with simple Nash-style models to explain why the United States bears a disproportionate share of support for global R&D and why contribu-

tions by the rest of the world (ROW) are positive but smaller. Empirically, they combine 2018 OECD-country price indices for brand-name originator drugs from RAND with country totals and brand shares to construct country-level contributions under two marginal-cost (MC) assumptions: (i) a lower bound using Turkey’s relative price index (14.2% of the US price index), and (ii) 24% of the US net price (derived from generic-entry price declines adjusted for US rebates). They find the US accounts for about two-thirds to three-quarters of total OECD contributions; many ROW countries contribute well above zero; and contributions scale strongly with national GDP, consistent with “exploitation of the large by the small.” The theory-to-empirics linkage is clear and the analysis is policy-relevant; however, there are important measurement and identification caveats that constrain the strength of inference. The central empirical construct— $(P_i - MC)/P_i$ multiplied by brand revenues S_i —relies on strong simplifications. MC is treated as a single constant across countries and molecules, expressed as a fixed fraction of the US price index. This ignores international heterogeneity in supply-chain and distribution margins, taxes, regulated pharmacy markups, and confidential rebates/discounts—factors that differ systematically by country and class. The authors acknowledge this and offer two MC scenarios, but both remain US-anchored and country-invariant; the lower bound (Turkey’s price index) is likely too low (different baskets, distribution costs), while 24% (CEA-based) generates negative “contributions” for a few countries (Turkey, Greece, Estonia), signaling misspecification rather than true below-MC pricing. A more thorough sensitivity analysis (e.g., MC ranges by country-income group or distribution-cost adjustments) and uncertainty propagation would strengthen credibility. The reliance on aggregate price indices raises compositional concerns. RAND’s indices depend on matched baskets and weighting choices; applying $(P_i - MC)/P_i$ at the index level implicitly assumes a homogeneous mix across countries and that index arithmetic aligns with average margins. If the therapeutic mix differs meaningfully, the contribution measure may embed composition bias. Relatedly, the S_i revenues (total pharma sales $\times$ brand share) may be measured at different price “levels” (ex-manufacturer vs retail) across countries; if contributions use ex-manufacturer MC but retail-level S_i , this would overstate quasi-rents. Clarifying the pricing level for both components and aligning them would reduce measurement error. The cross-country regressions relate $\log(\text{contribution})$ to $\log(\text{GDP})$, with very high R^2 ($\sim 0.85\text{--}0.90$) and elasticities >1 . While suggestive of the Olson–Zeckhauser “exploitation” pattern, this relationship is close to mechanical: contributions scale with spending and price, both of which scale with GDP and population. Using nominal US\$ contributions alongside real PPP GDP from Penn World Table further complicates interpretation; unit mismatches can influence elasticities. The inference that elasticities >1 constitute decisive evidence for exploitation is weakened when country dummies (US, Japan, Switzerland, Germany) are included; precision falls and the elasticity’s excess over 1 is less robust. A more discriminating test would examine contribution shares (contribution/GDP) vs GDP, contributions per capita vs GDP per capita, or decompose contributions into price- and volume-driven components with controls for pricing institutions (reference pricing, external reference pricing breadth, HTA cost-effectiveness thresholds), disease burden and age structure, insurance coverage, and launch-lag/reimbursement stringency. The bargaining/value-based pricing discussion is informative, yet the empirical section does not directly connect institutional features to contribution levels beyond a few industry-dummy controls. The theoretical model makes simplifying assumptions (homogeneous drugs, identical per-capita valuations, constant R per drug, interior/corner solutions), which are fine for exposition but limit direct empirical map-

ping. The paper argues marketing is complementary to R&D and thus part of the quasi-rent that supports the public good; however, not distinguishing marketing from R&D/profit may overstate the portion of rents that truly finances innovation. Some marketing may be business-stealing rather than socially informative. The claim that US prices are, on average, below US willingness-to-pay per QALY and thus even the US underpays is plausible but rests on broad literature estimates; a quantitative welfare simulation linking contributions to innovation elasticities (e.g., number and quality-adjusted new drug approvals) would substantively strengthen policy conclusions. Robustness is limited. Beyond the two MC scenarios, the paper does not propagate uncertainty from price indices, brand shares, or rebates into contribution estimates or into the GDP elasticities. Alternative specifications are mentioned (linear rather than logs; adding out-of-pocket shares and cost-effectiveness thresholds) but not fully documented (sources, variable definitions, full outputs). Code and data processing are not shared; replication is feasible from public sources but would be facilitated by posted scripts. Despite these limitations, the core findings are credible in direction: (i) ROW contributions are meaningfully above zero and thus free riding is partial, not complete; (ii) the US contributes a dominant share; and (iii) larger economies contribute disproportionately in absolute terms. The paper's value lies in reframing the pricing debate toward contributions to a global public good, quantifying country-level contributions with transparent, if stylized, assumptions, and empirically illustrating the size-contribution gradient. To improve, the authors could: (1) align price levels (ex-manufacturer vs retail) and adjust for confidential rebates outside the US; (2) model MC as a distribution varying by country-income group and therapeutic area and provide wide sensitivity bands; (3) analyze contributions per capita and as shares of GDP, and explicitly decompose contributions into volume and price components; (4) incorporate institutional covariates (reference-pricing intensity, HTA thresholds, reimbursement inclusion rates, launch lags) to test bargaining/value-based hypotheses; (5) link contributions to innovation outcomes using published elasticities to simulate welfare effects of counterfactual ROW price increases; and (6) release code and detailed data construction. Overall, the work is clearly argued, timely, and policy-relevant, with informative empirical patterns, but its quantitative conclusions depend on simplifying assumptions about costs and prices that merit more thorough sensitivity and institutional controls.

i GPT-5.2 Pro Assessment

This manuscript frames cross-country differences in branded pharmaceutical prices as differences in national “contributions” to a global public good (pharmaceutical R&D), using public-goods/alliance theory (Samuelson/Lindahl; Olson-Zeckhauser; Bergstrom-Blume-Varian) and an empirical implementation based on OECD price indices from a RAND report. The core empirical construct is “contribution” = $(\text{price} - \text{marginal cost}) / \text{price} \times \text{branded-originator sales}$, interpreted as quasi-rents that fund/incentivize innovation. The most important limitations stem from (i) the mapping from observed revenues to innovation incentives, and (ii) the mechanical nature of the main empirical relationship. First, the paper equates contemporaneous quasi-rents on all branded-originator sales in 2018 with a country's contribution to global innovative effort, but the incentive-relevant object is expected, risk-adjusted global profit streams for marginal R&D projects (and for particular therapeutic areas) rather than the single-year cross section of markups on a broad stock of products at mixed life-cycle stages. This construction also blurs profits vs. quasi-rents (e.g., it does not net out fixed

costs beyond marginal cost, such as regulatory compliance, pharmacovigilance, and country-specific commercialization costs), does not address the division of surplus along the supply chain (manufacturers vs. wholesalers/pharmacies/PBMs), and does not incorporate how revenues translate into R&D budgets (which may be weakly linked at the margin in the short run). Second, the paper’s headline empirical finding—log contributions are strongly explained by log GDP with elasticity > 1 —appears partly mechanical because the dependent variable is built from drug sales that are themselves strongly increasing in country size/income (and likely scale more than proportionally with GDP due to health spending elasticities, demographic structure, and pricing institutions). In other words, regressing a spending-based construct on GDP will almost inevitably yield very high R^2 ; this does not, by itself, discriminate among the paper’s proposed strategic models (interior Nash contributions, corner solutions/free-riding, bargaining, or “value-based pricing”), nor does it identify whether observed pricing patterns are equilibrium responses to public-good incentives versus other determinants (market power, insurance design, reference pricing, regulation, launch sequencing, exchange rates, therapeutic mix, and measurement artifacts in cross-country price indices).

A major identification threat is that the empirical analysis is essentially descriptive and does not attempt to isolate variation plausibly attributable to the public-goods mechanism. The paper interprets positive estimated “contributions” in most countries as evidence against the common claim that ROW prices are near marginal cost, but this inference depends crucially on the marginal cost assumption and on what “near MC” means in practice. The two MC proxies used are both fragile: (1) taking Turkey’s price index as MC assumes (a) the same drug basket/quality and (b) negligible markups in Turkey; but Turkey’s low index could reflect different product mix, distribution costs, exchange-rate/PPP issues, lower willingness-to-pay, regulated margins, and/or delayed access to newer drugs. (2) deriving MC as 16% of US list price from generic-entry erosion, then “correcting” by a single average rebate factor, assumes the generic price converges to marginal cost, that average erosion generalizes to the branded-originator basket used by RAND, and that rebates/discounts translate proportionally; all are contestable. The paper’s own results show that small changes in MC can flip “contributions” negative for some countries, which highlights that the level estimates are not robust. In addition, assuming a common MC across countries is especially problematic because their MC includes distribution and other downstream costs that can vary substantially with wages, geography, channel structure, and regulation; yet those are precisely the elements that can differ across OECD markets.

The measurement of prices and revenues is another key vulnerability. The RAND report provides price indices designed for international comparison, but the manuscript’s transformations (inverting indices relative to the US; combining with total sales and “brand share”) implicitly assume that the price index aligns with the revenue base used for sales and that the brand-originator spending shares are comparable across systems. Differences in formulary coverage, launch lags, negotiated confidential discounts, and the composition of “brand-name originator” baskets can produce index differences that are not simply “the same drugs priced differently.” This matters because the paper interprets price differences directly as differences in contributions to innovation incentives. The analysis also ignores quantities/access: if low-price countries have reduced availability or delayed launch, then lower spending may reflect restricted access rather than “free-riding,” and the welfare implications change.

On the theory side, the paper provides a standard public-goods exposition and useful intuition (interior vs corner equilibria, exploitation of large by small), but the link from the stylized

model to the constructed empirical measure is underdeveloped. In the model, “contribution” is a choice variable of a country (e.g., an implicit tax/subsidy for R&D), whereas in practice prices are outcomes of complex bargaining and regulation interacting with heterogeneous demand, supply, and institutional design. The paper discusses bargaining and value-based pricing, but the empirical section does not attempt to test between them (e.g., using measures of health technology assessment stringency, reference pricing regimes, launch timing, or bargaining power). The “exploitation hypothesis” test (elasticity > 1) is also not a clean test of the Olson–Zeckhauser mechanism because it conflates (a) demand for pharmaceuticals and health spending with (b) intentional strategic free-riding on R&D.

In terms of contribution and relevance, the paper addresses a policy-relevant question (why US branded prices are higher; how global R&D may be financed) and offers a coherent narrative challenging “full free-riding.” However, because the empirical strategy is largely accounting-based and the key quantities are sensitive to strong assumptions, the evidentiary strength for the policy recommendation (press ROW countries to pay higher prices to move toward a global optimum) is limited. A more compelling contribution would require either stronger structural grounding (e.g., a bargaining model mapped to observables) or quasi-experimental evidence linking cross-country price changes to R&D outcomes, plus a welfare framework that treats access, equity, and differential pricing explicitly. Replicability is moderate-to-low: the data sources are public, but the paper does not provide code, detailed extraction steps, or diagnostics (influence/leverage, alternative baskets, or uncertainty propagation from the MC assumptions).

Gemini 2.0 Flash Assessment

This paper examines pharmaceutical pricing and R&D through the lens of global public goods theory, arguing that the current system leads to suboptimal R&D investment. The authors challenge the notion that ROW (Rest of World) countries are entirely free-riding on US pharmaceutical R&D, presenting empirical evidence that these countries do contribute, albeit less than the US. They develop a theoretical framework based on Nash non-cooperative models and Lindahl pricing to analyze international price differences and their impact on global R&D. The authors use data from the RAND report to estimate countries’ contributions to the global public good, measured as the proportion of sales revenue exceeding marginal costs. They find that GDP size is a significant determinant of national contributions, suggesting that larger economies contribute more. The paper conducts regression analysis and sensitivity tests to support its claims. A key issue is that US policies and higher drug prices incentivize pharmaceutical innovation, a global public good, but ROW’s cost-minimization objectives lead to lower contributions. The core argument is that the US overpays, while ROW underpays, for branded pharmaceuticals, resulting in a global underinvestment in R&D. This work is limited by the use of aggregate data and simplifying assumptions, such as identical marginal costs across countries and a focus on OECD nations and new branded drugs. Internal consistency is a concern, as the theoretical models might not fully align with the empirical analysis and the measures of ‘contribution’ are not perfect. The study also assumes that pharmaceutical R&D has major global benefits. The robustness of the findings to alternative model specifications is not fully explored, and the policy implications, while mentioned, are not deeply analyzed. The paper could benefit from discussing alternative policy instruments and their effectiveness. Additionally, there are replication barriers because the source of the cost estimates used is

not clearly explained and the steps taken to ensure integrity of data are not mentioned.

⚠️ GPT-5.2 Pro: Key Issues Identified

1. 1) The dependent variable (“contribution” = quasi-rents on all branded-originator sales in a single year) is not clearly the incentive-relevant object for R&D, which depends on expected global profit streams for marginal projects; life-cycle stage, risk, and therapeutic heterogeneity are ignored.
2. 2) The main empirical result (log contribution strongly explained by log GDP with elasticity > 1) is plausibly mechanical because the contribution measure is built from drug spending/sales, which naturally scales with GDP; this does not identify public-goods strategic behavior.
3. 3) Marginal cost (MC) is weakly identified: using Turkey’s low price index as MC and using generic-entry price erosion (then adjusting by an average rebate) both rely on strong assumptions and can materially change levels and even signs of contributions.
4. 4) Assuming a common MC across countries is particularly problematic because the paper’s MC includes distribution and downstream costs that vary across systems (wages, geography, pharmacy/PBM margins), confounding cross-country markups.
5. 5) Cross-country price indices may reflect differences in drug mix, launch timing, reimbursement/coverage, and confidential discounts; interpreting them as “same products, different prices” risks mismeasurement of both prices and implied quasi-rents.
6. 6) The analysis treats lower spending/markups as free-riding, but does not incorporate quantity/access differences (e.g., launch lags, reimbursement refusals), which could mean lower prices coincide with reduced consumption of newer drugs.
7. 7) Theoretical constructs (Lindahl optimum; Nash interior/corner equilibria; bargaining) are not mapped to testable implications beyond the GDP correlation; the empirical work does not discriminate among the proposed models.
8. 8) The “exploitation hypothesis” test (elasticity > 1) is not a clean test of Olson–Zeckhauser in this setting because it conflates general health/pharma spending elasticities with strategic under-contribution to R&D.
9. 9) The construction of national branded-originator revenues from total sales $\times$ brand share assumes comparability across countries and alignment with the RAND price index basket; potential inconsistencies are not audited.
10. 10) Outlier sensitivity and leverage are not thoroughly assessed (the US and Japan dominate totals; negative/zero contributions are dropped in logs), which can affect coefficient estimates and inference in small samples.
11. 11) The policy conclusion that ROW countries should pay more is not supported by a welfare calculation that explicitly weighs dynamic innovation gains against static access/affordability losses and distributional concerns under differential pricing.
12. 12) Replication barriers remain: no shared code, limited detail on data wrangling/transformations from the RAND tables, and no uncertainty propagation for MC assumptions or index measurement error.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

B.10 Green et al. 2025

B.10.1 Ratings Comparison

Metric	Claude Sonnet 4	GPT-5 Pro	GPT-5.2 Pro	Gemini 2.0 Flash	Human
Overall	72 [65–78]	80 [70–88]	72 [58–84]	50 [30–70]	57 [37–71] (n=2)
Claims	75 [68–82]	82 [72–90]	73 [58–86]	60 [40–75]	60 [40–78] (n=2)
Methods	78 [72–84]	74 [62–84]	68 [52–82]	55 [35–70]	32 [20–46] (n=2)
Adv. Knowledge	70 [62–77]	77 [67–86]	76 [62–88]	50 [30–70]	55 [28–82] (n=2)
Logic & Comms	74 [68–80]	85 [76–92]	80 [66–91]	70 [60–80]	60 [44–78] (n=2)
Open Science	82 [76–87]	91 [84–97]	82 [65–93]	75 [60–85]	80 [73–90] (n=2)
Global Relevance	68 [60–75]	86 [76–94]	86 [74–95]	70 [50–85]	89 [62–96] (n=2)
NA	—	—	—	—	4 [3–4] (n=2)
NA	—	—	—	—	2 [2–3] (n=2)
NA	—	—	—	—	89 [62–96] (n=2)

GPT-5 Pro Assessment

This meta-analysis asks which interventions most durably reduce meat and animal product (MAP) consumption, restricting to randomized controlled trials (RCTs) with behavioral outcomes measured at least one day after treatment and adequate sample size. The dataset comprises 35 papers, 41 studies, 112 interventions, and about 87,000 participants. The main finding is a small pooled standardized mean difference (SMD = 0.07, 95% CI [0.02, 0.12], 0.082). Publication-bias adjustments (selection model and a “worst-case” non-affirmative-only analysis) bring the pooled estimate near zero. Studies aiming specifically to reduce red and processed meat (RPM) show larger effects (SMD = 0.25, 95% CI [0.11, 0.38]), but substitution to other animal products is unmeasured, so impacts on net MAP remain unclear. Effects are consistently small across theoretical approaches (choice architecture, persuasion, psychology, combined persuasion+psychology), with only two choice-architecture RCTs meeting criteria. Strengths include: (i) a well-motivated focus on behavioral outcomes beyond immediate post-intervention measures, addressing compensation risks; (ii) explicit minimum power criteria; (iii) use of robust variance estimation (RVE) to handle dependent effect sizes from multi-arm designs; (iv) heterogeneity-aware interpretation (e.g., percentages of true effects above thresholds); (v) publication-bias diagnostics using both selection modeling and a conservative non-affirmative-only estimate; and (vi) strong computational reproducibility (R Markdown, containerization, publicly archived code and data). The narrative is balanced, avoids overclaiming, and situates results against prior reviews that relied more on non-randomized designs or attitudinal outcomes.

Key limitations, several of which could materially affect precision or small differences in estimated magnitude, are as follows. First, inclusion criteria and search: three inclusion decisions were made after data collection began (separating RPM studies as a distinct analytic category; excluding studies designed to induce substitution from one MAP type to another; excluding interventions with no participant agency, such as meals served without choice). These are defensible but should ideally be pre-specified or accompanied by sensitivity analyses using alternate inclusion rules to show robustness. The search strategy is multi-pronged and appro-

priate for an interdisciplinary topic with grey literature, but the review appears limited to English-language sources and does not state that a protocol was preregistered; both factors raise risk of omission or selection flexibility. A PRISMA diagram is referenced, with details in the Supplement, but explicit protocol registration and dual-review screening would further strengthen credibility.

Second, data extraction and coding were conducted by the first author alone. Given the complexity of effect-size conversions and moderator coding across diverse designs and measures, the lack of dual coding/inter-rater reliability checks increases risk of extraction or classification error. The paper codes open-science features (pre-analysis plan, open data), but does not report inter-coder agreement or independent verification for study characteristics, risk-of-bias assessments, or effect-size calculations.

Third, outcome selection and measurement: the rule to take the longest follow-up per intervention is conceptually consistent with the paper's emphasis on sustained behavior change, but it introduces heterogeneity in follow-up lengths across studies and may attenuate impacts relative to immediate measures. Sensitivity analyses using alternate selection rules (e.g., pre-specified primary outcomes if declared, or earliest vs. longest follow-up) are not reported. Many outcomes are self-reported and sometimes coarse (e.g., binary reductions), which invites social desirability bias and reduces measurement precision. The paper highlights these issues but does not provide subgroup analyses by measurement type (objective vs. self-report), which would be informative for robustness and practical inference.

Fourth, effect-size computation: the authors primarily use Glass's Δ standardized by the control-group standard deviation; they refer to the "pre-treatment control group standard deviation," but many included RCTs may not have baseline measures—clarification would help. When effect sizes must be converted from other statistics (regression coefficients, z , z), the formulas are standard, but conversion complexity increases the risk of error. A notable, idiosyncratic decision is coding "unspecified null" results as $SMD = 0.01$. Even though small, this positive value could bias pooled estimates and publication-bias diagnostics upward; sensitivity analyses setting these to 0 or excluding them would be advisable.

Fifth, handling of clustering and dependence: the dataset includes both individually randomized and cluster-randomized trials (minimum ten clusters). It is not explicit whether intraclass correlation (ICC) adjustments were used when studies did not report cluster-corrected variances. If cluster-induced variance inflation is not accounted for, standard errors could be underestimated and cluster trials overweighted. RVE appropriately handles dependence from multiple arms sharing a control and applies a small-sample correction, but it does not fix misspecified within-study variances. A sensitivity restricting to individually randomized trials or to cluster trials with ICC-adjusted SEs would bolster robustness.

Sixth, modeling, bias assessment, and moderators: RVE is appropriate for multiple dependent effects, and the heterogeneity metrics are well used. However, the selection-model publication-bias analysis (implemented in metafor) typically assumes independence; it is not clear whether a single effect per study was used for that analysis or whether dependence was otherwise addressed. The funnel plot and non-affirmative-only approach are helpful complements, but an explicit sensitivity using one effect per study for bias modeling would clarify robustness. Moderator analyses are usefully cautious (no causal interpretation claimed), but some strata are very small (e.g., only two choice-architecture studies; adolescents); pooled subset estimates can be unstable. A reporting anomaly appears in Table 2 ("Multi-region" $SMD = 0.21$ with $CI [0.21, 0.21]$), implying zero SE; this likely reflects a coding or reporting issue and should

be checked.

Seventh, interpretation and policy relevance: the “unsolved problem” conclusion is supported by the small pooled effect and near-zero bias-corrected estimates. The larger average effects for RPM-targeted interventions are clearly documented, but the key question of substitution to other animal products is not directly measured in eligible studies; the text notes this appropriately while at times speculating that substitution is likely. A focused review of trials measuring multiple categories (if any) or a clearer statement that none qualify would sharpen the evidence base. For practitioners, translating SMDs into more interpretable units (e.g., proportion of meals containing MAP, grams/day) and reporting absolute differences where possible would enhance decision utility. External validity is limited by the preponderance of university/cafeteria contexts and high-income regions; the paper acknowledges this.

Overall, this is a careful, well-executed synthesis that raises the evidentiary bar by prioritizing randomized designs, behavioral outcomes with at least short-run persistence, and rigorous bias diagnostics. The primary vulnerabilities affecting precision are single-extractor coding, some post-hoc inclusion rules, potential under-adjustment for clustering in effect-size variances, the independence assumption in one of the bias models, and the unusual treatment of unspecified nulls. None seem likely to overturn the qualitative conclusion of small average effects on net MAP, but they justify conservative uncertainty around the exact magnitude and some subgroup contrasts. The open-science and computational reproducibility practices are exemplary, and the paper makes a valuable, policy-relevant contribution by clarifying how much weaker the evidence looks once stricter criteria are imposed and by highlighting important gaps for future RCTs (e.g., price changes, defaults with longer-run outcomes, objective consumption tracking, and substitution effects).

i GPT-5.2 Pro Assessment

This manuscript conducts a theory-guided meta-analysis of randomized controlled trials (RCTs) aiming to reduce meat and animal product (MAP) consumption, requiring (i) randomized assignment, (ii) consumption outcomes measured at least one day after treatment begins, (iii) minimum sample/cluster sizes, and (iv) a pure no-treatment control. The core finding—an overall pooled SMD around 0.07 with sensitivity analyses suggesting results near zero under plausible publication bias—follows logically from the dataset and analytic approach, and the paper is unusually explicit about how relaxing rigor thresholds increases estimated effects. The strongest contributions are (a) focusing on behavioral outcomes rather than intentions/attitudes, (b) trying to guard against short-lived “one-meal” effects and compensatory behavior by requiring at least a one-day lag, (c) handling dependence from multi-arm studies using robust variance estimation (RVE), and (d) incorporating multiple publication-bias sensitivity approaches alongside a “worst-case” subset analysis. The manuscript is also relatively candid about the limits of moderator/meta-regression inferences and about weaknesses in the underlying primary studies (self-report, coarse outcomes). That said, several design and inference choices materially affect credibility and interpretation.

The most consequential threat to the quantitative conclusions is outcome and effect-size construction. The authors extract “one outcome per intervention,” defined as the net MAP (or RPM) measure with the longest follow-up. This choice is defensible to prioritize durability, but it is also a strong form of outcome selection that can change effect estimates systematically (e.g., longer follow-ups often attenuate effects). The paper does not present a parallel analysis

using alternative decision rules (e.g., primary outcome as declared by the study, pre-registered primary endpoint when available, or averaging across follow-ups with a multivariate model), nor does it quantify how sensitive results are to the “longest follow-up” rule. Closely related, the meta-analysis uses standardized mean differences across a mix of outcome types (self-reported servings, frequency counts, binary reduction indicators, purchase data, etc.) with multiple conversion pathways (from regression coefficients, z-scores, eta-squared), sometimes using Glass’s Δ and sometimes pooled SDs when control SDs are unavailable. This heterogeneity in outcome scaling and conversion can introduce additional measurement error and non-comparability, especially when effects are very small (SMD 0.07) and thus sensitive to modest conversion differences. The paper does not clearly quantify the share of estimates requiring non-preferred conversions, nor does it provide sensitivity analyses restricting to “highly comparable” outcome types (e.g., objective purchasing/administrative outcomes only, or continuous consumption quantities only).

A particularly important decision is how “insufficient information” nulls are handled: if the text reports a null but there is not enough information to compute an SMD, the authors code an “unspecified null” as 0.01. This imputation can bias the pooled estimate in either direction depending on the true underlying effects, but because the overall estimated effect is itself close to zero, even small systematic coding choices matter. Moreover, imputing 0.01 rather than 0 (or multiple-imputation over plausible ranges, or excluding such effects while checking robustness) embeds an arbitrary positive bias. The paper should more fully justify this choice, report how many effects are “unspecified nulls,” and show robustness to alternative codings (0, ± 0.05 , or sampling from an uncertainty distribution).

Study selection and coding procedures are another major source of uncertainty. The search process is extensive and tailored (reviews, citations, author outreach, Google Scholar terms, AI search, and nonprofit databases), which is a strength for gray literature coverage. However, it is less transparent than a conventional database-query systematic review: the exact queries, screening protocol, and inter-rater reliability are not described in detail in the main text, and inclusion/exclusion decisions were “primarily made by the first author.” Single-coder extraction increases risk of missed eligible studies, inconsistent coding, and subtle biases in borderline decisions. The paper also reports that three inclusion decisions were made after data collection began (e.g., defining RPM as a separate category, excluding “substitution-inducing” studies, excluding interventions leaving no room for voluntary choice). Some of these are conceptually sensible, but post hoc eligibility refinements can create researcher degrees of freedom unless clearly preregistered or accompanied by sensitivity analyses that show how conclusions change under alternative inclusion rules.

The classification into four theoretical approach categories is useful descriptively, but the inferential use of those categories is limited by sparse data (especially choice architecture: only 2 studies, 3 estimates) and by non-exclusive categorization (studies can occupy multiple categories). The authors appropriately avoid certain meta-regressions because of category overlap, yet they still present pooled estimates by theory and by persuasion type that could be over-interpreted by readers as comparative performance. Given the wide CI for choice architecture (driven by tiny k), the manuscript’s narrative claims about the relative evidence base for choice architecture should be framed even more cautiously: the result is less “choice architecture is (in)effective” and more “few qualifying long-lag RCTs exist under these definitions.”

The publication-bias analyses are a strong feature, but they also rely on assumptions that are only partly met in this literature. Selection models and significance-based methods pre-

sume selection on statistical significance and sign; yet much of the included evidence may be in gray/nonprofit reports where publication incentives differ, and where selective reporting could occur at the outcome/analysis level rather than the study level. The “worst-case” non-affirmative subset is informative as an extreme bound, but it is not a realistic model of reporting when analysts can choose among multiple outcomes and follow-ups. Relatedly, the paper’s risk-of-bias assessment appears relatively coarse (self-report vs objective; preregistration/open data; publication status). It does not fully engage with standard RCT bias domains (allocation concealment, attrition/missingness, differential measurement, compliance, clustering issues), which could help interpret heterogeneity and small effects.

Interpretation-wise, the conclusion that “meaningfully reducing MAP consumption is an unsolved problem” is plausible given the small pooled SMD and the sensitivity analyses, but it risks overstating what the meta-analysis can establish. First, SMDs do not map cleanly to policy-relevant units (grams/week, meals/week, CO₂e, cost-effectiveness). A small SMD could still correspond to a meaningful absolute reduction in some contexts (e.g., cafeteria purchase defaults) or could be highly cost-effective if cheap and scalable. Second, the inclusion criteria (pure controls, minimum lags, voluntary choice) intentionally focus on a subset of interventions; that is valuable for internal validity but may exclude structural or supply-side interventions that are practically important even if they reduce “choice.” Third, the RPM vs MAP distinction is an important conceptual advance, but the manuscript’s discussion of substitution remains speculative because most RPM studies do not measure non-RPM MAP; readers should be reminded that the 0.25 pooled effect pertains to RPM outcomes only, not net MAP.

Overall, the paper is a credible, well-motivated synthesis that pushes the field toward behavioral outcomes, longer follow-up, and more conservative inference. Its main vulnerabilities are (i) outcome/effect-size construction choices (especially imputed nulls and longest-follow-up selection), (ii) single-coder screening/extraction with post hoc eligibility refinements, and (iii) limited comparability across heterogeneous consumption measures. These issues do not negate the central message that durable behavior change effects are, on average, small under current evidence—but they widen uncertainty about the exact magnitude, about which intervention classes might work in specific settings, and about how to translate SMDs into decision-relevant impact estimates.

i Claude Sonnet 4 Assessment

This meta-analysis examines interventions to reduce consumption of meat and animal products (MAP) through a systematic review of 35 papers comprising 41 randomized controlled trials with 112 interventions and approximately 87,000 subjects. The study employs rigorous inclusion criteria requiring RCTs with at least 25 subjects per group, measurement of consumption outcomes at least one day post-intervention, and pure control groups. The authors find a small overall pooled effect (SMD = 0.07, 95% CI: [0.02, 0.12]), indicating limited effectiveness of existing interventions. Key methodological strengths include comprehensive search strategy across multiple sources, robust statistical analysis using robust variance estimation methods, and careful attention to publication bias through multiple sensitivity analyses. The study makes an important distinction between interventions targeting all MAP versus only red and processed meat (RPM), finding larger effects for RPM-focused interventions (SMD = 0.25) but noting the critical limitation that these studies don’t measure potential substi-

tution to other animal products. The authors appropriately acknowledge that their stricter methodological criteria likely explain smaller effect sizes compared to previous reviews that included studies with weaker designs or non-behavioral outcomes. However, several limitations compromise the findings: relatively small sample of eligible studies limits statistical power, many outcome measures remain coarse or self-reported raising social desirability concerns, the meta-regression analyses cannot establish causal relationships between study characteristics and outcomes, and the fundamental challenge of measuring long-term dietary behavior change persists. The theoretical categorization into four approaches (choice architecture, persuasion, psychology, and combined) is reasonable but somewhat arbitrary, and the paper would benefit from clearer discussion of mechanisms underlying the small observed effects.

i Gemini 2.0 Flash Assessment

This meta-analysis investigates the effectiveness of interventions aimed at reducing meat and animal product (MAP) consumption. It synthesizes evidence from 35 papers, encompassing 41 studies and 112 interventions, with a total sample size of approximately 87,000 subjects. The authors categorize interventions into four main strategies: choice architecture, persuasion, psychology, and a combination of persuasion and psychology. The primary outcome is the standardized mean difference (SMD) in MAP consumption after intervention. The authors found that the pooled effect of all interventions is small (SMD = 0.07, 95% CI: [0.02, 0.12]). Interventions targeting red and processed meat (RPM) consumption showed larger effects (SMD = 0.25, 95% CI: [0.11, 0.38]). The study acknowledges limitations, including relatively few studies meeting stringent methodological criteria and potential for substitution effects (reducing RPM but increasing other MAP). The authors also address potential publication bias and conduct sensitivity analyses. The analysis is well-structured, with a clear explanation of methods and a detailed exploration of potential moderators, like study population, region, publication era, and delivery method.

However, several issues are apparent.

First, the analysis relies on studies with heterogeneous methodologies and outcome measures, potentially affecting the validity of the pooled effect sizes. While the authors used robust variance estimation (RVE) to account for dependence between observations, it is not clear how effectively this addresses the underlying heterogeneity in study design and measurement. Relatedly, some of the included studies used self-reported consumption data, which is susceptible to social desirability bias and recall errors. Moreover, the meta-analysis relies on effect sizes extracted by a single author, raising concerns about potential extraction bias and the need for independent verification. A lack of information on the specific types of interventions used and their implementation challenges limits the practical applicability of the findings.

The literature review is adequate but doesn't delve deeply into the theoretical underpinnings of the interventions or explore the nuances of behavior change strategies. The discussion section could benefit from a more in-depth examination of the implications of the findings for policy and practice. For instance, the authors could discuss the potential for combining different intervention strategies or tailoring interventions to specific populations.

Furthermore, the sensitivity analyses, while mentioned, are not thoroughly discussed, and their impact on the overall conclusions is not fully clear. The authors acknowledge limitations related to study-level moderators, but they do not explore potential interactions between moderators or conduct subgroup analyses to identify specific populations or contexts where

interventions are more effective. Finally, the authors could provide more detailed recommendations for future research, including specific areas where more rigorous evaluation is needed and suggestions for improving the design and measurement of interventions. Given these limitations, there is a lot of uncertainty around the main results.

GPT-5.2 Pro: Key Issues Identified

1. 1) Outcome selection rule: extracting only the longest-follow-up consumption outcome per intervention is a consequential choice that may systematically shrink estimates and is not accompanied by sensitivity analyses using alternative rules (e.g., prespecified primary endpoints or multivariate follow-up modeling).
2. 2) Imputation of “unspecified nulls”: coding outcomes reported as null but lacking sufficient statistics as $SMD=0.01$ is arbitrary and can materially affect a pooled effect this close to zero; the count of such cases and robustness to alternative codings are not clearly presented in the main text.
3. 3) Heterogeneous outcome constructs: pooling across very different consumption measures (binary self-report reduction, frequency counts, purchasing data, servings) and multiple effect-size conversion pathways risks non-comparability and additional error; limited sensitivity analyses restrict to more homogeneous outcome types.
4. 4) Single-coder screening and extraction: inclusion/exclusion decisions and data extraction were primarily conducted by one author, increasing risk of missed studies, inconsistent coding, and bias; inter-rater reliability is not reported.
5. 5) Post hoc eligibility refinements: several inclusion decisions were made after data collection began (e.g., RPM category definition and exclusions for “substitution-inducing” or low-agency interventions), creating researcher degrees of freedom unless accompanied by preregistration or systematic robustness checks.
6. 6) Sparse evidence for choice architecture under inclusion criteria: only 2 studies (3 estimates) qualify, yielding extremely wide uncertainty; narrative comparisons across theoretical categories can be over-interpreted despite low k .
7. 7) Limited risk-of-bias assessment: the coded bias indicators (self-report/objective, preregistration/open data, publication status) do not fully address common RCT bias domains (attrition, allocation concealment, contamination, compliance, clustering/ICC handling).
8. 8) Publication-bias modeling assumptions: selection models and significance-based approaches assume selection on $p < 0.05$ and sign, but selective reporting may occur at the outcome/analysis level and incentives differ across gray vs journal literature; this complicates interpretation of “bias-corrected” means.
9. 9) Dependence and weighting: RVE is appropriate for multi-arm dependence, but results may still be sensitive to how multiple interventions from the same study are represented; additional checks (e.g., per-study aggregation under alternative models) would strengthen robustness.
10. 10) Limited translation to decision-relevant units: effects are reported mainly as SMDs, with minimal mapping to absolute reductions (e.g., meals/week, grams, CO₂e) or cost-effectiveness, making “meaningfully reducing” hard to interpret for practitioners.
11. 11) RPM vs MAP inference gap: the larger RPM pooled effect is clearly identified,

but substitution to other MAP is unmeasured in most RPM studies; discussion of substitution remains speculative and should be framed as a key evidence gap rather than an implied conclusion.

12. 12) Generalizability constraints: requirements for pure controls and voluntary choice, while improving interpretability, may exclude policy-relevant interventions (institutional defaults/portioning) and thereby limit external validity for real-world implementation contexts.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

B.11 McGuire et al. 2024

B.11.1 Ratings Comparison

Metric	GPT-5 Pro	GPT-5.2 Pro	Gemini 2.0 Flash	Human
Overall	83 [75–90]	72 [60–82]	75 [60–85]	86 [80–92] (n=2)
Claims	78 [70–86]	70 [56–82]	80 [70–90]	89 [82–96] (n=2)
Methods	76 [65–84]	66 [50–78]	70 [55–80]	88 [82–92] (n=2)
Adv. Knowledge	82 [72–90]	74 [58–86]	70 [60–80]	88 [80–92] (n=2)
Logic & Comms	88 [80–94]	78 [66–88]	85 [75–90]	80 [75–88] (n=2)
Open Science	62 [50–75]	58 [40–72]	70 [50–85]	83 [76–88] (n=2)
Global Relevance	90 [82–96]	88 [75–95]	80 [70–90]	93 [90–95] (n=2)
NA	—	—	—	4 [4–4] (n=2)
NA	—	—	—	93 [90–95] (n=2)

GPT-5 Pro Assessment

This is a comprehensive, decision-focused cost-effectiveness evaluation of two lay-delivered psychotherapy providers in sub-Saharan Africa, combining three evidence streams (a systematic meta-analysis of LMIC psychotherapy RCTs; charity-related causal studies; and charities’ M&E pre-post data) with multiple internal/external validity adjustments, household spillover estimates, and explicit sensitivity analyses. The report’s strengths include: a large, carefully constructed LMIC psychotherapy evidence base with multilevel meta-analysis and time-decay modeling; an explicit risk-of-bias screen; transparent treatment of extreme outliers; a publication-bias adjustment; conversion of SD-year effects to WELLBYs; explicit external-validity adjustments (group vs individual, lay delivery, dosage); a reasoned, if tentative, spillover model; and extensive robustness checks and GRADE-based quality ratings. The writing is clear, assumptions are declared, and major uncertainties are foregrounded. The authors also conduct site visits and engage with external reviewers. Relative to typical applied research, this is unusually thorough. However, several core choices are necessarily judgmental and introduce identification and robustness risks.

Key issues and potential threats: 1) Weighting of evidence sources. The combination of “Bayesian-informed” weights (based on statistical uncertainty) with subjective adjustments (largely informed by GRADE ‘relevance’) is reasonable but ultimately ad hoc. The M&E pre-post series borrows duration from the general meta-analysis, creating dependence between sources. A unified hierarchical Bayesian framework or formal data-fusion approach could reduce arbitrariness and more transparently propagate uncertainty.

- 2) Long-term follow-ups. The handling of a few very long-run follow-ups (4 effect sizes) is critical. The authors exclude these from the core model, then multiply the total effect by 1.54 to represent a 50–50 weighting between models with and without the long-term data. This is a pragmatic but ad hoc fix; results may be sensitive to a small number of studies with atypical follow-up horizons and to unobserved attrition or selective reporting. A more formal sensitivity model (e.g., leave-one-/leave-subset-out, attrition-adjusted decay priors) or pre-registered decision rule would strengthen credibility.
- 3) Publication bias adjustment. Averaging across multiple methods (including some known to perform poorly under high heterogeneity) provides a central factor of ~ 0.69 for the general meta-analysis, but the choice to average methods—and the chosen set—remains judgmental. Given multilevel dependence and time-moderation, only one method natively fits the modeling structure; triangulation is sensible, but additional design-based checks (p-curve sensitivity, selection models fit to subsets, robustness to method exclusion) would be valuable. The practical effect seems conservative overall, but the approach is still somewhat arbitrary.
- 4) Dosage adjustment. The log-based dosage factor $\ln(\text{attended}+1)/\ln(\text{intended}+1)$ is plausible (given diminishing returns) and more defensible than linear proportionality, but still heuristic. The dose–response moderator in the meta-analysis was imprecise, and attendance data are sparse. For Friendship Bench, extremely low attendance (mean 1.12/6 sessions) drives a large discount and is the largest uncertainty. The authors cite single-session therapy literature and the structure of PST to argue non-negligible effects at low dosage; however, direct dose–response evidence in this population and modality is underdeveloped. More detailed dose–response studies or instrumental-variable analyses exploiting operational variation would help.
- 5) Spillovers. The 16% household spillover ratio combines (i) a naïve average of two higher-quality RCTs’ spillover estimates ($\sim 12\%$) and (ii) a pathways-based synthesis ($\sim 21\%$) that adds non-RCT evidence and demographic weighting. The authors explicitly label this “very low” quality and impose a wide assumed distribution in Monte Carlo (95% CI 0–50%). Still, spillovers account for about one-third of total effects, so better identification (e.g., RCTs powered on household outcomes, pre-registered plans, longitudinal sibling/partner modules) would substantively reduce uncertainty.
- 6) Pre-post pseudo-synthetic control. Using pooled control trajectories from other RCTs on the same scales to adjust pre-post M&E results is innovative but fragile. Control groups differ (often “enhanced usual care”), populations are not matched, and parallel-trends are untested. The authors down-weight these estimates and apply a replication discount and response-bias factor, but the construction remains speculative. Validation against trials with both between- and within-person data would be informative.

- 7) Country-of-origin adjustment (Iran). The model includes an Iran indicator and down-weights the intercept for studies from that country due to unusually large effects and general concerns. While this likely moves in the right direction, it risks overcorrection if context or program mix legitimately explains higher effects. A pre-specified country- or region-level random-effects structure, leave-one-country-out checks, or covariate adjustment for study characteristics (beyond a country dummy) would reduce reliance on a coarse binary.
- 8) SWB/MHa/WELLBY mapping. Converting mixed MHa/SWB SD-year effects into WELLBYs via a fixed global Cantril Ladder SD=2 is consistent with the authors' framework and supported by their claim that MHa does not overestimate SWB effects. Still, construct validity and scale comparability in these specific LMIC populations remain an open question; additional crosswalk validation—especially within the same samples—would bolster confidence.
- 9) Heterogeneity and uncertainty propagation. Heterogeneity is substantial in the general meta-analysis (I^2 0.15). The authors opt against prediction intervals, arguing the use case is not a new-study prediction but a charity-level effect after external validity adjustments. Nonetheless, many adjustments (e.g., replication, response bias, dosage, adult/adolescent uplift) are applied as fixed scalars with limited uncertainty modeling. Broader probabilistic treatment of adjustment parameters could widen credible intervals and better reflect true uncertainty.
- 10) Inclusion and interpretation of higher risk-of-bias FB trials. Two of the four FB-related RCTs are overall rated “high” RoB, yet included with justification that the main bias likely attenuates effects and that these trials are highly relevant. This is reasonable but unconventional; sensitivity analyses that re-weight by RoB level or replace those estimates with bounds would mitigate concerns about upward bias.
- 11) External validity moderators. Adjustments for group format and lay delivery rely on moderator coefficients from the LMIC meta-analysis. These may be confounded by unobserved differences (e.g., modality mix, program intensity, comorbid conditions). Explicit multivariable moderator models, shrinkage, or regularization and transparent robustness of these coefficients would help.
- 12) Costs and counterfactuals. Cost per client is transparently computed from budgets and clients treated; for partners, the adjustment for counterfactual activity reduces cost-effectiveness appropriately. Details on partner counterfactual estimation and uncertainty are mostly relegated to an appendix; more explicit checks (e.g., historical baselines, difference-in-differences on partner sites) would strengthen this component.
- 13) Reproducibility and data access. The analysis is implemented in R with an external appendix and a summary spreadsheet, but code availability is not clearly linked here and the M&E datasets are private, limiting full replication. The meta-analytic dataset appears carefully extracted and double-checked; public release of code and de-identified extraction sheets would materially improve reproducibility.

Net assessment: The report is methodologically sophisticated and unusually transparent for applied charity evaluation, with many sensible corrections and honest acknowledgement of

fragile steps. Several core elements (long-term effects, dosage, spillovers, evidence weighting) remain judgment-heavy and should be treated as provisional. The main claims—that both programs look substantially more cost-effective than cash transfers under preferred assumptions, and remain so under most but not all plausible alternatives—are fairly represented and subjected to sensitivity tests. Further high-quality, directly relevant RCTs (especially for adult group IPT in current StrongMinds operations), better dose–response evidence, improved spillover identification, and a more formalized weighting/combination framework would substantially increase confidence. Overall, this is a strong and useful contribution to wellbeing-centric cost-effectiveness assessment, with clearly signposted uncertainties and room for methodological tightening.

i GPT-5.2 Pro Assessment

This report is an unusually detailed, decision-oriented cost-effectiveness evaluation of two large-scale lay-delivered psychotherapy programs in Africa, expressed in WELLBYs per \$1,000. Its strongest features are (i) a broad systematic review/meta-analysis backbone (84 RCT “interventions”, multilevel meta-analysis, multiple follow-ups/outcomes), (ii) explicit separation of evidence sources (general RCT evidence, charity-related RCT evidence, charity monitoring pre–post), (iii) an unusually explicit catalogue of internal/external validity “adjustments”, and (iv) extensive sensitivity/robustness reporting. However, the headline cost-effectiveness claims rely on a chain of modeling and judgment calls where several key steps are only weakly identified by data, are highly assumption-dependent, and sometimes combine heterogeneous constructs in ways that can readily shift results by multiples. The most important vulnerabilities are: (1) the construction of “total effect over time” from a linear decay model and a triangular area-under-the-curve formula, with ad hoc handling of a small number of influential long-term follow-ups; (2) the conversion from standardized effects (SD-years) on mixed mental health and subjective wellbeing instruments into WELLBYs via a single global SD mapping; (3) the use of non-causal charity pre–post data corrected by a “pseudo-synthetic control” derived from control arms of other trials, which is not a standard identification strategy and risks substantial residual bias; (4) subjective weighting across the three evidence sources, justified as an “unsolved problem,” which is central to the final point estimates and especially consequential for StrongMinds given one low-effect RCT; and (5) the inclusion of household spillovers based on very sparse evidence, acknowledged as “very low” quality, yet contributing a large fraction (~30–40%) of total benefits.

On the general evidence: the report’s systematic review/meta-analysis appears carefully executed (double extraction checks, RoB assessment with two rounds, multilevel modeling to address dependence across outcomes/follow-ups), and the decision to trim extreme effect sizes ($g > 2$) and exclude “high” RoB studies is defensible for a decision report. That said, the effect-size distribution is highly heterogeneous and the report introduces several strong filters/adjustments with limited external validation: (i) outlier removal by a hard threshold ($g > 2$) can interact with small-study effects and can change perceived heterogeneity; (ii) an “Iran indicator” is used to downweight large effects from one country based on suspicion of bias and broader concerns about paper integrity—this may be prudent, but it is an unusually strong, partly conjectural adjustment that could also remove genuine contextual differences; and (iii) publication bias correction is implemented by averaging multiple methods (some known to behave poorly under heterogeneity), which is transparent but still not clearly cali-

brated to the estimand given multilevel dependence and time-moderation.

The total-effect-over-time modeling is a major driver of cost-effectiveness. The baseline assumption is linear decay until the effect hits zero, then integrating the triangle. While this is a reasonable first approximation, psychotherapy effects plausibly decay nonlinearly, may plateau, or may show maintenance/booster dynamics; the linear model can over- or underestimate tails. The handling of a small set of very long follow-ups is particularly influential: the report excludes four long-term effect sizes from the core time model but then applies a bespoke multiplicative “time adjustment” (1.54) intended to represent a 50/50 compromise between models with and without these points. This is candidly described as lacking clear precedent. Because duration multiplies into total WELLBYs, this decision can materially change cost-effectiveness and should be considered a structural uncertainty, not merely a robustness check.

The charity-specific evidence is treated thoughtfully but remains the largest substantive uncertainty. For Friendship Bench, the major concern is dosage: the charity’s reported mean attendance of 1.12 sessions out of up to 6 drives a large downward adjustment via a chosen log dose-response function. The justification that early sessions may be especially therapeutic/psychoeducational is plausible, but the exact mapping from attended sessions to effect size is weakly evidenced, and the $\log(+1)$ functional form is essentially a normative prior. For StrongMinds, the core tension is between a single low-effect RCT (implemented via a partner on adolescents with low compliance and pandemic overlap) and much larger non-causal M&E pre-post improvements; the report argues for limited relevance of the RCT and gives it 20% weight, but this is ultimately a judgment. The additional upward “relevance” adjustments to the low-effect RCT (compliance LATE uplift, adult-adolescent adjustment from an external database, partner-type adjustment) are internally coherent, yet increase dependence on assumptions about what differs between study and current operations.

The M&E pre-post component is both potentially valuable and methodologically fragile. The proposed “pseudo-synthetic control” uses control-arm change from other LMIC psychotherapy trials on similar scales to subtract spontaneous remission and related trends. This does not ensure exchangeability between charity clients and those external control arms (different baselines, recruitment, measurement contexts, regression-to-the-mean dynamics, seasonality, interviewer effects, and differential survey incentives). The report partially acknowledges this by giving low weights and applying large generic discounts (replication 0.51 and response bias 0.85), but these are themselves not estimated in the charity context and could either overcorrect or undercorrect.

Finally, the overall conclusion that these programs are $\sim 5\text{--}6\times$ as cost-effective as cash transfers depends on the WELLBY framework choices: the assumed mapping from SD-years to WELLBYs (using a global SD 2 on a 0–10 ladder), treating affective mental health scales and SWB scales as exchangeable (supported by a separate report rather than directly validated here), and including household spillovers with wide uncertainty. These choices are defensible in an applied welfare context but should be interpreted as producing a “model-based decision metric” rather than a tight empirical estimate. Overall, the report is a strong example of transparent, impact-oriented synthesis, but its quantitative point estimates should be taken as moderately uncertain and sensitive to a few non-standard modeling/elicitation decisions that are only partially anchored by data.

Gemini 2.0 Flash Assessment

This paper presents an in-depth cost-effectiveness analysis of two charities, Friendship Bench and StrongMinds, which deliver lay-delivered talk psychotherapy in Africa. The core metric is the cost-effectiveness of interventions and charities based on their impact on subjective wellbeing, measured in terms of wellbeing-adjusted life years (WELLBYs). The analysis uses three sources of evidence: (1) a meta-analysis of 84 randomized controlled trials (RCTs) of psychotherapy in low and middle-income countries (LMICs); (2) RCTs of programs related to the charities (4 for Friendship Bench and 1 for StrongMinds); and (3) Monitoring and Evaluation (M&E) pre-post data from the charities themselves. The authors estimate the initial effect and duration, adjust the total effect to account for internal and external validity concerns, and estimate household spillovers. They then calculate a final effect estimate for each charity by combining the three estimates from different evidence sources, using informed subjective weights, and calculate the cost-effectiveness. Several updates were performed in this version of the analysis, including the extraction of additional small sample studies, a second risk of bias evaluation, the inclusion of a recent working paper, an updated weighting system, the addition of charity monitoring and evaluation (M&E) pre-post results as an additional source of evidence, site visits, and updated program implementation details. The authors assess the overall quality of evidence using a stringent GRADE-adapted criteria. Key limitations include the very low attendance/dosage with Friendship Bench, the fact that the only RCT of the StrongMinds programme (Baird et al., 2024) is only partially relevant, and the subjectivity involved in weighting the different evidence sources. They find both charities are cost effective, though this depends greatly on what weighting one gives to the Baird et al paper.

GPT-5.2 Pro: Key Issues Identified

- 1) Total-effect-over-time relies on a linear decay assumption and triangular area-under-curve; psychotherapy effects may decay nonlinearly or have long tails, making total WELLBYs highly assumption-sensitive.
- 2) Handling of influential long-term follow-ups is ad hoc (exclude from core time model, then apply a 1.54 multiplicative compromise); this is a structural modeling choice without clear precedent and materially affects results.
- 3) The WELLBY conversion (SD-years $\rightarrow$ WELLBYs via a single “SD=2 on 0–10 ladder” mapping) and combining affective mental health (MHa) with SWB outcomes embed normative/construct assumptions that can shift cross-intervention comparisons.
- 4) Charity M&E pre–post effects are “corrected” using a pseudo-synthetic control from other trials’ control arms; exchangeability is not established, so residual bias (regression to mean, measurement context, incentives, selection/attrition) could be large.
- 5) Subjective weights across evidence sources (general RCTs vs charity-related RCTs vs M&E) are central to the headline estimates; the procedure is transparent but remains a major source of discretion, especially for StrongMinds where the RCT conflicts with other evidence.
- 6) Household spillovers contribute ~30–40% of total benefits but are estimated from

- very sparse and heterogeneous evidence (few interventions, one dominant study) and acknowledged as very low-quality; the spillover prior strongly affects totals.
7. 7) Dosage adjustment for Friendship Bench is driven by extremely low reported attendance (1.12 sessions) and a chosen log(+1) dose-response form; empirical support for this functional form and parameterization is limited.
 8. 8) The “Iran study” adjustment treats larger effects as likely biased based on contextual suspicion; while potentially prudent, it risks discarding true contextual heterogeneity and is not validated against independent indicators of study quality.
 9. 9) Publication-bias adjustment uses an average of multiple correction methods (some known to perform poorly under heterogeneity); the resulting discount is not clearly tied to the estimand under multilevel dependence and time moderation.
 10. 10) The replication (0.51) and response-bias (0.85) discounts for M&E data are generic and not estimated in the charity context; they may double-count some biases or miss others, and materially alter the M&E contribution.
 11. 11) StrongMinds RCT relevance assessment is plausible but partly speculative (pilot/partner/adolescents/COVID); the decision to downweight rather than treat as a serious negative update is consequential and could be contested.
 12. 12) Cost modeling includes counterfactual partner adjustments for StrongMinds; these assumptions are important to \$/WELLBY and would benefit from clearer empirical grounding and uncertainty propagation.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

B.12 Peterman et al. 2025

B.12.1 Ratings Comparison

Metric	GPT-5 Pro	GPT-5.2 Pro	Gemini 2.0 Flash
Overall	82 [74–89]	74 [60–85]	65 [45–80]
Claims	81 [73–88]	69 [54–82]	70 [50–85]
Methods	78 [70–86]	71 [56–84]	60 [40–75]
Adv. Knowledge	83 [75–90]	77 [63–88]	75 [60–85]
Logic & Comms	76 [68–84]	72 [57–85]	80 [70–90]
Open Science	64 [50–78]	55 [38–71]	75 [60–85]
Global Relevance	92 [85–97]	86 [75–94]	85 [70–95]

i GPT-5 Pro Assessment

This is a well-conceived, pre-registered (PROSPERO) systematic review and meta-analysis that synthesizes 1,307 effect sizes from 93 randomized evaluations covering 218,828 women across 45 LMICs to assess the impacts of social safety nets (SSNs) on women’s economic

achievements and agency. The search and screening procedures follow PRISMA standards, and the analysis uses standardized mean differences (Hedges' g), robust variance estimation (RVE) to address dependence among multiple estimates per study, and several robustness checks (multi-level random-effects with clustered SEs, exclusion of influential effects via Cook's D , and orchard plots). Pooled effects are small-to-moderate and statistically significant overall and across the two domains (economic achievements and agency), with suggestive differences by SSN modality: robust positive impacts for UCTs, productive asset transfers, and social care services, smaller effects for CCTs, and mixed or null effects for in-kind transfers; public works appear positive but less precisely estimated. The study is policy-relevant, broad in scope beyond cash transfers, and adds value by compiling recent RCT evidence and providing cross-modality comparisons and a narrative review of cost-benefit assessments. At the same time, several important limitations temper interpretation and external validity. First, between-study heterogeneity is very high (I^2 typically ~80–95%, $p > 0.02$). Although the authors report sensitivity checks and meta-regression, the explanatory power of moderators is low and prediction intervals are not reported; hence the pooled point estimates may conceal wide dispersion of context- and design-specific effects, limiting generalization. Second, outcome construction and comparability raise concerns. The paper pools a broad set of indicators under “economic achievements” and “agency,” but agency constructs (e.g., decision-making, autonomy, voice) are measured heterogeneously, often via self-report instruments with known issues (e.g., ambiguity about jointness of decisions, context dependence, social desirability). The limited availability of aggregate agency indices and the small number of studies for some subdomains (e.g., aspirations, leadership) reduce precision. Third, the standardization strategy and effect size computation choices are not fully transparent in the excerpt: many outcomes (e.g., labor force participation) are binary, yet all estimates are converted to SMD/Hedges' g . The manuscript does not detail (in the main text) the conversion formulas for binary outcomes, imputation of standard errors when needed, or how multiple follow-up waves within studies were handled beyond RVE (e.g., whether impacts were averaged or modeled distinctly). This affects interpretability of magnitudes and complicates translation into policy-relevant units. Fourth, while the RVE approach is appropriate for dependent effects, key modeling choices (e.g., assumed within-study correlation parameter, small-sample correction variant, sensitivity of results to alternative assumptions) are not reported in the main text. Multi-arm trials with shared control groups can lead to over-weighting without appropriate adjustment; while RVE helps, explicit handling of shared controls (e.g., splitting controls or modeling covariance structures) is not described. Fifth, the meta-regression covariates appear coarse and may be measured with error, limiting detectability of design effects. For example, “value of economic benefits” is summarized as total USD per arm without clear PPP or inflation adjustments, without normalization by duration or baseline consumption, and without distinguishing frequency/timing—critical determinants of real purchasing power and behavior. Classification of “plus” components is broad (training/information, economic, health/protection, etc.), and there is no harmonized coding of intensity or fidelity; “gender-targeting” is prevalent (75%), leaving little variation to detect effects; key features like recipient identity (who in the household receives the transfer), payment modality, or operational details (e.g., payment reliability, proximity) are not analyzed. Sixth, the time dimension is only coarsely incorporated (12 months intervention, 12 months post-intervention), despite wide variation in follow-up (0–140 months) and likely non-linear dynamics (fade-out vs accumulation). Seventh, there are internal inconsistencies that need correction: the reported pooled effect for public works ap-

appears as 0.127 in one section and 0.273 in another; the 95% CI for decision-making includes an implausible upper bound (e.g., 0.036–0.37 likely a typographical error); and the cost-benefit section reports a negative BCR (−1.98), which is not meaningful for a benefit-cost ratio (BCRs should be non-negative)—this may reflect a mislabeling of NPV or another metric. Eighth, assessment of publication bias is not described in detail; given high heterogeneity and dependent effects, conventional funnel asymmetry tests are unreliable. No selection models, p-curve, or PET-PEESE results are reported in the main text, leaving small-study or selective reporting concerns insufficiently addressed. Ninth, the inclusion of only experimental studies strengthens causal inference but may exclude informative quasi-experimental evaluations, and the language coverage (English/French/Spanish) yielded no non-English studies despite extensive Latin American literature—raising the possibility of language or database filtering effects. Tenth, while the authors note a quality assurance measure (Table S4), a standard risk-of-bias framework (e.g., RoB 2) and its integration into weighting or sensitivity analyses is not detailed; downweighting or stratification by risk-of-bias could strengthen robustness. Eleventh, reusability and replication: the manuscript states replication files are available in supplementary materials and will be deposited publicly if accepted; however, the data extraction form is not publicly available, and it is not clear whether a fully reproducible database with coding rules (indicator mappings, SSN classifications, decisions on multiple arms/follow-ups) is provided. Without the extraction instrument and a transparent codebook, independent verification and extension are constrained. Lastly, the cost-benefit synthesis, while useful, is narrative and based on a small and modality-skewed subset (mostly UCTs/assets), with diverse methodologies and minimal incorporation of gender-specific benefits; as the authors note, these likely understate benefits relevant to women’s outcomes. Overall, the analysis is rigorous and timely, the conclusions are cautious where warranted, and the practical relevance is high; addressing the internal inconsistencies, clarifying effect-size conversions and RVE assumptions, improving the treatment of transfer values and time dynamics, reporting prediction intervals, and enhancing open-science materials (especially extraction instruments and coding protocols) would further strengthen credibility and utility.

i GPT-5.2 Pro Assessment

This manuscript reports a large, pre-registered systematic review and meta-analysis of experimental evaluations of social safety nets (SSNs) in LMICs, focusing on women-specific measures of economic achievement and agency. The main result is a small but statistically significant pooled standardized effect (Hedges’ g 0.11 overall; 0.11 for economic achievement and 0.10 for agency) across 1,307 effect sizes from 93 studies (115 publications) in 45 countries, using robust variance estimation (RVE) to address within-study dependence. The work’s core strength is the breadth of coverage (modalities beyond cash, multiple outcome families, many contexts) paired with a reasonably standard modern meta-analytic workflow (PRISMA reporting, PROSPERO protocol, sensitivity checks with multilevel models, influence diagnostics). However, the credibility and interpretability of the pooled estimates are constrained by (i) extremely high heterogeneity (I^2 often ~80–95% and τ^2 often >0.02), (ii) outcome and intervention diversity that makes a single standardized mean difference hard to map onto actionable policy magnitude, and (iii) several transparency and internal-consistency issues that make it difficult to verify key steps without full supplements and deposited replication materials.

First, the central inferential challenge is the combination of very high between-study heterogeneity with relatively modest meta-regression explanatory power. When heterogeneity is at this level, the pooled mean effect can be a weak summary for decision-making unless the paper clearly characterizes the dispersion of true effects and the conditions under which effects are larger or smaller. The manuscript acknowledges heterogeneity but largely continues to interpret the pooled mean as general evidence that “SSNs empower women.” That conclusion is directionally plausible, but it risks overstating external validity: in a setting where true effects plausibly vary widely by program design, baseline constraints, measurement, and implementation quality, a small positive average does not imply most programs meaningfully improve women’s outcomes, nor does it identify which marginal SSN investments are best.

Second, the analysis aggregates a very broad outcome set—ranging from labor force participation, earnings/hours, savings, assets, expenditures, and debt to “voice,” autonomy/self-efficacy, and decision-making—into common standardized units after sign harmonization. This is common in umbrella meta-analyses, but it introduces interpretability risks: (a) standardized effects depend on outcome variance, which varies substantially across constructs and measurement instruments; (b) some outcomes are mechanically closer to transfers (e.g., savings/assets/expenditures) while others are more structural (e.g., leadership, aspirations), so pooling can conflate “direct resource effects” with deeper empowerment; (c) measurement validity for agency indicators is known to be challenging (ambiguity about jointness, reference periods, and construct comparability), which can attenuate or distort effect sizes in ways that differ by survey module and context. The paper correctly flags measurement limitations for decision-making, but does not fully propagate those limitations into interpretation of modality comparisons or policy implications.

Third, while the authors use RVE to handle multiple effects per study, the paper (as presented here) does not make fully explicit several implementation choices that can matter materially: the assumed within-study effect size correlation (ρ) in correlated-effects RVE, whether and how small-sample corrections were applied, and whether sensitivity to ρ was checked. Because many effects come from the same studies and intervention arms, these assumptions can influence standard errors and, to a lesser degree, pooled estimates. The study reports additional robustness using multilevel random-effects with clustered SEs and Cook’s D trimming, which is reassuring, but readers still need the exact analytic specification to evaluate dependence handling.

Fourth, there are apparent internal inconsistencies/possible typos in reported numerical results that warrant careful reconciliation. For example, the pooled effect for public works is described with different magnitudes in different places (e.g., ~ 0.127 in some summary statements/figures versus ~ 0.273 in the narrative), and at least one confidence interval appears implausibly wide or misprinted (e.g., decision-making CI upper bound shown as 0.37 given a point estimate around 0.087). These inconsistencies may be innocuous transcription errors, but in a meta-analysis they undermine confidence unless systematically resolved and traceable to code output.

Fifth, the moderator analysis is limited by multiple testing and low effective degrees of freedom for some subgroup cells (especially modalities with few studies, e.g., social care, public works, Europe/Central Asia). The paper notes that the Europe/Central Asia result is driven by one study, which is good practice, but the broader approach—running many separate regressions and extended models—raises the probability of chance findings without adjustment or a clear pre-specified moderator hierarchy. Additionally, several substantively important moderators (transfer size relative to baseline consumption, payment frequency, recipient iden-

tity, conditionality intensity/enforcement, childcare availability, baseline gender norms, local labor demand) are either proxied coarsely (e.g., terciles of USD value) or not included, limiting the ability to inform “how to design SSNs for empowerment” beyond modality labels.

Sixth, inclusion is restricted to RCTs. This improves internal validity on average but narrows policy coverage: many large national SSNs are evaluated quasi-experimentally, and excluding them can bias the evidence base toward smaller, more intensively supported pilots and toward contexts where randomization was feasible. The manuscript also reports that despite multilingual searching, it did not identify French and Spanish experimental studies; if true, this may indicate either genuine evidence scarcity or search/retrieval limitations, but either way it affects representativeness.

Seventh, the cost-benefit component is necessarily descriptive (only ~25 papers) and heterogeneous in methods; most studies do not incorporate women-specific benefits, and comparisons across programs are not harmonized. The manuscript treats this appropriately as narrative, but its policy takeaway (“smart investments”) should be framed cautiously, given that many BCRs depend heavily on assumptions about persistence and valuation of benefits, and because household-level consumption/asset benefits may not track women-specific empowerment gains. Finally, open science and replicability are currently partial: the paper states replication files are available as supplementary materials and will be deposited publicly upon acceptance, but this (and the non-public extraction form) limits immediate auditability. Given the scale of coding (1,307 effects) and the sensitivity of inclusion/exclusion and outcome mapping decisions, full public release of the extraction sheet, codebook, and code is particularly important here.

Overall, this is a substantial and potentially field-shaping synthesis showing that, on average, SSNs modestly improve women-specific economic outcomes and some agency dimensions, with weaker evidence for conditional and in-kind modalities. The main threats are not obvious fatal flaws but rather (1) the interpretability of a small pooled SMD under massive heterogeneity, (2) limited traction on design moderators, (3) some reporting inconsistencies, and (4) incomplete reproducibility until full materials are deposited.

i Gemini 2.0 Flash Assessment

This paper presents a systematic review and meta-analysis examining the impact of social safety nets (SSNs) on women’s economic achievements and agency in low- and middle-income countries (LMICs). The analysis encompasses 1,307 effect sizes from 115 publications, representing 93 studies and 218,828 women across 45 LMICs. The authors find a statistically significant positive pooled effect of SSNs on both economic achievements and agency. The effects are particularly strong for unconditional cash transfers (UCTs), social care, and asset transfers. Meta-regression analysis explores factors influencing these impacts, considering study, intervention, and outcome-level characteristics. A cost-benefit analysis, based on a subset of studies, suggests that SSNs are generally smart investments, although gender-specific benefits are often underrepresented in cost-benefit calculations. The authors acknowledge limitations, including the heterogeneity of studies, potential for publication bias, and limited consideration of longer-term impacts. Overall, the study provides robust evidence supporting the role of SSNs in promoting women’s economic empowerment and agency in LMICs, highlighting the need for gender-sensitive design and evaluation. However, there are several issues that temper my confidence in the work. The meta-regression results in Table 2 do not seem

to be statistically significant. It is also unclear to me how they are addressing the issue of multiple hypothesis testing. The RVE estimator is intended to correct for standard errors for dependency within studies that report on multiple estimates per outcome type, but it's not clear that they fully address autocorrelation and clustering. Furthermore, there is not much discussion of limitations of the specific studies included in the meta-analysis. There is very high heterogeneity between the studies, which they mention, but I would like a deeper dive into the implications of that, and why we should expect these interventions to be comparable in the first place. It's also important to note the relatively small effect sizes that they are estimating.

⚠️ GPT-5.2 Pro: Key Issues Identified

1. 1) Extremely high heterogeneity (I^2 often ~80–95% and $\tau^2 > \sim 0.02$) means the pooled mean effect may be a weak policy summary; the paper should more strongly emphasize dispersion/prediction intervals and limits to generalization.
2. 2) Several apparent internal inconsistencies/typos in reported magnitudes and intervals (e.g., public works pooled effect reported with different values; at least one implausible confidence interval) need reconciliation and traceability to code output.
3. 3) Outcome harmonization across very different constructs (savings/assets/expenditures vs autonomy/voice/decision-making) using standardized effects risks conflating mechanisms and complicates interpretation; stronger justification and construct-validity discussion is needed.
4. 4) RVE implementation details are not fully transparent in the main text (e.g., assumed within-study correlation ρ , small-sample corrections, sensitivity to ρ), which can matter with many dependent effect sizes per study.
5. 5) Moderator/meta-regression analysis appears underpowered for key subgroups and vulnerable to multiple-testing false positives; a pre-specified moderator hierarchy and/or correction strategy would strengthen inference.
6. 6) Modality comparisons (UCT vs CCT vs in-kind, etc.) may be confounded by correlated design features (targeting rules, recipient identity, enforcement intensity, complementary programming, baseline contexts) that are only coarsely measured; causal interpretation of modality differences should be more cautious.
7. 7) Restricting inclusion to RCTs improves internal validity but may reduce representativeness for large-scale national programs commonly evaluated quasi-experimentally; this could bias conclusions about real-world SSN performance.
8. 8) The evidence base is heavily concentrated in certain regions (notably sub-Saharan Africa) and in cash modalities; conclusions about underrepresented modalities (social care, public works, fee waivers) and regions are therefore fragile.
9. 9) Agency measurement is known to be noisy and context-dependent; the paper notes some limitations (e.g., decision-making ambiguity) but does not fully assess how measurement heterogeneity may attenuate or distort pooled results.
10. 10) Cost-benefit evidence is sparse and not harmonized; most studies omit women-specific benefits, making any overall “value for money” implication highly uncertain and potentially a lower bound in unknown ways.
11. 11) Open-science posture is incomplete at present (public repository only “upon accep-

tance”; extraction form not public), limiting immediate reproducibility and external auditing of coding decisions.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

B.13 Weaver et al. 2025

B.13.1 Ratings Comparison

Metric	GPT-5 Pro	GPT-5.2 Pro	Gemini 2.0 Flash
Overall	90 [85–95]	86 [78–92]	75 [60–85]
Claims	88 [82–94]	84 [74–91]	80 [70–90]
Methods	85 [78–90]	88 [80–94]	75 [65–85]
Adv. Knowledge	89 [83–95]	83 [72–90]	70 [55–80]
Logic & Comms	90 [85–95]	87 [79–93]	85 [75–90]
Open Science	62 [50–72]	73 [58–86]	80 [70–90]
Global Relevance	95 [90–98]	93 [85–97]	80 [70–90]

GPT-5 Pro Assessment

This is a strong cluster-randomized evaluation of maternal cash transfers implemented through India’s anganwadi (ICDS) system in Jharkhand, with unusually rich measurement along the causal chain from income to child outcomes. The core identification strategy is credible: random assignment at the AWC (cluster) level after registration, high compliance (administrative records indicate >99% transfers executed), good balance on pre-determined covariates, and no differential attrition across treatment and control in any round. Sector fixed effects account for the stratified design; standard errors are clustered at the AWC level. The setting—government-run delivery at scale, in a sample designed to be near-representative of the state—enhances external validity. Results on intermediate outcomes are compelling and policy-relevant: transfer recipients increased food expenditure (11–21% in IHS units), maternal/child calories and dietary diversity, and maternal nutrient consumption. The paper carefully characterizes use of cash (no increase in “sin” goods; non-food outlays rose proportionately), and the marginal propensity to spend on food and to consume calories is compared to recent estimates for in-kind PDS expansions; while not a like-for-like causal comparison, the magnitudes are plausibly similar. Intra-household allocation evidence is unusually direct, based on ingredient-level 24-hour recall and individual-specific portions; mothers start from lower absolute consumption than other adults, and their gains are at least proportionate, implying improved equity. The study is transparent about implementation realities (recurrent payment delays, limited IVR message delivery, banking frictions), increasing the relevance of ITT estimates for policy. The primary substantive limitation is the lack of average gains on target children’s anthropometrics despite clear nutrition gains. The heterogeneity analysis

suggests larger WAZ gains where sanitation is better, consistent with the WASH literature; however, sanitation measures are post-treatment and observational. Although treatment is orthogonal to sanitation and the authors show no treatment effect on sanitation, attributing effect moderation causally to sanitation remains tentative because unmeasured correlates (wealth, behaviors, local health systems) could co-vary with sanitation. The residualization exercise (LASSO-selected observables) strengthens this pattern, but cannot fully rule out omitted variables; the paper is appropriately cautious and frames this as suggestive evidence. The sibling anthropometrics (WAZ +0.11 SD at Y3) are a notable positive spillover, consistent with household-wide consumption gains; the lack of sanitation heterogeneity in siblings aligns with greater WASH sensitivity among younger children, but alternative explanations (selection of the “random” sibling, differential measurement error) should be discussed further. Measurement is a major strength and also a source of constraints. The ingredient-weighted, individual-specific 24-hour recall is a best-practice approach for at-home foods; yet it captures one day only (vulnerable to day-of-week/seasonal variation), excludes breastmilk for infants (the paper is clear that child calorie measures are solid foods only), and cannot capture out-of-home snacks perfectly, though the team uses approximations. Year 2 measurement was disrupted by COVID; detailed calories/nutrients rely on the smaller Y2 field subsample, while phone surveys provided only dietary diversity and some behaviors. The authors address this with a Y3 in-person round and show high Y2–Y3 anthropometric correlations, but precision for some Y2 outcomes is limited. The ASQ-3 functional development gains (0.12 SD at age 3) are important and consistent across subdomains; however, ASQ is parent-reported and enumerators/parents may not have been blinded to treatment. The authors preempt concerns by noting effects on more-objective items and the one-year lag since transfers ended, but some risk of experimenter demand or social desirability bias remains; future work could incorporate direct child assessments or structured observation for a subsample. Internal consistency is generally good: food spending and consumption effects are mutually reinforcing; increased visits to formal providers (and more withdrawals “for medical expenses”) fit with the design’s women-controlled accounts. One interpretive tension is the claim that empowerment may grow over time (citing larger maternal calorie gains in Y2), while the empowerment index shows gains in Y1 but no significant change in Y2; readers would benefit from a more systematic mediation analysis or a decomposition distinguishing “mobility/transaction ability” from intrahousehold bargaining constructs. Statistical practice is sound overall, but there are two caveats: (i) multiple hypothesis testing is not explicitly adjusted despite many outcomes and heterogeneity analyses; the pre-analysis plan and index construction mitigate concerns, yet adjusted q-values or FWER/FDR corrections would strengthen inference, especially for secondary and heterogeneity results; (ii) pooling across two implementation phases implies different survey months (seasonality), and although the authors report no differential effects by phase and include sector fixed effects, explicit month/phase-by-treatment robustness checks would be reassuring for food consumption outcomes that are seasonal. Interference/spillovers across AWCs within sectors are a theoretical possibility (three treated and three control AWCs per sector), but likely limited given registration closed pre-randomization and transfers paid to named accounts; still, a short discussion or bounding analysis of cross-AWC contamination would be useful. The comparison to PDS effects is informative but rests on different samples/periods and assumptions (inflation adjustments, linearity); readers should view it as suggestive rather than definitive. Policy interpretation is balanced: the paper neither overstates anthropometric results nor ignores the real gains in diets, equity, medical care-seeking,

and functional development; the WASH complementarity is positioned as a plausible mediator and a hypothesis for policy coordination rather than a proven causal interaction. Open science practices are mixed: the trial is pre-registered and analysis plans are cited; variable construction is documented in detail; however, availability of analysis code and de-identified data are not clearly stated, which could limit reproducibility. Constructively, the authors could: add multiplicity adjustments (or at least report sharpened q-values) for key families of outcomes and heterogeneity; include survey-month or phase-by-treatment robustness; report randomization inference for main outcomes; probe potential within-sector spillovers; present sensitivity of the sanitation interaction to alternative WASH measures (e.g., using only neighbor latrine use) and to bounding (e.g., Oster/Altonji) with added covariates; and, if feasible, release code and a replication package. Despite these limitations, this is a careful, policy-relevant RCT that advances evidence on maternal cash transfers in India, clarifies income–nutrition links, highlights intra-household equity improvements, and broadens the focus from anthropometrics to functional child development, while opening an important agenda on coordinating cash with WASH to translate diet gains into growth.

i GPT-5.2 Pro Assessment

This manuscript reports a large cluster-randomized evaluation of an unconditional but “framed” maternal cash transfer program implemented through the public ICDS/anganwadi system in Jharkhand, India. The design (480 anganwadi centers randomized 1:1, ~2,400 registered pregnant women) is credible and unusually strong for assessing ITT impacts at policy-relevant scale; compliance appears very high with essentially no control contamination, stratification and clustering are appropriate, balance and attrition checks are reassuring, and the study is supported by trial registry and multiple pre-analysis plans. The paper’s main empirical contribution is its detailed, individual-level measurement of food intake (ingredient quantities plus allocations to mother/child), allowing the authors to track multiple links in a causal chain from transfers to nutritional intake to child outcomes. The headline findings—sizable increases in food spending, calories, diet diversity and maternal nutrient index; narrower intra-household gender gaps in food consumption; no average impacts on child HAZ/WAZ for the target child; suggestive sanitation moderation for WAZ; and improvements in age-3 functional development (ASQ) plus WAZ gains for older siblings—are, on their face, broadly plausible and well aligned with the pattern of intermediate outcomes.

The most important limitations are not about internal validity of the core ITT estimates, but about (i) interpretability of mechanisms and measurement, (ii) robustness of key secondary claims (especially sanitation interactions and “empowerment over time”), and (iii) some avoidable design/implementation constraints that interact with COVID-era data collection. First, while the intervention is described as unconditional, it is not “pure cash”: payments were made into women’s accounts and accompanied by nutrition-oriented framing and (imperfectly delivered) IVR messaging. This is central to external validity: the estimates should be interpreted as the effect of a bundled policy package typical of Indian maternal transfers rather than a generic household income shock. Relatedly, the paper argues that messaging had limited marginal impact, but the evidence is indirect (heterogeneity by having provided a phone number, with imperfect “treatment-on-treated” for IVR receipt). A more direct analysis using administrative delivery logs on call attempts/connection or an instrument for actual listening exposure would strengthen the claim that effects are primarily due to cash rather than

information.

Second, several key measures are vulnerable to reporting/measurement issues. The intensive 24-hour dietary recall with ingredient measurement is a strength, but it still relies on accurate recall of all dishes, correct mapping of utensils to volumes, and correct partitioning of cooked food shares across members; classical error would attenuate effects, but differential error (e.g., treated respondents reporting more “nutrition-appropriate” foods) could bias results. The paper partially alleviates this by showing consistent increases in food spending and some persistence in year-3 food-item consumption, but there is limited triangulation against objective biomarkers (e.g., anemia/hemoglobin) or administrative purchase/withdrawal records beyond self-reports. Similarly, the child development outcome is parent-reported (ASQ-derived), which raises concerns about experimenter demand or differential optimism; the authors discuss why demand effects may be limited, yet the ASQ remains a subjective outcome without direct assessor-based testing. The magnitude (0.12 SD) is modest and plausible, but stronger reassurance would come from interviewer-blinded modules, validation subsamples, or comparison to objectively observed tasks.

Third, the paper’s causal chain logic is persuasive, but several intermediate-channel interpretations risk overreach. The “increasing empowerment over time” narrative rests heavily on the larger maternal calorie effect in year 2 versus year 1 and on suggestive differences with other members; however, year-2 nutrient/calorie measurement is based only on the truncated in-person subsample (fieldwork halted by COVID), and the remainder relies on phone-based diet diversity rather than quantities. This creates a risk that year-2 comparisons mix true dynamics with sample composition (e.g., districts reached pre-COVID) and mode effects. The authors do provide checks (e.g., comparing year-1 effects within the year-2-measured subsample) that help, but the empowerment-over-time interpretation is still weaker than the main ITT claims on year-1 nutrition and year-3 outcomes.

Fourth, the sanitation heterogeneity is potentially important but is also the most interpretively delicate part of the paper. The sanitation index is measured post-treatment (year 3) and includes some enumerator observations plus respondent perceptions about neighbors. The authors show it is uncorrelated with treatment assignment, supporting internal validity of interaction estimation, and they test that sanitation does not moderate impacts on nutrition, supporting the “sanitation as mediator” story. However, because sanitation is not randomized, the interaction could still reflect other correlated area characteristics (health service access, food markets, baseline wealth, disease ecology) that affect growth responses to improved diets. The paper’s residualization exercise (using LASSO-selected covariates) is a helpful sensitivity check and even strengthens the interaction, but it cannot rule out unobserved confounding or measurement error in sanitation. Moreover, the interaction results are more compelling for WAZ than HAZ; given HAZ’s slower-moving nature, the timing of measurement (post-transfer) could matter, but the paper also relies on the fact that year-3 anthropometrics correlate strongly with year-2. Overall, the sanitation moderation should be framed as suggestive evidence consistent with prior work rather than a definitive estimate of complementarity.

Fifth, the COVID disruption complicates the narrative coherence across years. Some key outcomes were not measured in year 2 for much of the sample (anthropometrics, calories, depression), and non-food spending in year 1 is missing due to a survey form error. These gaps do not undermine the core year-1 nutrition and year-3 child outcomes, but they weaken the ability to characterize trajectories, persistence mechanisms, and how spending patterns map into longer-run outcomes. The paper does present year-3 “persistence” measures, but these

are narrower (selected food items, diet diversity) than the year-1 caloric/nutrient measures. Despite these limitations, this is a high-quality applied micro paper: the experimental design is strong; the measurement strategy is unusually informative; results are internally consistent (spending → intake → some developmental outcomes, with plausible nulls on anthropometrics); and policy relevance is substantial given scale of maternal transfers in India. The main recommended improvements are to (i) tighten causal language around sanitation moderation and empowerment dynamics, (ii) deepen robustness/triangulation for parent-reported ASQ and dietary measures (or explicitly bound demand/measurement concerns), (iii) clarify multiple-testing control across large outcome families and heterogeneity analyses, and (iv) improve replication readiness via clear public code/data availability statements and, where privacy prevents sharing, provide synthetic/replication packages with detailed construction scripts.

i Gemini 2.0 Flash Assessment

This paper presents experimental evidence on the impact of a maternal cash transfer program in Jharkhand, India, focusing on gender equity and child development outcomes. The study utilizes a large-scale randomized evaluation across 480 public early childhood care centers (AWCs), with half receiving monthly cash transfers of Rs. 500 to registered mothers for two years. The research design is strong, with pre-registered analysis plans and a focus on intent-to-treat (ITT) estimates to address compliance. The study meticulously collected detailed consumption data, nutrient intake measurements, and child development outcomes at ages one and three.

The paper demonstrates several positive impacts: significant increases in household food expenditure and nutritional intake for mothers and children, improvements in intra-household equity, and gains in children's functional development (cognitive, gross motor, and fine motor skills). The authors find that cash transfers increased caloric intake, dietary diversity, and nutrient consumption, while also narrowing gender disparities in food consumption. However, the study finds no significant gains in standard child anthropometric measures (height- and weight-for-age z-scores), suggesting that the transfers did not improve physical growth outcomes. The authors explore heterogeneity in treatment effects, finding suggestive evidence that sanitation conditions may mediate the translation of improved nutrient intake into child anthropometric outcomes. They also analyze the effect of messaging on nutrition and find limited independent impact. The research contributes to the literature on cash transfers, nutrition, gender, and child development, and has direct policy implications for the expansion of cash transfer programs in India and other LMICs.

However, the study also exhibits some limitations. The lack of baseline data necessitates reliance on balance tests using pre-existing characteristics, which may not fully capture pre-intervention conditions. The COVID-19 pandemic affected data collection in the second year, requiring phone surveys and limiting the ability to measure certain outcomes (e.g., anthropometrics). The study acknowledges that the transfers were often delayed due to government fiscal processes, which is important for interpreting the ITT estimates. The findings regarding anthropometric measures are nuanced, as the average treatment effects are insignificant, but there is evidence of positive effects in areas with better sanitation and among older siblings. Further research is needed to explore the longer-term impacts of the intervention and to rigorously test the complementarities between income and sanitation in reducing child stunting.

Internal consistency is generally strong, with clear connections between the research questions, methods, results, and policy implications. The authors transparently discuss the limitations of their study and acknowledge potential confounding factors. Overall, the paper presents a well-designed and executed study with valuable insights for policymakers and researchers interested in maternal and child health in developing countries. However, the lack of impact on anthropometric measures and the potential influence of unobserved confounders limit the strength of the claims and the generalizability of the findings.

⚠️ GPT-5.2 Pro: Key Issues Identified

1. 1. Clarify that the estimated effects are for a bundled intervention (cash delivered to women + nutrition framing/IVR), not a generic income shock; claims about “unconditional cash” should be qualified accordingly.
2. 2. Sanitation heterogeneity relies on non-random, post-treatment sanitation measurement; interaction estimates should be presented as suggestive and vulnerable to confounding by correlated area characteristics (health access, markets, disease ecology), despite balance on treatment and residualization checks.
3. 3. Year-2 calorie/nutrient outcomes are based on a COVID-truncated in-person subsample; comparisons across years (e.g., “empowerment grows over time”) may be partly driven by sample composition and survey mode changes.
4. 4. Child functional development is measured via parent-reported ASQ items; differential reporting/experimenter-demand effects cannot be fully ruled out without more objective assessor-based measures or validation subsamples.
5. 5. Dietary intake measurement, while unusually detailed, still depends on 24-hour recall and complex allocation/portion measurement; the paper could better assess robustness to measurement error and potential differential misreporting.
6. 6. Evidence that IVR messaging had little effect is indirect (based on having registered a mobile number); stronger tests using call delivery/connection data or instruments for actual exposure would improve interpretability.
7. 7. The manuscript analyzes many outcomes and heterogeneities; it should more explicitly define primary families, address multiple-hypothesis testing (beyond indices), and distinguish confirmatory vs exploratory findings.
8. 8. Post-treatment sanitation index includes respondent perceptions of neighbors and enumerator observations; reliability and potential measurement error (including enumerator subjectivity) deserve more discussion and sensitivity analyses using alternative sanitation proxies.
9. 9. The paper infers that intermediate-channel changes (knowledge, empowerment, AWC services) explain little via cross-sectional correlations; this “explained share” approach is not causal and may be misleading if interpreted too literally.
10. 10. Non-food spending was not measured in year 1 due to a survey form error; this limits the welfare/accounting interpretation (budget shares, non-food mechanisms) during the key period of transfer receipt.
11. 11. External validity beyond Jharkhand and beyond ICDS-linked registrants is good but not complete; the sample is slightly wealthier than the state average and depends on engagement with AWCs, which may differ in other states/settings.
12. 12. Replication readiness is unclear from the manuscript text excerpt: despite reg-

istry/PAPs, the paper should state whether code/data (or restricted-access procedures) are available and provide full construction scripts for complex nutrition and sanitation indices.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

B.14 Williams et al. 2024

B.14.1 Ratings Comparison

Metric	Claude Sonnet 4	GPT-5 Pro	GPT-5.2 Pro	Gemini 2.0 Flash	Human
Overall	68 [55–78]	87 [78–93]	82 [74–89]	70 [55–80]	50 [35–70] (n=2)
Claims	65 [52–75]	82 [75–88]	78 [68–86]	75 [60–85]	30 [18–42] (n=2)
Methods	70 [58–80]	75 [65–85]	79 [69–87]	65 [50–75]	25 [12–38] (n=2)
Adv. Knowledge	75 [65–83]	85 [75–92]	86 [77–93]	70 [55–80]	55 [45–80] (n=2)
Logic & Comms	72 [62–80]	90 [84–96]	88 [80–94]	80 [70–90]	58 [46–70] (n=2)
Open Science	78 [70–85]	62 [50–75]	67 [55–78]	75 [65–85]	62 [52–72] (n=2)
Global Relevance	82 [75–88]	92 [85–97]	90 [82–96]	85 [75–95]	75 [65–85] (n=2)
NA	—	—	—	—	5 [5–5] (n=2)
NA	—	—	—	—	4 [3–4] (n=2)
NA	—	—	—	—	75 [65–85] (n=2)

i GPT-5 Pro Assessment

The paper presents a high-resolution, pantropical model of the potential for natural forest regeneration on deforested lands, trained on observed regrowth (2000–2016) and biophysical/socioeconomic covariates, with predictions provided at 30 m resolution and country- and biome-level summaries. The main strengths are: (i) the explicit use of observed natural regrowth (rather than assumed potential) to train the model; (ii) high spatial resolution outputs that align with implementation needs; (iii) an extensive and carefully curated predictor set; (iv) a transparent methodological narrative with sensitivity checks; and (v) an explicit, policy-relevant translation into above-ground carbon sequestration potential. The manuscript is well written, the logic is generally careful, and the discussion acknowledges critical caveats about additionality, permanence, leakage, and socioeconomic feasibility. However, several issues limit the certainty and the directness of the policy inferences. First, the foundational training dataset (natural regrowth vs plantations vs open) has known omission errors in humid biomes (producer’s accuracy by area for the natural regrowth class is low, $18.7\% \pm 5.4$ when area-weighted, despite good user’s accuracy), implying the model may learn from an incomplete sample of positive instances. This likely biases the signal toward conditions associated with more easily detected or larger regrowth patches (0.45 ha) and nearer to existing forests, potentially underrepresenting smaller, more diffuse, or harder-to-detect regrowth farther from

edges. Second, the modelling and validation strategy provides encouraging overall accuracy (87.9% on a large, balanced validation set), but accuracy is reported on an equal class-balance sample, not on true prevalence; calibration diagnostics (e.g., reliability curves, Brier score) are absent, and the RF vote fractions are treated as probabilities without post-hoc calibration. As a result, the “probability” values integral to the weighted-area calculation may not be well calibrated, and the expected area of regeneration derived by summing probabilities could be biased. The authors partially address spatial overfitting by examining performance versus distance from training points; accuracy declines to ~81% at 2–2.5 km then rises, indicating some spatial autocorrelation effects and distributional shifts. A more conservative spatial cross-validation (e.g., block CV, ecoregion/biome folds) and per-region confusion metrics would better characterize generalization, particularly in Southeast Asia where performance is noted to be lower but not fully quantified. Third, the decision to base the final predictive model on biophysical variables alone is defensible for mapping biophysical suitability that is more stationary and higher resolution, but implies that outputs reflect “environmental potential” rather than realized potential under current socioeconomic constraints. This choice should reduce temporal brittleness but risks overestimating near-term realizable regeneration in areas with persistent anthropogenic pressures, land tenure constraints, or perverse incentives; the authors flag this clearly, but some readers may still misinterpret the maps as forecasts of spontaneous regrowth. Fourth, input data are heterogeneous in resolution (e.g., 30 m tree cover, 300 m land cover, ~1 km climate/NPP, 250 m soils). Although the output is at 30 m, predictions in many places are effectively controlled by coarser inputs, which is visible in map patterns and recognized by the authors. This mismatch affects local prioritization, and users should not overinterpret fine-scale heterogeneity where coarse covariates dominate. Fifth, the uncertainty characterization for area estimates is not commensurate with the underlying modelling uncertainty. Reported “confidence intervals” for the total area with potential (e.g., 214.78–215.22 Mha) are implausibly narrow and appear to reflect only numerical aggregation nuances rather than parameter or model uncertainty; the methods do not explain how these intervals were derived or whether they incorporate any resampling, calibration, or propagation from training data/classifier variance. Similarly, the carbon sequestration ranges (23.4 Gt C; range 21.1–25.7) seem to inherit variation from Cook-Patton et al. but do not appear to propagate classification uncertainty, probability calibration error, downscaling error (from 1 km to 30 m with a uniform 0.09 ha-per-cell scalar), or the acknowledged risks of non-persistence, reclearance, leakage, and implementation constraints. The downscaling of 1 km carbon accumulation to 30 m by simple proportional scaling ignores within-pixel heterogeneity; coupled with uncalibrated probabilities, this introduces additional, unquantified uncertainty in spatially explicit carbon sums. Sixth, temporal assumptions deserve caution: climate inputs reflect 1970–2000 normals; predictions labeled “near future (2030)” adopt 2015/2018 land/forest layers but do not incorporate climate change, altered fire regimes, or policy shifts. The discussion notes that climate change may impede or enhance regeneration, but the model is essentially a static extrapolation of 2000–2016 associations. Seventh, domain definition is liberal (excluding water/urban/bare/sparse vegetation), reducing subjective constraints but admitting some misclassified or marginal lands; this is mitigated by relying on the model to downweight low-probability areas, though it places more weight on model calibration and variable quality. Eighth, interpretation of partial dependence is appropriate, but the dominance of “distance to forest” and “local forest density” likely reflects both true ecological processes and the detection/training biases discussed above; the strong empirical proximity effect (98%

of high-potential cells within 300 m of forest) is intuitive but may be somewhat amplified by the training signal. Ninth, regional performance heterogeneity (e.g., lower accuracy in parts of Southeast Asia) is acknowledged but not fully characterized (no region-specific confusion matrices, precision/recall, or error maps for users). Tenth, reproducibility is good but not ideal: large, tiled prediction rasters are openly archived on Zenodo; inputs are public; however, code is “available on request” rather than archived with a DOI, and reproducing the full pipeline may require substantial compute and careful versioning of large geospatial datasets. Overall, this work substantially advances practical restoration planning by providing an empirically grounded, high-resolution map of biophysical potential for natural regeneration and a transparent, usefully cautious narrative. Its main limitations are under-quantified uncertainty, probability calibration issues, heterogeneity in predictor resolutions, and the intentional omission of socioeconomic constraints in the final predictive model, which together mean that the headline area and carbon numbers should be seen as indicative upper bounds under conducive management and protection, rather than forecasts of realized outcomes. Future improvements could include spatial block cross-validation, probability calibration and uncertainty propagation (e.g., bootstrapping, ensemble models), publication of code with an executable workflow, region-specific performance reporting, and scenario analyses incorporating climate and policy trajectories.

i GPT-5.2 Pro Assessment

This manuscript delivers a high-resolution (30 m) pantropical map of the “potential for natural regeneration” on currently non-forested lands within three tropical/subtropical forest biomes, trained on observed natural regrowth between 2000–2016 from an earlier remote-sensing product and extrapolated to “current (2015)” and “near future (2030)” conditions. The main technical strength is the combination of a very large training/validation sample, a broad covariate set, and a transparent ML pipeline (random forest with variable selection, partial dependence/importance summaries, spatially-aware validation diagnostics). The main limitations, however, stem from how the target variable is defined and measured, how the negative class/domain is constructed, how uncertainty is handled (or not handled) when converting probabilistic outputs into headline area and carbon totals, and how “potential” is interpreted for policy. First, the dependent variable is “natural regrowth” operationalized as vegetation >5 m tall over a 2000–2016 window, derived from Landsat-based tree-cover change plus a plantation-versus-natural classifier from prior work. This definition is meaningful for some restoration questions but does not directly correspond to “successful forest restoration” in ecological or social terms: it misses early-stage regeneration below 5 m; it implicitly requires a minimum growth rate over ~16 years; it is sensitive to the detection and classification limits of Landsat-era products; and it does not guarantee persistence beyond 2016. The Methods acknowledge that the upstream regrowth mapping omitted many humid-biome patches and had low producer’s accuracy on estimated area in the humid biome (dropping markedly when area-adjusted), implying systematic omission that can bias both training labels (false negatives) and geographic patterns of “where regeneration occurs.” The authors argue that user’s accuracy is high when patches are detected, but for potential mapping, omission-driven label noise is consequential because it can teach the model that certain environments rarely regenerate when in fact regeneration occurred but was not mapped.

Second, the construction of “non-regeneration” locations is not a clean counterfactual of “lands

that could have regenerated but did not.” The negative class is drawn from a broad “available domain” that excludes water/urban/bare/sparse classes, existing 2000 tree cover, mapped regrowth patches, and forestry activity, but otherwise includes lands that may have been continuously cultivated, actively maintained pasture, repeatedly cleared, or otherwise socioeconomically constrained. This is partly intended (they want the model to learn empirical low-probability conditions rather than impose assumptions), but it also means the model is predicting an amalgam of (i) biophysical capacity to regenerate, (ii) land-use pressure/management regimes correlated with biophysical proxies, and (iii) remote-sensing detectability. This matters for interpretation: the paper repeatedly frames outputs as “biophysical conditions can support natural regeneration” and emphasizes biophysical-only predictions, yet the labels embed human decisions (continued agriculture, fire management, reclearance) that are not fully separable from biophysics. Dropping socioeconomic covariates because they are “subject to change” may improve map stability, but it also increases the risk that biophysical variables serve as proxies for omitted human drivers (e.g., proximity to intact forest captures both seed source and governance/land-use intensity gradients). Relatedly, the update of forest-distance/density covariates to 2018 tree cover while the response is based on 2000–2016 regrowth introduces temporal mismatches and potential incoherencies acknowledged in Methods; this may be acceptable pragmatically, but it further complicates causal/biophysical interpretation.

Third, performance reporting leans heavily on a single “accuracy” figure (~87.9%) with balanced classes. Accuracy under class balancing can overstate real-world utility when the true prevalence of regeneration is low and when decision-makers care about calibration (probabilities) and specific error tradeoffs (precision/recall) rather than overall accuracy. The manuscript notes spatial autocorrelation concerns and provides a distance-to-training diagnostic; nonetheless, the validation still uses randomly drawn points without explicit spatial blocking, and accuracy varies substantially (notably poorer in parts of Southeast Asia). The paper would be more convincing with additional metrics (AUC/PR-AUC, sensitivity/specificity across thresholds, calibration curves/Brier score), clearer reporting of regional performance distributions, and explicit discussion of how miscalibration would affect the “expected area” and carbon aggregation.

Fourth, the headline quantitative claims—215 Mha “has potential” and 23.4 Gt C over 30 years—are presented with extremely narrow confidence intervals for area (e.g., 214.78–215.22 Mha), which appear to reflect only deterministic arithmetic or sampling variability rather than the dominant uncertainties: label error in the response map, model specification/transfer uncertainty, spatially structured errors, predictor data uncertainty (many covariates are 250 m–1 km and resampled to 30 m), future non-stationarity (climate change, fire regimes, land-use dynamics), and uncertainty in carbon accumulation surfaces. The paper does acknowledge non-stationarity qualitatively and notes mixed covariate resolutions, leakage, and permanence issues, but the quantitative uncertainty communicated in the headline area figure is not commensurate with these acknowledged uncertainties. Similarly, the carbon calculation down-scales a 1 km carbon accumulation layer to 30 m via resampling and a fixed area factor, which effectively assumes sub-kilometer homogeneity (and ignores covariance between regeneration probability and within-pixel carbon potential). The “maximum potential” framing is clear in places, but several passages risk being read as actionable sequestration totals rather than conditional expectations under strong assumptions (regeneration occurs, persists, is protected from fire/reclearance, and is additional).

Finally, while the paper is explicit that it does not incorporate feasibility, opportunity costs, equity, food security, biodiversity objectives, or policy instruments, the Discussion moves quickly from “potential” to suggestions for offsets and incentive targeting. That is not inappropriate, but users could easily over-interpret the map as indicating where natural regeneration is “available” or “low-cost” in a practical sense. The most important missing robustness for practitioner use is an explicit coupling (even coarse) to competing land uses and deforestation/reclearance risk, and clearer guidance on decision thresholds and how to use probabilities (versus the provided >0.5 binary product). Despite these limitations, the work is a substantial mapping contribution and likely to be a valuable input layer for restoration prioritization, provided that readers treat it as a first-stage biophysically-informed (but label- and land-use-entangled) probability surface with limited quantified uncertainty rather than a precise estimate of realizable restoration area or credits.

i Claude Sonnet 4 Assessment

This research paper presents a machine learning-based analysis to map global potential for natural forest regeneration in tropical regions, estimating 215 million hectares with regeneration potential that could sequester 23.4 Gt C over 30 years. The study builds on previous remote sensing work by Fagan et al. (2022) to develop predictive models using biophysical and socioeconomic variables. While the work addresses an important policy question and uses sophisticated methods, several significant limitations emerge. First, there are substantial data quality concerns - the underlying natural regeneration detection has very low producer accuracy when based on estimated area (18.7% in humid biomes), which undermines confidence in the training data quality. The paper acknowledges this but doesn't adequately address how this affects model reliability. Second, the validation approach shows evidence of spatial autocorrelation effects that likely inflate accuracy estimates, with performance dropping from 87.9% to 81.4% at intermediate distances. Third, there are methodological inconsistencies in how ‘available for regeneration’ areas are defined, with liberal inclusion criteria that may not reflect realistic restoration contexts. Fourth, the carbon sequestration estimates rely on applying external carbon accumulation rates uniformly across heterogeneous landscapes without accounting for local variation in growth potential. Fifth, the temporal assumptions are problematic - the model assumes 2000-2016 conditions will persist to 2030 and beyond, ignoring climate change, land use pressures, and policy changes. Sixth, there are internal inconsistencies in data sources and resolutions that create coherency issues between predictor and outcome variables. The paper makes important contributions to restoration planning but overstates the precision and reliability of its estimates given these fundamental data and methodological limitations.

i Gemini 2.0 Flash Assessment

This paper presents a spatial model for natural regeneration potential in tropical forested countries, aiming to identify areas where forests can recover effectively with minimal intervention. The authors leverage remote sensing data and machine learning to estimate the potential for natural regeneration across tropical forest biomes. They estimate that 215 million hectares have the potential for natural forest regeneration, representing an above-ground carbon sequestration potential of 23.4 Gt C over 30 years. The study highlights the need for

targeted restoration initiatives, emphasizing the importance of natural regeneration in achieving national and global environmental agendas. The paper is well-written and addresses an important topic with significant implications for climate change mitigation and biodiversity conservation.

However, several issues warrant attention. First, the model relies on historical data (2000-2016) and assumes that these conditions will persist into the future (2030). This assumption may not hold, given ongoing climate change and land-use pressures. The authors acknowledge this limitation but do not fully explore the potential impacts of changing environmental conditions on natural regeneration potential. For example, increased drought and fire risk could significantly reduce the area suitable for natural regeneration, leading to an overestimation of carbon sequestration potential.

Second, the model incorporates a range of biophysical and socioeconomic variables, but the relative importance of these variables is not thoroughly investigated. The authors mention that the model containing only biophysical variables was selected because these factors are less subject to change and have a higher spatial resolution than socioeconomic variables. However, this decision could limit the model's ability to capture the complex interactions between human activities and natural processes. Additionally, the accuracy of the predictions varies considerably across different regions, with the lowest accuracies occurring in portions of South-east Asia. This spatial heterogeneity suggests that the model may not fully capture the local factors influencing natural regeneration potential.

Third, the study does not adequately address the potential trade-offs between natural regeneration and other land uses. While the authors acknowledge that some management actions may be required to facilitate natural regeneration (e.g., excluding grazers, fire protection), they do not fully explore the potential impacts of these actions on local communities and economies. For example, restricting grazing could have negative consequences for livestock farmers, while promoting forest regeneration could reduce the availability of land for agriculture. A more comprehensive assessment of the social and economic implications of natural regeneration is needed to ensure that restoration initiatives are equitable and sustainable.

Fourth, the paper lacks a thorough discussion of the uncertainties associated with the carbon sequestration estimates. The authors provide a range of 21.1-25.7 Gt C for the above-ground carbon sequestration potential, but the uncertainties associated with below-ground biomass and the long-term persistence of carbon sinks are not fully addressed. Additionally, the authors do not explicitly consider the potential for carbon leakage, which could offset the benefits of natural regeneration in one area by increasing deforestation in another.

Fifth, while the authors state that "Robust, natural regeneration certification schemes may also be needed - without them, issuing carbon credits will be challenging...", the authors have not provided detail or support for this call to action. The role of Indigenous stewardship is not included in this paper, and there is a risk that the use of this data will exclude Indigenous Peoples from benefits and participation.

Finally, the paper could benefit from a more detailed discussion of the policy implications of the findings. The authors mention the need for targeted restoration initiatives and the potential for carbon offsets, but they do not provide concrete recommendations for how governments and other stakeholders can promote natural regeneration at scale. A more actionable policy framework would enhance the practical relevance and usefulness of the study.

⚠ GPT-5.2 Pro: Key Issues Identified

1. 1. The target variable (“natural regrowth” as >5 m vegetation gain from 2000–2016) is a partial proxy for restoration success and likely misses early regeneration and slow-growing systems; it also does not ensure persistence beyond 2016.
2. 2. The upstream regrowth map used for labels has acknowledged humid-biome omission and low producer’s accuracy for area-adjusted estimates, creating systematic false negatives that can bias model training and geographic patterns.
3. 3. “Non-regeneration” points are sampled from a broad domain that likely includes lands under active management or repeated clearing; thus the model mixes biophysical capacity, human land-use decisions, and detectability rather than isolating biophysical potential.
4. 4. The paper reports high balanced-class accuracy but provides limited information on calibration and threshold-dependent tradeoffs (precision/recall), which are central when aggregating probabilities into expected areas and when using binary maps for targeting.
5. 5. Validation uses random splits with a post hoc autocorrelation-distance diagnostic but not fully spatially blocked cross-validation; performance may be optimistic and appears regionally variable (notably weaker in parts of Southeast Asia).
6. 6. The extremely narrow confidence intervals on the headline area estimates are not credible as full uncertainty bounds because they omit major sources of uncertainty (label noise, transfer/non-stationarity, predictor error/resolution mismatch, and model specification uncertainty).
7. 7. Predictor rasters have mixed spatial/temporal resolutions (30 m to ~1 km; climate normals 1970–2000; NPP 2000–2015; burn area 2001–2017), yet outputs are presented at 30 m; this can create visually fine-scale artifacts and overconfidence in local precision.
8. 8. Updating key forest covariates to 2018 while training on 2000–2016 outcomes introduces temporal inconsistency; combined with known version differences in GFW products, this may yield local incoherence and affect transfer to 2015/2030 predictions.
9. 9. Dropping socioeconomic covariates for prediction stability may increase omitted-variable bias, with biophysical variables acting as proxies for land-use intensity/governance; this complicates the “biophysical conditions can support” interpretation.
10. 10. Carbon accumulation estimates rely on resampling a 1 km carbon potential layer to 30 m with implicit within-km homogeneity and without propagating uncertainty or accounting for covariance between regeneration probability and carbon potential.
11. 11. The map is framed as a first-stage input, but discussion of offsets/targeting could be misread as identifying realizable, additional, and permanent sequestration opportunities; clearer operational guidance on use and limits would reduce misuse.
12. 12. Code is only available “on request,” limiting reproducibility and independent auditing of modelling choices (variable selection, sampling, preprocessing, and aggregation).

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C Other Papers

C.1 Abbink et al. 2025

C.1.1 Ratings Comparison

Metric	Claude Sonnet 4	GPT-5 Pro	Gemini 2.0 Flash
Overall	72 [65–78]	74 [60–86]	65 [50–80]
Claims	75 [68–82]	73 [60–85]	75 [60–85]
Methods	70 [62–77]	70 [55–82]	65 [50–75]
Adv. Knowledge	78 [72–85]	78 [65–88]	70 [55–80]
Logic & Comms	74 [68–80]	82 [72–90]	80 [70–90]
Open Science	68 [60–75]	35 [20–55]	70 [50–85]
Global Relevance	82 [76–87]	85 [75–92]	80 [65–90]

i GPT-5 Pro Assessment

This paper reports a well-executed incentivized field experiment in 10 low-income urban neighborhoods in Nashik, India, eliciting willingness to pay (WTP) for 5 kg of rice versus varying cash amounts from predominantly female respondents. Strengths include a clear, policy-relevant question; an incentive-compatible design with real redemption; careful piloting; transparent description of sampling (listing then household-level random sampling within slums); symmetric redemption logistics (vouchers for cash and rice, same shop); verification that all vouchers were redeemed; and multiple robustness checks for the construction of WTP/deadweight loss (DWL). The empirical analysis appropriately uses household-clustered errors and random-effects models with slum and round effects. The headline findings—that average WTP exceeds market value (implying a “deadweight gain,” DWG), that infra-marginal status and female headship strongly predict DWL/DWG, and that commitment motives and intra-household bargaining likely drive observed behavior—are interesting and relevant to the cash-versus-kind debate. However, several issues temper the strength of the conclusions and point to places where the paper could be strengthened.

First, the concept and language of “deadweight gain” require careful framing. The measure here is the individual respondent’s WTP relative to market price; welfare in the standard unitary model suggests cash weakly dominates in-kind for marginal beneficiaries and indifference for infra-marginal ones. The paper shows that when the decision-maker’s control over cash differs from control over in-kind, WTP for rice can exceed market value, giving a private DWG to the respondent. The authors acknowledge that household-level welfare may differ, and their conceptual framework clarifies that “DWG” can reflect intra-household redistribution or commitment value rather than efficiency gains in the standard sense. Still,

readers may interpret “deadweight gain” as a system-level welfare statement; the paper would benefit from consistently emphasizing that this is a decision-maker-centric metric and not a general-equilibrium welfare claim.

Second, the construction of WTP/DWL is necessarily discrete and involves assumptions for corner cases. For single-switch respondents, using the midpoint of the switch interval is standard, and the authors also present lower/upper-bound and interval-regression robustness. For rice-only and cash-only respondents, assigning WTP=550 and 25 (and testing alternatives 650/0) is sensible but still somewhat ad hoc; top-coding and bottom-coding materially influence the distribution tails and, given their frequency (rice-only 13–15%, cash-only ~33%), can meaningfully affect average DWL/DWG. Reporting partial-identification-style bounds on average DWL across plausible codings would make the headline results more transparent; the paper notes the overall DWG is likely an underestimate given the upper cap at Rs 500, but a formal bounded-mean presentation (and sensitivity to different upper cash caps) would be stronger.

Third, monotonicity and switching behavior. The design resembles a multiple price list but does not enforce monotonicity; 3.5% of observations had multiple switches and were excluded. This exclusion is standard in elicitation work, but additional sensitivity analyses would be valuable: (i) include multiple-switchers with monotonicity-imposing corrections; (ii) show that key coefficients (female headship, marginality) are robust when multiple-switchers are included using appropriate methods (e.g., interval regression allowing non-unique switch regions). The appendix suggests interval regression was tried for single-switchers; expanding that to include multi-switchers would address selection concerns, however small.

Fourth, an important internal inconsistency emerges between the main-text description and the instrument screenshots: the paper states respondents were “offered increasing cash amounts,” whereas Appendix E shows the choice order in decreasing cash amounts (500, 400, 350, ..., 50). Order effects and potential framing/anchoring matter in discrete choice lists. The paper should clarify the actual order used in the field, justify it, and (ideally) show that results are not sensitive to order. If decreasing order was indeed used, the cognitive load and starting-point anchoring could bias switching patterns and complicate interpretation of the “single-switch” categorization.

Fifth, classification of marginal vs infra-marginal respondents is based on baseline market rice purchases relative to the 5 kg experimental dose. This is practical, but potentially time-varying and measured with error; round-specific shocks to consumption or supply could change status. Because marginality is central to both theory and results, the paper should discuss measurement error implications and, where possible, show robustness to alternate thresholds (e.g., 4–6 kg) or to using round-specific measures if available. Relatedly, defining infra-marginality with respect to market rice alone abstracts from PDS quantities and any intertemporal shifts in household storage or consumption; the authors partly address this by controlling for round effects and showing limited price variation, but explicit sensitivity analysis on the marginality threshold would help.

Sixth, the interpretation of round effects. The probability of choosing cash rises over rounds; the authors suggest learning and/or within-household renegotiation after windfalls. This is plausible and consistent with round-by-round heterogeneity (effects concentrated among women in male-headed households), but it remains speculative without direct evidence on budget renegotiation. Moreover, attrition and respondent identity stability are not fully documented. The instrument states that if the prior respondent is absent, “identify any other

adult,” which could alter the decision-maker from round to round. The paper would benefit from: (i) round-by-round attrition analysis, (ii) a check on whether the same respondent answered across rounds, and (iii) robustness restricting to respondents observed in multiple rounds and/or those consistently responsible for rice purchases. Without this, dynamic interpretations should be offered more tentatively.

Seventh, the commitment-device evidence relies on post-draw self-reports of the “primary reason,” which are potentially subject to post-rationalization and social desirability. The binary commitment measure is derived from select reasons (“cash will get spent on less useful things” and “hard to control how cash will get spent”); the authors show robustness when including “we are running short of rice,” but the subjectivity remains. It would strengthen the argument to triangulate with baseline behavioral measures (e.g., intra-household control indices) in the main regressions rather than only in descriptive tables, while acknowledging endogeneity concerns.

Eighth, external validity and generalizability. The context is specific: 10 slums in one city, a commodity central to Indian diets, a very familiar in-kind program (PDS), and redemption through known local shops. These features make the experiment clean and credible but limit generalization to other geographies, goods, and delivery infrastructures. The paper is appropriately cautious, but the policy section could more explicitly note that the strength of the DWG likely depends on intra-household control, resale frictions, and the salience/fungibility of the in-kind good.

Finally, transparency and replicability. The paper describes the design in detail and includes screenshots of the instrument and vouchers, plus numerous tables and appendices. However, there is no link to pre-analysis plans, code, or data. Given the relative simplicity of the design, sharing de-identified data and code would substantially improve reproducibility and enable meta-analysis. Minor points: a logistic or probit mixed model in addition to the linear probability model would be a useful robustness check for the binary choice model; the paper controls for slum and round effects, but presenting slum-level heterogeneity could illuminate implementation or social-context differences; and testing sensitivity to alternative cash upper bounds (e.g., simulated extrapolation) would bound the likely magnitude of overall DWG. Overall, despite the caveats above, the paper provides credible, novel evidence on how intra-household bargaining and commitment considerations affect preferences for in-kind versus cash, and it makes a useful contribution to an important policy debate.

i Claude Sonnet 4 Assessment

This paper presents an incentivized field experiment testing deadweight losses from in-kind transfers through eliciting willingness to pay for rice versus cash in low-income Indian households. The core finding—that in-kind transfers generate deadweight gains rather than losses on average—contradicts standard economic theory and warrants careful scrutiny. The experimental design is generally sound with proper incentivization and voucher redemption, but several issues arise. First, the measurement approach relies on discrete cash choices creating intervals for WTP rather than precise values, leading to somewhat arbitrary assignment of WTP values for ‘rice-only’ (Rs 550) and ‘cash-only’ (Rs 25) respondents. Second, while the authors control for several alternative explanations (transaction costs, quality, trust), the interpretation relies heavily on female headship as a proxy for bargaining power, which may conflate other household characteristics. Third, the exclusion of multiple-switch respondents

(3.5%) without thorough robustness checks raises questions about selective analysis. Fourth, the theoretical framework in Appendix D, while intuitive, makes strong assumptions about intra-household budget allocation that may not hold in practice. The cross-round variation in choices suggests potential learning or adaptation effects that complicate interpretation. The policy implications about offering choice between cash and in-kind transfers are reasonable but require more careful consideration of implementation costs and complexity.

i Gemini 2.0 Flash Assessment

This paper investigates the presence of deadweight losses or gains associated with in-kind transfers, specifically focusing on food subsidies. The authors conducted an incentivized field experiment in India where low-income respondents were offered a choice between a free quantity of rice and varying amounts of cash. Contrary to standard economic theory, the headline results indicate a deadweight gain on average, although there is variation across respondents. The research explores potential explanations for this, including infra-marginality, commitment considerations, and intra-household bargaining dynamics. A striking contrast emerges: women in female-headed households exhibit deadweight losses, while women in male-headed households experience deadweight gains. The study highlights that the greater bargaining power of women within households increases the propensity to choose cash over rice. The authors use a well-designed experiment, which they acknowledge, controls for transaction costs, quality of product, trust considerations, and insurance against price volatility.

However, the study also suffers from limitations.

First, the analysis is restricted to female respondents to mitigate any confounding effects related to the respondent's gender, which reduces the sample size and potentially introduces selection bias, though an appendix suggests results are similar for the full sample.

Second, the experiment was conducted in a specific context (low-income urban neighborhoods in Nashik, India) which might limit its external validity to other regions or populations.

Third, the WTP measure is not truly continuous due to the discrete cash offers and the WTP measure is constructed from a panel with a small number of treatments; sensitivity to alternative assumptions about WTP bounds (or other methods of imputation), such as interval regression or other methods of dealing with grouped data, may have been warranted. Fourth, while the authors control for several factors, there may be other unobserved variables that could influence respondent choices and deadweight loss, such as social networks, access to credit, or other household characteristics.

Fifth, the authors acknowledge the limited temporal variation in rice prices during the experiment, which restricts their ability to analyze the insurance value of in-kind transfers. The framing of the choices as within a single, final round is potentially problematic, especially given the subsequent discussion of commitment devices and intrahousehold bargaining.

Sixth, though all vouchers were redeemed, no details are given of the process used to verify this.

Seventh, more details could be provided about the process and training associated with the listing exercise and the shopkeeper participation.

Eighth, the authors acknowledge some evidence of learning or renegotiation over rounds but the analysis of these elements is not fully convincing, with no clear explanation given for differing learning rates between male- and female-headed households, and a reliance on statistical significance.

Ninth, the theoretical framework has potential, but also presents the conceptual problem that gains or losses only accrue to the decision maker, and this is not a household-level welfare analysis.

These issues temper the confidence in the headline results, especially concerning the magnitude and interpretation of deadweight gains and losses.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.2 Aghion et al. 2017

C.2.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	83 [75–90]	65 [45–80]	86 [75–95] (n=2)
Claims	78 [70–86]	60 [40–75]	—
Methods	75 [65–84]	70 [50–85]	75 [58–88] (n=2)
Adv. Knowledge	81 [72–89]	65 [45–80]	86 [72–92] (n=2)
Logic & Comms	88 [82–94]	80 [70–90]	58 [45–75] (n=2)
Open Science	47 [35–60]	50 [30–70]	88 [90–100] (n=2)
Global Relevance	70 [58–78]	70 [50–85]	91 [82–100] (n=2)
NA	—	—	4 [4–5] (n=2)

i GPT-5 Pro Assessment

This agenda-setting paper models AI primarily as an extension of automation in a task-based framework and explores implications for growth, factor shares, and the possibility of singularities. The core theoretical contribution is to embed automation in a CES aggregator with elasticity of substitution below one (<0), leveraging a Baumol “weak links” mechanism to reconcile extensive automation with bounded capital shares and to generate rich transitional dynamics. The analysis shows: (i) in Zeira-style task automation, τ (the weight of automated tasks) rises with automation, which tends to raise capital’s share and growth via capital accumulation; (ii) when tasks are gross complements (<0), price effects counteract this by pushing expenditure away from increasingly cheap automated goods, so aggregate capital shares need not explode even with near-complete automation; (iii) under particular paths for automation (e.g., a constant fraction of remaining tasks becomes automated), the model can deliver asymptotic balanced growth with capital shares well below 100%, despite $\tau \rightarrow 1$. Useful comparative statics include a knife-edge condition for constant factor shares and a regime-switching simulation illustrating periods of rapid automation followed by pauses, aiming to mimic observed fluctuations in capital shares and growth. However, several substantive limitations temper the claims. Most crucially, automation dynamics τ are entirely exogenous; the main results hinge on specific paths of τ rather than arising from explicit microfoundations of R&D incentives,

costs of automation, or the discovery of new tasks. This creates identification-like concerns for interpretation: different plausible θ processes would yield very different long-run behavior, and the paper does not narrow these with data or structural estimation. The elasticity restriction $\epsilon < 0$ is structurally pivotal: it generates the “capital-depleting, labor-augmenting” twist of automation and the Baumol constraint, but the paper does not empirically validate $\epsilon < 1$ for the relevant aggregation of tasks, nor does it explore the robustness of results to alternative substitution patterns (beyond noting that $\epsilon > 0$ reverses some conclusions). The macro closure uses a constant investment rate rather than intertemporal optimization; while convenient and transparent, it limits welfare analysis and could alter transition dynamics relative to standard Euler-equation behavior. Labor supply is fixed; demographic dynamics enter only in later idea-production sections as S or n without explicit household side. In extending AI to idea production, the paper provides clear derivations: a one-time increase in μ raises levels but not long-run growth if $\epsilon < 1$, while continuous automation can raise growth by effectively growing “research effort per task.” These results depend on conditions like $BK/(CS) \rightarrow \infty$ and the assumption that the idea CES also has elasticity $\epsilon < 1$; these are plausible but untested and leave open questions about parameter magnitudes needed to match historical growth stability. The singularity section is logically careful in distinguishing Type I (unbounded but finite-time growth rates) from Type II (finite-time blow-up) and shows that full automation of ideas with $\epsilon > 0$ implies a mathematical singularity; with Cobb-Douglas, a threshold condition $\epsilon > 1$ can deliver Type II singularities even without complete automation. Yet these findings depend on simplified closures (e.g., constant S , constant savings rate), omit resource, thermodynamic, or other physical constraints, and abstract from costs and feasibility of automating essential “Baumol tasks.” To their credit, the authors explicitly discuss these objections (essential non-automatable inputs, fishing-out/burden-of-knowledge reducing μ , and hard natural-law constraints), and provide an appendix showing how super-fast GPT arrival ($\mu \rightarrow \infty$) can depress human R&D incentives and drive growth to zero in a Schumpeterian setting—nicely balancing the earlier singularity scenarios. The firm-level discussion (market structure, platforms, sectoral reallocation, internal organization, self-employment) is insightful but largely conceptual; it surfaces plausible mechanisms (e.g., imitation risks, inverted-U competition, knowledge diffusion vs business-stealing, delegation/monitoring tradeoffs) without formal modeling or empirical tests in this paper. The empirical section is intentionally light-touch and descriptive: capital shares by industry (KLEMS) show broad increases outside services since 2000; motor vehicles (a heavy robot adopter) exhibits rising capital shares in several countries, but cross-industry changes in robots vs capital shares (2004–2014) show little correlation. The authors properly caution that multiple forces (markups, concentration, unions, composition) confound simple links. That said, empirical evidence remains insufficient to discipline key parameters or to validate the CES/Baumol mechanism quantitatively; filtering choices ($HP = 400$) are reported but code and exact construction details are not provided, impeding replication. Figures from simulations depend on parameters not fully enumerated; the lack of shared code limits reproducibility. Overall, the paper’s strengths are clarity of conceptual framing, transparent derivations, and a coherent synthesis of automation, structural change, and AI-in-ideas, including careful discussion of boundary cases and objections. Primary weaknesses are exogeneity of automation and technology processes, lack of microfoundations for θ or At , limited quantitative calibration or identification, and minimal open-science support. The work advances the research agenda by sharpening questions and offering tractable models that can be extended to endogenize automation, embed optimizing households, incorporate

costs and feasibility constraints, and engage with data. It is persuasive as theory-building and agenda-setting rather than as an empirically validated or policy-calibrated analysis. Future work should (i) endogenize automation and task discovery (R&D races, costs, policy), (ii) estimate cross-sector substitution elasticities relevant to the “weak links” channel, (iii) calibrate transitional dynamics to match capital share and growth facts, (iv) test distinctive predictions (e.g., capital-depleting/labor-augmenting twist) using detailed input-output/task data, and (v) provide code and data workflows to facilitate replication and extension.

Gemini 2.0 Flash Assessment

This paper explores the potential impacts of artificial intelligence (AI) on economic growth, utilizing models and historical data to examine how AI might affect automation, productivity, and income distribution. The authors frame AI as the latest stage in a longer process of automation, referencing previous work by Zeira (1998) and Acemoglu and Restrepo (2016) to structure their analysis. They investigate several aspects, including how AI-driven automation might influence economic growth, capital and labor shares, and the possibility of singularities or superintelligence scenarios. The paper also considers the role of firm-level dynamics, market structure, and innovation incentives in mediating the relationship between AI and economic growth, and offers an analysis of sectoral evidence regarding capital shares and automation. The paper is well-written and presents a structured agenda for understanding the multifaceted relationship between AI and economic growth, contributing to the academic field by posing critical questions and refining existing models, but it is somewhat limited in scope. The authors use existing economic models (e.g., Zeira, Acemoglu-Restrepo) and adapt them to incorporate AI, offering a theoretical exploration, but more empirical validation and exploration of the assumptions could be useful.

Here are some specific points:

- **Theoretical Focus:** The paper primarily uses theoretical models to explore the potential effects of AI. While these models are valuable for understanding possible mechanisms, there is limited empirical validation to assess the real-world relevance of the model assumptions and predictions.
- **Simplifying Assumptions:** To make the models tractable, the authors often rely on simplifying assumptions (e.g., constant automation rates, exogenous technological change). The impact of these assumptions on the robustness and generalizability of the results is not always fully explored.
- **Endogeneity:** While the paper touches on endogeneity (e.g., automation driven by research effort), many factors are treated as exogenous. A deeper consideration of the endogenous nature of technology adoption, innovation, and market structure would strengthen the analysis.
- **Data and Measurement:** The empirical section uses capital share data and robot adoption as proxies for automation. These measures may not fully capture the nuances of AI-driven automation, which can affect various aspects of production and innovation beyond simple factor shares.
- **Limited Novelty:** The paper builds extensively on existing economic models, adapting them to the context of AI. While this approach provides a useful framework, the incremental novelty and impact beyond existing literature is not always clear. The modeling choices are relatively standard given the research questions, but this is well motivated.

- **Policy Implications:** The paper does not delve deeply into policy implications. While it raises important questions about the future of work and income distribution, more explicit discussion of policy options and their potential consequences would be beneficial.
- **Singularity Scenarios:** The discussion of singularities, while interesting, remains highly theoretical and speculative. The paper acknowledges the limitations and objections to these scenarios, but a more critical assessment of their plausibility and relevance would be helpful.
- **Communication of Evidence:** The sectoral evidence presented in Section 6 is somewhat descriptive and does not offer strong support for the theoretical models. The paper acknowledges the complexity of disentangling the effects of automation from other economic forces, but a more rigorous empirical analysis would be valuable.
- **Replicability:** The paper provides clear descriptions of the models and data sources, which would facilitate replication. However, the code and detailed data used in the simulations and empirical analysis are not provided, which limits the ability of other researchers to reproduce and extend the work.

In summary, this paper offers a valuable theoretical framework for analyzing the potential economic effects of AI. However, its reliance on simplifying assumptions and limited empirical validation raise questions about the robustness and real-world relevance of its conclusions. Addressing these issues would improve the paper’s contribution to the field.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.3 Akram et al. 2017

C.3.1 Ratings Comparison

Metric	Claude Sonnet 4	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	72 [63–80]	87 [82–92]	80 [65–90]	70 [50–78] (n=1)
Claims	68 [59–76]	86 [80–91]	85 [75–95]	—
Methods	75 [67–82]	80 [72–87]	80 [70–90]	70 [50–90] (n=1)
Adv. Knowledge	78 [70–84]	88 [83–93]	75 [60–85]	70 [50–90] (n=1)
Logic & Comms	76 [69–83]	84 [78–90]	85 [75–95]	70 [50–90] (n=1)
Open Science	73 [66–79]	35 [25–50]	75 [60–85]	70 [50–90] (n=1)
Global Relevance	82 [76–87]	90 [85–96]	85 [75–95]	70 [50–90] (n=1)
NA	—	—	—	2 [1–3] (n=1)
NA	—	—	—	2 [1–3] (n=1)
NA	—	—	—	70 [50–90] (n=1)

This paper uses a large and thoughtfully designed saturation RCT to study direct and general-equilibrium effects of subsidizing seasonal migration among landless households in rural Bangladesh. Strengths include: (i) randomized variation in offer intensity across 133 villages (roughly 14% vs 70% of eligibles), creating plausibly exogenous village-level labor supply shocks; (ii) multiple complementary data sources (six-round high-frequency household labor modules, endline income survey, two employer surveys, repeated shopkeeper price surveys, and a 2016 follow-up on re-migration and food security); (iii) careful attention to both individual and market-level outcomes (migration, earnings by location, wages, hours, employer costs/profits, and prices); and (iv) theory-based interpretation linking saturation, strategic complementarities in migration, and village labor market slack. The main findings—substantial increases in migration at high saturation, higher urban earnings, increases in rural agricultural wages and in-week work availability for those at home, modest food price effects (especially fish), and few signs of harmful spillovers on non-beneficiaries—are generally well supported by the experimental variation and consistent across data sets. Nonetheless, some aspects would benefit from clarification, additional robustness, or alternative checks. First, identification of employer wage effects relies on retrospective employer reports by fortnight; recall may be noisy despite enumerator protocols, and wages exclude in-kind (food) payments, which could bias measured wage changes if the composition of compensation adjusted during shocks. Providing robustness that includes estimated in-kind values or validating against any contemporaneous logs would strengthen claims. Second, the employer profit results are fragile: costs rise and one profits specification is significant, but differencing pre/post (2013 to 2015) reduces significance, and yields show no systematic change. The paper is appropriately cautious, but interpretation should highlight the uncertainty and potential recall/measurement error in employer accounting. Third, the intra-household labor substitution null (no increased labor by non-primary members) is informative, but power and measurement deserve discussion: the high-frequency module tracks all household members, yet the “non-primary worker” category might include heterogeneous participants with limited participation margins in a short window; power calculations or confidence intervals around substantively meaningful minimum detectable effects would contextualize the null. Fourth, the mechanism evidence for strategic complementarity (network ties, traveling with companions) convincingly supports risk/coordination channels, but is based on a restricted 2013 network sub-sample (n 998) that may not be fully representative; selection into that sub-sample and any differences with the main sample should be detailed. Relatedly, while the differential take-up to identical offers across high- vs low-saturation villages is compelling, the paper does not fully decompose cost-sharing vs risk-sharing vs destination job-finding channels; additional heterogeneity analysis (e.g., by prior migration experience, wealth/credit proxies, or distance to destinations) or leveraging “can rely on” vs “know but can’t rely on” ties as instruments for own migration would bolster mechanism claims. Fifth, some IV specifications in the high-frequency data have relatively weak first stages (e.g., F 5.6 when instrumenting own migration using high-intensity assignment only), and standard errors are large; results relying on these should be interpreted cautiously and, where feasible, supplemented by intent-to-treat (ITT) contrasts or village-level two-stage regressions that harness stronger first stages (as done elsewhere). Sixth, multiple-testing concerns are non-trivial: many outcomes (wages across sectors, prices for 12 items plus an index, several labor and income outcomes) are examined; the few statistically significant food price changes (e.g., fish at 10% level) could be false positives without

adjustments. Reporting false discovery rate (FDR) or familywise error rate corrections, or at least highlighting which results remain under BH/FDR, would improve interpretability. Seventh, the “best-week” comparison for primary workers’ home earnings is intuitively compelling but risks selection on timing; an alternative would be to use week-by-village variation in the fraction absent and instrument it with randomized offer saturation interacted with week-of-season to isolate exogenous slack. Eighth, while the paper argues villages are closed labor markets during the lean season, only modest evidence (distance to other treated villages unrelated to outcomes) is provided; additional checks on cross-village labor flows and sensitivity to excluding villages near treatment borders would be useful. Ninth, the construction of village emigration rates uses denominators (eligible counts) that were imprecise in 16 villages; sensitivity analyses restricting to the 117 villages with precise listings are helpfully reported, yet it would be good practice to show robustness to alternative denominators or re-weighting. Tenth, there is a minor internal consistency issue to clarify: in the wage IV table, the right-hand variable is labeled as the 2015–16 migration rate though the wage reference period is late 2014; if this is a label error (most likely), it should be corrected; if not, the timing/exclusion restriction needs justification. Eleventh, the endline income measure uses a five-month recall; while triangulation with the high-frequency measures shows coherence in magnitudes, a more explicit reconciliation (e.g., converting both to common weekly terms, sanity checks on variance ratios, and comparisons by treatment arm) would enhance credibility. Twelfth, real-income results are not explicitly netted out by the measured local price changes; given the reported Laspeyres index increase of about 2.7% at high saturation, providing real-income changes would help quantify welfare gains. Thirteenth, code/data availability is not indicated; replication would benefit from public de-identified data, survey instruments, pre-analysis registration details (beyond trial registration), and code. Finally, external validity is naturally limited to seasonal/circular migration contexts with similar slack, agrarian seasonality, and urban demand; the paper acknowledges this and notes destination-side effects are not studied. Overall, the design is strong for detecting village-of-origin equilibrium effects, the empirical patterns are internally coherent, and conclusions are appropriately cautious where evidence is weaker. Addressing the above issues (especially measurement/recall, multiple testing, timing consistency, and additional robustness to in-kind wages and real-income calculations) would further solidify the paper’s contributions.

i Claude Sonnet 4 Assessment

This paper examines the general equilibrium effects of seasonal migration subsidies in rural Bangladesh through a large-scale randomized controlled trial. The study addresses an important question about spillover effects of migration programs, which is highly relevant for development policy. The experimental design is sophisticated, randomly varying the intensity of treatment across 133 villages and collecting rich high-frequency data on labor market outcomes. The authors find that migration subsidies create positive spillovers through increased village wages and work availability, with migration decisions showing strategic complementarity. The identification strategy is strong, exploiting random assignment of different treatment intensities across villages. However, there are several concerns: (1) The timing of the intervention was delayed due to political disturbance, potentially affecting the interpretation of results, (2) Some key results rely on relatively small samples (e.g., employer survey with ~400 observations), which may limit statistical power, (3) The assumption that villages constitute

closed labor markets may not hold perfectly, and (4) Some robustness checks show sensitivity to specification choices. The wage effects, while statistically significant, are based on employer reports with potential recall bias despite efforts to minimize this. The food price effects are small and mixed, with increases in some items offset by decreases in others. The paper makes important contributions to understanding migration spillovers and provides policy-relevant insights, but some findings rest on relatively weak statistical foundations and would benefit from additional robustness testing.

Gemini 2.0 Flash Assessment

This paper examines the impact of a seasonal migration subsidy program in rural Bangladesh on labor market outcomes, food prices, and food security. The authors conducted a randomized controlled trial across 133 villages, varying the saturation of subsidy offers to landless households to induce temporary out-migration during the lean season. The study design is rigorous, employing high-frequency surveys, endline surveys, and employer surveys to capture a comprehensive picture of the program's effects. The authors also leverage existing social network data to examine spillover effects. A key finding is that the migration subsidies increased beneficiaries' income and generated positive spillovers, including increased agricultural wage rates and available work hours in the villages. The study also finds evidence of strategic complementarity in migration decisions, with higher density of subsidy offers increasing individual take-up rates. Food prices increased slightly, driven by increases in protein prices but offset by decreases in non-tradable food prices. The authors provide a theoretical framework to interpret the empirical results, modeling the village labor market with endogenous migration. Policy implications are discussed, suggesting a re-thinking of restrictions on internal mobility and informing the choice between rural development programs and investments in infrastructure to improve connectivity. This research makes several important contributions. The RCT design and extensive data collection allow for relatively strong causal claims about the effect of the intervention. The focus on spillover effects is also valuable, as this is an area often overlooked in migration studies. Finally, the global relevance of the study is high, as seasonal migration is a common phenomenon in developing countries, and the findings can inform policy decisions related to poverty reduction and labor market interventions. However, there are some potential limitations. First, the generalizability of the findings may be limited by the specific context of rural Bangladesh. Second, there are assumptions in the simple logic that employers don't react to the treatment in the short-run by changing their production function. This is probably a reasonable assumption in the short run, and we collected data from employers to check whether it is true. The departure of migrants may lower the demand for food in the village, whereas the extra income they bring back could increase the demand for food. If food markets are not well integrated across space, then food prices could either decrease or increase as a result. With integrated markets, supply would adjust, and there should not be any detectable change in food price.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.4 Alatas et al. 2019

C.4.1 Ratings Comparison

Metric	Claude Sonnet 4	GPT-5 Pro	Gemini 2.0 Flash
Overall	75 [65–85]	78 [70–86]	65 [50–80]
Claims	72 [62–82]	82 [74–90]	70 [55–85]
Methods	78 [68–88]	82 [72–90]	65 [50–80]
Adv. Knowledge	80 [70–90]	76 [66–86]	70 [55–85]
Logic & Comms	85 [75–92]	86 [80–92]	75 [60–90]
Open Science	55 [45–65]	45 [30–60]	70 [50–85]
Global Relevance	85 [75–93]	72 [58–82]	80 [65–95]

i GPT-5 Pro Assessment

This paper reports a large, carefully designed social-media field experiment in Indonesia that randomizes how immunization messages are delivered via Twitter and links online diffusion to offline beliefs and reported behaviors. Core online identification is strong: by randomizing whether a celebrity directly authors a tweet versus retweets an otherwise identical message composed by an ordinary user, and exploiting the platform feature that downstream users see the originator but not intermediate retweeters, the authors separate endorsement effects from mere reach. They further randomize “credibility boosts” (source citations) for non-sensitive messages and the timing of influencer activity, and, in Phase III, exogenously force some ordinary followers to retweet to address selection into retweeting. The Twitter feed was chronological during the study window, reducing concerns about algorithmic ordering. Online outcomes (likes/retweets) are observed for the universe of relevant followers, and analyses use Poisson models with appropriate fixed effects, clustering, and robustness checks. The main online findings—substantial endorsement effects ($1.7\times$ more downstream engagement for celebrity-authored messages vs. masked celebrity retweets), reduced diffusion when sources are cited ($\sim 25\text{--}50\%$ drops depending on configuration), and concave “complex contagion-like” returns to multiple exposures—are well-supported by design and analysis, with plausible mechanisms and a formal model in the appendix framing why seemingly “more credible” sourcing can reduce sharing.

Identification threats and limitations mostly concern measurement and external validity rather than internal validity for the online results. First, downstream exposure paths are inferred using the baseline follower network and shortest-path logic; while the authors scrape retweet cascades, the precise path can be ambiguous when multiple routes exist. This introduces measurement error (especially if follow/unfollow dynamics changed between the baseline network snapshot and tweeting), likely attenuating effects but not obviously biasing treatment contrasts given randomization. Second, for the F2 endorsement test, in theory F2s should not directly follow the celebrity; the paper’s schematic implies this restriction, but the implementation details on enforcing it for analysis are only implicit. Third, the organization-level analyses have limited power (only 7 orgs), and those results are appropriately relegated to the appendix. Fourth, the negative effect of source citation could partly reflect mechanical or UX frictions (e.g., links occupying characters or changing tweet aesthetics) in addition to sig-

naling; while the randomization to source for non-sensitive tweets addresses many confounds, more granular checks (e.g., link vs handle, character counts) are not reported. Fifth, while Poisson is standard for count outcomes, overdispersion is likely; the use of robust (clustered) SEs mitigates but does not fully address distributional misspecification; showing Negative Binomial robustness would strengthen confidence.

The “shape of exposure” analysis cleverly uses exogenous same-day exposure variation from multiple celebrities followed, with user and date fixed effects. This supports a concave increase in retweeting from the first additional exposures (1→2 and 1→3) and flattening thereafter. However, it focuses on same-day doses among users following 3 celebrities; longer-window accumulation and heterogeneous user attention patterns are not explored, so extrapolation beyond the tested dose window is limited.

The offline component offers suggestive evidence that randomized exposure to the campaign (via the phase assignment of celebrities) increased awareness of the hashtag and immunization discussions on Twitter and improved knowledge on a salient fact (domestic production, linked to halal concerns), with mixed results on other knowledge items. Reported increases in knowing neighbors’ immunization status and in network members’ immunization behavior are intriguing but should be interpreted cautiously. The offline sample is recruited via promoted tweets among followers of participating accounts, skewing urban and likely more digitally engaged than the general population; this limits external validity. The exposure instrument uses “actual exposure” conditional on “potential exposure” (number of followed study celebrities and their planned tweet counts), which is sound given random phase assignment, but relies on a baseline follower snapshot; if follow/unfollow behavior changed, exposure mismeasurement may attenuate effects. Self-reported outcomes (own and network behaviors) introduce recall and social desirability concerns; the paper acknowledges that increased reporting could reflect more conversations rather than true behavior change, and randomization-inference p-values are often marginal (frequently between 0.05 and 0.15). There is no administrative verification of immunization nor geospatial matching to facility records, which would strengthen claims. Multiple hypothesis testing is partly acknowledged (Bonferroni style for the knowledge set), but a more systematic adjustment across families of outcomes would be welcome. Power is limited for own-behavior outcomes given the low probability that respondents had a child of vaccinating age during the study period.

Other design notes: “Sensitive” tweets always included sources; source effects are therefore estimated only on the randomized subset of non-sensitive tweets (correctly handled in analysis). Celebrities could veto messages but reportedly did not, helping maintain exogeneity. The endogeneity of observed retweet counts driving further diffusion is addressed with a Phase III manipulation varying the number of forced retweets, with null effects on downstream retweeting. Clustering at the original-tweet level for online regressions and at the “combination of celebrities followed” for offline regressions is reasonable; randomization-inference provides additional comfort given complex dependence. The conceptual model in Appendix A is useful to interpret counterintuitive sourcing results and to clarify when endorsement and credibility might have offsetting effects through signaling incentives.

Communication is generally clear, tables are interpretable, and the alignment between claims and evidence is careful—especially the differentiation between strong online causal results and more tentative offline effects. However, policy prescriptions in the conclusion (e.g., “send messages without citing credible sources”) arguably overreach, given ethical and long-run trust considerations and context-specificity; a more nuanced framing distinguishing message

diffusion from belief accuracy and downstream welfare would be advisable.

Open science and reproducibility are mixed. The trial is pre-registered (registry ID cited) and IRB-approved. Methods and randomizations are described in notable detail. However, the Twitter Firehose data are proprietary and may be hard to share; the paper does not state that code and data are publicly available. Replication without author-provided materials would be challenging. Providing de-identified derived datasets (e.g., tweet-level outcomes and assignments) and code would substantially improve reproducibility.

Overall, the paper delivers unusually credible causal evidence on how influencer identity and message features affect online diffusion at national scale, and it links these to suggestive offline changes. The main limitations lie in the offline measurement strategy, external validity, and reproducibility constraints inherent to platform data. The work is valuable to both researchers and practitioners designing public-health campaigns, but results on sourcing and offline behaviors should be interpreted within the study's context and evidentiary bounds.

i Claude Sonnet 4 Assessment

This paper presents a large-scale Twitter experiment in Indonesia examining celebrity endorsement effects in an immunization campaign. The authors recruited 46 celebrities/organizations and 1,032 ordinary users to randomly tweet immunization content, allowing causal identification of endorsement versus reach effects. The experimental design is sophisticated, exploiting Twitter's retweet structure where followers-of-followers (F2) can be randomly blinded to celebrity involvement. Key findings include: celebrity endorsement increases likes/retweets by 70%, but source citation reduces diffusion by 50%, and multiple exposures show concave effects. The paper also finds suggestive evidence of offline impacts through a phone survey. While the experimental design is innovative and the scale impressive, several concerns emerge. The identification strategy for separating endorsement from reach effects relies on potentially endogenous F1 retweeting decisions, though robustness checks with forced retweets partly address this. The offline survey analysis suffers from limited statistical power and relies heavily on self-reported measures that may be influenced by social desirability bias. The theoretical framework in Appendix A, while providing intuition, doesn't clearly predict the observed negative effects of source citation. Multiple hypothesis testing is not consistently addressed across outcomes. Data availability appears limited - the paper mentions code and data being available 'as reasonably possible' but doesn't provide clear replication materials. The external validity may be limited to the specific Indonesian Twitter context in 2015-2016. Despite these limitations, the paper makes important contributions to understanding social media information diffusion and provides policy-relevant insights about celebrity endorsements versus source credibility in public health campaigns.

i Gemini 2.0 Flash Assessment

This paper investigates the impact of celebrity endorsements on social media, specifically focusing on a nationwide Twitter experiment in Indonesia promoting vaccination. The authors partnered with 46 high-profile celebrities and organizations, totaling 7.8 million followers, to randomly tweet or retweet content promoting immunization. The study leverages the structure of Twitter's retweet chains to disentangle reach versus endorsement effects. The experimental design involves randomizing several aspects of the tweets, including the originator (celebrity

vs. ordinary user), the inclusion of a source citation, and the timing of the tweets. They then measure online reactions (likes and retweets) and offline outcomes (beliefs, knowledge, and behavior) via a phone survey. The main findings suggest that celebrity endorsements significantly increase engagement (likes and retweets), but the inclusion of source citations decreases diffusion. Moreover, multiple exposures to the campaign have a non-linear effect, with diminishing returns after a few exposures. The phone survey results suggest that the campaign increased awareness of immunization and influenced knowledge and reported behavior related to immunization within individuals' networks. The paper contributes to the literature on information diffusion, social influence, and public health campaigns. The setting and experimental design are a notable strength. The fact that they use real celebrities and real followers is also a strength. However, the analysis relies on observational data and survey responses, which may be subject to biases (selection, social desirability, recall bias), and they rely on LASSO for variable selection, which is not always stable. The offline effects are relatively modest. The theoretical model is very abstract and has almost no contact with the data analysis. The magnitudes of the effects are relatively small, and some of the interpretations rely on subtle nuances of social dynamics, which may not be robust. There is a risk of over-interpreting statistically significant results given the complexity of the social context and the numerous tests performed.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.5 Alcott et al. 2024

C.5.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	86 [78–92]	75 [55–85]	86 [73–96] (n=2)
Claims	82 [72–90]	80 [60–90]	—
Methods	78 [68–86]	70 [50–80]	78 [58–87] (n=2)
Adv. Knowledge	89 [84–95]	70 [50–80]	32 [28–38] (n=2)
Logic & Comms	90 [85–95]	85 [75–95]	78 [70–82] (n=2)
Open Science	85 [78–92]	80 [60–90]	96 [92–100] (n=2)
Global Relevance	80 [72–88]	70 [50–80]	12 [6–20] (n=2)
NA	—	—	4 [4–5] (n=2)
NA	—	—	4 [4–5] (n=2)
NA	—	—	20 [14–26] (n=2)

GPT-5 Pro Assessment

This paper develops a clear sufficient-statistics framework to evaluate the welfare effects of nudges in imperfectly competitive markets with taxes and endogenous prices, and implements

it with two incentivized online experiments on fuel-economy and sugary-drink labels. The theoretical contribution is strong and well integrated with the empirical sections. The main empirical finding—that labels reduce average demand for the more harmful good but can reduce total surplus by increasing the variance of choice distortions, especially when taxes are optimal or pass-through is low—is carefully stated as model-based and parameter contingent. Nonetheless, several identification and external-validity issues deserve attention, and some assumptions underlying the empirics are strong relative to the weight the welfare formulas place on treatment-effect heterogeneity and its covariance with bias.

Key theoretical points and potential limitations: - The welfare formulas rely on smooth demand and a constant conduct parameter (η), and neglect higher-order terms in the demand elasticity with respect to price and nudge intensity. These are plausible approximations but are not tested empirically. - The price-impact result $dp/d\tau = (1-\eta)E[\epsilon]$ (and its multi-good analogue) hinges on the condition $\eta \cdot dD/d\tau = -\eta \cdot E[\epsilon] \cdot dD/dp$. This is reasonable in cases (e.g., constant elasticity, homogeneous η), yet its validity is not empirically verified for the settings studied, and nudges could plausibly shift elasticities in ways that violate this restriction. - The sufficient-statistics welfare expressions are for marginal consumers; the empirical implementation substitutes population-level statistics, justified by precision gains and potential generalizability as marginal sets vary. This is a useful approximation, but the sensitivity of welfare signs to these moments suggests that bias from employing population moments over marginal ones could be nontrivial in some applications.

Experimental design and measurement concerns (cars): - Bias is defined as the difference between baseline relative WTP and a “true” relative value constructed from WTP if gas is free minus self-reported fuel costs. This choice is internally consistent with the incentives but raises measurement-error risks: (i) WTP-if-gas-free is itself elicited and could embed idiosyncratic error; (ii) miles driven and gas prices are self-reported; (iii) the assumption that depreciation is age-driven (not mileage) is convenient but contestable; (iv) the “true” value omits other ownership cost differentials (e.g., maintenance, performance), potentially conflating preference heterogeneity with bias. - The incentive scheme is nonstandard and cognitively demanding; although the authors mitigate this (explanations and emphasizing “answer honestly”), understanding might remain imperfect (they cite this literature). If participants do not fully grasp payoffs, elicitation noise may inflate treatment-effect variance. - Off-MPL responses are imputed to the median of off-MPL values and outliers are removed using a wide, four-variable top/bottom 5% screen. These steps improve precision for variance estimation but can bias estimates of $\text{Var}[\epsilon]$ and $\text{Cov}[\epsilon, \tau]$, the pivotal welfare statistics. Appendix checks (keeping all or only 1% trimming) show similar point estimates but with much larger uncertainty, underscoring fragility of the variance components to data cleaning. - The experimental context is hypothetical for vehicles; real purchasing constraints, financing, search, and dealership interactions are absent. This likely inflates salience and could both raise mean effects and alter heterogeneity relative to the marketplace.

Experimental design and measurement concerns (drinks): - Bias is constructed from nutrition knowledge and a self-control proxy via an external mapping (η , compensated elasticity) from prior work, assuming linearity and unconfoundedness. This is plausible but strong; misspecification directly affects the level and dispersion of η and thus $\text{Cov}[\epsilon, \tau]$. The externality is assumed homogeneous; in reality health externalities likely vary (e.g., by BMI, comorbidities, insurance spillovers). Both choices simplify estimation but potentially mischaracterize true distortion variance. - Only ~22% of participants had an MPL choice realized; labels were

shown on screens but not printed on shipped products. This reduces real-world correspondence and may affect the relationship between preferences, bias proxies, and realized consumption. - The sample is recruited via targeted social media ads, less representative than the probability panel used for cars; the sample has higher education and lower average self-control than national averages. No weighting is used. These choices are acceptable for an illustrative experiment but limit generalizability and could affect heterogeneity patterns.

Econometric specification and identification of heterogeneity: - Treatment-effect variance is identified from the treatment-control difference in the dispersion of baseline-to-endline WTP changes, assuming equal elicitation noise and normality of β , γ , δ , and ϵ . This normality assumption is convenient but not tested. The approach also presumes independence between measurement error in β and γ (Assumption 1); in the cars study, β uses baseline WTP, creating potential mechanical correlation with baseline-endline change. The authors acknowledge this and provide an alternative covariance check only for drinks. The resulting $\text{Cov}[\beta, \gamma]$ for cars—central for the welfare signs—may be biased toward zero. - The multi-level mixed-effects model with random slopes is a reasonable choice for precision, but given the centrality of heterogeneity, additional nonparametric or robust alternatives (e.g., deconvolution, randomization-based dispersion tests, or IV approaches for bias proxies) would further strengthen claims.

Policy parameters and sensitivity: - The pass-through rate and relative markup are calibrated from adjacent literatures (hybrid subsidies, customer rebates, soda taxes). These are thoughtful choices but imperfect analogues for the specific product pairs. Results are highly sensitive to α and β ; the paper appropriately provides sensitivity figures over wide ranges. Nonetheless, estimated welfare magnitudes are small; sign reversals can occur for plausible parameter shifts, underscoring the need for local calibration in applied use. - Psychic costs are measured in drinks and suggest large negative utility for graphic warnings; the main welfare tables set $I=0$, while narrative explains how including I would further worsen graphic warnings. Presenting headline welfare numbers both with and without I would clarify policy conclusions.

Robustness and transparency: - The paper offers numerous robustness and clarity checks: label-specific results, trimming checks, “marginal consumer” subsamples, rich sensitivity plots over $\text{Var}[\beta]$, $\text{Cov}[\beta, \gamma]$, β , γ , and $E[\beta]$. Registration is documented; lack of a pre-analysis plan is justified by structural components; replication files are promised. These are strong transparency practices.

Overall judgment: The theory is a valuable, portable contribution clarifying when average effects are irrelevant and elevating distortion variance and targeting as central statistics. The empirical demonstrations are thoughtfully designed and incentivized but inherently limited in external validity and reliant on strong distributional and measurement assumptions precisely where the welfare sign is determined (heterogeneity moments). The authors are appropriately cautious about generalizability. The work meaningfully advances how we evaluate nudges and provides a template for future applied measurement, while the quantitative welfare conclusions from these specific experiments should be interpreted cautiously and as illustrative rather than definitive.

i Gemini 2.0 Flash Assessment

This paper presents a theoretical framework and experimental evidence on the welfare effects of nudges, such as simplified information and warning labels, in markets with taxes and

endogenous prices. The authors argue that welfare depends not only on average treatment effects, but also on how the nudge affects the variance of choice distortions. They implement the framework with experiments evaluating automotive fuel economy labels and sugary drink health labels. The paper’s results clarify that average treatment effects are an incomplete and potentially misleading measure of welfare, and that measuring heterogeneity in treatment effects is crucial. The authors find that labels decrease purchases of low-fuel economy cars and sugary drinks, but may decrease welfare, because they increase the variance of choice distortions. The paper develops a theoretical framework for evaluating the welfare effects of nudges using public finance sufficient statistic approaches and applies the framework to novel randomized experiments evaluating automotive fuel economy labels and sugary drink health labels.

The paper is well-written and clearly structured, and the authors provide a thorough discussion of the theoretical framework and experimental design. The paper also provides a comprehensive literature review and discusses the implications of the findings for policy. However, the paper has some limitations. First, the experimental design is somewhat artificial, and it is unclear whether the results would generalize to real-world settings. Second, the sample sizes are relatively small, which may limit the statistical power of the study. Third, the authors’ estimates of the variance of treatment effects are somewhat imprecise, which may limit the reliability of the welfare analysis. Fourth, the paper assumes that the nudges only affect valuations of a single product, which may not always be the case in real-world settings. Finally, the authors don’t address distribution effects, that is, the intervention might have different welfare implications for different groups.

Overall, the paper is a valuable contribution to the literature on nudges and welfare. The paper’s theoretical framework and experimental evidence provide new insights into the welfare effects of nudges, and the paper’s findings have important implications for policy. However, the paper’s limitations should be taken into account when interpreting the results.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.6 Angrist et al. 2023

C.6.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	92 [87–96]	75 [60–85]	86 [79–94] (n=2)
Claims	90 [85–95]	80 [65–90]	80 [72–88] (n=1)
Methods	85 [78–91]	75 [60–85]	82 [74–89] (n=2)
Adv. Knowledge	93 [88–97]	75 [60–85]	92 [84–99] (n=2)
Logic & Comms	88 [82–93]	80 [65–90]	82 [73–88] (n=2)
Open Science	40 [25–55]	70 [55–80]	68 [56–81] (n=2)
Global Relevance	96 [92–99]	85 [70–95]	92 [88–98] (n=2)
NA	—	—	4 [4–5] (n=2)

NA	—	—	4 [4–5] (n=2)
NA	—	—	92 [88–98] (n=2)

i GPT-5 Pro Assessment

This manuscript reports a coordinated set of randomized trials of phone-based instruction in primary mathematics across five low- and middle-income countries during school closures, with additional analyses of delivery models (NGO vs government), ancillary outcomes (beliefs, non-cognitive skills), a natural shock (typhoon), and multiple measurement-validity checks for phone-based assessments. On the core questions—can low-tech, phone-delivered targeted instruction raise learning during emergencies and scale through government systems?—the design and evidence are strong, with large, precisely estimated intent-to-treat effects pooled across sites and consistent positive effects in each country. The study’s breadth (five countries, >16,000 households), clear pre/post structure, and high engagement rates (70–80%) bolster credibility and policy relevance. Still, several limitations and threats merit attention.

Internal validity is generally strong for the primary learning outcomes. Random assignment, balance checks, and high, non-differential follow-up (~80%) reduce selection concerns. The authors appropriately cluster standard errors at the unit of randomization (household or school-grade) and show low ICCs in cluster-randomized arms. However, mixed randomization units across arms and countries complicate pooled inference; while clustering by assignment unit is correct, readers would benefit from complementary hierarchical/multi-level models to better reflect site-level heterogeneity and to separate within- and between-site variance. Pooled OLS with country fixed effects and an alternative weighting scheme are presented; adding a random-effects meta-analytic specification would more transparently quantify cross-site variation and shrinkage.

Measurement via phone-based assessments is a common concern; the paper addresses this unusually well. Time caps, solution explanation checks, and ensuring enumerators assessed students they did not tutor limit differential help and expectancy bias. Validation exercises are convincing: phone vs in-person assessments for the same students in Kenya show no systematic differences; back-checks show strong reliability; randomized item variants at a given proficiency level show equivalence; and effort tasks indicate no treatment-induced test-taking effort effects. Residual risks remain (household assistance beyond enumerator detection; mode familiarity effects aligned with treatment), but the presented checks substantially mitigate them.

The typhoon analysis in the Philippines is a useful quasi-experimental addition but should be interpreted with more caution than the RCTs. While treatment is orthogonal to typhoon exposure, “affected” status might correlate with unobservables not captured by the included controls. Baseline equivalence on observables between affected and non-affected students is reassuring but not dispositive. The results nonetheless support the paper’s broader claim that the phone-call modality remains effective amid compounded shocks.

Findings on government vs NGO delivery (Nepal and the Philippines) are valuable for scalability. Both delivery models yield statistically indistinguishable effects of similar magnitude. This directly addresses common “voltage drop” concerns when moving from NGO-led pilots to government implementation. One nuance is that in Nepal government teachers served students outside their own regions, whereas in the Philippines teachers called students in their schools/grades; these implementation differences could matter for external validity to standard public systems, although the direction of bias is unclear. The embedded randomization

of teacher participation in Nepal and the resulting shifts in self-reported beliefs and practices (e.g., increased targeting to learning level; increased parent engagement; higher desire to teach) are promising, but they rely on self-reports and lack observed classroom practice or post-program student outcomes; persistence is unknown.

External validity is unusually well supported for this literature through multi-country replication, multiple languages, and implementation by both NGOs and government. Still, two caveats limit generalization: (i) the approach requires phone access; households without phones (a non-trivial share in some settings) are not captured, potentially excluding the most marginalized; the authors note high mobile penetration relative to internet but do not quantify coverage in the study samples versus the full population; (ii) the Kenya site is a large private-school network and includes grades 1–2 (vs grades 3–5 elsewhere) with partial school reopening during implementation; these differences likely contributed to smaller effects there and illustrate context sensitivity. The heterogeneity analysis is concise; stronger gains among children of less-educated caregivers are plausible and policy-relevant. Minimal gender or baseline-level heterogeneity likely reflects the targeted, one-on-one pedagogy.

Statistical reporting is mostly careful and transparent. The main effects are large (0.30–0.35 SD pooled for phone+SMS) and robust across specifications; SMS-only effects are modest overall (0.08 SD) and concentrated in the most disrupted contexts (Uganda, Philippines). There are many secondary outcomes (beliefs, demand, non-cognitive measures); no multiple-hypothesis correction is applied, and several reported p-values hover around conventional thresholds. The main learning results would survive any reasonable correction, but the manuscript should either pre-specify families of outcomes with adjustments or present false discovery rate-adjusted q-values. Relatedly, the primary learning metric “average level” (0–4 from innumerate to division, then standardized) is coarse; complementary analyses by item/proficiency and additional competencies partly address this. Durability of effects remains unknown; endline is “a few months after” program end, which is helpful, but longer-term follow-ups would strengthen claims about resilience and cost-effectiveness over time.

Implementation and monitoring are clearly described, with credible evidence that targeting accuracy improved across trials (“learning curve”), plausibly contributing to larger effect sizes over time. Variation in dosage (8–16 weeks, differing teacher-student ratios) is noted but not exploited analytically; dose-response analyses would add value. Potential spillovers to non-treated students or siblings are not discussed in depth; however, household-level randomization and the absence of effects on caregiver labor outcomes suggest minimal harmful externalities. The “crowd-out” checks indicate net increases in caregiver educational activities without evident labor-market tradeoffs, likely due to the low time burden.

Cost-effectiveness is central and promising, but methods could be more transparent. The average cost per child (~\$12) is reported without detailed cost components by country, perspective (provider vs social), or sensitivity analyses (e.g., tutor wages, supervision, telecommunications). The conversion to LAYS (3.4 per \$100) and the “up to four years per \$100” claim are plausible and align with prior frameworks, but a table showing country-specific costs, effects, uncertainty, and sensitivity to key assumptions would improve credibility and comparability. As presented, the ranking among top interventions is suggestive rather than definitive.

Open science and reproducibility fall short of best practice: the paper does not state a pre-analysis plan, nor does it provide public replication files or data access information. Given the policy salience, sharing de-identified data (where ethical) and code, or at least a detailed

replication appendix, would be a major contribution and enable meta-analysis. Ethical approvals and consent procedures for phone-based child assessments are not detailed and should be clarified.

Overall, this is a well-executed, highly policy-relevant set of trials that moves the literature on “education in emergencies” beyond small, context-specific pilots. The core causal claims are well supported; the scalability through government systems is a major contribution; and the measurement work on phone assessments is careful. Key areas to strengthen include transparent cost-effectiveness accounting, longer-run follow-up, multiplicity handling for ancillary outcomes, and open-science practices. Even with these caveats, the work is likely to be very useful to policymakers and practitioners and merits a strong publication venue focused on development economics and education policy.

i Gemini 2.0 Flash Assessment

This is a well-executed study that investigates the impact of phone-based tutoring interventions on learning outcomes in five different countries (India, Kenya, Nepal, Philippines, and Uganda) during COVID-19 related school closures. The research uses large-scale randomized controlled trials (RCTs) to evaluate the effectiveness of remote instruction for primary school children, comparing interventions delivered by NGOs and government teachers. The study also examines the impact of SMS messages as a standalone intervention. The strengths of the paper include the rigorous experimental design, the large sample size across multiple contexts, the analysis of cost-effectiveness, and the examination of scalability within government systems. The findings suggest that phone call tutorials can lead to substantial learning gains, are cost-effective, and can be effectively implemented by both NGO and government teachers. The authors also provide evidence on the disruptive effects of school closures and the potential of phone call tutorials to mitigate learning losses during emergencies. However, there are some potential limitations and areas for improvement. The study focuses primarily on numeracy skills, and it is unclear whether the findings generalize to other subjects or skills. There might be heterogeneity in the implementation fidelity and context-specific factors across the five countries, which could affect the generalizability of the results. The long-term effects of the interventions are not examined. While the authors do provide validity checks for phone-based assessments, it would be helpful to have more details on the specific measures used and potential biases. I would also be interested in seeing more direct evidence of the mechanisms through which the interventions impact learning outcomes, as well as the potential for spillover effects on other students or teachers. Despite these limitations, this is a valuable contribution to the literature on education in emergencies and provides practical insights for policymakers and practitioners.

In terms of threats to identification, there is a possibility of selection bias, as participation in the study may not have been random. The extent of this bias is unknown. The reliability of the data could also be a concern, as there may have been errors or biases in the collection or reporting of data. The use of phone-based assessments may have also introduced biases, as some students may have been more comfortable or familiar with this mode of assessment than others. Regarding data limitations, the study focuses primarily on numeracy skills, and the long-term effects of the interventions are not examined. The generalizability of the findings may also be limited, as the study was conducted in specific contexts and the results may not be applicable to other settings or populations. Regarding misinterpretations and internal

inconsistencies, there may be inconsistencies in the reporting or analysis of data, which could lead to misinterpretations of the results. It is also important to consider the potential for confirmation bias, as the authors may have been more likely to interpret the results in a way that supports their hypotheses. In terms of missing robustness, there may be a lack of robustness in the analysis, as the results may be sensitive to changes in the assumptions or methods used. In terms of replication barriers, the study provides a detailed description of the methods used, but it may be difficult to replicate the study exactly due to contextual factors or data availability.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.7 Arora et al. 2023

C.7.1 Ratings Comparison

Metric	Claude Sonnet 4	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	72 [65–78]	78 [68–86]	75 [60–85]	70 [66–78] (n=2)
Claims	70 [62–76]	77 [66–85]	80 [70–90]	68 [62–80] (n=2)
Methods	68 [60–75]	73 [62–82]	75 [60–85]	58 [54–67] (n=2)
Adv. Knowledge	76 [70–82]	87 [78–93]	70 [55–80]	68 [66–72] (n=2)
Logic & Comms	78 [73–83]	83 [75–90]	85 [75–95]	66 [64–76] (n=2)
Open Science	82 [78–87]	35 [20–50]	70 [50–80]	75 [68–80] (n=2)
Global Relevance	74 [68–80]	72 [60–82]	60 [40–75]	70 [67–80] (n=2)
NA	—	—	—	4 [4–5] (n=2)
NA	—	—	—	3 [3–4] (n=2)
NA	—	—	—	70 [67–80] (n=2)

i GPT-5 Pro Assessment

This paper asks how three components of public science—abstract knowledge (publications), human capital (PhD graduates), and university/public-sector inventions (patents and patent-proximate outputs)—affect corporate R&D and innovation in incumbent, publicly traded U.S. firms. The core empirical strategy builds firm-specific, time-varying measures of exposure to each component and instruments them with Bartik-style shifts based on predicted federal agency R&D budgets. Predicted budgets are generated from the political composition of congressional appropriations subcommittees, then mapped to fields, advisors, and patent subclasses using Dimensions funding linkages, NPL citations, and an original advisor–dissertation mapping. Exposures to public knowledge are weighted by firms’ historical publishing across OECD subfields; human capital exposures are weighted by textual similarity (SPECTER) between dissertations and the firm’s patents; public invention exposures are weighted by firms’ historical patent shares across subclasses. Across patents, publications, AMWS scientists,

R&D, and Tobin's Q, the 2SLS results consistently suggest: (i) public knowledge per se has little effect; (ii) public invention tends to reduce corporate R&D, research outputs, and value; (iii) human capital increases all three. The paper adds extensive robustness and heterogeneity analyses (frontier vs follower firms; industries), and alternative measures (e.g., patent-cited publications, dissertation-based variants), as well as Poisson CF-IV checks.

The identification approach is ambitious and generally well-motivated, but several credible threats remain. First, the exclusion restriction for the political-instrumented agency budgets is not fully secured. Appropriations subcommittee composition may move with election cycles and macro/sectoral conditions that simultaneously affect incumbent firms' R&D, value, and labor markets (e.g., sector-specific demand, regulation, defense cycles), especially given large, well-known agency-specific shifts (NIH vs DoD) that track sectoral fortunes. While year and firm fixed effects are included, there are no industry-by-year fixed effects; this leaves scope for spurious correlation if agency budget shifts correlate with industry shocks that also drive incumbents' outcomes. The paper partially addresses related concerns by controlling for direct federal awards to the focal firm and includes agency-exposure controls, but more demanding falsification/placebo tests (e.g., using "irrelevant" agencies; lag-lead timing tests; sensitivity to including industry $\times$ year FE where feasible; leave-one-agency-out rotations) would strengthen the case.

Second, the shift-share design hinges on the "shares" being plausibly predetermined. For public knowledge, shares depend on firms' past publishing across OECD subfields, which many incumbents do infrequently; exposure for non-publishing firms may default to near zero, potentially attenuating estimates for knowledge spillovers or confounding composition effects. For human capital, the shares are derived from textual similarity between dissertations and the firm's patents, defined by 5-year cohorts. Although the human-capital variable is instrumented, the instrument also uses the same similarity-based weights. This raises a selection concern: if firms' past patent portfolios (which shape the weights) reflect unobserved, persistent shocks that also affect outcomes, the instrument may not fully purge endogeneity. That said, the reliance on predicted (rather than actual) agency budgets and advisor-level funding windows tied to the start of training mitigates reverse causality. Still, additional diagnostics (e.g., reweighting with alternative, outcome-independent shares; leaving out own-firm patents; using only earlier time windows for shares) would help.

Third, measurement choices may bias the "no effect of pure knowledge" conclusion. The public knowledge measure is coarse (OECD subfields, global publications) and likely noisy relative to firms' true absorptive scope, especially for non-publishing incumbents. The path from science to use typically runs through codification steps (e.g., high-impact, field-bridging articles) and tacit channels; knowledge relevance might be better proxied through non-patent literature cited by firms' patents, field-technology concordances, or co-citation-based proximity measures that do not depend on firm publishing. The paper does implement NPL-based measures for invention and some alternative constructions, which is a plus; further triangulation for knowledge (e.g., science-of-science salience weights, high-impact subsets) would test the robustness of the "little or no response" result.

Fourth, the interpretation that public invention reduces firm R&D due to downstream competition, rather than as a cheaper external input, relies heavily on the sign for firm value. This is plausible but not definitive. The Tobin's Q variable exhibits extreme dispersion (mean 34, SD 688; median 1.7), suggesting severe outliers and potential measurement error. Although the dependent variable is logged, winsorization or robustness to alternative value/profit mea-

asures (e.g., operating margins, ROA, cash flow, or market-adjusted returns) would materially strengthen that conclusion. Some results in the text suggest profits also decline with public invention, but the specification and robustness for profits are not fully detailed here; emphasizing these results, and reconciling Q with profit findings, would improve credibility.

Fifth, the extensive use of Dimensions linkages (funders-to-pubs; NPL citations) and the advisor mapping from PQDT to Dimensions introduces potential matching errors and coverage shifts over time (e.g., secular increases in funding acknowledgments). The reliance on proprietary data and high-performance computing (e.g., SPECTER similarity at scale) creates replication barriers. The paper documents procedures and runs extensive robustness checks (including alternative measures for human capital and public invention, and federal lab-only variants), which is commendable. Still, full code release, clearly versioned data extracts, and sensitivity to alternative parameterizations (e.g., depreciation rates, cohort windows, similarity thresholds) would enhance reproducibility and external scrutiny.

Sixth, multicollinearity and just-identified 2SLS settings limit diagnostic testing. In joint models, the three endogenous regressors are correlated (by construction and empirically). With one instrument per endogenous variable, overidentification tests are unavailable; weak-ID tests are reported and look strong, but reporting first-stage partial R² and sensitivity to removing one component at a time (beyond what's already shown) would help separate channels. The CF-IV Poisson replication is a strength, but further nonlinear checks (e.g., QML Poisson with firm-specific trends, or GMM with pre-trends) could consolidate the count-outcome findings.

On external validity, the focus is on incumbent, publicly listed U.S. firms with patents and R&D, emphasizing the corporate core of business R&D. This is appropriate for the research question but excludes private incumbents and smaller R&D performers; the results should be interpreted as effects on established firms. The heterogeneity analyses (frontier vs follower; industry splits) provide useful nuance; more structured decompositions clarifying whether frontier firms face thinner supplies of relevant public inventions (versus higher returns to internal R&D) would be valuable.

Overall, the paper's data construction is ambitious and carefully described; the findings are plausible, internally consistent across outcomes, and robust to numerous alternative measures and specifications. The largest remaining concerns are (1) the strength of the exclusion restriction for the political-instrumented Bartik shocks; (2) possible measurement and selection biases in the knowledge and human-capital exposure constructions; (3) interpretation of value/profit effects with noisy Q and limited alternative value measures; and (4) replication barriers given proprietary data and heavy compute. Addressing these with additional falsifications, sensitivity to more aggressive fixed effects, alternative value outcomes, and greater open-science materials would materially strengthen the paper's causal claims and policy relevance.

i Claude Sonnet 4 Assessment

This paper examines how different components of public science—knowledge (publications), human capital (PhD scientists), and inventions (patents)—affect corporate R&D. The authors use an instrumental variable approach, exploiting political composition of congressional appropriations subcommittees to predict federal agency R&D budgets, which then serve as instruments for firm-specific exposure to public science. The main findings are: (1) abstract public

knowledge has little effect on corporate R&D, (2) public inventions (university patents) reduce corporate R&D, suggesting competition rather than complementarity, and (3) human capital (PhD scientists) increases corporate R&D. Several methodological concerns arise. First, the identification strategy relies on political composition predicting agency budgets, but the first-stage relationships are relatively weak and the exclusion restriction may be questionable—political factors could affect corporate R&D through channels other than public science funding. Second, the measures of ‘relevant’ public science are complex constructions involving multiple weighting schemes and similarity algorithms (like SPECTER), introducing potential measurement error that could bias results. Third, the interpretation of university patents as purely competitive rather than complementary inputs is not well-established theoretically or empirically—the negative effects could reflect other mechanisms. Fourth, while the paper controls for direct federal funding to firms, there may be other confounding channels through which political variables affect corporate outcomes. The heterogeneity analysis by firm frontier status is interesting but relatively limited. The paper makes a valuable empirical contribution to understanding science-industry linkages, but the causal interpretation should be viewed with some caution given the identification challenges and complexity of the measurement approach.

Gemini 2.0 Flash Assessment

This paper examines the relationship between public science (knowledge, human capital, and inventions) and corporate R&D. It uses an instrumental variable approach, exploiting firm-specific exposure to changes in federal agency R&D budgets driven by the political composition of congressional appropriations subcommittees, to address endogeneity concerns. The study finds that: (1) abstract public knowledge (publications) has little effect on corporate R&D; (2) public inventions (university patents) reduce corporate R&D; and (3) human capital (PhDs) positively affects corporate R&D. These effects vary across firms and industries, with firms on the technology frontier responding differently than followers. The paper contributes to the literature by focusing on established firms and differentiating the impact of different components of public science on corporate R&D. A key strength is the rigorous econometric approach using instrumental variables to address endogeneity. The use of textual similarity measures and bibliometric linkages to connect federal funding to publications, dissertations, and patents is also novel and valuable. However, some limitations exist: First, the reliance on publicly traded firms may limit the generalizability of the findings to smaller or private firms. Second, while the instrumental variable approach helps address endogeneity, the identifying assumptions may still be debated. The assumption that federal funding is driven by political factors and not technological ones may be difficult to fully validate. Third, the use of historical data (1980-2015) may limit the relevance of the findings to current innovation dynamics, given the increasing role of startups and open innovation models. Fourth, the study does not fully explore the mechanisms through which public inventions compete with corporate R&D, or why firms have limited ability to absorb abstract ideas. Fifth, while the paper considers heterogeneity across firms and industries, more nuanced analyses of specific technology sectors or types of innovation could provide additional insights. Finally, although several robustness checks are performed, there may be additional unobserved factors that could influence the results. These threats suggest some caution in interpreting the magnitude and precise estimates, even if the main qualitative results appear relatively robust. The summary statistics

presented in Appendix Table C16 show substantial differences between industries, suggesting that a single national-level analysis could obscure important heterogeneity and potential omitted variable bias. It is also unclear whether or not the methods are likely to generalize to international contexts, which would represent a substantial strengthening of the paper.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.8 Bahar et al. 2022

C.8.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	80 [72–88]	75 [60–85]	75 [68–84] (n=2)
Claims	78 [70–86]	80 [70–90]	—
Methods	75 [65–84]	70 [55–80]	76 [68–84] (n=2)
Adv. Knowledge	82 [72–90]	75 [60–85]	78 [70–85] (n=2)
Logic & Comms	84 [76–92]	85 [75–95]	79 [72–85] (n=2)
Open Science	30 [15–45]	70 [50–80]	69 [56–72] (n=2)
Global Relevance	88 [78–95]	80 [70–90]	77 [79–95] (n=2)
NA	—	—	3 [3–4] (n=2)
NA	—	—	4 [3–4] (n=2)
NA	—	—	78 [82–95] (n=2)

i GPT-5 Pro Assessment

This paper leverages a large administrative census of undocumented Venezuelan migrants (RAMV) linked to Colombia’s formal firm registry (RUES) to estimate the causal effect of a 2018 residency regularization (PEP) on entrepreneurship. The identification strategy is the paper’s centerpiece: migrants’ permission to apply for PEP was released in batches determined by national RAMV census numbers; the authors argue that day- and city-specific rainfall shocks affected migrants’ within-week registration timing and thereby their national queue position relative to ex post application-window thresholds. They develop an “RVI” (running-variable instrument) framework inspired by composite-treatment designs: (i) model the mapping from precipitation to census numbers conditional on date and municipality fixed effects and observed family traits; (ii) generate, by permuting weather realizations within city×week, a distribution of each migrant’s counterfactual census numbers; (iii) from this, compute an expected application-window length (expected instrument) and use the realized minus expected window length as quasi-random variation in an IV/2SLS setup (fuzzy RD). Using this, they estimate that PEP receipt raises the probability of formal firm registration by 1.2–1.8 percentage points (relative to a baseline of about 0.16%), with effects accruing over 3–4 years and bringing annual creation rates close to those of natives by 2022. A difference-

in-differences panel exercise suggests the visa effect is about twice as large as the “physical relocation” effect on firm creation. Heterogeneity analyses show similar relative magnitudes for sole proprietorships vs sociedades and for employer vs non-employer firms; effects are present across baseline employment statuses.

Strengths: (1) The setting is unusually well-suited for credible identification and policy relevance; batch access to applications is tied to an administratively arbitrary national sequence and the policy announcement appears unanticipated, reducing scope for strategic behavior. (2) The methodological contribution is clearly articulated and well-motivated: leveraging noise in the running variable to extend RD-like designs beyond tight windows is elegant and potentially broadly applicable. (3) The authors provide several supportive diagnostics: no visible bunching at thresholds; stacked RD balance on observables near cutoffs; placebo year-specific effects pre-PEP are essentially zero; first-stage strength is substantial (partial F-statistics 50–263); and the timing pattern (effects growing over time) argues against pure formalization of pre-existing firms. (4) Outcome measurement draws on the comprehensive formal business registry; the legal distinction between personas naturales and sociedades is well-explained, and linking to legal representatives for sociedades is sensible.

Key issues and threats to identification: (A) Exogeneity of the rain-based RVI relies on the assumption that, conditional on date and municipality fixed effects and observables, the residual variation in rainfall timing within city $\times$ week affects firm creation solely through its impact on application-window timing (and hence PEP uptake). While plausible (and balance tests between “lucky” and “unlucky” migrants are encouraging), residual concerns remain that differential responsiveness to rain (e.g., time/income constraints, risk tolerance, diligence) could correlate with unobservables predictive of entrepreneurship. The authors partly address this by controlling for the expected instrument and mean running variable and profiling compliers, but a more formal sensitivity analysis (e.g., bounding exercises allowing for correlated heterogeneity in rain-responsiveness) would strengthen claims. (B) The mapping from precipitation to census numbers is estimated with rich fixed effects and interactions; however, the monotonicity and stability of this mapping across heterogeneous municipalities and dates are assumed rather than tested exhaustively. External validation or cross-validation (e.g., train-test splits by cities or weeks, or alternative functional forms including non-parametric fits) would increase confidence that the counterfactual census-number distributions are well-approximated. The use of 200 weather permutations seems adequate but sensitivity to the number of permutations and to alternative weather metrics (e.g., intensity, cumulative rainfall, binary rain/no rain, storms) would be informative. (C) The key step of re-ranking predicted census numbers within each simulation to match the realized distribution is pragmatic but somewhat ad hoc; presenting results without re-ranking (which the authors state are similar) would be a valuable robustness table. (D) Clustering and inference: the stochastic variation plausibly resides at least at the city $\times$ day level (or day level, given national sequencing). The paper alternates between clustering by family (first stage) and by date of census completion (second stage). A consistent clustering strategy that nests the shock’s level (e.g., two-way clustering by city $\times$ day and family, or city $\times$ day only, with wild-cluster bootstrap for the relatively small number of unique days) would bolster inference credibility. Reporting permutation or randomization-inference p-values using the same city $\times$ week permutation scheme would further align inference with the design. (E) The LATE interpretation could be foregrounded more consistently. Some language (e.g., “bringing [migrants] to parity with natives”) risks overgeneralizing beyond compliers—those whose PEP uptake is sensitive to the application-

window timing induced by rain-driven queue noise. Scaling implications for the full eligible population should be presented clearly as LATEs with appropriate caveats.

Data and measurement considerations: (F) The match to RUES relies on valid Venezuelan IDs; the analytic sample is about 332k of 442k RAMV registrants. This selection likely affects external validity (those with IDs may be more formal or higher-resource). While unlikely to bias identification given the within-sample design, descriptive checks comparing matched vs unmatched registrants (where possible) would help. (G) The outcome captures formal firm registration; given the prevalence of informality in Colombia, the results reflect changes at the upper tail of business formalization. The authors argue that the time profile is inconsistent with mere formalization of pre-existing businesses, which is persuasive, though direct evidence on informal activity (if any proxies exist) would strengthen this inference. (H) The panel decomposition of migration vs visa effects assumes that all pre-PEP firms by migrants are net new, explicitly biasing toward overstating the relocation effect. The finding that PEP's effect is still larger is credible but conservative specification choices (e.g., excluding the immediate pre-PEP period, alternative lags, event-study plots with pre-trend tests) should be summarized in the main text.

Scope, mechanisms, and policy relevance: (I) The mechanisms posited (greater legal certainty, access to banking, reduced deportation risk, better contracting conditions) are plausible; the paper could bolster this by showing downstream indicators if available (e.g., bank account access, credit uptake, wage employment transitions), or by leveraging heterogeneity (e.g., larger effects in sectors requiring formal contracting or in municipalities with better financial access). (J) Spillovers and general equilibrium effects are not addressed (e.g., displacement or competition with native entrepreneurs), which is reasonable for scope but should be noted as a limitation when interpreting welfare implications. (K) The heterogeneity analyses are informative; some estimates are imprecise, but broader patterns support the main story.

Reproducibility and transparency: (L) The administrative data (RAMV, RUES) are not public; code and replication packages are not indicated. Given the novel identification and multiple modeling choices (e.g., mapping function, permutations, re-ranking), sharing code and detailed instructions for authorized users would substantially improve reproducibility. Consistency in standard error clustering and providing additional robustness tables in the main paper (currently some claims are relegated to text) would also help.

Overall, this is a strong, policy-relevant paper with a creative design, careful diagnostics, and plausible effects of economically meaningful magnitude. The main remaining uncertainties concern the precise exogeneity of the rain-based within-week assignment and the alignment of inference with shock-level clustering. These do not overturn the paper's contribution but should temper the certainty of causal claims; additional robustness and clearer LATE framing would solidify the case.

i Gemini 2.0 Flash Assessment

This paper examines the impact of granting residency visas to Venezuelan immigrants in Colombia on their entrepreneurship, exploiting a natural experiment where undocumented immigrants were made eligible for a resident visa in 2019. The authors use administrative data linked to the complete formal business registry to study the 2018 policy shift in Colombia that made nearly half a million Venezuelan undocumented migrants eligible for a resident visa. To establish causal estimates, they develop a novel extension of a regression discontinuity de-

sign. Their design uses variation in the running-variable (coming from rain) to instrument for migrants' choices to apply for visas. Immigrants who receive the visa increase their economic activity in the form of higher entrepreneurship by a factor as high as 12, bringing it to parity with native Colombians four years later. This research question is valuable and important given the scale of undocumented immigration and the potential benefits of regularization. The authors address the key identification challenge—the endogeneity of legal status—with a novel instrumental variable approach. They use local weather shocks (rainfall) as an instrument, arguing that rainfall affects the timing of registration for the census that determined visa eligibility, thus creating exogenous variation in the census number assigned to each migrant. They introduce a novel extension to the regression discontinuity design. They provide a clear explanation of the policy context, data, and empirical strategy. The paper also includes a range of robustness checks and extensions to the main analysis, including panel data estimations, heterogeneity analyses, and tests for pre-trends. Despite these strengths, some potential limitations and areas for further scrutiny include: 1) The validity of the weather shock instrument: the assumption that rainfall only affects registration timing and not other factors related to entrepreneurship needs careful justification. Are there potential confounding factors related to weather? 2) The local average treatment effect (LATE): The IV estimates apply to compliers, those whose visa application decision is affected by rainfall-induced delays. External validity depends on the characteristics of compliers, which may not be representative of all migrants. The paper addresses this by comparing compliers to always-takers but further discussion is needed. 3) The use of formal firm registration as the outcome variable: it is important but may not capture the full scope of entrepreneurial activity, especially given that many Venezuelan immigrants may operate in the informal sector, particularly early on. 4) The interpretation of the effect on formal firm creation: while the authors argue that the increase in firm registration reflects new firm creation rather than the formalization of existing informal firms, it is hard to be certain. 5) The limited discussion of statistical power: Given the multiple analyses, it would be useful to discuss the power of the tests and the potential for Type II errors. 6) The assumptions about functional form: the mapping from weather to the number could depend on some parametric assumption.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.9 Banerjee et al. 2022

C.9.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	92 [87–96]	65 [55–75]	90 [78–94] (n=2)
Claims	90 [85–95]	75 [65–85]	—
Methods	89 [84–94]	60 [50–70]	85 [60–98] (n=2)
Adv. Knowledge	93 [88–97]	70 [60–80]	90 [78–94] (n=2)
Logic & Comms	88 [83–93]	80 [70–90]	85 [60–98] (n=2)

Open Science	66 [50–80]	40 [25–55]	80 [48–88] (n=2)
Global Relevance	93 [88–98]	80 [70–90]	90 [80–100] (n=2)
NA	—	—	5 [4–5] (n=2)
NA	—	—	5 [4–5] (n=2)
NA	—	—	95 [72–100] (n=2)

i GPT-5 Pro Assessment

This manuscript introduces and applies Treatment Variant Aggregation (TVA), a principled procedure to reduce a large policy space in factorial designs by pooling “dosage variants” that appear empirically equivalent and pruning ineffective options, and then estimating the best policy while adjusting for the winner’s curse. The application is a large cross-randomized trial in Haryana, India, spanning 75 unique policy bundles combining incentives, SMS reminders, and network ambassadors. The paper is ambitious and carefully executed overall, but several assumptions and implementation choices merit scrutiny, and a few design features could affect inference or replication.

Methodology and identification. TVA rests on strong but transparent assumptions: (i) a “beta-min” condition so marginal effects are either appreciably nonzero or zero (not local-to-zero); (ii) $K < n$ with controlled growth; (iii) homoskedastic errors for the LASSO/Puffer support recovery proofs; and (iv) an RCT structure delivering near-orthogonality exploited by the Puffer transform to restore irrepresentability. The authors are clear that the theoretical support recovery is pointwise and not uniform, and that winner’s-curse inference nests both model selection and asymptotic regression normality in practice, a combination not fully proven (they rely on a plausibility argument and simulations). These caveats are important: in empirical contexts with many small, nonzero interactions, TVA may over-pool; conversely, very conservative penalization can under-detect true marginal effects. The paper mitigates these concerns through extensive simulations, including several “approximate sparsity” regimes, showing TVA’s relative robustness versus alternatives, and through sensitivity and bootstrap checks in the application. Still, the beta-min and homoskedasticity assumptions are not guaranteed in practice; heteroskedasticity and clustering are likely, and while the implementation uses a PufferN variant and cluster-robust inference post-selection, the formal support recovery result remains under homoskedasticity.

Design and estimation details. A notable identification concern is the clustering level. Incentives’ presence/absence is randomized at the PHC level; incentive type/dose is at sub-center; ambassadors are at village; SMS saturation is at sub-center then child-level. The paper states it clusters at the sub-center (SC) level as the highest treatment level, but presence/absence of incentives is at PHC. Not clustering at the highest relevant randomization unit (PHC) may understate standard errors for policy bundles that include the incentives vs. no-incentives margin. Multiway or the highest-level clustering would be preferable. The outcome is village-month measles shots; district-time fixed effects are included and villages are population-weighted, which is sensible, but potential serial correlation and within-PHC correlation warrant careful clustering.

Interference and spillovers. The ambassador arm intentionally exploits within-village diffusion. The unit of outcome is the village, which weakens SUTVA concerns within villages. However, inter-village spillovers (e.g., shared sub-centers/PHCs or residents traveling) could bias estimates if exposure crosses village boundaries. The authors do not present direct tests

for cross-village spillovers. This is likely modest given village-based sessions, but remains a potential identification threat.

Outcome measurement and substitution. Administrative data were used both to trigger reminders/incentives and to measure outcomes. The authors conduct validation audits showing high agreement and run checks suggesting little substitution from other venues; this is a strength. Nonetheless, measurement is endogenous to the intervention workflow, which always raises the possibility of behavior that influences recording (though audits mitigate this concern substantially).

TVA specification choices. The admissible pooling rules and treatment-profile mapping need researcher judgment. In the application, slope and flat incentives are treated as distinct “types” that cannot pool; trusted vs random vs information hubs are similarly constrained; trusted-info-hub is treated as a higher “dose” of info hub. These choices are sensible and largely pre-motivated, but they are consequential: alternative admissible lattices could produce different pools. The paper partially addresses this through clear rules and simulations, but some degree of researcher degrees-of-freedom remains. The penalty selection is explicitly conservative (mapped to an extremely small nominal Type I error in a backward-elimination analogue), which controls false positives but may miss effective variants and reduce power. The authors present an “admissible ” range and show winner’s-curse-adjusted estimates are stable within it, which is reassuring.

Winner’s curse adjustment. The hybrid estimator from Andrews et al. is appropriate in spirit, but the paper acknowledges the theoretical gap when both model selection conditioning and asymptotic normality are jointly present. The practical performance (simulations and bootstraps) looks good: the best-policy effect for immunizations per dollar is very stable (best selection in 96% of bootstraps), while the immunizations-per-month best policy is somewhat less stable (77% of bootstraps). The adjusted CIs remain informative; still, readers should view the hybrid inference as approximate.

Cost-effectiveness. The paper reports that the most cost-effective policy (information hubs plus SMS, no incentives) raises immunizations per dollar by 9.1%. It is not fully clear what cost components are included (e.g., one-off costs of seed identification surveys, messaging platform overheads, vs marginal costs only). A detailed costing appendix or a decomposition (fixed vs variable, program vs research cost) would help practitioners assess implementation tradeoffs and transferability.

Communication and consistency. The logic is generally clear; the Hasse diagram exposition and the mapping from marginal effects to pooling are well executed. One inconsistency to flag: the text sometimes implies SC is the highest treatment assignment level for clustering, yet incentives vs. no-incentives is randomized at PHC. Clarifying this and, if feasible, re-estimating with clustering at the highest level (or multiway clustering) would strengthen confidence. Figures and tables are well labeled; simulation design choices are adequately explained.

Open science and replicability. The procedure is described in detail, with proofs, simulation code logic, and pooling algorithms outlined; however, it is not clear if replication code and the administrative data (de-identified aggregates) are publicly posted. Given PII and government partnership, data sharing may be constrained, but posting fully worked replication code with synthetic or simulated data and detailed instructions for re-running TVA on new datasets would enhance replicability. A software implementation of TVA (package or scripts) would be a valuable contribution.

Bottom line. Substantively, TVA identifies meaningful complementarities that a limited-factorial design might have missed: information hubs amplify incentives and reminders, and some bundles perform better on either levels or cost-effectiveness. Methodologically, TVA is a useful addition for policy selection in large factorials under plausible sparsity. The main caveats concern the strength of assumptions for exact support recovery, the incomplete theory for post-selection winner's-curse inference, sensitivity to admissible pooling rules and penalty choice, and clustering at the appropriate randomization level. Even with these limitations, the work is rigorous, transparent about assumptions, empirically careful, and practically valuable.

i Gemini 2.0 Flash Assessment

This paper introduces a methodology called Treatment Variant Aggregation (TVA) to address the challenge of selecting the most effective policy bundle from a large set of potential combinations, particularly in the context of randomized controlled trials (RCTs). The core idea is to pool policy variants that have similar impacts and prune those deemed ineffective, thereby reducing the dimensionality of the problem and improving estimation. The method involves re-representing a fully-saturated factorial regression in terms of incremental dosages, utilizing a Hasse diagram lattice to guide pruning and pooling, applying a Puffer transformation to address multicollinearity, using LASSO for variable selection, estimating OLS on the selected policies, and adjusting for the winner's curse. They apply the method to data from an RCT in Haryana, India, testing interventions to stimulate demand for immunization. The RCT involved cross-randomizing incentives, SMS reminders, and local ambassadors (information hubs), resulting in 75 policy combinations across 915 villages. They find that the most effective policy combines incentives, ambassadors as information hubs, and reminders, increasing immunizations by 44%. The most cost-effective policy uses information hubs and SMS reminders but no incentives, increasing immunizations per dollar by 9.1%.

Several potential issues affect this work:

1. **Oversimplified Assumptions:** The method relies on assumptions about the nature of treatment effects (e.g., minimal and realistic assumptions on the inferential environment, that incremental dosage changes have meaningful effects, and assumptions 1-6), which may not always hold in practice. In particular, the assumption that small effect sizes are ignorable (local-to-zero) could be problematic if some potentially useful interventions have effects that are too small to be reliably detected with the sample size and methods used.
2. **Multicollinearity Mitigation:** While the Puffer transformation is employed to address multicollinearity, the effectiveness of this approach depends on the specific structure of the RCT design and may not fully eliminate the issue. This is potentially more problematic as K increases to higher levels.
3. **Non-Uniform Convergence:** The authors acknowledge that the convergence in distribution is not uniform (in the parameter space), raising concerns about the reliability of post-selection inference, although they argue that non-uniformity does not have much bite in their setting.
4. **Homoscedasticity Assumption:** The theory assumes homoscedastic errors, which may not be realistic in practice, especially in contexts with varying levels of access to

healthcare and awareness of immunization benefits.

5. **Sparsity Assumption:** The method relies on sparsity assumptions (i.e., that only a small number of policies have meaningful impacts). This may not be reasonable in settings where many interventions have small but non-negligible effects.
6. **Generalizability:** While the study is conducted on a large scale in Haryana, India, the specific policy combinations and their effectiveness may not be generalizable to other contexts with different cultural, economic, and healthcare systems. They acknowledge that one key question is whether the information hubs can effectively transmit information in areas where the messengers might be more effective with commercial messages.
7. **Omitted Interaction Effects:** The authors acknowledge that the method may not capture all relevant interactions between interventions, and that there is a danger of paring down the number of treatments a priori because combined interventions work better than each in isolation.
8. **Weighting:** The weighting scheme could be an area for further improvement.
9. **Lack of Transparency:** The data that could be used to assess replicability and reproducibility is not provided.

These issues suggest caution in interpreting the results and highlight areas for further research to improve the robustness and applicability of the TVA methodology.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.10 Banerjee et al. 2023

C.10.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	88 [80–93]	75 [65–85]	84 [79–98] (n=2)
Claims	87 [79–93]	80 [70–90]	—
Methods	85 [77–92]	80 [70–90]	90 [76–98] (n=2)
Adv. Knowledge	92 [86–97]	70 [60–80]	89 [68–98] (n=2)
Logic & Comms	82 [74–90]	85 [75–95]	78 [55–90] (n=2)
Open Science	62 [45–78]	80 [70–90]	60 [45–88] (n=2)
Global Relevance	84 [75–92]	80 [70–90]	65 [42–85] (n=2)
NA	—	—	4 [14–5] (n=2)
NA	—	—	4 [3–5] (n=2)
NA	—	—	88 [77–99] (n=2)

This is a large, carefully implemented village-level RCT of a universal cash program, comparing three delivery regimes (12-year UBI, 2-year UBI, and a PV-equivalent lump sum) after roughly two years, when cumulative receipts were similar across arms. Internal validity for intent-to-treat contrasts against control is strong: randomization at the village level with stratification, very high survey response rates (97% for households), close adherence to assigned payment schedules, and baseline balance checks all support credibility. The paper is transparent about its limited *ex ante* power for cross-arm differences at this early horizon and reports one-sided tests motivated by a simple forward-looking lumpy investment model; nevertheless, the frequent emphasis on “borderline” differences ($p = 0.10\text{--}0.15$) and one-sided alternatives elevates Type I error risk given the extensive outcome battery. The pre-analysis plan with pre-specified aggregated outcomes reduces but does not eliminate multiplicity concerns; many tables report numerous unadjusted *p*-values across related outcomes, and there is no systematic correction for multiple testing at the family level. The primary “macro” village outcomes (firm counts, revenues, costs, net revenues) combine a non-agricultural enterprise census with household-reported agricultural enterprise aggregates scaled to village totals. A substantive measurement choice is that “enterprises with missing data [are] assumed to be zero”; because enterprise response rates differ modestly by arm (slightly higher in ST/LT than control), this coding can induce upward-biased treatment effects in treated arms if nonresponse correlates with positive outcomes. At minimum, robustness using (i) weights or imputation based on observed covariates and (ii) specifications restricted to responding enterprises would help bound this risk. Relatedly, “net revenues” exclude household labor costs, so they overstate profits for welfare accounting and should be interpreted as value added net of intermediate inputs, not economic profits. Wage effects rely on “prevailing” wage reports from village elders (available in ~60% of villages) and respondent reservation wages; both are informative but subject to reporting error and selection (only villages with a prevailing wage are included), so wage results should be read cautiously and triangulated with alternative measures when possible. The price analysis is underpowered: only one price round, limited cross-market treatment variation, and relatively small exposure contrasts mean the non-rejection of price changes does not tightly rule out meaningful inflation; external corroboration with denser designs (e.g., cited studies) is appropriate, but the claim that revenue increases reflect volumes rather than prices remains only weakly supported in this sample. The large increases in self-assessed land values (exceeding transfer magnitudes in some arms) are particularly fragile: markets are thin with few transactions, the valuation is self-reported, and amenity changes and in-migration plausibly confound perceived values; the paper notes these caveats, but stronger validation—e.g., hedonic indices using observed investments, satellite-based proxies, or independent market quotes—would help distinguish valuation shifts from real price changes. In-migration is an important mediator: both LT and LS arms show significant increases in household counts, which can dampen wage effects, stimulate retail expansion, and alter per capita outcomes; while the paper documents these flows, it does not decompose the share of treatment effects on enterprise activity and wages attributable to in-migration versus resident behavior (e.g., shift-share or mediation analysis could add clarity). The simple model of forward-looking households facing lumpy investment and savings/credit frictions is useful for organizing cross-arm expectations, but it is not structurally estimated or calibrated to match moments; it serves as an interpretive device rather than a tested mechanism. Some mechanism claims remain suggestive: for example, the interpretation of smaller depression reductions in LS as

“stress” from one-off risk is plausible but not directly measured; a richer mental health battery or measures of perceived risk/variance would strengthen the case. The study excludes code and data access details; although a pre-analysis plan and clear outcome construction exist in the text, replication will be difficult without public codebooks, data, and scripts—especially given the complexity of aggregation (e.g., scaling household agricultural enterprises to villages, Conley SEs, trimming rules). The cross-randomized planning/savings nudges are not analyzed here; while random assignment ensures no bias, reporting their (null or not) interactions could clarify heterogeneity in savings and enterprise formation and reduce unexplained variance. The treatment is universal within villages and individual-based (all adults), which is a strength for external validity to “universal” policies relative to household-targeted transfers, but generalization outside rural Kenya with mature mobile money infrastructure is limited; the ability to borrow against transfers and operate retail supply chains may differ markedly in settings without these features. On outcome construction, the authors appropriately add migrant outcomes back to sending households for relevant totals and note that results are not sensitive to omitting them; nonetheless, this procedure relies on elicited shares for some migrants and may introduce noise—sensitivity checks with alternative imputations would be useful. Spatial autocorrelation is addressed with Conley SEs, strata FE are included, and baseline covariate adjustments (including adult-count interactions) are appropriate; however, given likely cross-village spillovers (markets, labor), the design is not optimized to detect them (as acknowledged), so village-level ITT estimates should be interpreted as “treatment plus within-village equilibrium effects,” not pure partial equilibrium responses. Overall, the empirical narrative—substantial non-ag retail expansion, occupational reallocation from wages to self-employment without net labor supply reductions, and improved food diversity/protein intake—is consistent across data sources and robustly documented. The cross-arm pattern—LS LT > ST for local economic aggregates despite equal PV delivered to date—is substantively important and policy-relevant, though precise cross-arm inference is limited by power and multiple testing. The paper significantly advances knowledge by credibly studying a long-horizon universal transfer at scale and comparing tranche structures; the main threats to interpretation are measurement/aggregation choices (especially missing-as-zero for enterprises and self-assessed land values), underpowered price measurement, and limited transparency for full replication. Strengthening robustness around enterprise nonresponse, presenting family-wise error adjustments for key outcome families, and offering replication code/data would materially improve confidence in some headline magnitudes, but the qualitative conclusions are well-supported and important for policy design.

Gemini 2.0 Flash Assessment

This paper presents an analysis of the short-term impacts of a Universal Basic Income (UBI) experiment in rural Kenya, comparing a long-term UBI to shorter-term transfers (either as a stream of small payments or a lump sum). The research leverages a field experiment where communities received either a long-term UBI, short-term UBI, or a lump sum transfer, with a control group for comparison. The study examines various economic outcomes, including enterprise activity, labor supply, consumption patterns, and asset accumulation, as well as welfare-related outcomes such as well-being and mental health. A theoretical framework is also presented to help interpret the empirical findings. The study appears well-executed, with a large sample size and attention to experimental integrity. The main findings suggest that

long-term transfers led to substantial economic expansions, particularly in the non-agricultural sector, without reducing overall labor supply. The lump sum transfers had similar, if not larger, economic impacts, while the short-term transfers had noticeably smaller effects. These results are largely consistent with a model of forward-looking lumpy investment, subject to savings and credit constraints. The paper is clearly written and addresses a policy-relevant question with a rigorous experimental design. However, it focuses on short-term effects, which may not fully capture the long-term dynamics of UBI. Also, while the paper discusses potential mechanisms, more direct evidence on these mechanisms would strengthen the conclusions. Given the nature of RCTs, general equilibrium effects may be difficult to extrapolate and are not fully explored.

Here are some key issues detected in the manuscript:

1. **Short-Term Focus:** The study's analysis is limited to the short-term impacts (approximately two years) of the UBI intervention. This is a relatively short period to assess the long-term effects of UBI, which is intended to be a sustained intervention. Some outcomes, such as educational attainment or long-term asset accumulation, may take longer to materialize.
2. **General Equilibrium Effects:** While the study acknowledges the potential for general equilibrium effects, such as changes in prices or wages, the analysis of these effects is limited. The study does not reject the null hypothesis that consumer prices were unaffected, but the estimates are not precise enough to rule out meaningful appreciation or depreciation. More comprehensive analysis of general equilibrium effects would provide a more complete picture of the intervention's impacts.
3. **Mechanism Identification:** The paper proposes a theoretical framework to explain the observed results, focusing on savings constraints, credit constraints, and increasing returns. However, the evidence for these mechanisms is largely indirect. More direct evidence, such as data on savings behavior or access to credit, would strengthen the conclusions.
4. **External Validity:** The study is conducted in rural Kenya, which may limit the generalizability of the findings to other contexts. The specific economic conditions, social norms, and institutional environment of rural Kenya may influence the impacts of UBI. Further research is needed to assess the external validity of the results.
5. **Limited Power for Heterogeneity:** The study acknowledges that the impacts of UBI may be heterogeneous, depending on individual characteristics or circumstances. However, the power calculations suggest that the design was not expected to be powered for two-sided tests of the null of equality between treatment arms. More power may be needed to explore heterogeneity in treatment effects.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.11 Barberio et al. 2022

C.11.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	68 [55–78]	55 [35–75]	50 [35–65] (n=2)
Claims	70 [58–80]	70 [50–85]	—
Methods	62 [45–72]	60 [40–75]	50 [35–60] (n=2)
Adv. Knowledge	73 [60–84]	70 [50–85]	48 [30–60] (n=2)
Logic & Comms	78 [68–88]	75 [60–90]	45 [30–60] (n=2)
Open Science	32 [20–45]	70 [50–85]	55 [40–70] (n=2)
Global Relevance	86 [75–95]	80 [65–95]	70 [60–85] (n=2)
NA	—	—	2 [2–4] (n=2)
NA	—	—	2 [2–3] (n=3)
NA	—	—	60 [50–75] (n=2)

i GPT-5 Pro Assessment

This paper builds a Monte Carlo model of a hypothetical “vaccine megafund” investing in 120 preclinical mRNA vaccine candidates across 11 EIDs (plus “disease X”), simulating correlated clinical development, manufacturing, and revenues under stochastic outbreaks. It reports a negative expected annualized return (−6% p.a.) and NPV (−\$9.5B at 10% discount), with clinical trials driving 94% of costs; price per dose is the dominant determinant of financial viability, while increasing PoS, enlarging the portfolio, and allowing challenge trials do little to change the economics. The study is timely, well written, and transparent about many key assumptions, and it improves on earlier megafund work by explicitly simulating outbreak timing and a costed mRNA manufacturing chain. However, several modeling choices and omissions limit interpretability and external validity, and they also create replication barriers. First, the demand side is highly stylized. Outbreaks are modeled as independent annual Bernoulli events with fixed probabilities and fixed “average infections,” yet the mapping from these infections to vaccine demand is not consistently used. In Table 1, each EID has an “average infections” parameter, but later the model assumes a fixed 10 million doses for any local EID outbreak and 1 billion for disease X, irrespective of the nI values or population-at-risk, coverage, uptake, and number of doses per person. This disconnect risks internal inconsistency; it likely biases results (arguably optimistically on revenue for many EIDs, given low infection counts like MERS/West Nile), although the central qualitative conclusion (private returns are unattractive) would only be strengthened with more conservative demand. The social-impact metric “epidemics prevented” appears to function as “an EUA-ready vaccine exists when the outbreak occurs,” which does not account for epidemic growth dynamics, time-to-deployment, acceptance, or two-dose/booster schedules; it may therefore overstate outbreaks “prevented” versus “mitigated.” Second, competition and cannibalization are handled asymmetrically. The paper reasonably assumes no incremental demand from having multiple portfolio vaccines approved for a given EID (full cannibalization within-platform), but it does not model external competition from other modalities or non-portfolio developers, which would further erode market share and revenue. Conversely, the decision not to implement portfolio optimization or dynamic sequencing (e.g., stage-gating phase 3 programs conditional on early results) likely inflates costs; the discussion acknowledges this but does not quantify it. Third, the clinical development module mixes realistic and idealized elements. The choice to defer phase 3 un-

til an outbreak reduces modeled costs but also delays availability—potentially inflating the “prevented epidemics” count given roll-out lags. The human challenge trial scenario permits replacing a large phase 3 with an 8-month, \$12.5M “phase 3 via HCT,” which is ethically and regulatorily optimistic and unlikely to support licensure alone for most pathogens; treating this as an upper bound is sensible, but then its limited impact on returns reinforces the demand-side constraint. The PoS inputs draw on Project ALPHA and are further boosted (by a technology factor for six antibody-mediated EIDs), yet the disease-specific applicability and magnitude of such boosts are uncertain; importantly, even with optimistic PoS, revenues remain the limiting factor—a useful, robust insight. Fourth, the correlation structure is novel but under-specified. Correlating trial outcomes via a proprietary “biological similarity” distance (taxonomy/qualitative/quantitative/protein sequence features) and mapping $\text{distance} = 1 - d$ is plausible as a heuristic, but the construction, feature weighting, preprocessing, and validation are not reproducible from the text, and the within-pathogen correlation is simply set to 0.8. The authors partly address correlation uncertainty via an equi-correlation sensitivity, but a more granular exploration (e.g., perturbing the bespoke matrix, stress tests near non-PD boundaries) would strengthen robustness claims. Fifth, the manufacturing and supply-chain model is informative but simplified. It uses reasonable literature-based costs and capacities (e.g., 30L, 5 g/L; 65 g/dose; \$0.27 fill-finish; \$1 supply chain), but it omits volatility and constraints in critical raw materials (LNP lipids, enzymes), cold-chain logistics in LMICs, batch failure rates, and capex ramp times for line build-outs (beyond a one-time “production line” fixed cost). These omissions affect timing/capacity to meet sudden outbreak demand and could reduce revenues relative to modeled availability. Sixth, revenue modeling uses list prices rather than net realized prices, does not incorporate tiered pricing or procurement discounts, and holds prices constant across settings. The sensitivity analysis rightly shows that price is the key lever, but without an explicit payer/AMCs/stockpiling demand schedule, feasibility of the price points (\$69–\$100/dose break-even) remains uncertain, especially for LMIC procurement—an issue the authors note but do not model. Additional influential parameters not varied include the discount rate (fixed at 10%), outbreak probabilities/infection sizes (kept as point estimates), and the number of doses per regimen; all merit sensitivity analysis because these directly scale cash flows. Seventh, some claims need tighter framing to avoid misinterpretation. The statement that there is a 66% probability the next disease X pandemic will be prevented is ambiguous: if the annual pandemic hazard is 1%, the unconditional probability of a pandemic within 20 years is ~18%; a 66% “prevention probability” is presumably conditional on occurrence. Clarifying conditioning would avoid confusion. Finally, reproducibility is limited. Several key calibrations are based on “domain expert input,” but there is no public code, synthetic data, or a detailed appendix reconciling all parameter values, the correlation matrix construction, or the exact cash-flow mechanics (e.g., IRR calculation details). Despite these issues, the paper’s central qualitative conclusion—that absent public instruments (AMCs, stockpiles, guarantees), EID-focused mRNA megafunds are unlikely to attract private capital—is well supported by the model structure and the provided sensitivities. The work advances prior megafund analyses by integrating outbreak stochasticity, an explicit mRNA manufacturing cost model, and a disease-similarity-based correlation approach, and it clearly identifies price and late-stage trial costs as the economic bottlenecks. To strengthen the contribution, the authors could: (i) align dose assumptions to Table 1 nI and explore sensitivity to coverage, regimen, and uptake; (ii) vary outbreak probabilities and infection sizes; (iii) include discount-rate sensitivity; (iv) model external competition and tiered pricing/AMCs/stockpiles;

(v) implement a simple dynamic stage-gating policy; (vi) clarify the conditioning of “disease X prevention” probabilities; and (vii) release code and the constructed correlation matrix. Even with these caveats, the analysis provides credible, policy-relevant evidence that purely private financing is insufficient and that targeted public–private mechanisms are needed to sustain an EID vaccine ecosystem.

i Gemini 2.0 Flash Assessment

This paper explores the financial viability of investing in a portfolio of mRNA vaccine candidates for emerging infectious diseases (EIDs). The core idea is to use a ‘megafund’ approach, diversifying investments across multiple vaccine candidates to mitigate the high risk and low returns traditionally associated with vaccine development for EIDs. The authors simulate the financial performance of a hypothetical portfolio of 120 mRNA vaccine candidates targeting 11 EIDs, including a ‘Disease X’ representing a potential future pandemic. They calibrate their simulation parameters using expert input on mRNA technology and an extensive literature review. The paper’s primary finding is that such a portfolio, despite the advantages of mRNA technology and diversification, yields a negative average annualized return (-6.0%) and a negative net present value (-\$9.5 billion). Clinical trial costs are identified as the major cost driver (94% of total investment), while manufacturing costs are relatively low (6%). Sensitivity analysis reveals that the price per dose is the most critical factor for financial success, whereas other factors such as increased probability of success (PoS) due to mRNA technology, portfolio size, and human challenge trials do not significantly improve financial performance. The authors conclude that continued collaboration between government and private sectors is essential for creating a sustainable business model for vaccine development against EIDs. There are several potential shortcomings: The model relies heavily on assumptions, particularly regarding the probability and impact of EID outbreaks, which are inherently uncertain. While the authors calibrate their model with expert input and literature review, the validity of these assumptions directly impacts the results. The assumption of independence between vaccine development outcomes, while addressed through correlation analysis, might be overly simplistic. Real-world dependencies, such as shared manufacturing capacity or regulatory hurdles, could influence the portfolio’s performance. The model might not fully capture the dynamic aspects of vaccine development, such as the potential for technological breakthroughs or changes in regulatory policies. The analysis focuses primarily on financial metrics (return on investment, net present value) and may not fully account for the broader social and economic benefits of preventing EID outbreaks, such as reduced healthcare costs, increased productivity, and improved global health security. The authors acknowledge this limitation but do not fully integrate these factors into their analysis. The paper assumes a fixed portfolio of vaccine candidates and does not consider active portfolio management strategies, such as adding or removing candidates based on new information or changing priorities. Active management could potentially improve the portfolio’s performance but also introduces additional complexities and risks. The modeling framework is complex, involving numerous parameters and assumptions, which makes it difficult to fully understand the drivers of the results and to assess the robustness of the conclusions. The authors provide some sensitivity analysis, but a more comprehensive exploration of the parameter space would be beneficial. The study does not deeply explore alternative financing mechanisms, such as advanced market commitments or public-private partnerships, which could potentially improve the financial

viability of vaccine development for EIDs. While the authors acknowledge the need for collaboration between government and private sectors, they do not provide a detailed analysis of specific policy interventions. The model assumes perfect vaccine efficacy and distribution. In reality, vaccine effectiveness can vary, and logistical challenges can hinder timely and equitable distribution, particularly in low-resource settings. The authors do not explicitly address issues related to intellectual property rights and technology transfer, which can be critical for ensuring access to vaccines in developing countries. The model’s focus on mRNA vaccines may limit its generalizability to other vaccine technologies. While mRNA vaccines have shown promise, other platforms may be more suitable for certain EIDs or populations.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.12 Barker et al. 2021

C.12.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	78 [70–86]	65 [45–80]	75 [70–84] (n=2)
Claims	76 [68–84]	70 [50–85]	—
Methods	77 [68–85]	70 [55–80]	75 [82–94] (n=2)
Adv. Knowledge	74 [62–84]	75 [60–85]	62 [55–65] (n=2)
Logic & Comms	84 [76–92]	80 [70–90]	72 [62–82] (n=2)
Open Science	42 [28–58]	70 [50–85]	70 [80–95] (n=2)
Global Relevance	80 [68–90]	80 [70–90]	62 [40–60] (n=2)
NA	—	—	62 [48–52] (n=2)

GPT-5 Pro Assessment

This is a large, well-executed randomized trial of a group-based CBT program delivered by recent graduates to poor households in rural Ghana. The design, sampling frame, intervention description, and analytic approach are clearly described. The program comprised 12 weekly 90-minute sessions delivered to same-gender groups of about 10 participants, across 258 communities randomized into pure control, pure CBT, or mixed “full program” communities (other economic programs were slated for later and were not yet announced at the time of these surveys). The study reports endline outcomes 2–3 months after completion (about 7 months after baseline), focusing on mental and physical health, socioemotional skills, cognition, and proximate economic outcomes. Intent-to-treat estimates show statistically significant gains in mental health (mental health index +0.15 SD; reductions in K10 and psychological distress prevalence), physical health self-report (+0.15 SD), socioemotional skills (+0.27 SD across self-efficacy, grit, and self-control), cognition (+0.08 SD driven by forward/backward digit span; null on Raven’s and Stroop), and an economic outcomes index (+0.19 SD; notably

higher current and expected Cantril ladder rankings). Heterogeneity by baseline mental distress is limited; effects generally appear for both those distressed and not distressed at baseline. A key contribution is a LASSO-based heterogeneity analysis: using a holdout procedure on controls to predict endline K10, the authors show larger treatment effects on mental/physical health for those predicted to be vulnerable to deterioration, but no analogous heterogeneity for socioemotional/cognition or the economic index. This supports two channels the authors hypothesize: (i) therapy pre-emptively reduces vulnerability to future mental distress, and (ii) therapy directly improves bandwidth-like capacities (socioemotional and some cognitive skills) irrespective of mental distress status.

Important limitations and identification threats are mostly about measurement and scope, not randomization. First, the follow-up horizon is short (2–3 months post-intervention). This is a major constraint for claims about durable human capital and economic impacts. The authors acknowledge fade-out as a concern and suggest “top-up” possibilities, but current results do not establish persistence. Second, most outcomes are self-reported. Mental and physical health measures use validated instruments (K10; BRFSS-style self-ratings; days with poor health), but they remain subjective and susceptible to reporting or demand effects. Economic outcomes are particularly subjective (Cantril ladder current and five-year expectations, and days unable to do regular activities). There is no objective measurement of earnings, consumption, assets (beyond baseline covariates), or verified labor market behavior at endline. The modest cognition result relies on tasks (digit span) potentially influenced by motivation and testing engagement; null effects on Raven’s and a Stroop-like test temper the bandwidth claim for “cognition” per se. Without enumerator blinding or an active control, some portion of self-reported improvements could reflect demand, social desirability, or non-specific group support effects rather than CBT-specific content. The paper does not report enumerator-treatment separation or blinding procedures, and it does not include an attention-placebo group; thus causal attribution to CBT mechanisms, as opposed to generic supportive group interaction, is not fully isolated.

Third, although the main analysis uses indices and randomization inference, the paper evaluates many related outcomes. The authors mitigate “multiple outcomes” concerns partly via index construction (Kling–Liebman–Katz style), but they do not present formal familywise or false discovery-rate adjustments across all outcome families. The primary conclusions rely on indices with very small RI p-values, which is reassuring; still, with many measures and some borderline p-values (e.g., days with poor mental health $p = 0.10$), clarity on a pre-analysis plan or primary outcomes would strengthen credibility. Fourth, compliance and attendance are not reported, nor is a treatment-on-the-treated analysis. Without uptake data, we cannot assess the intensity-dose relationship or compute program-on-the-treated effects. Fifth, attrition is not discussed. The endline sample sizes are large (~7,4xx), but without explicit attrition rates and balance of attrition by arm, there is residual concern about differential follow-up. Sixth, the heterogeneity analysis via LASSO is carefully implemented with sample-splitting and CI adjustments; however, the predicted K10 risk score includes baseline mental and physical health measures and many related variables, which may blur whether it captures true “vulnerability” versus a broader baseline well-being composite. The authors do provide evidence that baseline K10 alone does not predict treatment effect heterogeneity, which helps, but further validation of the predicted “risk” construct (e.g., out-of-sample, alternative ML methods, or parsimony checks) would be valuable. They also examine a predicted “bandwidth” risk score and correctly find no heterogeneity consistent with their theory; however, the bandwidth con-

struct itself is derived from self-reports for socioemotional skills and a limited set of cognitive tests, raising questions about construct specificity and sensitivity.

Seventh, generalizability is promising but not guaranteed. Communities were accessible by road and within specific Ghanaian regions; the intervention used local college graduates with two weeks of training and weekly debriefs; fidelity, supervision intensity, and counselor adherence are not quantified (no session-audit measures reported). The gender targeting at the community level is well handled analytically (no heterogeneity by gender), but implementation modality may matter for scale-up (single-gender group dynamics, local norms). Eighth, spillovers could bias estimates if untreated individuals in treatment communities were affected; the paper examines “spillover control” vs. pure control and generally finds small, non-significant differences, which is reassuring. Ninth, a small design wrinkle is the analysis sample: in control households, both adult head and spouse are included; in treated households, only the treated individual is included (spouses excluded). This choice avoids contamination but changes the composition of the analysis sample; the ANCOVA with village fixed effects and large N likely mitigates bias, but it is a non-standard asymmetry worth noting.

Finally, the interpretive step from observed short-run improvements to framing CBT as a “core strategy for increasing human capital” feels stronger than the evidence currently supports. The paper convincingly shows near-term gains in self-reported mental and physical health and in socioemotional skills, with small gains in digit span and improved subjective economic outcomes. It does not yet establish changes in objective economic behavior or long-run human capital accumulation, nor does it provide cost-effectiveness metrics essential for policy prioritization. The absence of cost data is a notable gap for practitioners. On the plus side, the work’s ethics appendix is thoughtful, the randomization inference appropriately mirrors the multi-stage re-randomization procedure, and the comparisons restricted to pure controls support minimal spillovers. Open science practices are mixed: the intervention manual is said to be publicly available; however, there is no link to data or replication code, which hinders reproducibility and external scrutiny. Overall, this is a rigorous and policy-relevant study with strong short-term internal validity and clear presentation. The main uncertainties relate to durability, measurement (self-report), mechanisms (CBT vs. generic group support), and practical cost-effectiveness for scale-up.

i Gemini 2.0 Flash Assessment

This paper examines the impact of cognitive behavioral therapy (CBT) on human capital in a general population of poor households in rural Ghana. The study uses a randomized controlled trial and finds that CBT leads to improvements in mental and physical health, cognitive and socioemotional skills, and economic outcomes. The authors argue that CBT can improve human capital through two pathways: reducing vulnerability to deteriorating mental health and directly improving bandwidth, increasing cognitive and socioemotional skills. They find no evidence of heterogeneity by baseline mental distress, suggesting that the program is relevant for a general population, not just those with diagnosed mental health issues. The study also explores the mechanisms behind these effects, finding that CBT can improve the quality of decision-making by helping individuals better allocate their mental resources. They split the control group into training and testing samples, train a Lasso using k-fold cross validation in the training sample, and then predict endline distress in both the control testing set and treatment group. They show that while baseline mental health

distress does not predict heterogeneity in treatment effects, vulnerability to mental distress does. Specifically, mental and physical health impacts of the CBT program are statistically significantly larger for those who are predicted by their Lasso to be vulnerable to deteriorating mental health. This is important as a direct test of their vulnerability hypothesis and also because it indicates that the churn between mental distress states is not just measurement error.

Here are some potential issues with the paper:

1. **Short Time Horizon:** The endline data is collected only 2-3 months after the intervention. This short time frame raises questions about the long-term sustainability of the observed effects. It is unclear whether these improvements will persist over time or if they are merely temporary gains.
2. **Self-Reported Measures:** A significant portion of the outcome measures, including mental health, physical health, and economic status, rely on self-reported data. This introduces the potential for bias, as individuals may over- or under-report their experiences due to social desirability or recall issues. The authors acknowledge this limitation but do not fully address its implications.
3. **Lack of Heterogeneity Analysis:** While the authors argue that CBT is effective for a general population, they find limited evidence of heterogeneity by baseline mental distress. This raises concerns about whether the intervention is truly beneficial for everyone or if certain subgroups may benefit more than others. Further exploration of heterogeneity could provide valuable insights into the specific populations for whom CBT is most effective.
4. **Potential Spillovers:** The study design involves delivering CBT to groups within communities, which raises the possibility of spillover effects. The authors acknowledge this concern and attempt to address it by excluding spouses of treated individuals from the analysis. However, it is still possible that other community members may have been influenced by the intervention, which could bias the results.
5. **Limited Generalizability:** The study is conducted in a specific context (rural Ghana) and with a particular population (poor households). This limits the generalizability of the findings to other settings and populations. It is unclear whether the same results would be observed in different cultural contexts or with individuals from different socioeconomic backgrounds.
6. **Measurement Error:** Although the authors use a Lasso to predict vulnerability to deteriorating mental health and find heterogeneous treatment effects based on this prediction, measurement error may still be a concern. The reliance on self-reported measures and the potential for social desirability bias could introduce noise into the data, which could affect the accuracy of the vulnerability predictions.
7. **Weaknesses in Identification.** While the authors attempt to address endogeneity concerns through randomization, there may still be unobserved factors that confound the relationship between CBT and the outcomes of interest.
8. **Lack of External Validity.** The study focuses on a specific CBT program delivered by lay counselors in a low-resource setting. This limits the external validity of the findings, as

it is unclear whether the same results would be observed with different CBT programs or with more highly trained therapists.

9. Lack of Replication. Although the study uses a randomization inference procedure to test the robustness of the findings, it does not provide information on whether the results have been replicated in other studies or settings.
10. Practical Relevance. The study examines the impact of CBT on a range of outcomes, including mental and physical health, cognitive and socioemotional skills, and economic outcomes. However, it does not provide detailed information on how these improvements translate into practical benefits for individuals and communities. Additional research is needed to understand the real-world implications of these findings and to develop strategies for maximizing the impact of CBT interventions.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.13 Benjamin et al. 2023

C.13.1 Ratings Comparison

Metric	Claude Sonnet 4	GPT-5 Pro	Gemini 2.0 Flash
Overall	78 [70–85]	90 [83–95]	78 [65–88]
Claims	80 [72–87]	88 [80–94]	85 [75–92]
Methods	82 [74–88]	90 [82–95]	75 [60–85]
Adv. Knowledge	75 [67–83]	88 [78–94]	80 [70–90]
Logic & Comms	85 [78–90]	91 [86–96]	88 [80–95]
Open Science	88 [82–93]	67 [50–78]	85 [75–92]
Global Relevance	65 [55–75]	78 [68–88]	70 [55–80]

i GPT-5 Pro Assessment

This manuscript tackles a central and long-standing identification threat in self-reported well-being (SWB) research: heterogeneity in how respondents use rating scales. The paper’s core contribution is a clear theoretical framework that distinguishes “general” from “dimensional” scale use, proposes design principles for calibration questions (CQs) that aim to have the same objective answer for all respondents, and develops econometric estimators to adjust common SWB analyses (means, covariances with covariates, covariances across SWB items, variances) for general scale-use heterogeneity. The work is ambitious and largely convincing, but several assumptions are strong and the credibility of the adjustments hinges on their validity. Key identification rests on: (1) an approximate linear translation between individuals’ scale use (Assumption 1), which the paper supports with graphical evidence across demographic groups and more detailed individual-level evidence in a second survey; (2) “vignette equiva-

lence” generalized to all CQs (Assumption 2), meaning CQs identify a common state, which is most plausible for visual CQs but less so for vignettes due to projection/fill-in risks; and (3) “generalized response consistency” (Assumption 3), equating scale use for SWB items at a given “height” to average scale use across CQs at that height. The authors discuss these assumptions candidly, emphasize trade-offs between adjusting for general vs. dimensional scale use, and show when the adjustment matters most (e.g., dependence on the target item’s mean “height”). The empirical applications on a large MTurk sample (~3,358 after stringent QC) are compelling demonstrations, but generalizability is limited by non-representativeness and the MTurk environment. Several specific issues deserve emphasis. First, dependence on linearity (Assumption 1) may be violated for some respondents or items; while the group-level linearity is tight, individual-level linearity requires the richer Bottomless survey for clearer support. Sensitivity checks to moderate nonlinearity are desirable beyond the provided simulations; in the semi-parametric estimator, the $K=1$ polynomial for $E[w]$ could be too restrictive for some constructs. Second, the linchpin Assumption 2 is most credible for visual CQs, yet visual perception can systematically differ by device/screen/ambient light or observer characteristics (e.g., age, color vision), raising concerns about nonclassical “perception error.” The model allows for perception error (ϵ) and “trembling-hand” error (η) , but assumes independence from μ , w , and covariates (Assumption 4). Persistent device- or observer-specific differences could violate mean independence. The paper partly addresses this by modeling μ as persistent and by relying on large- I averages for CQs, but readers should recognize that systematic device heterogeneity or correlated visual acuity with demographics could bias adjustments if not balanced. Third, Assumption 3 (generalized response consistency) is the primary bridge from CQs to SWB; it is plausible for general response tendencies, but it produces only a partial correction if dimension-specific scale use matters for the particular SWB item. The authors’ evidence that general scale use explains roughly 3/5 of scale-use heterogeneity (with higher shares for vignette-only sets) is informative, yet the remainder may still be consequential in specific domains; applied researchers may want to supplement with dimension-tailored CQs if feasible. Fourth, in variance and covariance applications, response errors for SWB items are not separately identified from true variance, forcing an assumption that SWB response-error variances equal those for CQs (Table 5). This is likely optimistic for SWB (expect greater perception error on subjective constructs), so the “adjusted” variance should be interpreted cautiously—arguably as an upper bound for true SWB variance after correction. The authors are appropriately cautious and provide complementary evidence from short-run repeats suggesting transitory error explains only part of the correction; still, the strong identifying assumption remains a major caveat for inequality inferences. Fifth, the MOM estimator for $\text{Cov}(x, w)$ relies on a zero co-skewness condition between μ , x , and centered w (eq. 13). The paper smartly contrasts MOM to semi-parametric estimates that relax this condition and finds small differences, which boosts confidence. Still, contexts with strong tri-variate dependence (e.g., demographics that correlate with both true SWB and response style in non-linear ways) could challenge the MOM estimator. Sixth, the empirical finding that adjustments matter more for items whose means are far from the estimated center implies that the utility of the correction is application-dependent. This is a strength (the paper tells readers when to expect big changes) and a caution (overgeneralization is possible). Seventh, sample and implementation concerns: the main data are from US MTurkers—diverse but not representative—with extensive QC; extrapolation to population surveys should be tentative. Visual CQ performance could vary across devices, and the paper does not report

device-stratified diagnostics. Rounding on the 0-100 scale is acknowledged as a future issue; it may interact with response styles. The factor-analytic illustration shows large changes after correction, but this also reflects structural assumptions in the comprehensive MLE (joint normality of (γ, δ, w) , item-invariant error structure for SWB mirroring CQs) that may not hold for all SWB items. Eighth, replicability and open science: preregistration is provided and appendices are extensive, but it is unclear whether code and microdata are openly available; full computational replication seems feasible but nontrivial, given custom MLEs and bootstrapping. Notwithstanding these caveats, the paper's logic is carefully laid out, diagnostics are thorough, and validations (improved alignment of subjective 0-100 with objective units for height/weight, and results across alternative response scales) credibly support the main claims: general scale-use heterogeneity is real, sizable, partly persistent, and can materially bias standard SWB analyses; the proposed estimators can mitigate these biases, with effects that depend on item "height" and application. The work is likely to advance best practice in SWB research (and related survey contexts), particularly for analysts willing to include CQs and to apply the semi-parametric or comprehensive MLE adjustments. Key remaining uncertainties center on the validity of Assumptions 2–3 across contexts, the strength of the linearity approximation, the equality of error variances for SWB vs. CQs, and robustness to device/visual heterogeneity. Addressing these with device-level stratification, expanded sensitivity analyses (e.g., $K > 1$), field implementations in representative samples, and public code would further consolidate the contribution.

Claude Sonnet 4 Assessment

This paper presents a comprehensive framework for measuring and adjusting scale-use heterogeneity in self-reported well-being (SWB) data using calibration questions (CQs). The authors identify a significant methodological issue: people may use response scales differently, which can confound SWB analyses. They develop novel visual and vignette CQs to measure both dimensional and general scale-use heterogeneity, proposing three classes of estimators (method-of-moments, semi-parametric, and maximum likelihood) to correct for these biases. The empirical work uses a large MTurk sample (3,358 high-quality respondents) with 18 CQs and 33 SWB questions. Key findings include substantial scale-use heterogeneity correlated with demographics, with general scale use explaining roughly 3/5 of total heterogeneity. The adjustment substantially affects some analyses (e.g., SWB variance estimates are reduced by more than half) but has minimal impact on others (e.g., life satisfaction regressions). Strengths include the innovative visual CQs that avoid vignette-related concerns, rigorous theoretical framework with linear translation functions, comprehensive econometric methodology, and practical validation using objective measures. However, several limitations warrant consideration: the MTurk sample may not be representative; the linearity assumption for translation functions, while empirically supported, could be restrictive; the comprehensive MLE makes strong distributional assumptions; and some key assumptions (particularly Assumption 2 for CQs and Assumption 3 for response consistency) may be violated in practice. The method requires researchers to collect additional CQ data, and individual-level scale-use estimates are noisy with only 18 CQs. The paper makes important theoretical contributions but raises questions about when the substantial methodological complexity yields practically meaningful improvements to SWB research conclusions.

This paper addresses the critical issue of scale-use heterogeneity in self-reported well-being (SWB) data, a potential confound that can significantly impact the validity of research findings. The authors introduce a novel approach using calibration questions (CQs) to measure and adjust for both dimensional and general scale-use heterogeneity. They conducted a survey on MTurkers, and find that scale-use heterogeneity is substantial and correlated with demographics. The paper provides a theoretical framework and econometric methods for quantifying and adjusting for this heterogeneity, and applies these new estimators to several standard SWB applications. The key innovation lies in the use of CQs, which are designed to have the same objective answer across respondents, thereby enabling the identification of scale-use patterns. The paper distinguishes between dimensional scale use (specific to an SWB dimension) and general scale use (common across questions) and shows that adjusting for general scale-use heterogeneity can alter research conclusions in some instances.

Strengths:

- **Novel Approach:** The use of calibration questions (CQs), particularly visual CQs, is a novel and potentially powerful method for addressing scale-use heterogeneity.
- **Theoretical Framework:** The paper provides a clear theoretical framework that motivates the econometric methods.
- **Empirical Validation:** The empirical analysis demonstrates the existence of scale-use heterogeneity and its correlation with demographics.
- **Comprehensive Econometric Methods:** The paper develops several estimators to adjust for scale-use heterogeneity, catering to different SWB applications.
- **Practical Implications:** The paper illustrates how adjusting for scale-use heterogeneity can change the conclusions of SWB analyses.
- **Careful Preregistration:** The authors preregistered their sample exclusions and some analyses, enhancing the credibility of the research.

Weaknesses and Limitations:

- **MTurk Sample:** The use of an MTurk sample limits the generalizability of the findings to the broader population.
- **Linearity Assumption:** The assumption of linear translation functions may not hold in all cases.
- **Response Error Independence:** The assumption of response-error independence may be violated in certain contexts.
- **Small Number of CQs:** In the empirical setting, the individual-level estimator for scale-use-adjusted SWB is biased and noisy because the number of CQs is small.
- **Vignette Bias:** The vignette CQs may be subject to systematic biases if respondents project their own situations onto the hypothetical scenarios.
- **Zero Co-Skewness Assumption:** Some estimators rely on the assumption of zero co-skewness, which may not always hold.
- **Parametric Assumptions:** The comprehensive MLE estimator relies on strong parametric assumptions about the joint distribution of key variables.

Overall Assessment:

Overall, the paper makes a valuable contribution to the SWB literature by providing a novel and rigorous approach to addressing scale-use heterogeneity. While the limitations related to the sample and assumptions should be considered, the paper offers a solid foundation for future research in this area.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.14 BenYishay et al. 2024

C.14.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash
Overall	78 [68–88]	65 [50–80]
Claims	76 [65–85]	70 [55–85]
Methods	73 [60–83]	65 [50–80]
Adv. Knowledge	82 [72–92]	75 [60–90]
Logic & Comms	74 [64–85]	75 [60–85]
Open Science	45 [30–60]	70 [55–80]
Global Relevance	88 [78–95]	85 [70–95]

i GPT-5 Pro Assessment

This manuscript assembles a rich geospatial panel to study the long-run impacts of small-scale, river-pumped irrigation perimeters (and floodplain “mares”) rolled out across northern Mali since 1999. It links per-perimeter treatment timing to satellite-derived indices (NDVI, NDWI), very high resolution (VHR) imagery, geocoded DHS child anthropometrics, and ACLED conflict data, estimating effects with staggered-adoption two-way fixed effects (TWFE) and an event-study approach adjusted for heterogeneous treatment effects. The main results—durable increases in water availability and greenness on irrigated fields, improved child nutrition among nearby households, and reduced conflict incidence immediately around perimeters (offset by increases further away)—are plausible and policy-relevant. However, several identification, measurement, and reporting limitations temper the causal interpretation and pose replication challenges.

Identification for the remotely sensed agricultural outcomes is the strongest part of the paper: the staggered roll-out is well-suited to TWFE/event-study; the authors cite de Chaisemartin–d’Haultfoeuille estimators to address heterogeneity, and pre-trend diagnostics for NDVI/NDWI appear flat, supporting parallel trends. Yet there are outstanding concerns about time-varying local confounders. The text indicates polygon and region-by-year fixed effects (plus precipitation and temperature controls), while a key figure caption suggests grid-cell and year fixed effects. This discrepancy in unit of analysis and fixed-effects structure should be reconciled. If only year FE are used (rather than region-by-year), district- or

region-specific shocks (including evolving conflict, local economic changes, or programmatic co-interventions) could bias estimates; even with region-year FE, within-region time-varying shocks could matter. The paper asserts that timing is not correlated with preceding changes in weather, agricultural outcomes, or community health, but only satellite-based pretrends are clearly shown in the main text; analogous diagnostics for survey and conflict outcomes are not presented. The models do not appear to control for local conflict directly in the agricultural regressions; if conflict influenced both implementation sequencing and field use, residual confounding is possible.

Measurement choices for agricultural outcomes warrant more clarity. The construction of annual NDVI/NDWI uses a single-month (May) “growing season” composite and a “pre-rainy” November composite. The rationale for these months and their alignment with cropping calendars for both pump-based perimeters and floodplain systems is not fully justified; if phenology differs by system or year, single-month composites may not capture peak greenness or water availability, potentially attenuating or mis-timing estimated effects. The text states that site-level means are computed by aggregating pixels within each perimeter polygon; however, a figure caption refers to “annual measurements of 30 m × 30 m grid cells” and lists 16,410 observations for 942 perimeters, which seems inconsistent with the polygon-year aggregation described earlier and with the 1986–2021 time span. Relatedly, the core agricultural specification is described in the text with polygon FE and region-year FE, while the event-study caption mentions grid-cell and year FE. These internal inconsistencies in unit of analysis, sample size, and fixed effects hinder reproducibility and make it difficult to assess the exact estimand. The paper also translates a 32% increase in NDVI into an illustrative yield gain for rice; given the diversity of crop types and the imperfect mapping between NDVI and yield, this should be framed more cautiously or supported with additional calibration.

The DHS analysis linking irrigation completion to child nutrition outcomes is innovative but methodologically weaker. The design is cross-sectional, with children from five survey rounds within 6 km of perimeters assigned to 0–2, 2–4, and 4–6 km bands relative to the nearest project. The TWFE specification includes distance-band, region, and birth cohort fixed effects. Without region-by-year (or district-by-year) fixed effects or cluster fixed effects, region-specific temporal shocks correlated with irrigation timing could confound estimates. The sample size is modest (1,603 children within 6 km, unevenly distributed across bands), and the paper does not detail how overlapping exposure to multiple perimeters is handled. The authors note “no differential changes” in other household characteristics, but do not enumerate these covariates nor report balance or placebo tests. Potential spatial imprecision in household cluster locations (and how this might affect distance-band assignment) is not discussed. Event-time dynamics are only partially explored; pre-trend analyses around irrigation timing for these survey outcomes are not shown in the main text. Overall, while the nutrition findings are plausible and consistent with the agricultural results, residual selection and measurement error risks remain.

The conflict analysis uses perimeter-year panels with counts in 0–1, 1–5, and 5–10 km rings and district-by-year fixed effects. This is a reasonable design, but several issues arise: (i) pre-trend/event-study evidence is not shown; (ii) ACLED event geolocation quality can vary, making a 0–1 km ring potentially sensitive to positional error; (iii) events could be counted in multiple perimeters’ rings, inducing dependence and inflating effective sample size; and (iv) effects are reported as percentage-point changes without clearly stating baseline probabilities, making it harder to interpret magnitudes. Actor-specific analyses are mentioned but not

detailed in the main text. These factors make the conflict spillover results suggestive rather than definitive.

The ecological assessment relies on VHR image interpretation for 33 perimeters (2015–2018) across eight clusters and auxiliary NDVI/NDWI analyses on nearby buffers and riverbanks for the same sample. This provides useful qualitative context (e.g., evidence of local gully formation and algal blooms), but the sample is limited and likely non-representative across the two-decade program. The causal attribution of erosion features to irrigation versus natural river dynamics is explicitly acknowledged as unclear. The null findings on off-perimeter NDVI/NDWI changes are encouraging but should be interpreted cautiously given the small, recent cohort and the potential for seasonal and phenological misalignment discussed above. The manuscript’s communication is generally clear and balanced (notably acknowledging distributional trade-offs and muted results for floodplain valorization), but the aforementioned internal inconsistencies (unit of analysis, fixed effects, and sample sizes in text versus figure captions) reduce transparency. Details on standard error clustering, weighting, and how missing satellite years are handled are not fully elaborated in the main text. Focus group insights are helpful qualitatively but lack methodological detail (sampling, protocols), limiting their evidentiary weight.

Open-science and replication are constrained. The irrigation intervention data are “available upon reasonable request,” commercial VHR imagery must be purchased, and no analysis code is provided. Key empirical tables for satellite outcomes are relegated to the supplement, and some critical diagnostics (e.g., pre-trends for survey and conflict outcomes) are not shown in the main text. These factors, together with the internal specification inconsistencies, pose significant barriers to reproduction.

In sum, the study makes a valuable contribution by credibly documenting long-lasting on-field agricultural gains from small-scale pumped irrigation and by integrating multiple data sources to probe downstream effects on nutrition, conflict, and local ecology. The agricultural findings are the most convincing; the nutrition and conflict spillover results are compelling but would benefit from stronger diagnostic checks, clearer handling of spatial imprecision and overlapping exposures, and fuller reporting. Addressing the reporting inconsistencies and providing code and clearer replication materials would substantially strengthen the paper’s credibility and usefulness to practitioners.

i Gemini 2.0 Flash Assessment

The paper investigates the long-term impacts of small-scale irrigation projects on climate resilience, agricultural production, child nutrition, and conflict in Mali. The study uses a quasi-experimental design, leveraging the staggered rollout of irrigation interventions across nearly 1,000 locations over two decades and linking this to satellite imagery, survey data, and conflict event data. The core finding is that irrigation leads to sustained increases in agricultural production, improved child nutrition in nearby communities, and reduced conflict risks, although these gains are partially offset by worsening conditions farther away from the newly installed irrigation. The study uses two-way fixed effects (TWFE) models to estimate counterfactual outcomes, adjusting for time-invariant factors and common shocks. The authors address potential biases in TWFE models by incorporating recent advances in methodology to account for duration-dependent treatment effects. They also perform robustness checks by ensuring that the timing of irrigation is not correlated with preceding changes in weather,

agricultural outcomes, or community health.

However, several issues could affect the validity and interpretation of the findings. The study relies on remotely sensed data (NDVI, NDWI) and survey data, which may be subject to measurement error or bias. For example, cloud cover affects the availability of satellite imagery, and survey data on child nutrition may be subject to recall bias. The potential for ecological impacts due to irrigation is assessed visually using high-resolution satellite imagery on a subset of sites, which may not be representative of all irrigation sites and which is open to interpretation bias. It is difficult to ascertain whether these ecological effects are due to irrigation or the flow of rainwater in the river. The study acknowledges some negative spillover effects, such as declines in child nutrition in communities located farther from irrigation sites, but the mechanisms driving these effects are not fully explored. The analysis of conflict events relies on the ACLED dataset, which may suffer from reporting biases or inaccuracies. The study does not explore heterogeneity in the impacts of irrigation based on factors such as the type of irrigation technology, the crops grown, or the socio-economic characteristics of the communities involved. This limits the ability to draw targeted policy recommendations. While the study controls for various confounding factors, there may be unobserved factors that affect both the implementation of irrigation and the outcomes of interest. The study mentions that the project interventions were designed to be “conflict-sensitive” but does not provide detailed information on how this was achieved or whether it was effective. The generalizability of the findings may be limited by the specific context of Mali, which is characterized by political instability, climate vulnerability, and dependence on rainfed agriculture. The long-term sustainability of the irrigation projects is not directly assessed, although the study finds that agricultural production gains persist even a decade later.

In summary, while the study provides valuable evidence on the impacts of small-scale irrigation in a challenging context, the limitations related to data, methods, and context warrant caution in interpreting the findings and drawing broad conclusions. More detailed information on the implementation of the irrigation projects, the characteristics of the communities involved, and the mechanisms driving the observed effects would enhance the credibility and policy relevance of the study. An additional limitation is the discussion of alternative interpretations of the results and how they might affect the conclusions. For instance, the study finds that irrigation has a positive impact on agricultural production, but it also acknowledges that this may be due to other inputs or practices used by farmers on these plots once irrigation becomes available. Similarly, the study finds that irrigation reduces conflict risk, but it also acknowledges that this may be due to the rebels sparing project sites as they ameliorate the living conditions for the population by whom they are supported or tolerated. These alternative interpretations suggest that the causal mechanisms underlying the observed effects may be more complex than the study acknowledges, and that irrigation may not be the sole driver of the observed outcomes.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.15 Bergeron et al. 2024

C.15.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	85 [75–92]	70 [50–85]	40 [30–50] (n=1)
Claims	77 [68–86]	75 [60–85]	—
Methods	70 [60–80]	70 [50–80]	23 [12–34] (n=2)
Adv. Knowledge	83 [72–90]	75 [60–85]	25 [15–35] (n=1)
Logic & Comms	88 [82–94]	80 [70–90]	75 [65–84] (n=2)
Open Science	55 [40–70]	65 [50–75]	79 [70–88] (n=2)
Global Relevance	79 [68–88]	75 [60–85]	50 [40–60] (n=1)
NA	—	—	50 [40–60] (n=1)

GPT-5 Pro Assessment

This manuscript proposes and analyzes an evolutionary model in which “demotivating beliefs” (e.g., beliefs about envy, luck, witchcraft, disdain for competitive striving) can spread and persist when environments are partly zero-sum. It derives clear comparative statics linking the degree of zero-sumness (), assortativity (), and the evolutionarily selected belief intensity $\beta^* = \frac{1}{1 + \frac{1}{\alpha}}$. The model implies a divergence between material welfare (hump-shaped in β within a fixed α) and subjective well-being (typically decreasing in β). The empirical sections test several predictions using two original surveys from Kananga, DRC (N 200 in 2015; N 1,000 in 2019) and the World Values Survey (WVS). In the DRC, a multi-item index of zero-sum perceptions correlates positively with envy and traditional (witchcraft/ancestor) religious practice/intensity, and negatively with Christian belief intensity; within-country vignettes and external correlates (employment histories, local rainfall histories) are used to validate the zero-sum measure. In the WVS, a single zero-sum item correlates robustly with a broad set of demotivating beliefs and with lower income, savings, education, occupational status, and subjective well-being; using zero-sum deciles as a proxy for fixed α , the authors show income is hump-shaped in demotivating belief intensity within each decile, while happiness/financial satisfaction generally declines with β , consistent with the theory.

Key strengths: (i) Theoretical clarity with explicit comparative statics and proofs under a standard payoff-monotone dynamic; robustness checks consider alternative demotivation channels and the nonequivalence to altruism. (ii) Thoughtful, context-appropriate empirical constructs in the DRC, including multiple zero-sum items, PCA indices, vignettes, and validations using occupational and climatic histories. (iii) A comprehensive global test suite with the WVS spanning multiple proxies for demotivating beliefs and multiple welfare outcomes. (iv) Novel and policy-relevant insight: the predicted divergence between material and subjective well-being, and the possibility that demotivating beliefs can be short-run efficiency-improving yet long-run growth-suppressing via learning-by-doing.

Principal concerns and suggested improvements:

- 1) Construct validity and measurement equivalence of “zero-sum” and “demotivating beliefs.” The DRC’s 6-, 10-, and 12-item indices cohere well, but the WVS relies on one item

(“people can only get rich at the expense of others”). Cross-language and cross-culture measurement invariance is not investigated; latent variable modeling (e.g., multi-group IRT or anchoring vignettes) would bolster comparability. In DRC, some PCA loadings (e.g., “closeness to non-Christians” in the 2019 sample) are near zero, indicating noise in the witchcraft index; the Christianity index explains only ~35% of variance in 2019. Reporting reliability metrics (Cronbach’s alpha/omega) and sensitivity to dropping weakly loading items would be helpful. More discussion is also warranted on whether items like “humiliating to receive money without working” cleanly proxy demotivation rather than norms around deservingness or social transfers.

- 2) Identification and endogeneity. Most empirical results are associative. Zero-sum perceptions, demotivating beliefs, and outcomes likely co-evolve with socioeconomic status, institutions, or personality traits; reverse causality (e.g., low income leading to more zero-sum views and demotivation) is plausible. The paper commendably provides validation exercises (DRC vignettes; early-life rainfall; WVS early-life growth), but these are not used as instruments for causal identification. A 2SLS design at least for the WVS (instrumenting zero-sum with early-life growth or immigration/mobility proxies at the cohort-by-country level, with country and cohort fixed effects) would more directly address endogeneity. For the DRC, plausibly exogenous variation (e.g., crop-specific rainfall shocks or market openings affecting competitive intensity) could strengthen causal claims about .
- 3) The within- “hump-shaped” test. Grouping individuals into zero-sum deciles is an intuitive way to approximate fixed , but this risks residual confounding if zero-sum and demotivation share unobservables within deciles. The main evidence is graphical (binscatter-style), with no formal curvature tests. For each decile, parametric regressions of income on demotivation and its squared term (with decile fixed effects and controls) and joint tests for concavity and interior maxima would substantiate the claim. Showing that the estimated turning point moves in the direction predicted as rises (higher β_1 implying higher β_2^*), or at least reporting its consistency across deciles, would tie more directly to the model.
- 4) Mapping subjective well-being. The theory’s “subjective well-being” is the agent’s subjective payoff; the empirics use “happiness” and “financial satisfaction.” While the directional prediction is borne out, these measures are not exact matches to the model construct. Acknowledge and test robustness using broader life satisfaction where available (beyond financial satisfaction), and explore whether results hold when conditioning on income to separate the affective discounting channel from purely material channels.
- 5) Role and measurement of assortativity (ρ). The model’s $\beta_2^* = \beta_1 \cdot \rho$ central result hinges on ρ , but ρ is largely unmeasured in the WVS and only approximated in the DRC via network homophily across tribe/family/religion/church using aggregated relational data. This proxy may not capture the assortativity relevant for competitive interactions. The paper finds ρ is uncorrelated with zero-sum perceptions in DRC and that controlling for it does not change coefficients, but deeper analysis—e.g., testing predictions across subsamples with higher vs lower measured assortativity, or showing how the implied β_2^* tracks ρ —would more fully engage the core mechanism.

- 6) Statistical inference and multiple testing. Many outcomes and belief measures are analyzed. P-values are unadjusted; the sample sizes in WVS are very large, so even modest effects are significant. Effect sizes and standardized coefficients are sometimes reported but could be more systematically emphasized (e.g., percent of a standard deviation per one-SD change in zero-sum). Clustering choices vary; in WVS country-level clustering (or two-way clustering by country and wave) would be conservative given common shocks and sampling design; reporting results with survey weights consistently would improve external validity.
- 7) Generalizability and sampling: The DRC samples are carefully drawn for the study's aims but include targeted neighborhoods and ethnic restrictions (in 2019) that may limit external validity beyond pre-industrial urban contexts in Kasai. The text is careful about scope but could more explicitly discuss external validity.
- 8) Open science and reproducibility. The paper thoroughly documents survey instruments, construction of indices, and WVS variable definitions. However, it does not link to code/data for the DRC samples, and code for replicating WVS analyses is not provided. Posting replication packages (with do-files and anonymized DRC data or at least a synthetic dataset documenting transformations) would substantially improve reproducibility.
- 9) Historical illustration. The endogenous-technology extension and the Industrial Revolution discussion are interesting but necessarily speculative. The Google N-grams exercise is suggestive at best; positioning it explicitly as illustrative (not evidentiary) is appropriate, as done here.

Overall, the theory is sharp and well-executed; the empirical evidence is broad, careful in measurement for the DRC, and persistent across many WVS outcomes, but remains primarily correlational. Formal within-decile curvature tests, stronger causal designs exploiting cohort-level exogenous variation, and improved measurement invariance checks would materially strengthen the case. Even as-is, the paper makes a substantial, novel contribution at the theory–empirics interface on culture, competition, and development, with clear implications for how perceived zero-sumness may shape both economic behavior and well-being across contexts.

i Gemini 2.0 Flash Assessment

This paper explores the relationship between “zero-sum” beliefs, effort, and economic development, building on the anthropological concept of the ‘image of limited good.’ The core argument is that when people perceive their environment as zero-sum (i.e., one person’s gain is another’s loss), they tend to develop beliefs and behaviors that discourage individual effort and competition, even if it harms overall economic outcomes. The authors present an evolutionary model to formalize this idea, showing how ‘demotivating beliefs’ can emerge and persist despite reducing incentives for productive activity. The model predicts a hump-shaped relationship between demotivating beliefs and material welfare within a society, but a negative relationship between zero-sum perceptions and both material welfare and subjective well-being across societies. The paper then tests these predictions using data from two sources:

surveys from the Democratic Republic of Congo (DRC) and the World Values Survey (WVS). The DRC data are used to examine the relationship between zero-sum perceptions, envy, and witchcraft beliefs. The WVS data are used to test the predictions about material welfare and subjective well-being on a global scale. The empirical results generally support the model’s predictions. Several potential issues should be considered in evaluating the paper. First, the measurement of ‘zero-sumness’ relies on survey questions about perceptions, which may not accurately reflect the true environment. The authors acknowledge this limitation and attempt to validate their measures using various approaches, including comparing them to objective measures such as rainfall and economic growth. Second, the causal interpretation of the results is not always clear. While the model provides a theoretical framework for understanding the relationship between zero-sum beliefs and economic outcomes, the empirical analysis is largely correlational. It is difficult to rule out the possibility that other factors are driving both zero-sum perceptions and economic outcomes. Third, the generalizability of the findings may be limited. The DRC data come from a specific context, and the WVS data may not fully capture the nuances of zero-sum beliefs in different cultures. Internal validity is generally strong, with the authors carefully specifying their model, deriving testable predictions, and conducting robustness checks. However, there are some potential threats. For example, the authors use principal component analysis (PCA) to create indices of zero-sumness, envy, and witchcraft beliefs. The choice of which variables to include in the PCA and how to interpret the resulting components could affect the results. External validity may be limited by the specific contexts and measures used in the study, replicability could be a challenge given that the construction and interpretation of the key variables rely on subjective judgment and PCA. The evolutionary model makes strong assumptions about cultural evolution, and the empirical tests may not fully capture the dynamic processes described in the model.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.16 Bettle 2023

C.16.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	75 [63–86]	60 [35–80]	32 [5–55] (n=2)
Claims	72 [60–84]	70 [50–85]	—
Methods	63 [48–74]	65 [40–80]	10 [0–40] (n=2)
Adv. Knowledge	68 [52–80]	75 [60–90]	40 [20–75] (n=2)
Logic & Comms	74 [65–85]	80 [70–90]	30 [10–60] (n=2)
Open Science	44 [30–58]	50 [30–70]	10 [0–50] (n=2)
Global Relevance	82 [72–91]	90 [80–99]	60 [30–88] (n=2)

This methodological guide proposes a structured approach to adjust effect sizes from RCTs (and some quasi-experiments) for use in cost-effectiveness analyses by decomposing “discounts” into internal replicability (Type M magnitude inflation and Type S sign errors) and external validity. It synthesizes literature on power, publication bias, effect-size exaggeration, and cross-context heterogeneity, and provides stepwise guidance, sanity checks using published replication/meta-analytic estimates, and a worked example. The conceptual decomposition is sound and useful, and the focus on study power, pre-registration, and mechanistic transportability is well aligned with best practice. The document is generally careful about caveats and provides actionable advice for applied analysts.

Important issues and limitations:

- **Quantification via multiplicative discounts:** The core workflow combines several adjustments (power-related exaggeration, methodological flaws, researcher bias, and Hawthorne/special care) by multiplying “inflation factors” and inverting. This implicitly assumes independence among error sources and risks double counting correlated influences (e.g., low power and publication selection; or Hawthorne vs special care). The text gestures at these dependencies but does not provide a formal rule for dependence or caps to prevent over-correction. A decision tree or explicit dependency structure would increase reliability.
- **Calibration and external validation:** Although the guide cites replication shrinkage and publication-bias-adjusted estimates from several fields, the recommended headline ranges (e.g., “many RCTs will need a 50–60% Type M discount”) are not calibrated specifically to the intervention domains and outcome types most common in LMIC global health/development CEAs. The worked example assigns 10–20% adjustments for specific factors without an empirical basis. A retrospective calibration exercise—comparing discounted predictions to later replications or scale-ups across a portfolio—would help refine default values and quantify uncertainty.
- **Power/MDE guidance and a technical error:** Diagnosing underpowered studies is central here, but guidance is mixed. The document incorrectly states: “if the MDE is higher than your best estimate of the effect size, it is reasonably well-powered”; this inequality is reversed (a study is well-powered when the expected effect is larger than the MDE). The simple power plots shown (for two-sample t-tests) may mislead for cluster-randomized trials, unequal group sizes, binary outcomes, or non-normal data, which are common in this domain. Clearer warnings and links to cluster-adjusted power/MDE tools would prevent misuse. The suggestion to use the coefficient of variation as a noise heuristic can fail when means are near zero or can be negative.
- **Baseline priors and retrodesign:** Encouraging analysts to form priors and use retrodesign to quantify Type M and Type S is a strength. However, the procedure to construct priors is under-specified and the worked example uses ad-hoc multiplicative shrinkages (e.g., reducing the mean because a share of studies were null, plus a blanket inflation correction) that do not correspond to a coherent probabilistic prior. Providing a minimal Bayesian template (e.g., normal prior on standardized effects informed by related interventions) and code would improve consistency.

- Scope and transferability of evidence: Table 1 aggregates replication shrinkage across psychology, medicine, and experimental economics; the text notes heterogeneity but does not clearly delineate when each source should inform LMIC health/development RCT adjustments. For instance, Button et al.’s neuroscience exaggeration factors may not be representative for large-sample administrative outcomes, while behavior-change field studies may show larger inflations. The Coville & Vivalt development-econ estimates are closest in domain; more weight on these (and similar) sources would be appropriate.
- External validity quantification: The mechanistic framework (sample differences, Hawthorne, equilibrium, special care) is appropriate and the use of Vivalt (2020) as an upper bound is reasonable. Yet translating these categories into numeric discounts remains largely judgment-based. Where feasible, anchors from the literature (e.g., NGO vs government implementation gaps; effect dependence on baseline prevalence or access measures) could be formalized into default priors with uncertainty, rather than point discounts. A clearer process for avoiding overlap between internal Hawthorne effects and external implementation changes would also help.
- Type S adjustments: The guidance to restrict Type S adjustments to a minority of cases (plausible sign reversal, low power, early-stage evidence) is sensible and supported by the cited development-econ calculations. However, the estimation still depends on a prior for the true effect; without a shared library of priors by intervention family (or prediction-market-derived priors), analysts may produce highly variable estimates.
- Quasi-experimental designs: The brief section correctly emphasizes design assumptions (IV, RD) and cites within-study comparisons showing limited systematic bias, but there is no concrete checklist or quantification rubric akin to the RCT section. Given frequent reliance on quasi-experiments in CEAs, a more developed protocol would be valuable.
- Reproducibility and open resources: The guidance references external tools (R packages, JASP), a “discounts library,” and other links not embedded or archived in the PDF. No code, workbook, or templates are provided for the end-to-end workflow (e.g., power/MDE for clusters, retrodesign with alternative priors, dependency-aware discount aggregation). This constrains reproducibility and comparability across analysts.
- Communication and internal consistency: Overall, the writing is clear and well structured, and the internal vs external validity split is well motivated. There are minor typographical issues and the single substantive MDE inequality error noted above. Some statements are overly categorical (e.g., “all RCTs will need a Type M adjustment”) where “most RCTs, especially early/underpowered ones” would be safer. The worked example is helpful but relies on unreferenced percentages; clearer justification or ranges would help users calibrate.

In sum, this is a thoughtful, practice-oriented synthesis likely to improve CEA robustness by counteracting effect-size inflation and improving attention to transportability. Its main weaknesses are reliance on cross-field averages without domain-specific calibration, a few technical misstatements and simplifications in power diagnostics, subjective and potentially dependent adjustments aggregated multiplicatively, and limited open-science assets. The “work in progress” framing is appropriate; adding validated default priors and ranges, code/templates,

a dependency-aware aggregation scheme, and retrospective validation against realized replications or scale-ups would substantially strengthen both credibility and usability.

i Gemini 2.0 Flash Assessment

This document provides a framework for estimating discounts to be applied to effect sizes from RCTs and quasi-experimental studies in cost-effectiveness analyses (CEAs), particularly for global health and development charities. The author proposes separating discounts into ‘internal replicability’ (addressing Type M and Type S errors) and ‘external validity’ (generalizability to new contexts). It focuses on factors influencing effect size inflation, such as statistical power, publication bias, researcher bias, and methodological flaws. The guidelines aim to balance accuracy with practicality, acknowledging resource constraints in evaluating numerous interventions. The document includes guidance for researchers, notes on relevant papers, and a worked example, emphasizing the importance of power calculations and baseline estimates. The appendix provides an overview of key papers and their findings related to effect size inflation and generalizability. The author is transparent about limitations, calling this work a work in progress, noting that they have not ‘done a deep dive’, and indicating areas for future research, particularly quasi-experimental studies and the use of Bayesian methods.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.17 Bhat et al. 2022

C.17.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash
Overall	87 [80–93]	75 [55–90]
Claims	86 [78–92]	80 [65–90]
Methods	82 [74–90]	75 [60–85]
Adv. Knowledge	89 [80–95]	70 [50–85]
Logic & Comms	91 [85–96]	85 [75–95]
Open Science	58 [40–70]	80 [65–90]
Global Relevance	88 [78–95]	85 [70–95]

i GPT-5 Pro Assessment

This is a valuable, ambitious long-run follow-up to two randomized psychotherapy trials in Goa, India (HAP: n 495; THPP: n 280), with 76% tracking 4–5 years later. The paper combines standard clinical outcomes (PHQ-9), novel incentivized belief-updating tasks, preference measures, and economic outcomes, and pre-registers main families of outcomes with FDR adjustments within families. The headline results are: persistent reductions in depression from

HAP (0.23 SD; 13 pp remission at PHQ-9 10, 5 years later), no detectable long-run effects from THPP, increased self-assessed patience and altruism (indices), reduced overoptimistic updating and final overconfidence in a feedback task, improvement in sleep, and no detectable effects on employment, earnings, productivity, or consumption. The design and exposition are strong. Nonetheless, several limitations and identification or inference concerns remain.

Internal validity and attrition: The follow-up rate is respectable for a 4–5 year horizon (76%), and the authors show balance on baseline covariates and similar follow-up rates across treatment arms. However, there is no formal bounding or sensitivity analysis for selective attrition (e.g., Lee bounds or trimming), nor weighting by inverse probability of follow-up based on baseline characteristics. Since baseline PHQ-9 and other predictors plausibly correlate with both attrition and outcomes, the claim that attrition is ignorable rests on non-rejection of balance tests that may be underpowered. Attrition patterns differ by trial (HAP 79%, THPP 69%); further probing whether attritors differ by baseline severity, gender, or socioeconomic variables, and whether any treatment–attrition interactions exist, would bolster credibility. A wave-by-wave analysis (given 2018–2020 fieldwork truncated by COVID-19) could rule out time-varying composition.

Trial heterogeneity and pooling: The pooling of two distinct interventions/populations (HAP: adults with PHQ-9 15; THPP: pregnant women with PHQ-9 10) is clearly presented, with separate trial analyses. Still, several interpretive claims (e.g., psychotherapy affecting belief-updating “independently” of depression) hinge on similarities across trials despite only HAP showing long-run depression effects. The mediation analysis is informative but correlational: mediators (e.g., endline behavioral activation) are themselves post-treatment endogenous outcomes. These mediation findings should be framed as suggestive and not causal.

Measurement and inference in belief-updating: The bracelet task and continuous probability elicitation via beakers are innovative and thoughtfully implemented (incentive-compatible truncated log scoring; comprehension checks; enumerator blinding; gender-matched comparators; defined Bayesian benchmarks using empirical performance). Even so, measurement frictions are nontrivial: degenerate priors/posteriors, frequent non-updates (44% of observations), wrong-signed updates (18%), and 68% of participants having at least one “irrational” update. The authors provide multiple robustness specifications (dropping never-updaters, non-updates, wrong-signed updates), which is a strength, but the prevalence of measurement noise raises concerns that structural updating parameters (H , L) may be sensitive to filtering rules and numeracy. Comprehension scores were similar across arms, mitigating differential bias, yet the overall interpretability of asymmetric updating magnitudes can be fragile in low-numeracy contexts. Additionally, early participants’ comparators came from pilot participants rather than the full study population; the authors should document whether results are sensitive to omitting these early observations.

Construct validity in overconfidence: The “full-information benchmark” for being in the top half uses the empirical distribution and observed signals. This is reasonable but still relies on stable performance distributions across comparator draws and consistent task implementation across home vs. lab settings and sessions. The choice of top-half criterion and group-of-10 framing is standard, but readers would benefit from sensitivity to alternative benchmarks (e.g., using parametric fits or leave-one-out empirical distributions) and clear reporting on the number of early observations with pilot comparators.

Preference measures and multiple testing: The paper pre-specifies five families of primary outcomes and applies FDR within families; for secondary outcomes (employment, consump-

tion, empowerment, IPV), adjustments are not applied and the authors clearly label them exploratory. The preference findings depend mostly on self-assessments (GPS) rather than incentivized tasks; this is acknowledged. The nulls on incentivized altruism/patience/risk leave open whether psychotherapy primarily shifts self-perception, not behavior. Given many measures, the index approach (Anderson) is a strength; still, main inferences hinge on relatively modest standardized effects and multiple comparisons; the reported q-values appropriately temper claims.

Economic outcomes and power: The null effects on employment/earnings are discussed with context (low female LFPR; home-based task offered; reservation wage and take-up measures). Confidence intervals rule out moderate effects on employment (>5 pp). This reduces concerns about simply being underpowered, though for some outcomes CIs remain wide. The lack of effects on the application decision, despite reduced final overconfidence, is interesting; the authors' interpretation (binding constraints and/or reversion to priors in high-stakes choices) is plausible, and the auxiliary regressions relate take-up to beliefs, reservation wages, and risk, but the behavioral mechanism remains not fully pinned down.

Cost-effectiveness and months of depression averted: The 9-month estimate over 5 years is based on linear interpolation between points (3, 12, and 60 months), and assumes no effects beyond 5 years; it ignores uncertainty in that trajectory and potential decay dynamics. While "conservative," readers should see sensitivity analyses to alternative interpolations (e.g., exponential decay, spline fits) and confidence intervals around months averted. The cost per recipient relies on earlier accounting; restating uncertainty there would improve decision utility.

Demand and beliefs about treatment efficacy: The elicitation is clever and incentivized. However, using the authors' estimated treatment effect as ground truth for scoring introduces sampling-error-induced noise into payoffs, and the finding that treatment increases perceived efficacy in both effective (HAP) and ineffective (THPP) settings highlights patient difficulty in identifying efficacy. The inference is sound but should be framed recognizing that "true effects" are themselves estimated with error.

Deviations from pre-analysis plan: The paper clearly discloses deviations—no stratified DML sample-splitting by trial, adding the beliefs-about-therapy module, and upgrading "final" overconfidence to a co-primary measure with adjustments. These are reasonable and transparent. Still, reporting sensitivity of primary inferences to simple OLS without DML and to alternative ML learners (or making code available) would improve robustness claims in a relatively modest sample.

Replicability and materials: The paper documents tasks and provides extensive appendices and references to the registry and the Social Science Prediction Platform. However, there is no direct link to public code/data in the PDF. Given the complexity of the belief-updating analysis and DML pipelines, open materials would significantly enhance reproducibility.

Overall, this is a careful, policy-relevant study that substantially advances evidence on the persistence and cost-effectiveness of scalable psychotherapy for depression in an LMIC, adds credible behavioral science evidence on belief-updating and self-perception, and clarifies limits on long-run economic impacts in this population. The main threats concern attrition sensitivity, measurement noise in belief-updating, mediation interpretability, and incomplete open-science artifacts. The writing is clear and cautious about nulls, adjustments are appropriate, and the methodological choices are generally reasonable and well-justified.

This paper investigates the long-term effects of psychotherapy on depression, beliefs, and economic outcomes in a low-income setting in India. It revisits participants from two randomized controlled trials (RCTs) four to five years after the original interventions, comparing those who received psychotherapy to a control group. The study explores the persistence of mental health improvements, behavioral changes, and economic impacts.

The key findings include:

1. **Sustained Reductions in Depression:** The treatment group showed a statistically significant reduction in depression compared to the control group, with the Healthy Activity Program (HAP) intervention being the primary driver of this effect. The Thinking Healthy Program Peer-Delivered (THPP) intervention did not show significant long-term effects.
2. **Changes in Beliefs:** Psychotherapy influenced participants' beliefs about themselves, leading to reduced negative self-perceptions and more accurate self-assessments in a work domain. Specifically, the treatment group exhibited less overconfidence in a lab-experimental setting.
3. **Increased Self-Assessed Patience and Altruism:** Participants in the treatment group reported higher levels of patience and altruism, although incentivized measures did not show significant changes.
4. **No Significant Economic Impacts:** The study found no significant effects of psychotherapy on employment, labor supply, earnings, or consumption.

The paper addresses a significant gap in the literature by examining the long-term impacts of psychotherapy in a low-income country, using experimental economics methods to study behavioral changes and preferences. The authors also compare their findings with expert predictions, revealing that experts underestimate the long-run mental health effects of psychotherapy.

Potential issues include:

- **Heterogeneity in Treatment Effects:** The long-run improvement in mental health is almost entirely driven by one of the two interventions, the Healthy Activity Program (HAP); and there were significant differences in the types of participants targeted.
- **Limited Economic Impact:** The lack of observable effects on economic outcomes, the authors suggest, may be due to other barriers women face in these populations, which are likely a study population.
- **Generalizability:** Study participants are largely female (88%), and their views may or may not represent general population.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.18 Bhattacharya and Packalen 2020

C.18.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	58 [45–70]	75 [60–85]	70 [55–85] (n=1)
Claims	44 [30–58]	80 [65–90]	—
Methods	40 [25–55]	70 [55–80]	—
Adv. Knowledge	59 [45–72]	75 [60–85]	—
Logic & Comms	78 [68–86]	85 [75–95]	80 [70–90] (n=1)
Open Science	24 [10–40]	60 [40–75]	50 [40–60] (n=1)
Global Relevance	62 [48–75]	70 [55–80]	50 [40–60] (n=1)
NA	—	—	4 [3–4] (n=1)
NA	—	—	3 [2–4] (n=1)
NA	—	—	50 [40–60] (n=1)

GPT-5 Pro Assessment

This paper advances an incentive-based explanation for perceived scientific stagnation, arguing that the shift to citation-centric evaluation has tilted researcher behavior away from exploratory, high-variance projects toward safer incremental work, thereby reducing the flow of breakthroughs. It presents a stylized lifecycle model of ideas (exploration → breakthrough → incremental advance), uses vivid historical case studies (PCR, CRISPR, *Helicobacter pylori*) to underscore the importance of early “scientific play,” and proposes augmenting citation-based evaluation with measures of novelty (“edge factors”) derived from textual analysis of scientific papers (e.g., using UMLS to map terms to concepts and the recency of idea-vintages). The central claims are plausible and well-motivated, but the paper’s empirical and methodological support within this document is limited relative to the strength of its causal narrative. The main identification threat is that the link from the “citation revolution” to reduced exploration (and then to stagnation) is argued rather than causally demonstrated. The paper synthesizes secondary evidence (e.g., NIH conservatism, HHMI grant effects, strategy choices in biomedicine) but does not provide new causal tests tying changes in citation-based incentives to changes in novelty pursuit, nor does it systematically quantify the relative contribution of citation incentives versus alternative drivers (e.g., rising costs of frontier tools, regulation, team science and specialization, publication bias, risk-aversion in funding, demographic shifts, or the growth and saturation of fields). The lifecycle model is intentionally conceptual: it employs schematic figures rather than formal structure, does not specify agent optimization, payoffs, or counterfactuals, and is not calibrated or validated against data dynamics (e.g., the temporal distribution of citations around breakthroughs or the prevalence and duration of low-impact exploration in different fields). Consequently, while the model is pedagogically useful, it does not yield falsifiable predictions or identify policy levers with estimated elasticities. The proposed novelty metric raises measurement concerns not fully resolved here. Defining novelty by the recency of words/phrases or concept-vintages risks synonym/homonym errors, field-specific terminology effects, and incentives to introduce new

labels without substantive conceptual advance; while the authors discuss UMLS mapping and embedding-based approaches as mitigations, this paper does not present implementation details, validation exercises, or error rates. The empirical “demonstration” (Figure 7) uses a single cohort (biomedical papers in 2001; N 191k), defines novelty as top-20% by idea-vintage recency, and shows the share of novel papers across citation percentiles. This is descriptive and informative but under-specified: there are no algorithmic details (tokenization, concept extraction and weighting, handling of multi-concept papers, normalization for subfields and document types), no uncertainty quantification, no sensitivity to thresholds (e.g., top-10%, top-30%), and no controls (journal, field, team size, funding, country, institutional prestige). As a result, while the figure supports the qualitative claim that novelty and impact are correlated but distinct, the magnitude, stability, and generality of that relationship remain unclear. External validity and scope are also limited by the empirical focus on one year and one domain; yet the paper generalizes to “science” broadly. The argument that academic search engines’ citation-forward interfaces materially steer researcher choices is plausible but is not tested here (e.g., via behavioral experiments, natural experiments in interface changes, or usage telemetry). On policy design, the paper makes a constructive case to add novelty metrics to evaluation dashboards but provides limited guidance on weighting between novelty and impact, field normalization, lag structures (novelty often pays off with delay), safeguards against gaming (e.g., gratuitous mention of new terms), auditing and transparency, or evidence that high “edge factor” portfolios predict future breakthroughs conditional on quality controls. It also does not quantify potential trade-offs (e.g., expected increases in failure rates, effects on early-career researchers with shorter horizons) or implementation costs for search engines and institutions. Communication is a strength: the writing is clear, the historical vignettes are engaging and relevant, and the schematic figures effectively convey the intended conceptual distinctions. However, the narrative occasionally adopts categorical language (“science has stagnated”) without providing new direct evidence; it relies heavily on cited literature for core empirical claims, rather than reproducing or extending those results. Replication and open science are weak in this document: there is no shared code or data for the 2001 PubMed analysis, no detailed appendix enabling re-implementation of the novelty pipeline, and no robustness tables; Figures 1–6 are schematic by design, and Figure 7 cannot be reproduced from the provided information. The contribution is therefore best seen as agenda-setting and synthetic: it organizes a coherent story, highlights a genuine misalignment in incentives, and proposes a tractable measurement direction that could be valuable if implemented, validated, and used judiciously alongside impact measures. To strengthen the case, the authors would need to (i) furnish causal evidence linking the rise of citation-based evaluation to declines in measured novelty and to downstream outcomes (e.g., difference-in-differences around institutional metric adoptions or search-engine interface changes; grant program reforms that reweight novelty), (ii) provide a thorough technical exposition and validation of the novelty metric across fields and time (including error analysis, sensitivity, and predictive validity for future impact), (iii) release code and replication data, and (iv) model or simulate policy counterfactuals to give decision-makers a sense of expected benefits, risks, and design parameters. As it stands, the paper persuasively frames an important problem and proposes a potentially useful measurement complement, but it stops short of the empirical and methodological depth needed to substantiate its strongest causal claims or to guide immediate policy adoption.

Gemini 2.0 Flash Assessment

This paper presents a compelling argument about the scientific stagnation in recent decades. The authors argue that new ideas no longer fuel economic growth as they once did. They propose a simple model of the lifecycle of scientific ideas, highlighting how changes in scientist incentives have contributed to this stagnation. They argue that the emphasis on citations as the primary metric for evaluating scientific contributions has shifted scientists' focus towards incremental science, discouraging exploratory projects that are essential for breakthroughs. The authors effectively use examples such as CRISPR and PCR to illustrate their points. They propose broadening the measurement of scientific productivity to include novelty metrics, which could encourage scientists to pursue more innovative and riskier projects. The paper's strength lies in its clear articulation of the problem, well-supported arguments, and constructive proposals for addressing scientific stagnation. However, the empirical validation of their model is limited, and the paper could benefit from a more in-depth exploration of the challenges in implementing novelty metrics. The paper is well written and easy to follow, but it can improve by providing more specific and actionable recommendations for universities, funding agencies, and academic search engines.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.19 Bruehlhart et al. 2021

C.19.1 Ratings Comparison

Metric	Claude Sonnet 4	GPT-5 Pro	Gemini 2.0 Flash
Overall	78 [68–85]	88 [82–93]	75 [60–85]
Claims	80 [72–87]	86 [78–92]	80 [65–90]
Methods	75 [65–82]	80 [72–87]	75 [60–85]
Adv. Knowledge	82 [75–88]	88 [82–94]	70 [55–80]
Logic & Comms	85 [80–90]	90 [84–95]	85 [75–95]
Open Science	70 [60–78]	72 [60–82]	70 [50–85]
Global Relevance	88 [82–93]	90 [84–95]	80 [65–90]

GPT-5 Pro Assessment

This study assembles a large multi-country dataset of helpline activity (about 8 million calls, with up to 2.2 million call-level observations) to describe high-frequency changes in distress during COVID-19 and to estimate partial associations between helpline metrics and epidemiological and policy variables. The empirical design combines descriptive event-study patterns around “outbreak” or shelter-in-place dates (Methods, eq. 1), pooled call-level linear probability models for topic shares (eqs. 2–4), a subnational US panel with state and week fixed effects

for suicide helpline volumes (eqs. 5–6), and a two-country daily panel for topic-specific call volumes (eq. 7). The data architecture is heterogeneous (some helplines provide conversation-level data on topics, others only aggregated time series; for the US suicide helpline the outcome is routed calls rather than answered calls), and the authors are generally careful to analyse each component with appropriate specifications and to discuss key limitations. The paper’s most policy-relevant claims—associations of stringency and income support with suicidality-related calls and other concerns, and the interpretation that income support likely alleviates distress—are largely framed as partial correlations, but the narrative occasionally reads causally (e.g., “implies”), despite identification limits that the paper itself acknowledges. The main internal threats are as follows. First, the policy regressions rely on observational variation that may be endogenous. In the US state panel, week fixed effects absorb national shocks, and state fixed effects absorb time-invariant heterogeneity, but state-by-week confounders (e.g., local economic shocks, mobility, school closures, test availability, media coverage, or health-system strain) that co-move with both non-pharmaceutical interventions (NPIs) and mental health could bias estimates. The Oxford policy indices are useful but aggregate; for US states, national policies supersede state policies in the constructed “total” index, which compresses cross-state variation and risks attenuation or conflation of state and national dynamics. The income support index is also largely national, limiting within-state identifying variation; this is particularly relevant when interpreting income support coefficients as distinct from time effects. The German/French daily topic regressions include detailed seasonality controls and helpline fixed effects but not national week fixed effects; thus, contemporaneous nationwide shocks may remain in the residual and bias associations with infection rates and the policy indices. Second, dynamics and lags are treated parsimoniously. Event-study plots document the timing around outbreak/SIP dates, but the panel regressions appear contemporaneous; psychological responses to infections or policies plausibly exhibit lead-lag structure (anticipation, adjustment, fatigue). Absent distributed-lag or event-window models in the panel regressions, coefficient signs and magnitudes may average over offsetting short-run and medium-run effects. Third, measurement and comparability issues deserve emphasis. Mapping heterogeneous helpline-specific taxonomies to common topic categories is pragmatic but induces non-classical measurement error; some helplines do not track certain topics, multi-topic coding is common, and topic salience may evolve with training or documentation rather than underlying distress. The authors mitigate this with helpline fixed effects and robustness checks, but the equivalence of constructs across helplines remains imperfect. Relatedly, the US outcome is routed calls, not completed conversations, unlike most other helplines; variation in answer rates or overflow handling could alter composition in ways not captured by call counts. The paper argues that unanswered calls are unfiltered and thus composition estimates are unbiased, yet practical triage (e.g., call-back queues, call timing, priority routing of certain numbers, chat vs voice) could still correlate with topic, especially under heavy load. Fourth, capacity constraints are acknowledged as a downward bias on recorded call volumes in the first wave, but the implicit assumption that topic shares are robust to capacity constraints is untested; if topics vary systematically by time-of-day or channel and answering probabilities covary with these, composition estimates may be affected. Fifth, the representativeness of callers is unknown and may change during the pandemic. The paper appropriately focuses on within-helpline changes, but time-varying selection (e.g., first-time versus repeat callers, age- or gender-specific changes in propensity to call) could confound topic shares; some descriptive facts on caller composition are reported, yet more formal checks (e.g., reweighting

or stratified analyses by caller novelty, or sensitivity bounds) would strengthen claims. Sixth, multiple testing and inference practices could be clearer. Across numerous topics, subgroups, periods, and geographies, the paper reports many coefficients and p-values. There is limited discussion of family-wise error or false discovery rates; occasional unconventional significance thresholds (e.g., $P < 0.016$) appear in figures without an explicit correction rationale, and the two one-sided tests (TOST) for “no change” in suicidality share employ an equivalence margin (± 0.35 percentage points) that is not substantively justified. While many effects are large and consistent, a more systematic multiple-comparisons framework and explicit equivalence margins would bolster credibility. Seventh, the interpretation of the negative association between infection rates and suicide-helpline calls invites alternative explanations beyond “displacement” by COVID-related concern (which is plausible and partly supported by movements in the Disaster Distress Helpline). For example, local surges could transiently reduce calling opportunities (illness, caregiving, competing stressors), or the denominator of distress shifts toward other channels (telehealth, online communities). Eighth, the paper primarily reports topic shares after the outbreak; given that total call volumes rose, relative declines in shares (e.g., violence, relationships) do not preclude absolute increases in those topic-specific counts. The German/French extended analyses partly address this by modelling topic-specific volumes, but a more explicit cross-helpline presentation of absolute topic counts alongside shares during the first wave would clarify whether some concerns rose in absolute terms while falling as a proportion. Ninth, in the US panel the state classification relies on phone numbers; mobile number portability can misclassify caller location. This adds noise and likely attenuates coefficients; however, the magnitude of misclassification is not quantified. Tenth, the event-study “week 0” definitions (first crossing of 1 per 100,000 cumulative infections vs first SIP) are defensible, but both dates can be noisy proxies for the perceived onset of the pandemic; sensitivity to alternative outbreak thresholds, within-country regional timings, or excess-mortality markers would be informative. Finally, on openness and reproducibility, the paper provides code and detailed equations, and the external data sources (JHU infections and OxCGRt indices) are public. However, helpline data are confidential and must be re-obtained via agreements with each provider; thus, full computational reproducibility is contingent on third-party approvals, and some definitions (e.g., internal routing, topic-coding rubrics) remain helpline-specific. Overall, the descriptive patterns (timing and magnitude of call surges, topic composition shifts toward fear and loneliness, stability across waves with added physical-health concern) are well-supported across multiple data sources and specifications. The policy associations are plausible and triangulated across the US suicide helpline and the German/French topic volumes, but should be read as correlational with potential residual confounding, limited within-unit identifying variation for some indices, and modest dynamic exploration. The paper’s contribution—a scalable, high-frequency lens on population distress with immediate policy relevance—is strong; further strengthening could come from explicit dynamic models, richer local controls (e.g., mobility, unemployment claims), instrumental or quasi-experimental designs where available, multiple-comparisons corrections, and more transparent justification for equivalence margins.

i Claude Sonnet 4 Assessment

This paper analyzes helpline call data from 19 countries during COVID-19 to study mental health impacts. The study is methodologically solid with a large dataset of 8 million calls and

exploits variation in timing of outbreaks and policies across jurisdictions to identify causal effects. The authors find call volumes peaked 35% above pre-pandemic levels six weeks after initial outbreaks, driven primarily by fear (including infection fears) and loneliness rather than traditional concerns like relationship issues, economic problems, or suicidal ideation. The panel data analysis convincingly separates effects of infection rates from policy stringency and income support, finding that more generous income support reduced distress-related calls while stricter containment measures increased them. However, several limitations constrain the findings' generalizability. The helpline caller population may not represent the broader public, potentially overrepresenting vulnerable groups with pre-existing mental health issues. Capacity constraints at helplines during peak periods mean recorded calls underestimate true demand, though the authors acknowledge this as a lower-bound estimate. The categorization of call topics, while reasonable, relies on mapping heterogeneous helpline classification systems into common categories, introducing measurement error. The analysis focuses heavily on the first wave with limited data for subsequent periods except for two helplines. Additionally, the paper doesn't adequately address potential changes in help-seeking behavior during the pandemic - people may have been more or less likely to call helplines due to isolation, stigma changes, or awareness campaigns. The finding that pandemic-related concerns 'replaced rather than exacerbated' underlying anxieties is intriguing but somewhat speculative given the data limitations. Despite these concerns, the paper makes a valuable contribution by demonstrating helplines as real-time mental health monitoring tools and providing policy-relevant evidence that financial support can mitigate mental health impacts of containment measures.

Gemini 2.0 Flash Assessment

This paper investigates mental health concerns during the COVID-19 pandemic using helpline call data from 19 countries. The study analyzes 8 million calls to identify trends in call volumes and the topics discussed. The main finding is that call volumes peaked six weeks after the initial outbreak, driven primarily by fear and loneliness. The study also examines the impact of containment policies and income support on suicide-related calls, finding that financial relief can alleviate distress triggered by lockdown measures. The authors compare the analysis of helpline calls with established approaches. They find that helpline data are well targeted on mental health concerns of a particularly vulnerable segment of the population.

Important strengths of this work include the large international dataset, the use of a panel structure to isolate the effects of local infection waves and policy measures, and the focus on a timely and policy-relevant issue. The study also acknowledges the limitations of helpline data, such as capacity constraints and potential selection bias. The authors make a good attempt to account for these limitations in their analysis.

However, some issues should be considered. First, the reliance on helpline data may introduce selection bias, as callers may not be representative of the general population. Second, the categorisation of call topics may be subjective and prone to measurement error. Third, the study does not address potential confounding factors, such as pre-existing mental health conditions or access to mental health services. Fourth, the study does not explore the long-term effects of the pandemic on mental health. Finally, the generalizability of the findings may be limited by the heterogeneity of the included countries and helplines.

In terms of advancing knowledge, the project contributes to the field by providing real-time information on mental health concerns during a crisis. The insights from the study inform our

beliefs about the effectiveness of interventions, such as financial relief, in alleviating distress. The project also adds useful value to other impactful research by providing a measurement tool that is broadly available and well targeted on mental health concerns. The logic and communication of the paper are generally clear and well-organized. The goals and questions are clearly expressed, and the reasoning is transparent. However, some sections could benefit from more detailed explanations and justifications. The authors accurately state the nature of their evidence and the extent to which it supports their main claims. Overall, the paper presents a valuable analysis of mental health concerns during the COVID-19 pandemic using helpline call data. While the study has some limitations, the strengths outweigh the weaknesses. The findings have important implications for policymakers and practitioners in the field of mental health.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.20 Buntaine et al. 2023

C.20.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	90 [84–96]	85 [75–95]	85 [78–90] (n=2)
Claims	88 [82–94]	85 [75–95]	—
Methods	86 [78–92]	80 [70–90]	76 [66–86] (n=2)
Adv. Knowledge	91 [84–97]	85 [75–95]	83 [76–89] (n=2)
Logic & Comms	85 [77–91]	90 [80–95]	74 [68–82] (n=2)
Open Science	64 [55–72]	70 [60–80]	59 [51–74] (n=2)
Global Relevance	93 [86–98]	90 [80–95]	91 [84–95] (n=2)
NA	—	—	5 [4–5] (n=2)
NA	—	—	5 [4–5] (n=2)
NA	—	—	90 [82–94] (n=2)

i GPT-5 Pro Assessment

This paper reports a large, carefully designed nationwide field experiment in China evaluating whether and how citizen appeals—filed via officially sanctioned channels—affect environmental enforcement and firm emissions. The design leverages real-time Continuous Emissions Monitoring System (CEMS) data for 24,620 major polluters and randomizes at the firm level between control, private appeals (to regulators or firms via several channels), and public appeals (Weibo posts tagging local environmental regulators). It further cross-randomizes prefecture-level treatment intensity (95% vs 70% of firms treated) to probe indirect/general equilibrium effects, and at the appeal level increases the visibility of some public posts by adding likes/shares. Main findings are robust and important: public appeals substantially re-

duce violation incidence (60% relative to a 0.9% daily baseline) and lower SO₂ and COD concentrations; private appeals yield smaller improvements. Mechanism evidence suggests public visibility shifts regulator priorities toward avoiding unrest, as shown by stronger regulator engagement (responses, length, and site visits) when social media visibility is experimentally amplified. The cross-randomized intensity suggests limited offsetting general equilibrium effects and possibly positive spillovers (a 3.5% reduction in ambient SO₂ at $p = 0.10$). The project is policy-relevant, well-motivated, and unusually scaled, with transparent pre-analysis steps and a suite of robustness exercises. Several issues deserve attention:

- **Interference and SUTVA.** By design, appeals are publicly observable and could plausibly affect non-targeted firms within a prefecture. The cross-randomization of treatment intensity (95% vs 70%) is a thoughtful solution, enabling a test of net spillovers. However, both treatment shares are high and close to each other; if spillovers are nonlinear or operate at finer geographic/industry networks than the prefecture, the design may have limited power to detect negative displacement. The ambient SO₂ result (3.5% decrease, marginal significance) points toward net benefits, but inference remains somewhat noisy.
- **Endogenous exposure to treatment.** Appeals are triggered by violations; firms differ in their frequency and persistence of violations. The main effects are intention-to-treat (ITT) using assignment $\times$ post; the paper acknowledges this and provides IV results in the appendix. Still, dosage variation (how many appeals a firm actually received, timing of first appeal) is partly endogenous to behavior post-assignment. This is mitigated by randomization and firm fixed effects; the event studies show no pre-trend differences, which is reassuring. Yet interpretation should keep in mind the reduced-form nature and that treatment exposure is conditional on violating.
- **Selection and verification of appealed violations.** Appeals are based on “verified” violations, with RAs removing suspected mechanical spikes/low-flow anomalies before filing. RAs were blinded to treatment, which limits bias, and core analysis uses official MEE data rather than scraped realtime data (good). However, the verification step introduces an additional human-judgment layer affecting which violations were appealed (and, by protocol, repeated violations within a week were not re-appealed). While this likely adds noise rather than bias, it could attenuate estimates and complicate precise dose-response interpretation. Providing inter-rater reliability statistics or replication of the verification criteria in code would further strengthen credibility.
- **Measurement and units.** The paper reports reductions in SO₂ and COD concentrations (Table 3 uses mg/m³ and mg/L; elsewhere the text references $\mu\text{g}/\text{m}^3$). The internal unit presentation could be clearer and absolutely consistent across text, tables, and figures. The authors do examine flows and find no offsetting changes, implying total emissions also fell; this is a strong and important check.
- **Mechanisms.** The Weibo visibility experiment supports the claim that public attention triggers greater regulator effort. However, regulator effort is measured (naturally) only in public-appeal arms; direct comparability to private appeals’ regulator responses is unavailable. The authors partially address alternative explanations: no evidence of CEMS manipulation (hours operated or implausibly low values), little support for “central oversight threat” or “media escalation” as main mechanisms, and no evidence that public appeals simply crowd-in many private appeals by others. The political-cycle evidence (larger effects when officials are later in term) is plausible but observational; confounding by contemporaneous policy cycles cannot be ruled out entirely.
- **Statistical inference and robustness.** Core results are precise; some general-equilibrium and ambient-air estimates are marginally significant (10%). The authors present multiple robustness checks: alternative violation definitions excluding low flow, alternative clustering, aggregation to week/month, and

distributional results (bins relative to standards). Adjustment for multiple hypothesis testing is not discussed; given the number of outcomes, this would help contextualize marginal results.

- External validity and sustainability. The eight-month window captures meaningful adaptation dynamics (event studies stabilize by week 20) but cannot establish long-run persistence or post-experiment decay. The context—CEMS transparency, social media penetration, and China’s institutional incentives—is distinctive; external validity to other settings will hinge on data transparency, regulator incentives, and social media credibility. Still, the core behavioral mechanism—public visibility making regulators more responsive—likely generalizes qualitatively.
- Implementation and replicability. The authors pre-registered (AEA RCT registry cited), describe scripts, and provide a data appendix. The main analysis uses official administrative data; however, access to the full CEMS and 12369 databases, as well as the daily violation identification/verification code, may be restricted, posing barriers to full reproduction. Sharing sanitized code for the main regressions and event studies would strengthen replicability.
- Constructive clarifications. It would be useful to: (i) report compliance/dosage by arm (e.g., mean number of appeals per firm conditional on violation) and the first stage for IV; (ii) unify units for emissions throughout; (iii) discuss any detection or moderation risk arising from purchased social engagement (visibility promotion) and how this might affect regulator interpretation; and (iv) provide more detail on power for the 95% vs 70% contrast.

Overall, the paper is methodologically strong, policy-relevant, and advances knowledge about social media-enabled accountability in environmental regulation at an exceptional scale. Minor clarifications and additional transparency around code/data access would further enhance confidence, but the main conclusions are well supported by the evidence presented.

Gemini 2.0 Flash Assessment

This paper presents a well-designed and executed field experiment in China to study the direct and indirect effects of citizen participation on environmental governance. The authors partnered with NGOs and utilized China’s Continuous Emissions Monitoring System (CEMS) data to examine how public and private appeals regarding pollution violations impact firm behavior and regulatory enforcement. The key innovation lies in the random assignment of firms to different ‘appeal’ treatments (private vs. public via social media) and varying the intensity of treatment across prefectures to assess general equilibrium effects. The use of real-world data and the focus on a crucial issue (environmental regulation in China) makes it highly relevant.

The study’s main finding is that public appeals, particularly those amplified through social media, lead to significant reductions in pollution violations and emissions compared to private appeals. This effect appears to be driven by a shift in local regulators’ priorities from economic growth towards avoiding public unrest, as evidenced by increased regulatory responsiveness to appeals with higher visibility. The experimental manipulation of social media visibility (adding “likes/shares”) provides convincing support for this mechanism.

While the paper is generally strong, several aspects could be strengthened:

1. Mechanism Understanding: The paper provides a strong case that the public appeals are so effective, because they tilt local regulators’ often competing goals away from facilitating economic growth and toward avoiding pollution-induced public unrest. But, it is important to know whether the public appeals have any impact on public satisfaction. If there is a channel to measure how the citizens feel about their appeals being responded

to, this would help in strengthening the argument that indeed the local regulators shift their efforts towards avoiding public unrest.

2. The study relies on CEMS data. While the authors mention the MEP exerts substantial effort to ensure the quality and authenticity of the CEMS data, further discussion of data validation and potential limitations would strengthen the robustness of the analysis. For instance, exploring potential data manipulation by firms, although the authors did not find strong evidence for this, is critical.
3. External Validity: While the Chinese context is important, more discussion on the generalizability of these findings to other countries and governance structures would improve the paper’s impact. In particular, how might the effectiveness of public appeals vary in countries with different levels of press freedom or government transparency?
4. Cost-Benefit Analysis: The authors briefly touch on the cost-benefit analysis of pollution appeals but state this is out of the scope of the paper. Given the policy relevance of the findings, a more detailed cost-benefit analysis, even with rough estimates, would be highly valuable. Quantifying the economic benefits of reduced pollution and comparing them to the costs of implementing the appeals system would provide actionable insights for policymakers.
5. Long-Term Effects: The study period is eight months, which limits the ability to assess the long-term sustainability of the observed effects. Do the reductions in violations persist over time, or do firms and regulators adapt their behavior? Addressing the limitations due to the time frame would improve the impact of the findings.
6. Heterogeneity: The effects should be measured based on the sectors. Does the sector/industry of the firm have any impact on how environmental violations are being handled? What is the difference between the industrial sector and the service sector, for example?

In summary, the paper makes a significant contribution to the literature on environmental governance and citizen participation. The experimental design is rigorous, the data is comprehensive, and the findings are policy-relevant. Addressing the areas for improvement mentioned above would further enhance the paper’s impact and generalizability.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.21 Carson et al. 2023

C.21.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	78 [68–87]	75 [60–85]	72 [55–85] (n=2)

Claims	80 [72–90]	80 [65–90]	66 [52–70] (n=2)
Methods	72 [62–84]	75 [60–85]	68 [59–78] (n=2)
Adv. Knowledge	70 [55–85]	70 [55–80]	76 [70–87] (n=2)
Logic & Comms	85 [75–92]	85 [70–95]	50 [40–62] (n=2)
Open Science	55 [40–70]	75 [60–85]	48 [32–62] (n=2)
Global Relevance	68 [50–80]	70 [55–80]	65 [58–74] (n=2)
NA	—	—	4 [3–4] (n=2)
NA	—	—	4 [3–4] (n=2)
NA	—	—	65 [58–74] (n=2)

i GPT-5 Pro Assessment

This paper investigates whether the aggregation rule used by a large biomedical funder (ranking by the mean of peer-review scores) aligns with scientists’ normative preferences over which projects to fund. Two discrete choice experiments with active, NIH-funded biomedical researchers are fielded: Study 1 elicits project rankings across experimentally varied distributions of reviewer scores attached to real (anonymized) abstracts; Study 2 elicits portfolio choices under budget shocks to test whether budget tightness changes tolerance for dissensus (variance). The main results are: (i) scientists place positive weight on the mean score and, conditional on the mean, also on score variance (interpreted as dissensus), with additional preference for right-skew; (ii) little evidence of asymmetric weighting of negative versus positive reviews; (iii) variance preference is stronger among domain “insiders” (measured by MeSH overlap); (iv) tighter budgets lead participants to drop higher-variance projects at the margin, while relaxed budgets do not symmetrically add variance; and (v) applying the estimated mean-variance preferences to external datasets that mimic NIH review produces nontrivial “reversals” of marginal funding decisions relative to the mean-only rule. The design is innovative and well-executed, but several limitations affect interpretation and external validity. First, the evidence concerns stated preferences in carefully controlled hypothetical tasks, not revealed preferences or realized scientific outcomes. Participants were given full distributions of 30 reviewer scores per project plus their mean and variance; this information set is richer and more explicitly moment-focused than typical practice (which emphasizes composite priority scores and narrative critiques). Explicitly displaying the variance and drawing attention to it may prime participants to “use” it—appropriate for eliciting normative aggregation preferences, but not necessarily predictive of behavior in situ. The mapping from these elicited preferences to welfare-relevant outcomes (e.g., scientific impact) remains untested, and the paper is careful to stop short of causal claims about outcomes; still, the policy reading (that weighting variance could change billions of dollars of decisions) should be interpreted as suggestive without an outcome link. Second, external validity and selection are concerns. The sampling frame covers five IRGs; response is 9.2% overall (313 in Study 1; 250 in Study 2). Although responder/attriter comparisons on observables are reported and largely balanced, substantial nonresponse bias is possible—especially if scientists with stronger views on dissensus or particular methodological tastes self-select into participation. Generalization to other NIH areas, other fields, or international contexts is uncertain. Third, although the abstraction randomization and MeSH-based proximity are convincing features, the “insider” effect depends partly on whether participants read abstracts. The paper uses a “hover” subgroup analysis as a placebo-style check; however, it appears based on self-reported hovering behavior (or

at least it's not unambiguously log-based), which could introduce measurement error. Also, some information about the projects (titles) is visible without hovering; title-only cues may be sufficient to induce perceived proximity. The MeSH overlap metric is reasonable, but measurement error in automated MeSH tagging of titles/abstracts and in mapping participants' grants to MeSH terms could attenuate or bias estimates of insider effects; the paper partially addresses this by standardization and fixed effects but cannot fully rule out residual confounding. Fourth, DCE design choices warrant scrutiny. The score distributions are synthetically constructed via a balanced incomplete block design using 10 reviewer ratings duplicated to 30; while this supports identification of moment preferences, it may produce atypical patterns relative to real NIH panels (e.g., number of raters varies by study section; rounding/anchoring; institutional discouragement of bimodality). The paper notes bounded-scale mechanics that tie mean and variance, but it would help to demonstrate that the experimental joint distribution of means/variances resembles real NIH data (where available) or at least the distributions in the external datasets subsequently used. The best-worst ranking is converted into three conditional logit choice sets; this imposes IIA and constant coefficients across ranks. The authors provide multinomial logit and G-MNL robustness checks showing qualitatively similar results, and they note weaker sensitivity to mean/variance in lower ranks—appropriate and transparent, but residual scale and substitution pattern concerns remain. Fifth, the interpretation of effect sizes merits careful calibration. The reported standardized coefficients indicate a modest practical effect of variance (on the order of low single-digit percentage points in choice probabilities per SD change), far smaller than the mean's effect but potentially consequential near the funding margin. Early in the paper, a statement about trading 0.1 points of mean for one unit of variance would benefit from explicit derivation, scale alignment, and sensitivity (e.g., to standardization and to how variance is computed from discrete 1–9 scores with 30 raters). Sixth, the budget experiment (Study 2) is informative but carries sequencing and path-dependence issues: participants first build a portfolio of four out of ten, then drop one under a cut, then add one under an expansion. Because “cut” always precedes “expand,” order effects and anchoring may drive asymmetry; a counterbalanced design or randomized order across participants would strengthen the inference that dissensus tolerance declines specifically under tighter budgets. Also, the limited number of tasks per participant may restrict power for granular heterogeneity analysis. Seventh, the application of estimated preferences to external datasets is a strength, but key implementation details are not fully spelled out. In particular, it is unclear whether means and variances were standardized within each external dataset before applying the weights, and whether the weights are directly transportable given differences in the number of raters, variance scaling, and rounding. The use of LOESS-averaged reversal rates and examples (e.g., tie-breaking) is persuasive, but a more formal uncertainty quantification for reversal rates (e.g., bootstrap CIs) would be valuable. Finally, openness and reproducibility are only partially addressed. The paper references a data appendix; however, it is not explicit in the manuscript whether individual-level choice data, randomization materials, the exact 54 score-distribution templates, the code for generating BIBD blocks, and analysis scripts are publicly available with sufficient documentation to enable full reproduction. Given the policy salience, releasing code and (de-identified) data, along with a pre-analysis plan (if any), would strengthen credibility. Overall, the work is clearly written, the main claims are well matched to the experimental evidence, and the analytic framework is appropriate. The identification strategy for moment preferences is strong within the DCE setting, and the heterogeneity, placebo, and sensitivity checks are thought-

fully designed. The principal caveats are external validity, selection, and the hypothetical nature of the tasks, along with incomplete transparency about code/data. The policy implications are promising but should be framed as normative preference elicitation rather than outcome-validated guidance until linked to realized scientific impact in future work.

i Gemini 2.0 Flash Assessment

The paper examines how scientists aggregate peer review scores when making funding decisions, contrasting this with the NIH's current practice of primarily using the average score. The authors hypothesize that scientists may value dissensus among reviewers (as measured by the variance of scores), potentially favoring more novel or radical projects. They conduct two discrete choice experiments with active biomedical researchers, presenting them with real project abstracts and experimentally assigned peer review scores. Participants are asked to choose which projects they would fund with their allocated budget. The experimental design allows the authors to test whether participants prefer projects with higher average scores or have preferences for other features of the score distribution, like dissensus. One experiment also examines the effect of budget constraints on dissensus tolerance, while another explores the role of expertise. The authors find that, on average, scientists do value dissensus, are more tolerant of dissensus when they have greater domain expertise, and are less tolerant of dissensus when budgets are tighter. They also find that incorporating scientists' preferences for dissensus would change marginal funding decisions for a non-trivial percentage of projects worth billions of dollars per year. However, the response rate is quite low which might bring into question the representativeness of the sample and the validity of the conclusions. The experiments simulate the funding process used by the NIH, but it may not fully capture the complexities of real-world decision-making. Additionally, using project abstracts and experimentally assigned peer review scores might be different than dealing with real grants in their own field of expertise. The statistical models used appear sound, and the authors conduct several robustness checks to ensure the validity of their results. These include accounting for factors such as skewness of scores and individual heterogeneity. They provide a transparent description of their data, methodology, and results, as well as replicability of their findings, by providing a data appendix and detailed explanations of their analysis. The conclusions are generally well-supported by the data. The topic has relevance to global priorities, as optimizing research funding allocation is crucial for advancing scientific progress and addressing global challenges. The findings have potential implications for funding agencies and policymakers, highlighting the importance of considering reviewer dissensus when making funding decisions. The authors communicate their findings in a clear and accessible manner, making the paper relevant to both researchers and policymakers.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.22 Caviola et al. 2022

C.22.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	84 [75–92]	65 [45–80]	71 [58–87] (n=2)
Claims	83 [75–90]	70 [55–85]	70 [57–82] (n=2)
Methods	78 [70–86]	60 [40–75]	70 [52–86] (n=2)
Adv. Knowledge	78 [68–88]	75 [60–90]	58 [48–72] (n=2)
Logic & Comms	85 [78–92]	80 [65–90]	71 [62–84] (n=2)
Open Science	92 [88–97]	90 [80–95]	90 [82–98] (n=2)
Global Relevance	72 [60–82]	70 [50–85]	65 [57–73] (n=2)
NA	—	—	4 [4–4] (n=2)
NA	—	—	4 [4–5] (n=2)
NA	—	—	65 [57–73] (n=2)

i GPT-5 Pro Assessment

This manuscript presents nine vignette-based experiments (N=5,776) probing lay population-ethical judgments along three axes: asymmetry between happiness and suffering, attitudes to creating new people, and tradeoffs between total vs. average welfare. The core empirical contributions are clear: (i) across several designs, respondents typically require a preponderance of happiness over suffering for a population or life to be net positive, with the asymmetry strengthened when intensities are described as “extreme”; (ii) when asked about adding a single new person to a neutral world, participants evaluate adding a very happy life as good and a very unhappy life as bad, approximately symmetrically; (iii) when comparing populations, participants display both totalist and averagist tendencies, with a marked intuitive pull toward average welfare that can reverse when prompted to deliberate. Strengths include careful operationalization of key philosophical contrasts, large samples with pre-registered designs in several studies, open data/materials/code, reasonable power analyses, and multiple conceptual replications. The paper is unusually transparent about limitations and alternative interpretations, and it connects the psychology to philosophical debates without overclaiming normativity.

Important limitations and threats to inference arise from construct validity, measurement, and generalizability. First, the core constructs (hedonic intensity, neutrality, average vs. total welfare) are presented on a -100 to +100 scale. Although the authors work to equalize intensity (Study 1c) and to clarify neutral at 0, this assumes acceptability of cardinal hedonic units with a zero point that marks a life “neither good nor bad.” The authors acknowledge that participants might interpret these scales idiosyncratically; indeed, comprehension failures and exclusions are substantial in places (e.g., Study 3c excludes ~26% for failing a basic scale check). While the main patterns often persist in re-analyses or subsets that pass checks, the magnitude of exclusions raises selection concerns (participants who better accept/understand the scale may differ systematically in moral reasoning). Second, the initial framing of “extreme happiness” (falling in love) versus “extreme suffering” (torture) may cue asymmetric intensity or salience, potentially inflating negativity effects in Study 1a; later studies mitigate this via symmetrical scales and an explicit equal-intensity assumption, yet residual differences in subjective interpretation may remain.

Third, the “thinking style” manipulation (intuition vs. reflection) relies on instruction-

only prompts without manipulation checks (e.g., response deadlines, latency measures, or CRT/Need for Cognition covariates). This invites concerns about compliance and whether observed differences reflect genuine processing mode changes or demand characteristics. The reported effects are modest in size and would benefit from converging evidence using time pressure, cognitive load, or pre-registered analytic plans tied to behavioral markers of deliberation. Relatedly, in several multi-dilemma designs, the number of hypothesis tests is large; while some ANOVAs include appropriate post-hoc adjustments, many one-sample or paired t-tests are unadjusted, increasing family-wise error risk. Effect sizes are generally reported, and the patterns are consistent across experiments, mitigating but not eliminating this concern.

Fourth, scope and external validity are limited. Samples are US MTurk/Prolific participants with standard quality screens; no cross-cultural or representative sampling is attempted. Given the normative and policy-relevant framing (population ethics, longtermism), cross-cultural replication would be valuable. Furthermore, the designs intentionally abstract away from resource constraints, inequality, ecological externalities, and identity considerations (e.g., “multiple Earth-like planets,” identical happiness distributions). This isolates axiological judgments but may limit applicability to real policy contexts in which those factors are salient. Notably, even after reminders about no overpopulation, many participants still preferred smaller very large happy populations (Study 3a), suggesting lingering real-world concerns or a general aversion to huge numbers that the instructions could not fully neutralize.

Fifth, some findings rely on participants’ ability to compute or conceptually grasp totals. In Study 3b, a substantial fraction could not identify equal totals despite correct average computations; nonetheless, averagist preferences remained when restricting to those who did compute totals correctly, which strengthens the claim but also highlights cognitive difficulty with “total” constructs. This cognitive difficulty likely contributes to the intuition/averagism effect and suggests that the observed “averagism” may, in part, reflect ease-of-evaluability rather than a stable, reasoned principle.

Sixth, internal consistency and interpretation nuances deserve care. The manuscript documents an asymmetry favoring suffering in fixed-population judgments (Studies 1a–c), yet symmetry for adding a single very happy/unhappy person (Studies 2a–b). The authors provide plausible explanations (global assessment of final states vs. local evaluation of changes, contrast with neutral baselines), but these hinge on task framing. Further work could test whether the addition symmetry persists when the added person’s welfare matches or falls below the existing population’s average, or when non-neutral baselines are used. Also, “mere addition” paradox-style tasks (Study 3d) elicit averagist choices even to the point of adding very unhappy people to improve averages; while the authors interpret this cautiously and show that reflection attenuates it, this striking result underscores how sensitive judgments are to presentation and to whether the comparison is framed as two alternative worlds vs. a change to one world.

Seventh, the paper’s normative implications are discussed responsibly but remain speculative given the measurement and sampling constraints. The authors mostly limit claims to descriptive moral psychology and appropriately separate axiological from deontic judgments; still, some readers may over-interpret the findings as endorsing policy stances. Where possible, the manuscript already warns against simple normative inferences and notes tensions with philosophical views (e.g., Repugnant/Sadistic conclusions). A further robustness step could be to preregister and replicate the most policy-relevant contrasts with stronger manipulation checks, diverse samples, and alternative measurement (e.g., choice-based conjoint, incentive-

compatible tasks where possible).

On reporting and transparency, the paper is exemplary: preregistrations for several studies, OSF repository with data/code/materials, explicit reporting of exclusions and checks, and acknowledgment of limitations and alternative mechanisms (negativity bias, proportion dominance, evaluability). Analyses are straightforward (t-tests/ANOVA, reliability of aggregated indices). Figures and tables are clear and consistent with text. Internal numbering and values appear coherent. The link to prior literature is careful and accurate, and the contribution—systematic empirical mapping of population-ethical intuitions—is novel and useful to both psychology and philosophy. Overall, despite the noted measurement and inference challenges inherent to this topic, the evidence base is strong enough to support the main descriptive claims, and the work provides a solid platform for follow-up research and for informing philosophical debate about the reliability and structure of population-ethical intuitions.

i Gemini 2.0 Flash Assessment

This paper investigates lay people's intuitions about population ethics, focusing on how people aggregate welfare across individuals. It explores three main questions: (1) whether happiness and suffering are weighed symmetrically, (2) whether the focus is on average or total welfare, and (3) whether only existing lives are considered or also potential lives. The research consists of nine experiments with a total of 5776 participants. The studies use online surveys, presenting participants with hypothetical scenarios involving populations with varying numbers of happy and unhappy individuals, and asking them to make judgments about the relative value of these populations.

The authors find that participants generally require more happy people to outweigh unhappy people for a population to be considered net positive. They also find a preference for both greater total welfare and greater average welfare, with a focus on average welfare sometimes leading to counterintuitive judgments. Participants do not consider the creation of new people as morally neutral, viewing the creation of happy people as good and unhappy people as bad. Here are some potential issues with the paper:

1. **Hypothetical Scenarios:** The study relies on hypothetical scenarios, which may not accurately reflect real-world decision-making. Participants may respond differently when faced with actual consequences.
2. **Oversimplification:** The scenarios simplify complex ethical considerations by focusing solely on happiness and suffering, potentially neglecting other relevant factors such as autonomy, justice, and rights.
3. **Cultural Bias:** The participants are primarily from the United States, which may limit the generalizability of the findings to other cultures with different values and beliefs.
4. **Measurement of Happiness and Suffering:** The study relies on self-reported measures of happiness and suffering, which can be subjective and difficult to quantify accurately.
5. **Statistical Power:** While the overall sample size is large, some of the individual experiments may have limited statistical power to detect small effects.
6. **Ecological Validity:** The artificial scenarios may not capture the nuances and complexities of real-world population ethical dilemmas.
7. **Lack of Nuance:** The study does not explore the reasons behind people's intuitions in depth, potentially missing important contextual factors.

8. **Limited Scope:** The study focuses on a narrow set of questions within population ethics, leaving other important issues unaddressed.
9. **Potential for Bias:** The authors’ own philosophical views may have influenced the design and interpretation of the experiments.
10. **Oversimplified Representation of Philosophical Theories:** The study simplifies complex philosophical theories (e.g., utilitarianism, averagism) and may not fully capture the nuances of these theories.
11. **Demand Effects:** Participants may guess the purpose of the study and alter their responses accordingly.
12. **Vague Definitions:** The definitions of ‘happiness’ and ‘suffering’ are not clearly defined, leading to potential misinterpretations by participants.

These issues suggest a need for caution when interpreting the results and highlight the importance of further research to explore the complexities of population ethical intuitions in more detail.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.23 Chuard et al. 2022

C.23.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	75 [65–85]	55 [35–75]	80 [65–90] (n=1)
Claims	70 [55–80]	60 [40–80]	—
Methods	68 [55–80]	50 [30–70]	77 [70–83] (n=1)
Adv. Knowledge	72 [60–85]	65 [45–85]	76 [64–85] (n=1)
Logic & Comms	78 [65–90]	75 [60–90]	91 [81–100] (n=1)
Open Science	45 [30–60]	70 [50–85]	96 [92–100] (n=1)
Global Relevance	70 [55–85]	70 [50–85]	86 [77–95] (n=1)
NA	—	—	3 [3–4] (n=1)
NA	—	—	3 [3–4] (n=1)
NA	—	—	95 [90–100] (n=1)

i GPT-5 Pro Assessment

This paper compares an epidemiological “dynamic distribution” (accounting) model, grounded in an SIR framework, to a standard economic reduced-form approach for quantifying the long-run human capital impacts of measles and the measles vaccine in the U.S. It is carefully conceived and makes good use of rich public data, and the side-by-side comparison is valuable. The core empirical finding is that the reduced-form strategy—interacting pre-vaccine measles

severity (mortality among ages 0–5, 1959–1963) with a pre/post indicator by birth cohort—robustly predicts improvements in adult education, income, welfare reliance, and health when using cohort fixed effects; while the epidemiological measure (state-by-cohort shares ever infected) does not produce robust or consistently signed long-run effects once cohort fixed effects are included, despite being strongly predictive of outbreak dynamics year-to-year. This suggests that variation in disease severity, rather than infection shares per se, better captures the margin relevant for long-run human capital.

Most important issues and limitations: - Identification and interpretation of the reduced-form design: The reduced-form is a two-period (pre/post) difference-in-differences in which the treatment intensity is state pre-vaccine measles mortality, implemented as $\text{Pres} \times 1(\text{pre } 1963)$ with state and cohort fixed effects. The coefficient is interpreted as a differential pre/post change by baseline severity. This is standard but still relies on parallel trends: that absent the vaccine, the association between severity and outcomes would have evolved similarly across states. The paper does not present an event-study or pre-trend tests to probe this assumption. Given 1963 is a single national “cut,” and many other changes in the 1960s (e.g., welfare programs, civil rights, schooling and health system changes) could correlate with pre-vaccine severity, demonstrating no differential pre-trends (or adding state-specific time trends or controls for other contemporaneous policies) would strengthen causal interpretation. - Construction of the epidemiological measure and possible attenuation: The dynamic distribution model requires estimating state-specific reporting rates (1950–1960) by the slope of cumulative cases on cumulative births, then dividing reported counts by these rates, and allocating infections across cohorts based on the stock of susceptibles. Two concerns could create non-classical measurement error: (i) reporting rates likely vary over time within states due to changes in surveillance and healthcare utilization; (ii) the model uses only national vaccination rates (no state-level roll-out heterogeneity), which compresses cross-state post-1963 differences. The paper offers robustness where reporting rate functions vary (linear trends, loess, spline) and changes initial susceptibility and age-specific infection risk ratios; estimates remain qualitatively similar. Nonetheless, the lack of state vaccination series is a first-order limitation that plausibly biases the “share ever infected” measure toward a common time pattern and limits credible cross-state variation for identifying long-run impacts. - Age-at-infection heterogeneity: The epidemiological model assumes age-independent infection risk among susceptibles (with a sensitivity allowing higher risk below age six). Because harms from measles plausibly vary with age at infection and with co-infections, the share-ever-infected may not capture the relevant severity margin. Without age-at-infection data, the model cannot adjust for this. This fits the paper’s conclusion (severity vs. infection shares), but also implies the epidemiological proxy is a noisy measure of the treatment of interest (harm). - Validation of the epidemiological model: The model’s predictive power for future outbreaks is a strength. However, the susceptible pool $S(t)$ is partially constructed from the same series used to evaluate predictions (measles cases), creating potential circularity. While the authors note the negative relation of $S(t+1)$ with cases is mechanical, and highlight predictive value of $S(t-1)$ across subperiods, an out-of-sample check or cross-validation (e.g., holding out some years or exploring city-level series) would more cleanly document predictive power independent of construction. - Outcome measurement and aggregation: Long-run outcomes are computed from the 2000 Census and ACS, aggregated to state-of-birth $\times$ birth-year $\times$ age cells and weighted by cell size, with age fixed effects. Birth year is imputed as survey year minus age; this induces heaping and some cohort misclassification. While this is typical, the paper could

quantify the likely bias (e.g., re-running on subsets with precisely measured ages or using coarser birth-year bins). Using state-of-birth exploits plausibly exogenous early-life exposure but does not address migration-driven selection into adult environments (e.g., different labor markets), which could be correlated with state-level measles severity; discussing how migration can bias estimates would be useful. - Confounding contemporaneous changes: The 1950–1980 birth cohorts span substantial national and state-level changes (education reforms, Medicaid introduction, economic restructuring). The paper’s reduced-form results are robust to cohort fixed effects, but does not include state-specific linear trends or controls for other health interventions. Presenting sensitivity with state-specific trends and additional controls (e.g., baseline non-measles child mortality, tuberculosis or polio burdens, school laws) would bolster claims that the interaction is picking up measles severity rather than broader correlated factors. - Mortality outcomes: The reduced-form measure strongly predicts higher measles-specific and overall ages 0–5 mortality in the pre period, aligning with expectations; the epidemiological measure shows an implausible negative association when cohort fixed effects are included. This highlights the likely measurement error and limited identifying variation in the epidemiological share-ever-infected. The paper is appropriately cautious in concluding that the severity proxy aligns with theory while the epidemiological measure is ill-suited for long-term effects in this setting. - Communication points: The definition of the “pre” indicator and the sign of coefficients can be confusing; explicitly presenting the DiD contrast (and plotting by severity deciles an event-study around 1963) would help readers interpret magnitudes as post- relative changes. The authors do a good job contrasting the time-series predictive value of the SIR-based measure versus its weakness for long-run human capital, but could be more explicit that the absence of state-level vaccination series likely contributes to the epidemiological measure’s limited success. - Open science and reproducibility: All data sources are public, but the paper does not provide a replication package or code. Given the complexity of the epidemiological accounting (reporting rate estimation, dynamic stock of susceptibles), code availability would be valuable for replication and adaptation. Overall, this is a clear and policy-relevant contribution: it shows why reduced-form designs focusing on severity proxies can outperform mechanically accurate infection-accounting for predicting long-run human capital, even in a setting—measles—where epidemiological dynamics are relatively tractable. The main remaining threats relate to the parallel-trends assumption in the reduced-form and measurement constraints in the epidemiological construction. Addressing these with additional robustness (event-study/pre-trends, state trends/controls, and, if possible, state-level vaccination uptake) and releasing code would further strengthen credibility and usefulness for practitioners.

i Gemini 2.0 Flash Assessment

This paper examines the long-term impacts of early-life measles exposure on human capital and health outcomes in the United States, comparing an epidemiological modeling approach with a reduced-form economic approach. The authors construct an epidemiological dynamic accounting model based on the susceptible-infected-recovered (SIR) framework to estimate measles infection shares across U.S. cohorts around the introduction of the measles vaccine. They then compare the results obtained from this model with those from a reduced-form approach that uses pre-vaccine measles mortality as a proxy for disease burden. The study finds that the reduced-form approach reveals improvements in educational, labor market, and health

outcomes among cohorts born after the vaccine introduction in states with high pre-vaccine measles mortality. However, the epidemiological modeling approach does not consistently detect such effects. This discrepancy suggests that the severity of measles infections, as proxied by mortality, is more relevant for long-term outcomes than the overall infection rate. The epidemiological model predicts future outbreaks well, but its measures of infection rates do not correlate well with long-term outcomes. The study concludes that focusing on disease severity may be more appropriate for measuring the long-term benefits of disease reduction. Despite the authors' efforts, the epidemiological model has weaknesses. The reporting rate estimation relies on the assumption that nearly everyone was infected with measles in the pre-vaccination era, and the subsequent allocation of cases across cohorts relies on further assumptions regarding age and infection risk. The identifying assumption is that controlling for a location fixed effect, the timing of measles outbreaks is independent of the true infection rates. In the robustness checks, the authors evaluate the extent to which alternative assumptions influence the dynamic distribution model. The alternative epidemiological modeling assumptions do not substantially alter the results, however, the epidemiological modeling approach remains weak due to cohort fixed effects.

Several potential issues may affect the validity and generalizability of the findings. First, the accuracy of the epidemiological model depends on the reliability of measles case counts and birth data. Underreporting of measles cases, especially in the early years, could bias the estimated infection rates and shares. The reporting rates are estimated rather than observed. The study lacks individual-level data on measles infections. Furthermore, birth data quality limitations might impact the precision of the cohort size estimations. Second, the study excludes several states due to missing data, which could affect the representativeness of the sample and limit the generalizability of the results to the entire U.S. population. Finally, the study focuses on measles in the context of the United States in the mid-20th century. The findings may not be directly applicable to other diseases, populations, or time periods with different epidemiological characteristics, healthcare systems, or socioeconomic conditions.

There are potential biases from omitted variable bias. The relationship between measles exposure and long-term outcomes could be confounded by other factors that vary across states and cohorts. While the authors control for age, birth cohort, and state of birth fixed effects, there may be other unobserved characteristics that influence both measles exposure and later-life outcomes. There is also a question of how representative these results are for other contexts, given the reliance on reporting rate proxies. The comparison of the epidemiological and reduced-form approaches relies on the assumption that the epidemiological model accurately captures measles infection rates, and it finds little effect from overall infection rates, suggesting they are not a key driver of long-term outcomes.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.24 Crawford et al. 2023

C.24.1 Ratings Comparison

Metric	Claude Sonnet 4	GPT-5 Pro	Gemini 2.0 Flash
Overall	72 [60–83]	74 [60–85]	65 [45–80]
Claims	75 [65–85]	72 [55–82]	60 [40–75]
Methods	68 [55–80]	68 [50–80]	60 [40–75]
Adv. Knowledge	78 [68–87]	70 [55–80]	70 [50–85]
Logic & Comms	80 [72–88]	80 [65–88]	80 [70–90]
Open Science	65 [50–78]	40 [25–55]	75 [60–85]
Global Relevance	85 [78–92]	85 [70–93]	80 [65–90]

i GPT-5 Pro Assessment

This paper conducts a systematic review and meta-analysis of observational studies linking children’s blood lead levels (BLL) to learning outcomes (IQ and standardized reading/mathematics scores), adds publication-bias corrections and study-level adjustments, triangulates with a small set of quasi-experimental studies, and simulates policy-relevant impacts of reducing BLL in low- and middle-income countries (LMICs). The empirical contribution is substantial and careful in many respects, but several methodological and interpretive issues limit precision and causal claims. On the review side, the search strategy is broad (three recent reviews plus a large Google Scholar sweep) and eligibility criteria are explicit; two independent extractors with adjudication reduce clerical error risk. Nonetheless, replicability would be greatly improved by sharing the extraction database and code; the paper provides neither, and the PRISMA-style flow diagram lacks full lists of excluded studies and reasons. The meta-analysis faces unavoidable harmonization challenges: many primary studies estimate linear BLL specifications or group contrasts, whereas the authors target a log-linear dose-response. They convert linear to log effects using the Linakis et al. algorithm (validated on NHANES and three other datasets), but this re-expression may be biased when exposures are highly skewed, as typical in LMICs; the validation sample may not generalize to all included contexts. They partially mitigate with meta-regression dummies (original log vs non-log and interactions with mean BLL), but the pattern that linear-model estimates tend to be larger persists, raising concern that residual conversion bias could inflate pooled effects. The weighting/variance approach is sensible (inverse-variance with robust variance estimation, RVE), but they assume within-study effect-size correlation = 1, an extremely conservative choice that is not subjected to sensitivity analysis across plausible rho values; this could distort standard errors and study weights. Heterogeneity is very large ($I^2 = 77\%$ for IQ and $>96\%$ for reading/math), and the meta-regressions explain little of it; only omission of parental education/IQ controls robustly predicts larger negative effects. The fully adjusted constant that targets LMIC, reading/math, log-specification, lag/average exposure, and publication-bias correction is -0.115 SD per $\ln(\text{BLL})$ with a large standard error (0.100), suggesting imprecision in the most policy-relevant estimate even if the direction is consistent. Publication bias is handled thoughtfully via multiple approaches (Egger, PET-PEESE, trim-and-fill, p-uniform*), with PET-PEESE used in the main specification, but all these corrections have known limitations in the presence of high heterogeneity and potential small-study effects; the funnel/p-curve diagnostics show evident selection, and while adjusted estimates fall only moderately, residual bias cannot be ruled out. The treatment of confounding is mixed: in-

clusion of study-level indicators for missing parental SES/education is helpful and attenuates effects, but this is a coarse adjustment that cannot recover within-study confounding; risk-of-bias tools for observational studies (e.g., ROBINS-I) are not applied. The coefficient-stability (Oster) analysis is constrained by the small number of studies reporting R^2 ; with only six estimates, assumed $\alpha=1$ and $R_{\max}=1.3\times R$ lead to an average one-third attenuation from unobservables—informative but fragile to assumptions and reporting gaps. The contrast with quasi-experimental evidence is valuable: across five natural-experiment settings, IV estimates are consistently larger than OLS, plausibly because of measurement error or different LATE populations; however, several of these studies use environmental proxies (soil/moss/pipe proximity) rather than BLL, complicating direct comparability to the meta-analysed estimand, and effect scales/units are not fully harmonized. The policy simulations are useful and clearly communicate orders of magnitude under log-linear assumptions, combining country BLL distributions (assumed log-normal per Ericson et al.) with the adjusted -0.12 SD per $\ln(\text{BLL})$ and World Bank HLOs; however, there is an apparent typographical inconsistency in the formula (the text subtracts 0.5 rather than $\ln(0.5)$), though the reported country gains align with the intended log-difference calculation. Assumptions of stable log-linear slopes across the entire exposure range and across countries, tests, ages, and school systems are strong; the meta-regression's mean-BLL association and known supra-linear dose-response at low levels suggest potential departures. The intervention evidence synthesis is informative but highlights thin causal evidence in LMICs outside the already completed leaded gasoline bans: targeted soil remediation at acute hotspots shows large BLL reductions (before–after designs), whereas broader parental education and regulation efforts yield modest average declines; cost data are sparse, and the paper stops short of a formal cost-effectiveness analysis for education outcomes. Other technical minutiae that could affect precision include imputation of p-values reported only as thresholds (taking half the threshold), conversion of percentile outcomes to z-scores under normality assumptions, substitution of population for sample SDs in some cases (the authors show small sensitivity), and potential between-study dependence when multiple publications draw on overlapping cohorts (addressed within-study via RVE but not across studies using the same cohort). Communication is generally clear, with helpful forest plots and a specification-curve analysis; the authors appropriately temper causal language (“if given a causal interpretation”) and situate estimates relative to prior meta-analyses. Overall, the central takeaways—observational associations around -0.2 SD per $\ln(\text{BLL})$ attenuating to about -0.12 SD after adjustments, negative across many specifications, quasi-experimental studies not contradicting and sometimes exceeding these magnitudes, and plausible education gains from BLL reduction—are credible in direction and plausible in magnitude, but precision and causality remain uncertain due to heterogeneity, harmonization risks, publication/selection bias, and limited strong-identification studies in LMICs. The work is highly relevant to global health and education policy, but actionable prioritization would benefit from stronger causal evidence in LMICs, explicit risk-of-bias grading, preregistration, and open materials for replication; a cost-effectiveness module focusing on educational gains (separate from health) would further increase usefulness for decision-makers.

i Claude Sonnet 4 Assessment

This meta-analysis examines the relationship between lead exposure and learning outcomes among children in developing countries. The authors conduct a systematic review covering

47 studies (286 estimates) examining associations between blood lead levels and IQ, reading, or mathematics performance. Their main finding is a benchmark estimate of -0.12 standard deviations per natural log unit of blood lead exposure after adjusting for observable confounds and publication bias. Several methodological issues are evident: First, the harmonization process across studies is complex and potentially error-prone, requiring extensive re-expression of coefficients from different functional forms (linear, log, discrete comparisons) into a common natural log metric using algorithms that the authors acknowledge are imperfect. Second, there is clear evidence of publication bias - funnel plots show asymmetric distributions with clustering of p-values just below significance thresholds, though correction attempts only moderately reduce estimates. Third, the quasi-experimental evidence is limited (only 5 studies) but consistently shows larger IV than OLS estimates, suggesting potential downward bias in observational studies due to measurement error or upward bias due to unobserved confounding. Fourth, the coefficient stability analysis suggests unobserved confounding could explain roughly one-third of the observational association. The policy simulations, while interesting, rest heavily on the causal interpretation of observational estimates and make strong distributional assumptions. The paper provides valuable evidence synthesis on an important global health issue, but the reliance on observational data, publication bias concerns, and methodological complexities in harmonization create substantial uncertainty around the magnitude of true causal effects.

i Gemini 2.0 Flash Assessment

This paper presents a meta-analysis investigating the relationship between lead exposure and children's learning outcomes, with a focus on developing countries. The authors systematically review and meta-analyze observational studies, adjusting for potential confounders and publication bias. They estimate a benchmark effect of a 0.12 standard deviation reduction in learning per natural log unit of blood lead. Recognizing the limitations of non-experimental data, they explore the magnitude of potential unobserved confounding and summarize findings from natural experiments. The research estimates that reducing lead exposure could significantly reduce the learning gap between rich and poor countries, highlighting the potential impact of targeted interventions and broader public health campaigns. The analysis also examines the sensitivity of estimates to various study characteristics, publication bias, and unobserved confounders, providing a comprehensive assessment of the evidence base.

However, several potential issues should be noted. First, the reliance on observational studies introduces inherent limitations regarding causal inference. While the authors acknowledge this and attempt to address potential confounding, the possibility of unobserved confounding remains a concern. The bounding procedures employed rely on assumptions about the degree of selection on unobservables, which may not hold in reality. Second, the re-expression algorithms used to harmonize effect sizes across studies with different functional forms are necessarily imperfect and may introduce bias, as acknowledged by the authors. The choice of the Linakis et al. (2021) method is justified based on its lower root mean squared error in a specific context, but its generalizability is uncertain. Third, the substantial heterogeneity observed between studies, even after controlling for various factors, suggests that the relationship between lead exposure and learning outcomes may be more complex than captured in the meta-analysis. The influence of specific contexts, populations, and methodologies may not be fully accounted for, which could affect the overall robustness of the findings. Finally,

the policy simulations rely on several assumptions, including the log-normal distribution of blood lead levels and the generalizability of the estimated effect size across different settings. These assumptions should be carefully considered when interpreting the policy implications of the study.

Specifically, the identification strategy suffers from reliance on observational data. While the authors are appropriately cautious about this limitation, the core estimates are still at risk of bias and misinterpretation. Also, the study depends heavily on meta-analytic techniques, which can obscure key differences in study design, population characteristics, and outcome measures. The combination of diverse studies might lead to an oversimplified understanding of a complex issue. Finally, the policy simulations, while interesting, are sensitive to underlying assumptions, and it is hard to know how much to trust the headline numbers without a firmer causal grounding for the core estimates.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.25 Denkenberger et al. 2022

C.25.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	62 [47–75]	60 [45–75]	62 [40–75] (n=3)
Claims	55 [40–70]	65 [50–80]	—
Methods	50 [35–65]	55 [40–70]	60 [45–75] (n=3)
Adv. Knowledge	63 [50–78]	70 [55–85]	60 [45–75] (n=3)
Logic & Comms	72 [62–82]	75 [60–90]	75 [52–85] (n=3)
Open Science	50 [35–65]	70 [55–85]	72 [45–85] (n=3)
Global Relevance	78 [65–90]	80 [65–95]	85 [65–92] (n=3)
NA	—	—	3 [2–4] (n=3)
NA	—	—	3 [2–4] (n=3)

i GPT-5 Pro Assessment

This paper presents a probabilistic, Monte Carlo-based comparison of the long-term cost-effectiveness of “resilient foods” (sunlight-independent food production strategies) versus artificial general intelligence (AGI) safety. The authors construct two parallel models for the resilient foods subproblem—one grounded in a small survey of global catastrophic risk (GCR) researchers (S-model) and another based on a single expert’s priors (E-model)—and combine these with an adapted version of a prior AGI safety cost-effectiveness model. The main quantitative conclusions (high probability that resilient foods funding, particularly at the immediate margin, is more cost-effective than AGI safety at an assumed \$3B margin) are directionally plausible under some longtermist views but rest on several strong assumptions and modeling

choices that introduce substantial theory and model uncertainty. The most critical issues are: (1) reliance on subjective priors with limited calibration and very small- n expert elicitation. The S-model uses a survey with 8/32 responses, including respondents affiliated with the implementing NGO and two authors, raising risks of demand characteristics, confirmation bias, and inside-view overconfidence; the E-model rests on one expert. The paper acknowledges these biases but they materially affect inputs that the importance analysis identifies as outcome-dominant (e.g., far-future loss from 10% agricultural shortfalls in S-model; nuclear war probability in E-model). (2) Parameterization of probabilities and impacts. The inadvertent US–Russia nuclear war probability priors (updated via Beta-binomial) produce annual means around 0.5–0.7%, which many readers will find high; they also exclude pathways such as intentional war and other dyads, which the authors state is conservative but creates ambiguity about how “full-scale” risk is represented. The 10% agricultural shortfall probabilities (1–8% per year S-model; 0.5–1.5% E-model) and long-term impact fractions are loosely grounded and differ by orders of magnitude between models, with limited external calibration. (3) Distributional choices. The paper often uses lognormal distributions for quantities that are bounded in $[0,1]$ (e.g., “% mitigation,” “% reduction in far-future potential”), which is theoretically awkward unless truncation is applied; the paper does not clearly document truncation or transformations. Where beta distributions are used (e.g., for some AGI inputs), the rationale is clearer. (4) Structural assumptions and independence. The two submodels (resilient foods and AGI) are assumed independent, even though real-world interactions are plausible (e.g., nuclear war could materially alter timelines for AGI development, or AGI could change resilience to food shocks). This independence assumption simplifies the analysis but could bias relative comparisons. (5) Treatment of marginal vs average cost-effectiveness and returns to scale. The conversion from average to marginal cost-effectiveness using a blanket “logarithmic returns” assumption is heuristic and applied differently across the domains, while the comparison is performed at different margins (\$100M for resilient foods vs \$3B for AGI). This may partially drive headline ratios; exploring alternative functional forms of returns and common margins would strengthen robustness. (6) Sensitivity and robustness. The paper supplies importance analysis and a simple “pessimistic-parameter” reversal test (needing simultaneous changes to 3–5 inputs to overturn the headline result), which is useful, but it does not examine correlated uncertainties, structural uncertainties, or robust decision methods. Because output distributions are highly skewed with heavy tails, ratio-of-means estimates are fragile; the probability-of-superiority metric is more decision-relevant here but still inherits the subjective input structure. (7) Moral hazard and behavioral responses. The moral hazard adjustment is included but justified with a very rough elasticity argument; it seems too narrow given the political and strategic complexity of nuclear postures and broader behavioral responses to “backup food” messaging. (8) Measurement of “far-future potential.” The core outcome is “percent of far-future potential saved per dollar,” which depends on strong longtermist assumptions (finite ceiling of value, nearly zero pure time discounting) and aggregation across heterogeneous existential pathways; while the authors discuss theory uncertainty and “cluelessness,” the results are highly sensitive to normative stance and to tail assumptions. (9) Replicability and openness. The conceptual model is well described and some Guesstimate links are provided, but full code and all data to reproduce Analytica results are not made publicly available within the paper; survey instruments are included as an appendix, but raw responses and elicitation protocols (e.g., calibration exercises) are not detailed, limiting reproducibility and independent re-elicitation. (10) Policy realism and imple-

mentation frictions. The resilient foods pathway depends on rapid repurposing and scale-up of technologies (SCP, cellulosic sugar, seaweed, greenhouses) under extreme stress; while prior work is cited, this paper could more explicitly model supply chain bottlenecks, governance constraints, equity in distribution, and time-to-scale under varying cooperation regimes. On the positive side, the paper is clear and transparent about many uncertainties, compares two distinct parameterizations to highlight input-sensitivity, and communicates policy-relevant implications: the near-term urgency of resilient food preparedness, potential co-benefits for 10% food shocks (with stronger near-term evidence), and the likelihood that both resilient foods and AGI safety are cost-effective at saving expected present lives compared to standard global health benchmarks. The logic is largely well structured, figures are clear, and limitations are explicitly acknowledged. Overall, the work advances the conversation on cross-risk portfolio optimization and provides a starting point that future research can strengthen by (i) improved expert elicitation (e.g., Delphi with calibration scoring), (ii) better-justified bounded distributions and correlations, (iii) alternative returns-to-scale assumptions and common-margin comparisons, (iv) structural dependence modeling between risks, (v) explicit modeling of implementation timelines and constraints, and (vi) open, reproducible code with full parameter documentation. Given the prominence of the headline claim and the high uncertainty regime, readers should weigh the conclusions as suggestive rather than definitive and place greater emphasis on near-term, more empirically anchored claims (notably the case for preparing for 10% agricultural shortfalls) while treating the far-future comparative results with caution.

i Gemini 2.0 Flash Assessment

This paper presents a cost-effectiveness analysis comparing investments in resilient foods (alternative food sources that are less dependent on sunlight) with investments in artificial general intelligence (AGI) safety, both aimed at mitigating existential risks to humanity. The core argument is that global agricultural catastrophes, such as nuclear winter or abrupt climate change, pose significant long-term threats that resilient food systems can effectively address. The analysis employs Monte Carlo models to estimate the cost-effectiveness of both resilient foods and AGI safety, using data from expert surveys and existing literature. It compares the relative benefits of these interventions in terms of increasing the long-term potential of humanity. The authors find that, within their model assumptions, resilient foods are likely more cost-effective than AGI safety at the margin, suggesting that current funding for resilient foods is disproportionately low. The paper also discusses the theoretical and model uncertainties inherent in long-term forecasting and sensitivity analyses. Overall, the paper seeks to inform prioritization of disaster risk mitigation measures by highlighting the potential of resilient foods as a high-impact intervention. However, there are several issues with the manuscript:

1. **Oversimplification of Complex Systems:** The Monte Carlo models appear to oversimplify complex sociotechnical systems, potentially leading to biased results. For example, the models treat AGI and resilient food development as independent, which may not be realistic given potential interactions (e.g., AGI could accelerate resilient food production).
2. **Subjectivity in Parameter Estimates:** The reliance on expert surveys introduces subjectivity in parameter estimates, particularly regarding the reduction in far future

potential due to various catastrophes. The accuracy and reliability of these expert opinions are difficult to ascertain, and there is a potential for response bias. The low response rate (8 out of 32) raises concerns about the representativeness of the survey results.

3. **Model Uncertainty:** There is significant model uncertainty, particularly regarding the choice of probability distributions and model structure. The authors acknowledge that different model structures could lead to different results, but they do not fully explore these alternative structures. For instance, an agent-based model might provide a more nuanced understanding of the complex sociotechnical systems involved.
4. **Theory Uncertainty:** The paper acknowledges that predicting the impact of any intervention on the far future involves inherent epistemic uncertainty. However, the discussion of cluelessness and the limitations of long-term forecasting could be expanded. The analysis does not fully address the potential for unintended consequences or the difficulty of assigning probabilities to extremely rare events.
5. **Limited Scope:** The paper focuses primarily on comparing resilient foods and AGI safety, while other existential risks, such as pandemics or asteroid impacts, are only briefly mentioned. A broader analysis that includes a wider range of risks and interventions could provide a more comprehensive picture of global priorities.
6. **Data Limitations:** The data used in the models are subject to various limitations, including the availability and quality of cost estimates, the accuracy of nuclear war probability estimates, and the difficulty of quantifying the long-term impacts of interventions. The authors acknowledge these limitations, but they do not fully address how they might affect the results.
7. **Moral Hazard Considerations:** The discussion of moral hazard is somewhat limited. The paper acknowledges that awareness of a food backup plan could potentially increase the likelihood of nuclear war, but it does not fully explore the implications of this effect.
8. **Justification of Model Assumptions:** The justification for some of the model assumptions is weak. For example, the assumption that the status quo of no inadvertent nuclear war will continue is not adequately supported.
9. **Robustness Testing:** While the paper includes a sensitivity analysis, it could be strengthened by more extensive robustness testing. For example, the authors could explore the impact of different probability distributions or alternative model structures on the results.
10. **Overclaiming:** The authors sometimes overclaim the robustness of their results, given the various uncertainties and limitations. For example, the claim that it can be said with 98%-99% confidence that funding resilient foods now is more cost-effective than additional funding for AGI safety is not fully justified.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.26 Epperson and Gerster 2024

C.26.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	82 [74–89]	80 [65–90]	89 [79–94] (n=2)
Claims	80 [72–88]	85 [70–95]	—
Methods	78 [70–86]	75 [60–85]	72 [50–85] (n=2)
Adv. Knowledge	78 [70–85]	75 [60–90]	80 [56–91] (n=2)
Logic & Comms	88 [82–93]	90 [80–95]	80 [56–91] (n=2)
Open Science	62 [50–75]	70 [50–85]	62 [42–71] (n=2)
Global Relevance	72 [62–82]	70 [50–85]	88 [66–96] (n=2)
NA	—	—	82 [59–94] (n=2)

GPT-5 Pro Assessment

This study investigates whether willful ignorance is consequential for moral behavior and how responsiveness to information varies by information preferences, using a laboratory experiment linked to real purchase data from university canteens. Its central claims are: (i) about 30% of participants avoid costless information about intensive farming; (ii) exposure to a 360° VR video about pig farming reduces immediate meat consumption both in the lab (voucher choice) and in the field (canteen purchases), with much stronger effects among information avoiders than seekers; and (iii) effects decay to near zero within about two weeks. The design is innovative and policy-relevant, and the empirical strategy is generally sound: willingness-to-pay (WTP) for information is elicited via an incentivized multiple price list over a wide range, one price is randomly implemented, and inverse probability weighting (IPW) recovers the average treatment effect (ATE) and conditional ATEs (CATEs) by WTP type. The major strengths are (a) a credible within-study link between lab and field behavior, (b) a carefully described ex-ante pre-registration with clear mapping and transparency about deviations, (c) extensive robustness checks (empirical vs theoretical propensity scores; weighted least squares with covariates; alternative outcome definitions; placebo windows; purchase frequency checks), and (d) a useful exploration of WTP-specific treatment effects and cost-effectiveness under different cost regimes. The paper’s main identification lever—randomizing the implemented price conditional on elicited WTP—supports unconfoundedness given WTP and overlap for the bulk of the sample (312/330 with WTP within support). Weights are not extreme, and post-weighting covariate balance is good, with only small residual imbalances. Several limitations and threats to interpretation remain. First, the “information” treatment is a vivid NGO-produced VR video with a narrative from the pig’s perspective; this likely combines informational content with affective/persuasive elements. The paper acknowledges this and frames it as a real-world intervention; however, the interpretation of effects purely as “information” should be read as “information plus affective narrative.” The outside option (re-watching a previously seen 360° video of the central bank) is designed to control for time-on-task and VR use, but it may be mildly aversive or boring for some; the authors note this could make their WTP measure an upper bound for WTP for the intensive farming video. This makes the avoider/seeker classification conservative but also means the control condition is not a neutral

“no exposure” state; the ATE is best interpreted as watching the farming video vs re-watching a neutral/non-moral video. Second, the external validity of field effects is bounded by the population (students at two German universities), the specific canteens studied, and the VR format. While the immediate field effects closely match lab magnitudes, behavior is observed only in canteens; the study shows no change in the propensity to purchase at canteens post-treatment, which reduces concerns about substitution away from canteens, but cannot rule out substitution of meat consumption outside the canteen. The outcome construction relies on classifying meals as certainly with or certainly without meat using menus; roughly 14% of meals are unclassifiable and excluded. The paper mitigates this with dual outcomes (certainly-with and certainly-without), shows consistent results, and separately analyzes terrestrial and aquatic meat, but some measurement error may remain. Third, dynamics are estimated via overlapping seven-day moving windows with day-level plots; this is intuitive but introduces serial dependence across windows; the reported confidence intervals do not adjust for this dependence (they are repeated cross-sectional IPW standard errors in each window), so visual inference across adjacent days should be cautious. Still, the pre-period placebo windows show no effects, and the pattern of decay is clear. Fourth, the sample size fell short of the pre-registered target due to COVID-19, which also truncated the post-experiment observation window; the study remains adequately powered for main ATEs but some heterogeneity contrasts (e.g., seekers vs avoiders in the field) are imprecise. The main cross-group difference is strongly supported in the lab ($p < 0.03$), while the field estimates show the same pattern but weaker significance; WLS adjustments reduce the avoider–seeker difference in some specifications, suggesting some sensitivity to controls when conditioning on limited samples. Fifth, the mechanism analysis is explicitly exploratory. The paper shows that both avoiders and seekers update beliefs similarly and have similar baseline beliefs, while avoiders anticipate and experience stronger negative affect (PANAS items) after the video. This supports an attention/affect-regulation account, but affect is self-reported, measured post-treatment, and causality is not established. Sixth, identification hinges on accurate WTP measurement; eight participants with inconsistent price list responses are excluded, and extreme WTP values beyond the implemented price range are outside the identified region. The estimands thus pertain to the majority whose WTP lies within support. Seventh, the cost-effectiveness exercise is helpful but rests on strong assumptions: program cost parameters (VR rental and incentive payments), the linear decay to zero by day ~ 12 , extrapolation of WTP-bin CATEs, and the specific cost function selected. The authors present alternatives in the appendix and the qualitative message (best cost-effectiveness slightly below zero cost to target marginal avoiders) seems robust; still, these are model-based calculations rather than direct experimental tests of alternative pricing regimes at scale. Additional smaller points: potential session-level correlation is not addressed via cluster-robust SEs (assignment is individual, so this is a minor concern); there may be experimenter-demand or priming effects due to pre-video questions about farming, although both treated and control participants undergo parallel questioning; and the outside video could have idiosyncratic effects (e.g., interest in the Bundesbank), although it is re-watched and described as largely uninformative/ not entertaining by most. On open science, the work is pre-registered with a clear PAP and includes detailed appendices; code and data availability are not clearly documented in the PDF. Field data are sensitive and likely cannot be shared, but a replication package (code, variable construction, and synthetic data or de-identified aggregates) would enhance reproducibility. Overall, the paper convincingly demonstrates that willful ignorance is behaviorally consequential in a real consumption

context and that responsive individuals self-select away from information at zero monetary cost, with short-lived impacts. The design is careful, the analysis is transparent and robust, and the policy discussion is grounded, though the persistence and generalizability of effects, as well as the separation between “information” and affective persuasion, remain important caveats.

i Gemini 2.0 Flash Assessment

This paper investigates the impact of willful ignorance on consumer behavior, specifically focusing on meat consumption. The authors hypothesize that consumers avoid information about intensive farming practices to mitigate moral conflicts arising from the dissonance between their values (animal welfare) and desires (cheap meat). The study uses a combination of laboratory experiments and field data from university canteens to test this hypothesis. The lab experiment involves eliciting participants’ willingness to pay (WTP) to watch a video about intensive farming and then observing their choices of meal vouchers (with or without meat). The field data tracks actual meat purchases in university canteens before and after the experiment. This design allows the researchers to identify the causal effect of information exposure on meat consumption, while accounting for individuals’ preferences for information. The core finding is that willful ignorance is consequential, but only in the short term. Information avoiders significantly reduce their meat consumption immediately after being exposed to information about intensive farming, but this effect fades within two weeks. The authors also find that information avoiders are more responsive to information than information seekers. The research is well-motivated by a relevant real-world problem. It addresses key policy questions regarding how to design effective information interventions to promote ethical consumption. The combination of lab and field data strengthens the external validity of the findings. The use of a WTP elicitation method to address self-selection bias is a solid and creative element. A potential limitation is the reliance on observational data in the field, which may be subject to confounding factors despite the use of IPW. However, the consistency of the lab and field results strengthens confidence in the findings. The two-week time horizon may also limit conclusions about long-term effects. There are a lot of appendices, which is good, but I did not check them all. It’s not clear to me whether the information in the video is objectively factual and neutrally presented, or if it is emotionally manipulative, sensationalized, or biased in some way (but I understand the authors reproduce an existing real-world information intervention). The authors attempt to measure the effects of “negative affect” on information avoidance and meat consumption, but the findings are based on exploratory analyses and may not be robust. A stronger test would require random assignment to emotional states. Overall, the paper is clearly written, well-structured, and methodologically sound. The findings contribute to the literature on behavioral economics, moral decision-making, and the design of effective information interventions.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.27 Fangwa et al. 2023

C.27.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	86 [77–92]	75 [60–85]	65 [55–74] (n=1)
Claims	84 [74–90]	80 [70–90]	—
Methods	80 [68–88]	85 [75–95]	60 [55–70] (n=1)
Adv. Knowledge	83 [72–90]	70 [55–80]	70 [55–75] (n=1)
Logic & Comms	89 [82–95]	80 [70–90]	55 [50–65] (n=1)
Open Science	45 [30–60]	70 [50–85]	45 [30–60] (n=1)
Global Relevance	91 [84–97]	85 [75–95]	80 [70–90] (n=1)
NA	—	—	4 [3–4] (n=1)
NA	—	—	4 [3–4] (n=1)
NA	—	—	55 [45–75] (n=1)

i GPT-5 Pro Assessment

This paper studies a randomized governance intervention in primary health facilities in the DRC that bundles performance-based incentives with auditing and feedback (A&F), comparing treated facilities to randomized controls that receive equivalent funding but (by design) no incentives or A&F. Using administrative panel data (SNIS) over 14 quarters and facility-by-quarter difference-in-differences with event-time estimates, the authors find sizable improvements in operating efficiency (services per employee) and in birth outcomes (lower shares of stillbirths and neonatal deaths) that materialize gradually several quarters post-intervention; they also provide qualitative interviews and descriptive quality-score trends (collected only for treated facilities) consistent with a learning mechanism. In auxiliary, non-experimental analyses, matched comparisons to facilities outside the program suggest that funding alone expands scale but does not improve efficiency or quality, whereas funding plus governance improves both scale and quality. The design is compelling, the exposition clear, and the main effects are plausibly causal for the bundle. At the same time, several issues deserve attention. First, measurement and reporting could be affected by the treatment itself: the auditing/feedback component and performance-contingent incentives may change recordkeeping intensity and data quality (for both the numerator “services” and outcome denominators), potentially biasing measured gains in productivity and mortality shares. While the “community verification” and auditor checks plausibly mitigate gaming and misreporting, the bundle explicitly includes auditing in treated facilities, but apparently not in controls, which may create asymmetric improvements in data quality. The paper partially addresses strategic patient selection and spillovers by examining contemporaneous mortality shares in hospitals, matched controls, and non-program facilities in the same district, finding no offsetting increases; however, stronger evidence that data definitions, reporting protocols, and SNIS integration are comparable across arms—and that auditors’ verification does not induce systematic measurement changes independent of true performance—would further bolster the results (e.g., audit-intensity measures, placebo outcomes less subject to recording changes, or sensitivity restricting to outcomes un-

likely to be affected by coding). Second, the description of what the control group actually receives is slightly ambiguous across sections. The general PDSS description suggests EUPs monitor, verify, and provide quarterly performance summaries; Table 1 and the experimental description state only treated facilities receive A&F and performance-based transfers, while controls receive average-matched lump-sum transfers without A&F. If any minimal verification or feedback reached controls (e.g., to validate disbursements), it should be documented precisely; if not, comparability of data quality remains a potential concern, as noted. Third, several regressors are counts analyzed in logs (e.g., numbers of doctors, nurses), yet baseline means and distributions indicate many zeros (doctors' mean is ~ 0.06 with a mass at zero). The paper does not state how zeros in these variables (and any other zero-prone outcomes) were handled in logarithms (e.g., +1 shift, inverse hyperbolic sine, or exclusion), which matters for interpretability and replication; this should be clarified and sensitivity analyses using IHS or count models should be reported. Fourth, implementation and compliance are not quantified: the paper does not report take-up or intensity of A&F at treated facilities, nor crossovers (if any) into the control; nor does it report whether all facilities received their scheduled visits each quarter (in a setting with known accessibility constraints). The analysis thus appears to estimate intent-to-treat; a transparent account of compliance, any missed visits, and robustness to excluding facilities with low audit exposure would help. Fifth, attrition and missingness over time are not discussed; explicit reporting of facility-quarter coverage, and checks that attrition (if any) is balanced across arms and not related to outcomes, would mitigate selection concerns. Sixth, while the event-study pre-trends are flat and the randomized assignment supports identification, there are only four pre-period quarters; although adequate, including alternative specifications with facility-specific linear trends (as a check) could be informative. Inference is based on bootstrapped SEs clustered at the facility level, which is reasonable; presenting standard cluster-robust SEs and, if feasible, wild bootstrap clustered by higher-level units (e.g., district) would probe sensitivity to spatial correlation. Seventh, the “outside facility” comparison is useful but correlational; although the nearest-neighbor matching and pair-by-quarter fixed effects are appropriate, the analysis remains vulnerable to unobserved differences and is presented as auxiliary. Robustness to alternative matching strategies (propensity-score, calipers, multiple neighbors) and balance on additional covariates, if available, would strengthen these claims. Eighth, mechanism evidence relies on interviews and on quality scores observed only for treated facilities; these are consistent with learning and process improvements but cannot be causally attributed, nor can they disentangle the contributions of incentives versus auditing versus feedback. The authors appropriately frame the bundle as an inseparable package; still, clarifying which elements are likely first-order, and exploring heterogeneity by baseline managerial/organizational capacity or by exposure to peer density (beyond what is already shown) would add nuance. Ninth, the mortality outcomes (stillbirths, neonatal deaths) are low-probability events; the paper mitigates variance via weighted least squares by births and reports MDES, which is good practice. Additional balance and robustness checks using exact or permutation inference at the facility level could be informative, and presenting raw counts alongside rates (e.g., total deaths per facility-quarter) may help readers assess magnitudes and variance. Tenth, the study's external validity is intentionally limited (primary care in the DRC), and the authors note this; any discussion of resource requirements and implementation fidelity to replicate the bundle elsewhere would be valuable for practitioners. Finally, reproducibility/open-science: data stem from administrative SNIS sources and program records, with restricted access; the paper references a data

appendix but does not indicate public availability of de-identified data or replication code. Providing code, full variable definitions, and a clear account of all transformations (especially for log outcomes with zeros) would substantially improve replicability. Despite these caveats, the core randomized comparison is well-executed, the dynamic patterns are plausible and consistent across outcomes, pre-trends look clean, multiple robustness checks are undertaken, and the policy relevance is high. The main causal claim—bundling incentives with auditing and feedback improves efficiency and birth outcomes relative to funding alone—is well supported within this context; the auxiliary claim that “funding alone” is not a substitute is suggestive but less definitive due to non-experimental comparisons.

i Gemini 2.0 Flash Assessment

This paper examines the impact of a governance program in the Democratic Republic of Congo (DRC) healthcare sector. The program, implemented by the World Bank, aimed to improve the quantity and quality of primary healthcare services through a combination of performance-based incentives, auditing, and feedback (A&F). The study uses a randomized controlled trial (RCT) design, where health centers were randomly assigned to a treatment group (receiving both funding and the governance treatment) or a control group (receiving only funding). The authors find that the governance treatment led to improvements in both operating efficiency and health outcomes (reduction in stillbirths and neonatal deaths). They also find that funding alone is not a substitute for the governance treatment. The paper contributes to the literature on governance, performance-based incentives, and non-profit organizations in low-income countries. Methodologically, the paper uses a difference-in-differences approach, which is justified by the RCT design. The authors conduct several robustness checks, including tests for pre-trends, contamination, and alternative interpretations. They also provide auxiliary evidence from health centers outside the PDSS program to support their findings. The paper is well-written and clearly presents the research question, methodology, and results. However, there are some limitations. First, the study is specific to the healthcare sector in a low-income country, and the findings may not be generalizable to other settings. Second, the study does not directly observe which operational changes are made at the treated health centers, and the analysis of the quality scores is merely descriptive. The authors acknowledge these limitations and suggest avenues for future research. The paper makes a valuable contribution to the literature on governance and performance-based incentives in low-income countries, and the findings have important implications for policy and practice. The study is well-executed and the results are robust to a variety of sensitivity analyses. The research question is relevant and important, and the findings are likely to be of interest to a wide audience of academics, policymakers, and practitioners.

Some potential issues:

- The study relies on administrative data, which may be subject to errors or manipulation. However, the authors address this concern by using auditing and feedback mechanisms to improve data quality.
- The study is specific to the DRC, and the findings may not be generalizable to other contexts. However, the authors argue that the DRC is a relevant case study due to its high infant mortality rates and weak healthcare system.
- The study does not directly measure the mechanisms through which the governance treatment affects health outcomes. However, the authors provide suggestive evidence

from interviews with program participants and analysis of quality scores.

- The paper could benefit from a more detailed discussion of the costs and benefits of the governance treatment. The authors provide some estimates of the VSL gains, but a more comprehensive cost-benefit analysis would be helpful.

Despite these limitations, the paper is a valuable contribution to the literature and provides important insights for policymakers and practitioners working to improve healthcare outcomes in low-income countries.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.28 Hahn et al. 2025

C.28.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash
Overall	94 [88–98]	65 [50–80]
Claims	93 [87–97]	70 [55–85]
Methods	92 [85–96]	60 [45–75]
Adv. Knowledge	94 [88–99]	75 [60–90]
Logic & Comms	95 [90–99]	75 [60–90]
Open Science	80 [70–90]	60 [45–75]
Global Relevance	96 [90–99]	75 [60–90]

i GPT-5 Pro Assessment

This paper makes a substantial and ambitious contribution by extending the MVPF framework to climate policy, harmonizing evidence across 96 rigorously evaluated policy changes, and introducing a new sufficient-statistics method to incorporate learning-by-doing (LBD) into MVPFs. The work is analytically careful, explicit about assumptions, and unusually transparent about how behavioral responses are translated into externalities and then monetized in a consistent way. The core conclusions—utility-scale wind and residential solar subsidies (especially when LBD is included) dominate other subsidies; nudges are highly context-dependent with large returns in dirtier grids; and fuel taxes/cap-and-trade are efficient revenue raisers—are well supported within the authors’ framework and sensitivity analyses. The contrast with cost-per-ton metrics, and the clear explanation of why those can mislead relative to MVPFs, is a valuable service to the literature and to policy audiences.

That said, there are several important issues and uncertainties that temper the strength and generality of the results and should guide interpretation and future work:

- 1) External validity and pooling across contexts. The harmonized “2020 baseline” MVPFs apply elasticities, pass-through rates, and other parameters estimated in diverse years

and locations to a common national 2020 setting. This is a practical and sensible way to facilitate comparability, but it relies on stability of demand elasticities, pass-through, counterfactual product characteristics (e.g., EV counterfactual MPG), and market markups across time and space. The paper does discuss constant elasticity versus semi-elasticity assumptions, but the internal validity of many reduced-form estimates does not guarantee external validity to the 2020 national context; results may be sensitive to unmodeled heterogeneity in prices, technology, consumers, and firm behavior.

- 2) Learning-by-doing assumptions and identification. The new sufficient statistics for LBD require isoelastic demand and a cost function with marginal costs isoelastic in cumulative production, plus perfect competition and price = marginal cost. The mapping from a marginal subsidy at time t^* to future price/quantity effects (via the “time shift” logic, the second-order ODE, and the t^* calibration using X_{init}/x_{init}) is elegant but imposes structure that may be violated. Crucially, the empirical learning curves used (e.g., Way et al. 2022) are not incontrovertibly causal LBD; they may reflect correlated R&D, scale economies, or selection, and the paper acknowledges this uncertainty by reporting results with and without LBD. However, the magnitude of the LBD effects—especially for residential solar and EVs—heavily influences key rankings. If learning is partially internalized by firms (capture through intellectual property or market power), or if only part of the product cost is subject to LBD (which the paper partially addresses for EV batteries), the DP/DE terms could be overstated. Further, the incidence of price externalities (DP) is global; the paper provides US-only versus global MVPFs for emissions, but the geographic incidence of price benefits is less clearly delineated and could merit further sensitivity analysis.
- 3) Emissions modeling and rebound. Translating behavior into emissions relies on EPA’s AVERT marginal emissions factors, combined with a long-run grid decarbonization path (Princeton REPEAT). AVERT provides regional annual averages; it cannot capture intraday or seasonal marginal emissions variation or evolving unit-commitment constraints. The assumed 20% electricity rebound (based on DOI MarketSim elasticities) is reasonable but uncertain and may vary by region/time, particularly under changing renewable penetration. The approach is consistent across policies, which is a strength, but residual model error could differentially affect categories (e.g., EV charging time profiles vs. wind displacement of marginal fossil units).
- 4) Valuation of local pollutants and VSL. The paper uses AP3 damages with a \$9.5m VSL (EPA 2010) and applies county-level damages for electricity and driving. These are state-of-the-art inputs, but the VSL vintage and other health valuation updates could shift results, particularly where local pollution co-benefits are a large share (wind vs. EV charging in dirtier grids). The authors do not systematically re-estimate AP3 with alternative VSLs or demographic adjustments; this is defensible but worth noting.
- 5) Treatment of private benefits and internalities. The baseline choice to not count private energy savings as a separate benefit (invoking the envelope theorem) is correct under optimization, and the paper shows alternative results when buyers misoptimize (e.g., weatherization). For nudges, the baseline assumes zero direct WTP for information/shame/pride, but alternative specifications using evidence from the literature are

presented. Still, these choices materially affect MVPFs in some categories and should be borne in mind by applied users.

- 6) Producer surplus and markup assumptions. The paper includes firm profit incidence via markups (e.g., gasoline retail markups ~27% pre-tax, utilities ~13% above economy-wide average). These markup calibrations are uncertain and may vary with wholesale/retail structure, vertical integration, and tax incidence; the authors offer a robustness that omits firm profits, which is useful and reveals limited qualitative changes (though nudges in clean-grid regions are especially sensitive to utility profits).
- 7) Gasoline tax MVPFs and non-carbon externalities. Accident and congestion externalities are modeled as per-mile averages; in reality, reductions in VMT may be concentrated off-peak or among lower-risk drivers, altering realized non-carbon benefits. This is a standard limitation of such calculations but merits mention as a potential bias. The paper explicitly sets aside labor-leisure interactions (complements/substitutes), which could slightly affect fiscal externalities.
- 8) Cap-and-trade analysis. The RGGI MVPF relies on a dp/dq inferred from 2009–2016 data; applying this to 2020 or forward assumes stable marginal abatement cost curves and low leakage. The paper is careful to provide in-context results and caveats for baseline extrapolation. For the California program, partial coverage of firms constrains inference; the authors present conservative and less-conservative bounds. These sections illustrate the MVPF logic well, but compared to other parts of the paper, the evidence is thinner and more context-dependent.
- 9) International policy MVPFs and SCC incidence. The very large US-only MVPFs (e.g., efficient cookstoves) hinge on the assumed US share of global SCC and the fraction of SCC flowing through market productivity (taxable) vs. non-market channels. The paper is forthright that these assumptions are pivotal and provides alternative scenarios. Users should treat these as illustrative upper/lower bounds rather than precise policy prescriptions.
- 10) Publication bias and uncertainty. The paper documents moderate publication bias and corrects via Andrews & Kasy; it also bootstraps category averages where SEs are available. This is a major strength. Still, the aggregate MVPFs reflect composite uncertainty (elasticities, pass-through, emissions, valuations, markups, LBD parameters, program costs). While the authors provide many sensitivity checks (including SCC ranges and with/without LBD), full probabilistic uncertainty propagation would further strengthen the conclusions.
- 11) Replication and open science. The paper states that code is available on GitHub and that online appendices provide policy-level details. Given the breadth of the exercise (96 policies) and the use of external models (AVERT, AP3, SCC paths), this is essential. Assuming the repository is complete and well-documented, replicability should be high; if not, re-creation would be burdensome. Clear version control and data provenance (especially for AVERT/AP3 layers and the parameters feeding the LBD module) will be important for future updates.

Overall, this is a rigorous, carefully reasoned, and impactful piece of work. The main limitations stem from necessary modeling choices when harmonizing across heterogeneous studies

and the contested nature of LBD causality/incidence. The authors are transparent about these issues and generally provide informative robustness checks. Policymakers and researchers can productively use these MVPFs—especially as relative orderings and for budget-neutral trade-off analyses—but should consult the sensitivity tables when decisions hinge on contested inputs (SCC incidence, LBD, markups, rebound, and counterfactual vehicle characteristics).

i Gemini 2.0 Flash Assessment

This paper presents a comprehensive welfare analysis of 96 climate-related policies in the US over the past 25 years, using the Marginal Value of Public Funds (MVPF) framework. It addresses a relevant question: identifying effective climate change mitigation strategies, specifically by estimating the welfare impact of historical policies across subsidies, nudges, and revenue raisers. The paper contributes methodologically by introducing a sufficient statistics approach to quantify the benefits of “learning-by-doing” effects, incorporating them into the MVPF framework. The analysis considers a range of environmental externalities, including greenhouse gases and local pollutants. The study uses data from various sources, including quasi-experimental and experimental evaluations of policy changes, EPA models, and integrated assessment models.

Some potential issues include:

- **Reliance on Existing Evaluations:** The MVPF estimates are derived from existing studies. The quality and consistency of these evaluations directly impact the robustness of the findings. The paper acknowledges this reliance but doesn’t fully address potential biases or limitations in the underlying studies. It may be useful for the authors to incorporate some measure of the confidence/quality of the source estimates into their own analysis/credible intervals.
- **Assumptions and Simplifications:** The MVPF framework involves several assumptions, such as those related to social welfare weights, incidence of the social cost of carbon, discount rates, and behavioral responses. While the authors explore the sensitivity of results to some of these assumptions, it’s crucial to acknowledge the potential impact of these choices on the overall conclusions.
- **Data Harmonization:** Combining data from different sources and studies requires harmonization, which may introduce errors or inconsistencies. The paper mentions using a consistent method to translate changes in behavior into emissions and valuing externalities, but more detail on the harmonization process would enhance transparency.
- **General Equilibrium Effects:** The analysis focuses on specific policy changes and may not fully capture broader general equilibrium effects. The authors acknowledge rebound effects but may not fully capture all potential interactions and spillovers across different sectors and markets.
- **Publication Bias:** The paper includes a discussion of publication bias and applies the methods of Andrews & Kasy (2019) to correct for it, finding that estimates are roughly two times more likely to be published if they cross a t-stat of around 2 and that this does not change their conclusions. As this is an observational study that depends upon past estimates, I believe this is a fundamental threat, and I am not sure the authors can wave it away so easily with such a simple correction, because the underlying distribution of study types etc. can itself be biased in ways that are hard to predict or test. It may be helpful for the authors to show what happens when they exclude the studies with

lower t-statistics entirely.

- **Learning-by-Doing:** The new approach to quantifying the benefits of “learning-by-doing” effects is methodologically interesting, but it relies on several assumptions, including isoelastic production and demand functions. The authors should acknowledge the potential limitations of these assumptions and explore the sensitivity of results to alternative specifications.
- **Global vs. US-Specific Benefits:** The analysis considers both global and US-specific benefits, but the relative weighting of these benefits may be subjective and could influence policy recommendations. A decision-maker’s perspective on the relative importance of global versus local benefits should be explicitly considered.
- **Replicability:** While the authors state that all code is available on GitHub, the documentation and ease of replication could be improved. Clear instructions on how to access and use the code would enhance the paper’s value and impact.

Overall, this paper provides a valuable contribution to the literature on climate policy evaluation. The MVPF framework offers a comprehensive approach to assess the welfare consequences of policy changes, and the inclusion of learning-by-doing effects is a novel methodological contribution. However, some limitations and assumptions should be acknowledged. If the authors provided better diagnostics for each of their past estimates, and incorporated these past diagnostics into their analysis, it would be much better. I believe the journal tier norms over-value the quality of work of this type, because in practice they give credit for novelty and cleverness over sound application and diagnostic checking.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.29 Haushofer et al. 2020

C.29.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	88 [82–93]	70 [55–80]	82 [65–85] (n=2)
Claims	89 [83–94]	75 [60–85]	—
Methods	87 [81–92]	70 [55–80]	90 [85–95] (n=2)
Adv. Knowledge	86 [79–92]	70 [50–80]	80 [60–90] (n=2)
Logic & Comms	90 [85–96]	80 [70–90]	78 [70–90] (n=2)
Open Science	68 [55–80]	75 [60–85]	60 [40–80] (n=2)
Global Relevance	93 [88–98]	80 [70–90]	95 [80–100] (n=2)
NA	—	—	4 [2–4] (n=2)
NA	—	—	4 [3–4] (n=2)
NA	—	—	88 [60–90] (n=2)

This is a large, well-powered, pre-registered cluster-randomized trial comparing unconditional cash transfers (CT; USD 1,076 PPP) and a five-session WHO Problem Management Plus (PM+) psychotherapy, delivered by trained community health workers, alone and in combination, with careful measurement of both economic and psychological outcomes about one year after intervention. The design (233 villages; 5,756 households; four arms plus village-level spillover samples; additional randomization of CT frequency at the village level) is clearly specified, and the main analytic approach (ANCOVA with baseline controls when available, village-clustered SEs, stratification variables, and false discovery rate correction across the five primary outcomes) is appropriate and transparent. Key strengths include: (i) strong internal validity via randomized assignment at village and household levels, low and non-differential attrition (4–5%), and documented baseline balance; (ii) thoughtful outcome construction (indices following Anderson 2008; careful definition of primary vs. secondary outcomes; PPP conversion), and pre-specification of analytic choices; (iii) attention to measurement integrity (high-frequency checks, back-checks, GPS verification, random spot checks, M-Pesa identity confirmation); (iv) explicit checks for experimenter demand using positive/negative demand prompts in pure control respondents, and multiple IPV measurement modes (standard questions, envelope task, “smiley” pictorial task); (v) a clearly explained cost analysis and an informative Bayesian “negative predictive value” framing for interpreting null PM+ results. The main findings are coherent and internally consistent: CT robustly improves consumption (+20%), asset holdings (+47%), revenue (+26%), and psychological well-being (+0.23 SD), with no clear effect on IPV; PM+ yields no detectable benefits on economic or psychological outcomes overall, nor among high-distress subgroups, and the combined arm closely mirrors CT alone. The randomized lump-sum versus weekly transfers comparison suggests weekly disbursement modestly outperforms lump-sum on consumption and revenue but not on assets or well-being. Spillovers for non-recipients within treatment villages are small and statistically weak.

Most important limitations and interpretive cautions are: (1) Measurement is largely self-reported for key economic and well-being outcomes; while demand was examined in the control group and showed negligible effects, this test does not fully guarantee the absence of reporting bias in treated groups. The convergence across many outcomes, the lack of PM+ effects despite equal potential for demand, and the multiple IPV measurement approaches mitigate but do not eliminate this concern. (2) For monthly consumption, detailed baseline data were not collected; hence ANCOVA cannot adjust for baseline consumption, potentially increasing variance (though randomization and baseline balance help). (3) The PM+ administrative compliance records indicate very high completion rates (95%) but a later phone resurvey suggests fewer respondents recalled all five sessions; this discrepancy could reflect recall decay or actual lower engagement/fidelity. The study treats PM+ ITT appropriately, yet true uptake/quality may have been lower than administrative data suggest. Relatedly, the observed heterogeneity by CHW indicates meaningful variation in delivery quality; although authors state CHW assignment to clients was random, details of that randomization are not deeply documented, and residual confounding by locality or client composition cannot be entirely excluded. (4) The frequency-of-transfers experiment was randomized at village level, not individual; the pre-analysis plan initially contemplated village fixed effects but the authors correctly note this was infeasible *ex post* given cluster-level treatment. The inference relies on village-level randomization without village FEs; while unbiased in expectation,

precision and potential chance village imbalances warrant caution, especially since only two outcomes show clear differences. (5) A separate digital financial services (DFS) study removed about 10% of respondents (among those unregistered for M-Pesa) after baseline; the main models include a DFS indicator and “overweight” non-M-Pesa respondents to preserve representativeness. However, the exact weighting scheme is not fully detailed here, which could be a replication friction and a modest threat to transparency. (6) Only five primaries are FDR-adjusted; most secondary and component outcomes are not multiplicity-adjusted, so individual significant findings there should be viewed cautiously. (7) The one-year follow-up precludes strong statements about very short-run PM+ effects; the moving-window analysis shows no PM+ impact even at 7 months, but earlier effects (<7 months) cannot be ruled out. (8) The IPV signals are mixed: an increase in explicit reports for PM+ that does not survive FDR, with no corroboration from envelope/smiley tasks; this remains a sensitive area where additional triangulation or qualitative work would help clarify mechanisms. (9) External validity is bounded: the PM+ program was delivered as a general-purpose intervention in a rural Kenyan setting; previous success in other contexts (e.g., targeted to survivors of gender-based violence, or more intensive protocols) limits direct generalization. (10) Open science and reproducibility: the pre-analysis plan and detailed appendix are strong positives, but the paper does not clearly state whether data and code are publicly available, and some analytic details (e.g., DFS weighting) are not exhaustively specified, which may impose some replication barriers.

Despite these limitations, the identification is sound and the evidence strongly supports the main claims: cash transfers reliably improve both economic and psychological outcomes at one year in this context; PM+ as deployed here does not produce detectable benefits; combining the two does not add to CT’s impacts; and weekly CTs may modestly outperform lump-sum for certain margins. The study substantially advances practical knowledge by providing a credible direct comparison of two salient intervention classes under the same design with attention to spillovers, heterogeneity, and costs. The methods are standard but applied rigorously; the reasoning and communication are clear, with careful acknowledgment of uncertainty, multiple testing, and alternative interpretations. The paper is highly useful to policymakers and funders concerned with mental health and poverty programs, but would benefit further from public release of replication files, fuller documentation of the DFS weighting, and, in future work, an earlier follow-up to pinpoint transient psychotherapy effects and deeper fidelity audits to parse CHW heterogeneity.

i Gemini 2.0 Flash Assessment

This paper examines the comparative impact of cash transfers and a psychotherapy program (Problem Management Plus, PM+) on psychological and economic well-being in rural Kenya. The study uses a randomized controlled trial (RCT) design, assigning 5,756 individuals to one of four conditions: cash transfer (CT), PM+, combined CT and PM+, and a control group. The primary outcomes include consumption, asset holdings, revenue, psychological well-being, and intimate partner violence. Data were collected at baseline and one year after the interventions. The study finds that cash transfers had significant positive effects on consumption, asset holdings, revenue, and psychological well-being, while the PM+ program showed no measurable effects on either psychological or economic outcomes. The combined treatment effects were similar to those of the cash transfer alone. The authors explore several

potential explanations for the null findings of the PM+ intervention, including lack of power, experimenter demand effects, and implementation differences. They conclude that the lack of a specific problem to address (unlike previous studies where PM+ was used for gender-based violence) may explain the lack of lasting impact. Cost-effectiveness analysis indicates that cash transfers are more cost-effective than PM+ in this setting. The study employs a pre-analysis plan and uses various robustness checks, including FDR-adjusted standard errors to account for multiple hypothesis testing and analyses of spillover effects and heterogeneous treatment effects.

The main concerns are:

1. **External Validity of PM+:** The authors acknowledge that PM+ has been effective in other contexts, specifically when targeting gender-based violence. The generalizability of PM+ in a non-specific context is questionable, and the paper could benefit from a more in-depth discussion of the conditions under which PM+ is likely to be effective.
2. **Measurement of Psychological Well-being:** The psychological well-being index relies on self-reported measures, which may be subject to biases such as social desirability or recall bias. While the authors address demand effects, further discussion of the limitations of these measures would enhance the credibility of the findings.
3. **Intimate Partner Violence (IPV) Measurement:** The increase in IPV in the PM+ treatment arm, although not robust to FDR correction, raises concerns. The authors appropriately acknowledge the potential for PM+ to increase IPV and call for future research to investigate this further. However, the measurement of IPV, which includes explicit self-reports and an “envelope task,” could be further scrutinized for potential biases or limitations.
4. **Long-Term Sustainability:** The study only assesses outcomes one year after the interventions. The long-term sustainability of the cash transfer effects and the potential for delayed effects of PM+ are not addressed. This limits the understanding of the interventions’ lasting impact.
5. **Heterogeneity Analysis Limitations:** While the authors conduct heterogeneity analysis, the lack of significant interaction effects for PM+ raises questions about whether the analysis was adequately powered to detect such effects or if the pre-specified dimensions of heterogeneity were the most relevant. The exploratory analysis of severely distressed participants provides some insights but remains limited.
6. **Spillover Effects:** The spillover effects are generally small and statistically weak. The authors acknowledge this but could provide a more detailed discussion of the potential mechanisms that might explain the limited spillover effects, such as the nature of social networks in the study area.
7. **CHW Fixed Effects:** The analysis of CHW fixed effects reveals variability in the effectiveness of PM+ delivery. While the authors acknowledge this, further exploration of the characteristics of more effective CHWs could provide valuable insights for improving the PM+ program.
8. **Cost-Effectiveness Assumptions:** The cost-effectiveness analysis relies on certain assumptions, such as the costs of remitting cash transfers. A more detailed justification

of these assumptions and a sensitivity analysis would strengthen the robustness of the cost-effectiveness findings.

9. **Multiple Hypothesis Testing:** While the authors use FDR-adjusted standard errors for primary outcomes, they do not correct for multiple hypothesis testing for secondary outcomes or individual variables. This could lead to an overestimation of the significance of some findings.
10. **Data Integrity:** The discussion on data integrity checks in the Appendix (A.1.) does not offer enough specifics on the criteria used to evaluate the back-checks and spot checks. This makes it difficult to evaluate the effectiveness of these checks.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.30 Hill et al. 2024

C.30.1 Ratings Comparison

Metric	Claude Sonnet 4	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	75 [65–83]	93 [88–97]	75 [60–85]	80 [70–90] (n=1)
Claims	70 [60–78]	86 [80–92]	80 [70–90]	82 [74–90] (n=1)
Methods	68 [58–76]	85 [78–91]	75 [65–85]	85 [78–92] (n=1)
Adv. Knowledge	78 [70–85]	92 [86–97]	75 [60–85]	85 [78–92] (n=1)
Logic & Comms	82 [75–87]	92 [88–96]	85 [75–95]	80 [71–89] (n=1)
Open Science	85 [78–90]	55 [40–70]	70 [50–80]	90 [85–95] (n=1)
Global Relevance	72 [62–80]	90 [82–96]	70 [50–80]	80 [70–90] (n=1)
NA	—	—	—	4 [4–5] (n=1)
NA	—	—	—	4 [4–5] (n=1)
NA	—	—	—	80 [70–90] (n=1)

GPT-5 Pro Assessment

This paper introduces a clear, scalable construct—“pivot size,” operationalized as 1 minus the cosine similarity between the distribution of knowledge sources (journals for papers; CPC codes for patents) cited by a focal work and the distribution cited in an author’s prior body of work—to quantify how far a researcher moves when producing a new work. Using Dimensions publications (1970–2020) and USPTO patents (1980–2015), the authors document a strong, monotonic “pivot penalty”: larger pivots are associated with substantially lower impact, measured primarily as field- and year-normalized hit rates (top 5% by citations). The pattern is robust across science and technology, across field and technology classifications at varying granularities, across impact measures (binary and continuous), within-person fixed-effect analyses, and over time (with evidence the penalty has steepened). The study also

offers quasi-experimental evidence using retractions (push shocks) and observational analyses of COVID-19 (pull shock), and explores mechanisms (novelty vs conventionality) and moderators (career stage, team size/new collaborators, funding). Communication is lucid; the supplementary materials document data sources, disambiguation, and multiple robustness analyses.

Important limitations and identification threats remain. First, construct validity of the pivot measure: it relies on reference distributions rather than content or topic semantics. Referencing behaviors vary by field, venue, and editorial norms; in computer science, conference proceedings appear as distinct “journals,” which can inflate measured pivots; in the arts/humanities, books play a large role, possibly weakening the signal. The median pivot values (e.g., ~ 0.60 for non-COVID 2020 papers) suggest the metric often classifies even moderate within-area steps as sizeable pivots; while relative comparisons are meaningful, absolute scale calibration is non-obvious. The team-level pivot is defined as the mean of member pivots; this may mask heterogeneity in specialist vs generalist roles within teams (minimum/maximum pivot or distributional features could be informative). The reliance on cited-journal distributions risks conflating changes in referencing practice with shifts in substantive research area. The authors partially address these issues via alternative field-based pivot measures (L1 fields), across multiple CPC depths for patents, and via correlations with field switches; but complementing citation-based pivot with text-embedding/topic-distance measures would strengthen construct validity.

Second, while the descriptive association is convincing, causal interpretation is delicate. The within-author fixed effects regressions mitigate person-level confounding, but time-varying unobservables remain (e.g., authors may pivot when their pipeline is weak, or when facing funding shocks, teaching load changes, or strategic repositioning). The differences-in-differences design using retractions is a thoughtful “push” instrument to induce pivots, yet the exclusion restriction for the 2SLS interpretation is questionable: retractions plausibly affect subsequent impact through channels other than pivot size (e.g., devaluation of related knowledge, stigma, changed reviewer priors, reduced demand for adjacent topics). Matching treated and control authors only on pre-event publication rates is narrow; additional matching or controls for field proximity, prior impact trajectories, and coauthorship networks would bolster parallel trends. The event-study patterns are consistent with a contemporaneous shock, but the design cannot fully isolate whether lower post-event impact reflects pivot difficulty per se versus the negative spillovers of retraction on nearby areas. The smaller psychology replication case is consistent but underpowered.

Third, COVID-19 analyses rely on expected journal hit rates as a proxy for impact in 2020; this is reasonable given short citation windows but remains a proxy, sensitive to journal selection dynamics, special issues, and pandemic-era editorial practices. The strong impact premium for COVID papers alongside a persistent pivot penalty is compelling descriptively; the within-author fixed-effects version helps, but confounding remains (e.g., urgency-induced variance in quality, topical hype). Longer-horizon realized citations will be important to confirm.

Fourth, data and measurement issues: author and inventor disambiguation are handled using provider algorithms; the paper reports 84.6% coverage for author IDs and restricts analyses accordingly. The manual validation of 10 high-pivot authors (and surname-frequency checks) is helpful but limited in scope; residual misattribution could still bias extreme pivots, especially in regions with sparser metadata or non-Latin scripts. For patents, the bimodal pivot distribution may partly reflect sparse prior art references or firm assignment practices; the au-

thors explore restrictions (e.g., inventors with 10 prior patents; prior art with 100 references), which attenuate but do not eliminate bimodality. The decision to include only works with 5 references is sensible, though it induces selection; robustness to higher thresholds (shown) is reassuring.

Fifth, mechanism evidence (novelty vs conventionality): the finding that high pivots correspond to greater “tail novelty” but lower “median conventionality” of cited combinations is consistent with creativity literature and provides a plausible channel. However, these measures (from SciSciNet) are themselves citation-structure-based; triangulation with content-based novelty/conventionality would increase confidence. In regressions that include novelty and conventionality, pivot retains a large negative association with impact; still, causality is not established.

Sixth, scope and generality: the pivot penalty holds in the large majority of subfields/classes, but some small or particular fields (e.g., visual arts, certain CS subfields) deviate; the authors attribute plausible reasons (venue structure, field idiosyncrasies). The steepening penalty over time could reflect increasing specialization, but also changes in database coverage, reference practices, and field fragmentation; the within-author analyses help but cannot fully rule out these alternative explanations.

Seventh, open science and reproducibility: the paper provides extensive methodological detail, clear variable definitions, and numerous robustness checks, but no public code repository is cited, and one main dataset (Dimensions) is commercial (albeit broadly accessible). This constrains full computational replication and reuse. The inclusion of precise query details (e.g., COVID query), clear definitions of hit-rate normalization, and regression specifications is a plus; still, provision of code and derivative data (e.g., binned results) would notably improve reproducibility and downstream reuse.

Overall, the paper presents a well-articulated, carefully executed, large-scale empirical investigation with a simple, transparent metric that travels well across domains. The central descriptive claim—that larger research pivots, on average, carry substantial impact penalties that have grown over time—is strongly supported and robust in many dimensions. The quasi-experimental components are informative but should be interpreted cautiously with respect to causal mechanisms. The study’s policy relevance is high: it offers actionable implications for researchers, teams, funders, and institutions regarding specialization, team composition, hiring (“acqui hires”), and portfolio diversification. Key avenues for strengthening the work include (i) complementing the citation-based pivot with content-based distance measures, (ii) deeper matching/controls in the retraction DiD (field and network controls; placebo checks), (iii) exploring team-role heterogeneity (min/max/variance of member pivots), and (iv) releasing code and replication materials. Despite these caveats, the paper substantially advances our understanding of adaptability in science and technology and will likely serve as a durable reference in the science-of-science literature and for science policy.

Claude Sonnet 4 Assessment

This paper introduces a ‘pivot penalty’ concept to measure how researchers’ impact declines when they shift away from their established research areas. Using millions of scientific papers and patents, the authors develop a cosine-similarity metric to quantify research pivots and demonstrate that larger pivots consistently lead to lower research impact. While the paper addresses an important question about research adaptability, it suffers from several signifi-

cant methodological and interpretive limitations. First, the pivot measure based on journal references is a crude proxy for genuine research direction changes, potentially conflating superficial shifts in citation patterns with meaningful intellectual pivots. The choice to focus on journal-level similarity rather than content-based measures raises questions about construct validity. Second, the causal interpretation is highly problematic - the authors cannot distinguish whether poor outcomes cause researchers to pivot (reverse causality) or whether pivots cause poor outcomes. The observational design with individual fixed effects only partially addresses selection concerns. Third, the COVID-19 analysis, while topical, fails to provide compelling causal identification since the pandemic simultaneously affected both pivot behavior and impact channels. Fourth, the paper lacks sufficient exploration of mechanisms beyond brief discussions of reputation and expertise theories. The authors don't adequately address whether the penalty reflects genuine learning costs, audience effects, or measurement artifacts. Fifth, robustness concerns include the arbitrary choice of three-year reference windows, limited exploration of field heterogeneity, and insufficient attention to how the pivot measure might capture noise rather than signal. The retraction analysis provides some quasi-experimental variation but the effect sizes seem implausibly large and the identifying assumptions questionable. Despite these limitations, the paper tackles an important question with impressive scale and presents intriguing descriptive patterns that could inform future research on scientific careers and innovation.

i Gemini 2.0 Flash Assessment

This paper explores the phenomenon of the 'pivot penalty' in scientific and technological research, where researchers who significantly change their research direction (i.e., 'pivot') experience a decline in the impact of their work. The authors introduce a measurement framework to quantify researcher pivots using citation data from millions of scientific publications and patents spanning several decades. Their core finding is that larger pivots are associated with lower impact, a trend that has been growing over time and is consistent across various fields and career stages. They investigate this penalty in the context of exogenous events, namely paper retractions (a 'push' factor) and the COVID-19 pandemic (a 'pull' factor). The analysis employs rigorous statistical methods, including difference-in-differences designs and regression analyses with fixed effects, to establish the robustness of the findings. The authors further explore potential mechanisms behind the pivot penalty, such as reputational effects, knowledge depth, and combinatorial novelty, as well as moderating factors like teamwork and funding. The work is ambitious in scope, using very large datasets and diverse methodologies to explore an interesting and important question about the adaptability of science and technology. The findings have implications for individual researchers, research organizations, science policy, and the overall capacity of science to address new challenges.

However, there are some potential limitations and areas for further consideration:

- **Measurement of Impact:** The paper primarily relies on citation-based metrics (e.g., hit rate, journal placement) to quantify impact. While these metrics are widely used, they are known to have limitations, such as field-specific citation norms and time-dependent effects. It would be beneficial to explore alternative impact measures or to conduct robustness checks using a wider range of metrics.
- **Pivot Measure Validity:** The pivot measure is based on cosine similarity of referenced journals/technology codes, which is reasonable, there could be uncaptured nuances. For

example, two papers referencing similar journals might still have very different conceptual content.

- **Causality:** The study uses quasi-experimental designs to examine the impact of exogenous shocks (retractions, COVID-19). However, establishing causality in these settings is always challenging. There may be other confounding factors that are not fully controlled for, which could affect the interpretation of the results.
- **Generalizability:** While the study covers a large dataset of publications and patents, there may be limitations to the generalizability of the findings. For example, the pivot penalty may vary across different types of research (e.g., basic vs. applied), different institutional contexts, or different geographical regions.
- **Interpretation of Mechanisms:** The paper explores several potential mechanisms behind the pivot penalty, but the evidence for each mechanism is not always conclusive. Further research is needed to better understand the underlying drivers of the pivot penalty and to identify effective interventions to mitigate its negative effects.
- **Data availability:** While data and code availability is not directly mentioned, the exclusive use of Dimensions, which is a proprietary database, might limit the replicability by researchers who do not have access to this database.

Overall, the paper provides strong evidence for the existence of a ‘pivot penalty’ in science and technology. The study makes a valuable contribution to the science of science by quantifying an important phenomenon and exploring its potential mechanisms and implications.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.31 Jack et al. 2022

C.31.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	87 [80–94]	80 [65–90]	88 [78–92] (n=2)
Claims	86 [78–92]	85 [75–95]	—
Methods	83 [75–89]	80 [70–90]	80 [68–85] (n=2)
Adv. Knowledge	92 [86–97]	75 [60–85]	80 [78–90] (n=2)
Logic & Comms	84 [77–90]	85 [75–95]	90 [84–91] (n=2)
Open Science	55 [40–66]	70 [50–80]	85 [72–88] (n=2)
Global Relevance	95 [90–99]	85 [75–95]	92 [85–100] (n=2)
NA	—	—	4 [3–5] (n=2)
NA	—	—	4 [3–5] (n=2)
NA	—	—	92 [90–100] (n=2)

This paper presents a well-designed field experiment testing whether partial upfront payments in a Payments for Ecosystem Services (PES) contract reduce crop residue burning in Punjab relative to a standard, entirely ex-post conditional PES. The design leverages village-level randomization across four treatment variants and a control in 171 villages (1,668 farmers), with careful pre-registration and stratification. The primary findings are compelling: contract take-up is high and similar across arms (~72%), but compliance and satellite-derived non-burning improve markedly under upfront PES (about 10 percentage points higher compliance than standard PES; 8–12 percentage points higher non-burning relative to control, depending on classification threshold). In contrast, standard PES appears to generate mainly inframarginal payments (no detectable reduction in burning vs control in the primary remote-sensing specification). The paper combines administrative monitoring data, endline surveys, and a novel remote-sensing classifier trained on PlanetScope and Sentinel-2 imagery, and it provides a transparent discussion of thresholds, accuracy (78–82%), and robustness over threshold choices. The cost-effectiveness exercise indicates that upfront PES can avert burning at roughly 2,700–4,050 per additional unburned acre and plausibly saves lives at costs orders of magnitude lower than a benchmark abatement option (scrubbers at coal plants). The study explicitly anticipates scale-up challenges and equilibrium responses.

The strongest features include: (i) a pre-registered randomized design with clear treatment contrasts that isolate timing/conditionality effects; (ii) a carefully implemented monitoring protocol and an intention-to-treat analysis using remote-sensing outcomes that cover all arms; (iii) multiple robustness checks on satellite classification thresholds and an assessment of potential differential misclassification; (iv) attention to take-up, compliance, and behavioral mechanisms (trust vs liquidity) with suggestive evidence that upfront payments raised trust; and (v) an applied cost-effectiveness analysis linking outcomes to policy-relevant health benefits. The narrative is generally careful in characterizing inference, and the claim that standard PES payments were largely inframarginal is well supported by the zero-effect finding on remotely-sensed burning for that arm.

Key limitations and issues to note: - Measurement and identification: The main environmental outcome relies on a machine-learning classifier trained with positive labels (burns) from both treatment and control, but negative labels (no-burn) only from monitored treatment farmers. Although the authors present distributional checks showing similar classifier scores by treatment status conditional on burn/no-burn, residual risk remains that negative labels specific to treatment farmers induce subtle differential misclassification. Classification accuracy of 78–82% is respectable, but the risk of non-classical error (and the possibility that behavior or residue management choices in treatment arms affect spectral signatures) remains a threat. Still, the direction and significance of the main treatment contrast (upfront vs standard) appear robust to classification thresholds and are consistent with independent survey evidence on baler uptake. - Compliance measurement and monitoring protocol: The contract compliance outcome requires farmers to initiate monitoring calls within a narrow time window; hence, it cannot be measured for the control group and may be attenuated by forgetfulness or transaction costs. The satellite-based outcome mitigates this, but the comparison between compliance and satellite results suggests some “money left on the table” under certain thresholds. The authors discuss this, but it complicates precise behavioral interpretation of internal compliance metrics. - External validity and sample selection: The sample is restricted to two districts and cooperative-member farmers screened to be more likely burners

(e.g., based on equipment use), which raises questions about generalization to the full population of farmers or to other states with different agronomy, enforcement, or air transport patterns. This sample choice is defensible for power and policy relevance to high-burning settings but should temper claims about statewide or national impacts absent additional evidence. - Endline survey attrition: There is modestly higher attrition in treatment vs control for endline survey outcomes; the authors appropriately apply Lee bounds. Since the main environmental outcomes are not affected by survey attrition, this is a limited concern; still, it reduces precision for equipment-adoption outcomes. - Mechanisms: Pre-specified heterogeneity tests using generalized, ex-ante indices of liquidity and distrust are largely null; ex-post endline questions indicate higher trust in payment under upfront PES, but these are post-treatment self-reports that may reflect justification bias. The liquidity mechanism remains hard to pin down (upfront amounts plausibly cover less than half the cost of baling), and the absence of differential self-reported liquidity constraints by arm suggests either measurement limitations or that trust dominated. The paper is appropriately cautious, but the mechanism evidence remains suggestive rather than conclusive. - Potential contemporaneous policy signals: A court-ordered PES-like policy was announced during the season (later suspended), and around 30% of farmers had heard of it. Randomization ensures balanced exposure, but this background could influence take-up, expectations, or monitoring calls in ways that are not fully explored. - Cost-effectiveness assumptions: The mortality benefit calibration uses national estimates (66,000 deaths/year) and assigns benefits per acre using a \$1M VSL. These choices are defensible for a first pass but entail considerable uncertainty (e.g., spatial variation in exposure, alternative VSL values consistent with Indian income levels, meteorology, and spillover patterns). Administrative and monitoring costs beyond PES payments are excluded or discussed qualitatively; the authors note this and highlight that payments dominate costs in their implementation, but remote-sensing at scale (or alternative monitoring regimes) could shift cost structures and contract risk. - Internal consistency: There appears to be a likely error in the text stating that 4 million metric tons of CO₂ equivalents equals 1% of annual global greenhouse gas emissions; that magnitude is orders of magnitude too small to be 1% globally. This does not affect the main results but should be corrected. - Open science and replicability: The pre-analysis plan and detailed protocols are strengths. However, replicability of the satellite-based outcomes may be limited by proprietary imagery (PlanetScope) and the absence of shared code/data within this working paper. The reliance on a companion methods paper is helpful, but providing code, model parameters, and field-level labels would significantly improve reproducibility and facilitate future meta-analyses.

Overall, this is a strong applied study with clear, policy-relevant findings and careful handling of main identification and measurement risks. The core conclusion—that adding an upfront, unconditional component to PES can materially improve environmental outcomes even with lower conditional incentives—appears well supported and important for contract design in low- and middle-income settings. Remaining uncertainties primarily concern mechanism identification, replicability of the remote-sensing pipeline, and the exact magnitude of cost-effectiveness under alternative benefit assumptions or at scale. These do not overturn the paper’s principal insights but point to valuable extensions and robustness work for a subsequent version.

Gemini 2.0 Flash Assessment

This paper presents an RCT evaluating the impact of Payments for Ecosystem Services (PES) contracts on reducing crop residue burning in Punjab, India. The study compares standard PES contracts (payment conditional on verification of no burning) with contracts that include a partial upfront payment, regardless of compliance. The research question is highly relevant to global priorities, as air pollution from crop burning is a major health and environmental concern. The study design is strong, employing a randomized controlled trial in 171 villages, with remote sensing data used to measure burning rates. The use of remote sensing data is a particular strength, as it provides an objective measure of burning, reducing reliance on self-reported data. The authors also considered the cost-effectiveness of the different contract designs relative to mortality benefits and alternative pollution abatement strategies. The credible intervals are a bit wider than I would expect, but the results show fairly consistently that upfront PES outperform standard PES. However, the mechanism through which upfront payments increase compliance remains unclear, as heterogeneous treatment effects based on pre-specified measures of liquidity constraints and distrust were largely uninformative. The paper also does not explore possible alternative behavioral explanations, such as farmers interpreting upfront payments as a sign of the program's commitment, or using the money to address other pressing needs. This work is relevant to global priorities because reducing air pollution aligns with improving public health. The insights into the design of PES contracts, particularly the effectiveness of upfront payments, are valuable for practitioners and policymakers seeking to reduce environmentally harmful behaviors in contexts with credit constraints and weak institutions. One potential area for improvement is that while the upfront payment size seems arbitrarily chosen, perhaps a more systematic approach that optimizes the upfront share, or takes into account farmer characteristics could be a topic for future research.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.32 Jaroszewicz et al. 2024

C.32.1 Ratings Comparison

Metric	GPT-5 Pro		Human
Overall	88 [82–93]	82 [72–92]	(n=2)
Claims	88 [83–93]	—	—
Methods	87 [82–92]	92 [88–98]	(n=2)
Adv. Knowledge	85 [78–91]	72 [60–85]	(n=2)
Logic & Comms	90 [85–95]	92 [88–98]	(n=2)
Open Science	68 [58–78]	88 [82–92]	(n=2)
Global Relevance	79 [71–86]	78 [70–84]	(n=2)
NA	—	4 [4–5]	(n=2)
NA	—	4 [4–5]	(n=2)

i GPT-5 Pro Assessment

This is a large, carefully executed, preregistered RCT testing one-off unconditional cash transfers (UCTs) of \$500 and \$2,000 versus control among 5,243 low-income US adults during 2020–21. The design is strong: randomization after a baseline survey; unequal allocation powered for costly arms; a prespecified family of four primary indices (financial, psychological, cognitive capacity, health) constructed via the Anderson (2008) approach; multiple post-treatment surveys at 1, 6, and 15 weeks; and rich administrative data (payments sent/received, bank balances, and categorized transactions) for 43% of participants. The headline finding—no positive effects of either UCT on any preregistered survey index, with small negative effects on financial, psychological, and health indices concentrated in more subjective items—is consistently documented and contrasted with incentivized expert and lay predictions, which were overly optimistic. The authors probe many potential explanations, with the most consistent evidence pointing to increased salience of unmet obligations among recipients (“money on the mind,” higher reported needs over the next 30 days, greater spending-decision stress, feeling more burdened by others’ needs), supported by mediation analyses and a simple model in which small windfalls induce active financial management that reveals unexpectedly large obligations and temporarily lowers utility. Identification and internal validity are well addressed. Baseline balance is shown; intent-to-treat is primary; treatment delivery is tracked (87–89% of UCT dollars reached external accounts) with bank data confirming rapid spending and large short-run MPCs. Although 90% (\$500) and 88% (\$2,000) of treated versus 80% of controls completed at least one endline survey, the paper devotes extensive analysis to attrition: predictors at baseline, selective-attrition tests, Lee/Horowitz–Manski bounds, inverted-bounds “how extreme would missingness need to be?” exercises, and multiple imputation tailored to TSCS data. These point estimates could be somewhat biased downward in some specifications, but the authors argue convincingly that meaningfully positive effects are unlikely (e.g., would require 0.4–0.7 SD gaps among missing participants in the opposite direction of observed effects). The lack of differential response rates between the two cash arms bolsters the “no marginal benefit of \$2,000 vs \$500” conclusion independent of attrition. Compliance issues are transparent. A small share did not pull funds from the platform; an alternative analysis conditioning on correctly recalling receipt amount (a proxy for attention/compliance) yields similar, slightly attenuated, results. The “push vs pull” delivery friction is a limitation but is appropriate for ITT and well-documented; effects on actual receipt are reported, and bank data corroborate spending uplifts on necessities. Outcome measurement is a mix of self-reports and administrative data. Self-reports carry familiar concerns (demand, recall, social desirability), but the use of pre-specified indices, administrative validation for some financial measures, and convergence across many items mitigate these risks. The post hoc categorization into “objective” vs “subjective” items (coder-based, not preregistered) is informative but exploratory; it strengthens interpretation yet should be weighted with caution. The mediation analyses are sensible but inherently correlational; the authors appropriately flag this. Multiple testing is handled carefully: the primary indexing reduces multiplicity, and both Benjamini–Hochberg and Westfall–Young corrections are reported in appendices (main text reports unadjusted p-values with flags where corrections differ). Analytical choices (e.g., Anderson-index weighting, baseline adjustment to improve power, 90% winsorization vs

planned 95% top-only—claimed to be identical in this data) are explained and accompanied by robustness checks (z-score indices, added covariates, dropping the wave with timing error, disaggregating by survey wave, and item-level analyses). The wave-6 payment timing error is transparently used to probe shorter-run effects (1–6 days after UCT), with no positive effects observed, and a within-subject difference-in-differences check around the error also yields nulls. Mechanism testing is extensive. Embedded experiments on strategic distortion (telling participants of a potential new program) and reference dependence (prompting comparisons to pre- vs post-payout dates) mostly return nulls; harmful spending is ruled out using bank categories (consistent with prior work); expectation-mismatch and “inference from charity receipt” are considered and largely discounted. The “salience of obligations” mechanism is supported by multiple measures and by a stylized model predicting short-run utility declines from small windfalls and potential attenuation with larger transfers. The model is not estimated and serves as a conceptual aid; its long-run prediction of improved lifetime utility lies beyond the data horizon, a clear limitation. External validity is mixed. The sample comprises applicants to a single national non-profit; many (43%) had received \$500 months before the trial; the context is the pandemic with concurrent US stimulus and benefit changes; the UCTs are one-off, moderate by high-income standards (small relative to US median annual income, but large vs recipients’ recent monthly earnings). The authors explore heterogeneity (20 moderators) but find little robust evidence. Unexamined heterogeneity by prior non-profit receipt could matter, as could interactions with local benefits or debt portfolios (credit reports are not used). Still, the design and documentation enable credible inference about one-off UCTs of these magnitudes in a similar US low-income context. Open-science practices are strong on preregistration and instrument transparency; the administrative nature and sensitivity likely constrain full data sharing; code/data availability is not clearly stated, which affects reproducibility. Communication quality is high: the paper is logically organized, figures/tables are clear, assumptions and caveats are explicit, and the link between claims and evidence is careful. Overall, this is a rigorous and policy-relevant null-result paper that helps recalibrate priors on one-off cash transfers in high-income settings and prioritizes transparency about limitations (attrition, measurement, context). The remaining uncertainties center on (i) the extent to which the negative short-run subjective effects would generalize beyond the pandemic/non-profit-applicant context; (ii) whether larger or repeated UCTs would reverse the short-run effects (as the model predicts); (iii) how much of the subjective decline is pure measurement (e.g., reporting/anchoring) versus genuine distress; and (iv) longer-run economic outcomes (debt reduction, delinquency, credit scores) that are unobserved here. Nonetheless, given the sample size, design quality, administrative validation, and thorough sensitivity analyses, the core findings (no detectable short-run benefits on pre-specified survey outcomes and plausible short-run subjective costs) are highly credible and informative for policy and research practice.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.33 Kirkland et al. 2023

C.33.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash
Overall	70 [58–80]	55 [40–70]
Claims	74 [64–84]	65 [50–80]
Methods	62 [50–74]	50 [35–65]
Adv. Knowledge	66 [55–77]	60 [45–75]
Logic & Comms	78 [68–88]	75 [60–90]
Open Science	57 [42–70]	70 [55–85]
Global Relevance	49 [36–60]	70 [55–85]

i GPT-5 Pro Assessment

This paper offers a large, multi-country investigation (41 university samples across 36 countries; $N=6,665$, mean age 21.6, ~63% female) into how generalized trust, perceived anomie, and economic inequality relate to moral expansiveness (MES). Strengths include: (i) a clearly articulated theoretical framing and preregistered analyses; (ii) a psychometrically strong MES ($=.92$) with cross-national scope; (iii) use of linear mixed models with a random intercept for country and explicit decomposition of within- and between-country components; (iv) careful control strategy that includes both individual- and country-level covariates; and (v) a set of convergent analyses showing robust within-country associations between higher generalized trust and larger moral circles, and between larger perceived wealth gaps and smaller moral circles, with mediation by generalized trust. That said, several issues constrain causal interpretation, generalizability, and the precision of between-country conclusions. The design is cross-sectional and entirely self-report, raising risks of common-method variance and preventing strong causal claims; to their credit, the authors frequently acknowledge this limitation and use cautious language, but the mediation model (bootstrapped multi-level mediation) still implies process inferences that cannot be established without temporal ordering or exogenous variation. The key predictors—generalized trust (single item), perceived wealth gap (single item), and perceived Gini (constructed from a distribution question)—are likely measured with nontrivial error and culture-specific response styles; a single-item trust measure also limits reliability and cross-cultural comparability. Although anomie is assessed with a 12-item instrument and reliability is good, the conceptual overlap between low trust and anomie’s “breakdown in social fabric” facet invites concerns about multicollinearity; the paper does not report diagnostics. The authors emphasize within-country effects (e.g., trust→MES $b\ 0.88$, $p<.001$; perceived wealth gap→MES $b\ -0.53$, $p=.002$), but effect sizes—expressed relative to the full MES range (0–90)—appear modest at the individual level; the manuscript would benefit from standardized coefficients or variance explained for focal models beyond the small marginal R^2 reported in controls ($.018$). Between-country relationships are weaker and often nonsignificant (e.g., trust→MES between effect), yet the mediation via generalized trust is reported as significant at both levels. Given a small effective number of macro units and the modeling choice to treat multiple subnational samples as separate “countries” (e.g., US North/South; UK nations; Canada language groups) while attaching identical national covariate values (Gini, GDP-PPP, prosperity, democracy, homicide), there is a risk of pseudo-replication and inflated precision for between-country estimates. Conceptually and statistically, the decomposition termed “within- vs between-country” is closer to “within- vs

between-sample-site,” because each university sample is the clustering unit; this should be made explicit, and a three-level model (individuals nested in sites nested in countries) or random slopes would more faithfully capture heterogeneity across sites and countries. The non-association between national Gini and MES is informative, but may reflect measurement mismatch (national Gini vs student samples), limited cross-country leverage with few unique country values, and attenuation from measurement error and heavy covariate adjustment at the macro level (prosperity, GDP-PPP, democracy, homicide possibly collinear). The authors’ inclusion of many controls is transparent but potentially overcontrolled: ideological variables (economic/social conservatism) plausibly lie on pathways linking societal context to moral expansiveness, so adjusting for them may underestimate total effects or introduce post-treatment bias. Additional identification threats include omitted cultural response styles (e.g., acquiescence) and translation equivalence; while the authors report measurement invariance checks in the supplement, details are not in the main text, and there is no discussion of response style corrections or anchoring vignettes. The MES task itself forces relative placement into four categories within concentric circles, which can induce reference-dependent judgments and may play differently across cultures; the paper acknowledges that entity-specific analyses (e.g., outgroups vs animals vs environment) could reveal divergent patterns but does not conduct these. Replicability is partly supported by preregistration and released code (OSF), but data are “available upon request,” which reduces reproducibility and slows cumulation; departures from preregISTRATION are noted but details are relegated to supplements. The logic and presentation are generally clear, with appropriate partitioning of within/between effects and transparent acknowledgment of limits; figures and tables are interpretable, and the main claims are characterized cautiously. Overall, the work advances descriptive knowledge on moral expansiveness’ correlates, especially the role of generalized trust and perceived inequality, but causal claims remain tentative, generalizability is limited by convenience student samples, and the between-country inferences are weakened by clustering and measurement choices. Future work using longitudinal or experimental manipulation of perceived inequality/trust, multi-item cross-culturally validated trust measures, entity-level MES analyses, random-slope multilevel (or multilevel SEM with latent constructs), and open data would substantively strengthen the evidential base and policy relevance.

i Gemini 2.0 Flash Assessment

This paper investigates the relationship between societal factors (economic inequality, generalized trust, and anomie) and moral expansiveness across 36 countries, using a large multinational dataset. The central claim is that high inequality will be related to smaller moral circles, mediated by low generalized trust and greater perceptions of anomie. The study uses a multi-level approach, incorporating country-level and individual-level measures, with appropriate controls. Hypotheses were pre-registered and the analysis code placed on OSF. Data were collected from 41 universities across 36 countries. Moral expansiveness was measured using a concentric circles approach, where participants indicated their moral concern for 30 entities. Economic inequality was measured using country-level Gini coefficients, perceived Gini coefficients, and perceived wealth gaps. Generalized trust was measured with a single item, and anomie was measured using a 12-item scale. Control variables included individual-level measures (conservatism, gender, age, social status) and country-level measures (prosperity index, GDP PPP, democracy index, homicide rate). The analysis uses linear mixed models

(LMM) to account for the nested nature of the data. The authors found that more generalized trust and lower perceptions of a breakdown in the social fabric were related to greater moral expansiveness. A greater perceived wealth gap between the rich and the poor was linked to more restricted moral circles, mediated by lower generalized trust. The authors appropriately acknowledge that causality cannot be determined, and that the data may reflect awareness of inequality because one has a large moral circle. They highlight the importance of within-country effects. The discussion acknowledges limitations, including the correlational nature of the study, the focus on overall moral expansiveness, and the use of university samples which may over-represent higher-income countries. The authors also acknowledge that the MES scale may be influenced by relative judgments of moral concern and the limited number of categories.

While the research question is clearly articulated and relevant, there are some weaknesses in the methodology and interpretation. First, the reliance on a single-item measure for generalized trust raises concerns about its reliability and validity. Second, some country-level measures may not accurately reflect the experiences of individuals within those countries. Third, the authors do not investigate the potential endogeneity of the key independent variables. Fourth, as the authors acknowledge, the sample is drawn from university populations and is therefore not likely to be representative of the general population. Fifth, the reliance on self-report measures is problematic given the potential for social desirability bias. Sixth, while the authors appropriately control for relevant covariates, there may be other confounding factors that have not been accounted for. The use of observational data also limits causal inference. Seventh, the use of a single-item measure for generalized trust is not optimal, as it may not capture the full complexity of this construct. Eighth, the findings may be limited by the specific cultural contexts included in the study, and it is unclear whether the results would generalize to other regions or populations. The authors attempt to include a broad range of countries, but the sample is still not fully representative of the global population. Ninth, the effect sizes are generally small, which raises questions about the practical significance of the findings.

In summary, the paper tackles an important research question using a large cross-national dataset. However, the limitations in the measurement of key constructs, potential endogeneity issues, sampling biases, and the correlational nature of the study undermine the strength of the conclusions. The study would benefit from more robust measures, careful consideration of potential confounding factors, and a more nuanced analysis of the relationships between societal factors and moral expansiveness.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.34 Kremer et al. 2020

C.34.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	76 [66–85]	65 [45–80]	80 [64–92] (n=3)

Claims	61 [49–70]	70 [50–85]	—
Methods	47 [35–58]	65 [40–80]	82 [68–94] (n=3)
Adv. Knowledge	80 [70–88]	75 [55–90]	68 [45–70] (n=3)
Logic & Comms	88 [80–94]	80 [65–90]	75 [55–90] (n=3)
Open Science	34 [22–46]	60 [40–75]	70 [30–70] (n=3)
Global Relevance	94 [88–98]	85 [70–95]	82 [55–88] (n=3)
NA	—	—	4 [2–4] (n=3)
NA	—	—	90 [70–100] (n=3)

i GPT-5 Pro Assessment

This is a concise, policy-oriented synthesis of advance market commitments (AMCs), combining conceptual arguments, selective quantitative descriptors of the pneumococcal pilot, and design recommendations. The paper clearly distinguishes “technologically distant” versus “technologically close” targets, motivating AMCs as a tool to mitigate both static pricing distortions and dynamic underinvestment driven by holdup and limited low-income purchasing power. It highlights a specific design insight—tying AMC payouts to capacity commitments rather than purely per-dose subsidies—to counter the risk that a monopolist with limited capacity captures funds without expanding supply. However, the formal modeling that underpins several key design claims is not developed within this text and is instead deferred to an external working paper and an online appendix, limiting verifiability and making it difficult for readers to assess assumptions, solution concepts, or robustness to alternative modeling choices (e.g., the implications of multiple potential suppliers, bargaining structures, or tender formats).

On empirical and evaluative claims, the paper provides useful program facts (e.g., an initial \$1.5 billion commitment; a \$3.50 per-dose cap that subsequently declined to \$2.90; entry of a third supplier reportedly at \$2; distribution in 60 of 73 eligible countries by 2016; annual distribution exceeding 160 million doses; near half of the target child population covered by 2018; and a delayed, partial adoption in India). It repeats the widely cited estimate of roughly 700,000 lives saved through pneumococcal conjugate vaccine (PCV) use and cites a cost-per-DALY of \$83 at initial prices from prior work. Importantly, the authors explicitly acknowledge the absence of a credible counterfactual for the AMC itself: there is no identification strategy for how much the AMC changed the timing, price trajectory, capacity expansion, or coverage relative to plausible alternatives (e.g., push funding, standard pooled procurement, or different tendering rules). The comparison to rotavirus vaccine roll-out—suggesting PCV coverage converged to global rates about five years faster and experienced less severe shortages—is informative but descriptive and confounded. Differences across diseases in regulatory timelines, clinical complexity, manufacturing scale-up challenges, cold-chain requirements, global demand patterns, and pre-commitment landscapes could all drive these patterns. Without design-based causal inference or structural identification that isolates the AMC’s marginal contribution, conclusions about the AMC’s impact on capacity and coverage remain suggestive rather than definitive.

The pricing/co-payment discussion is analytically interesting but rests on calibrations not shown here. The authors argue that with asymmetric losses (lives forgone if capacity or country uptake fails versus comparatively small donor savings from tougher price/co-pay settings), it can be optimal ex ante to set higher-than-expected marginal cost prices and lower-than-

expected country co-pays. They report that under a specific objective (health benefits maximization with savings redeployed to other highly cost-effective health uses), a \$3.50 cap could dominate a \$2 price even if the lower price would deter participation only 4 percent of the time; they also note sensitivity to assumptions about alternative uses of savings. This line of reasoning is conceptually sound, but the paper does not provide parameter values, code, or sensitivity tables in the main text, and the online appendix is not included here; reproducibility and independent scrutiny are therefore constrained. Similarly, the claim that holding back tenders may have facilitated entry by the Serum Institute is plausible but counterfactual and confounded by substantial push funding and evolving technological readiness; no structured analysis is provided.

The theoretical recommendation to tie subsidies to capacity is grounded in a holdup/bargaining logic and echoes standard mechanism-design intuitions. The pilot's implementation—awarding a fixed share of the AMC fund tied to committed annual supply—aligns with this recommendation. Yet the short format precludes exploration of edge cases: e.g., how capacity commitments interact with tender competition, renegotiation risk if realized demand or costs differ materially from expectations, or strategic misreporting of producible capacity. Likewise, the multi-firm versus single-firm design trade-off is laid out clearly but qualitatively; there is no model-based welfare comparison under realistic market structures (e.g., dynamic entry with capacity investment frictions, common shocks to input costs, or correlated supply failures). The political economy section helpfully notes incentives favoring technologically close targets and potential specification lobbying but remains conceptual, without empirical evidence on lobbying intensity, its influence on target selection, or downstream welfare effects.

Data and sourcing are transparent at a high level but thin for replication. Many core facts come from the GAVI AMC Secretariat and WHO/CDC price lists; figures and comparative price discounts appear in an online appendix not included here, and no data or code are provided. There is no formal statistical analysis in the main text, and the paper relies on external studies for cost-effectiveness. Internal consistency of the numbers presented is reasonable, and the narrative is careful not to over-claim causality; indeed, the authors repeatedly note evaluation challenges. Nonetheless, some sentences risk being read as attributing aggregate health gains to the AMC rather than to vaccine use more broadly (e.g., “saved an estimated 700,000 lives”) despite earlier caveats. The disclosure that the authors participated in the pilot's expert group is appropriate; however, potential confirmation or design-justification bias cannot be ruled out, especially where evaluative claims rely on internal program sources.

Communication quality is a strong point: definitions are clear, the theory-to-design logic is well signposted, and policy-relevant trade-offs (price/co-pay levels, supplier numbers, eligibility, timing of tenders) are succinctly presented alongside realistic constraints. For practitioners, the paper offers actionable design heuristics (e.g., err on the side of higher donor top-ups and lower co-pays under asymmetric losses; consider capacity-linked commitments; weigh benefits of multiple suppliers for long-run prices and reliability). However, for researchers or decision-makers seeking quantified estimates of the AMC's marginal impact, the piece falls short: there is no causal estimate of acceleration, no structural calibration of supplier entry/capacity decisions under alternative mechanisms, and no counterfactual budget-impact analysis comparing AMCs to other procurement or R&D-push designs. Making code and data for the calibrations and descriptive figures public, pre-specifying an evaluation design, and conducting robustness checks (e.g., placebo comparisons using other vaccines, controlling for

supply/demand confounders, triangulating from firm-level capacity investment data) would materially strengthen credibility and utility for meta-analysis and future AMC design. In sum, the article excels as a lucid, experience-informed conceptual review with plausible design guidance and high global relevance, but it provides limited causal evidence and reproducible analysis of the AMC's incremental impact.

i Gemini 2.0 Flash Assessment

This paper provides a concise review of advance market commitments (AMCs) as a tool to incentivize vaccine development and distribution, focusing on the pneumococcal vaccine pilot program. It effectively summarizes the economic theory behind AMCs, highlights the practical experience gained from the pilot program, and discusses key design issues for future AMCs. The paper addresses both static and dynamic inefficiencies in vaccine markets, emphasizing the importance of incentivizing R&D and ensuring adequate production capacity. The analysis incorporates insights from a theoretical model and empirical observations from the pneumococcal vaccine initiative. Several factors contribute to uncertainty in the evaluation. First, counterfactuals are inherently difficult to establish in this context; judging what would have happened without the AMC is speculative. Second, the paper acknowledges trade-offs and complexities in AMC design. Third, some of the analysis relies on specific assumptions (e.g., regarding alternative uses of savings). Fourth, the paper identifies important but difficult-to-quantify factors, such as the role of political considerations in target selection and the potential for industry lobbying to influence product specifications.

Some potential limitations and areas for improvement include:

1. **Limited Empirical Evidence:** While the paper references some statistics on the pneumococcal vaccine rollout, it primarily focuses on theoretical arguments and design considerations. The paper argues that “the high cost effectiveness of PCV implies that the AMC would have been worthwhile were there even a small chance that it sped up PCV adoption.” This is a weak argument that may not be supported by evidence.
2. **Counterfactual Challenges:** The authors acknowledge the difficulty of establishing a valid counterfactual to assess the cost-effectiveness of the AMC. However, further exploration of alternative scenarios and sensitivity analyses could strengthen the analysis.
3. **Generalizability:** The paper draws heavily on the pneumococcal vaccine experience. While valuable, the insights may not be directly transferable to all other vaccines or health interventions, particularly those targeting different diseases or populations.
4. **Limited Discussion of Equity:** While the paper touches on the importance of country co-payments and equitable access, it could further explore the distributional impacts of AMCs and potential strategies to address inequities in vaccine access.
5. **Data limitations:** Given that the evaluation depends on comparing against a counterfactual, this is inherently difficult, and I would have liked to see more about the data limitations.
6. **External Validity:** Given the focus on a specific vaccine (pneumococcal), it is unclear how well these results generalize to other types of vaccines or health interventions.

7. **Robustness:** The paper could benefit from a more detailed discussion of the robustness of the findings to different assumptions and parameter values, particularly regarding the cost-effectiveness calculations and the impact of co-payments.
8. **Replication and transparency:** This paper is a short overview; the analysis depends substantially on other papers, especially Kremer, Levin, and Snyder (2019). Because of this, I find it difficult to assess the analysis or robustness in detail.

Overall, the paper provides a valuable overview of AMCs and their application to vaccine development and distribution. It highlights the theoretical underpinnings, practical considerations, and ongoing challenges in designing and implementing effective AMCs. Addressing the limitations mentioned above could further enhance the paper’s rigor, generalizability, and policy relevance. More work could be done to better defend the analysis of costs/benefits.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.35 Kremer et al. 2022

C.35.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	90 [82–96]	60 [45–75]	68 [58–78] (n=2)
Claims	88 [80–94]	65 [50–80]	—
Methods	85 [76–92]	60 [45–75]	62 [52–72] (n=2)
Adv. Knowledge	92 [85–98]	70 [55–85]	72 [58–88] (n=2)
Logic & Comms	87 [80–93]	75 [60–90]	72 [60–85] (n=2)
Open Science	88 [75–95]	80 [65–95]	62 [50–75] (n=2)
Global Relevance	97 [92–99]	85 [70–95]	68 [58–78] (n=2)
NA	—	—	4 [3–5] (n=2)
NA	—	—	3 [3–4] (n=2)
NA	—	—	35 [12–58] (n=2)

i GPT-5 Pro Assessment

This manuscript presents a systematic review and meta-analysis of randomized controlled trials (RCTs) of water quality interventions in low- and middle-income countries, focusing on all-cause under-5 mortality and associated cost-effectiveness. The central empirical contribution is the aggregation of mortality data—often underpowered or unreported in individual trials—across 18 studies (27,316 participants, 518 total deaths) drawn from 52 eligible RCTs identified via a transparent search protocol that extends and replicates prior reviews. The authors obtained mortality data by contacting study teams when not publicly reported, which

is a major strength for statistical power but raises selection concerns: mortality availability depends on whether it was collected and preserved, and author responsiveness. The paper uses several diagnostic checks (Egger's and Andrews-Kasy tests, association between diarrhea effects and mortality data availability, and simulations adding hypothetical null short studies) and finds no significant evidence of publication bias, while candidly acknowledging low power of formal tests given the small number of mortality studies. The "missing data not at random" risk remains the single most important limitation, although the simulations suggest the main result is unlikely to vanish unless a large number of sizeable null studies were missed. Methodologically, the meta-analysis is carefully executed, combining frequentist and Bayesian random-effects models on log-odds-ratios, with robustness to fixed-effects formulations. The frequentist REML estimate of heterogeneity snaps to zero (a known tendency with few studies), whereas the Bayesian model infers moderate but imprecise between-study heterogeneity (log-OR SD 0.24; wide CrI), appropriately carried into posterior predictive effects used for cost-effectiveness. The per-study effect extraction is mostly rigorous: when individual-level data (ILD) were available, logistic mixed models accounted for clustering; when ILD were unavailable for clustered designs, the analysis adjusts effective sample sizes using an ICC derived from the subset with ILD (1.8%). This ICC imputation is reasonable but imperfect; ICCs can vary by context and endpoint, and the reliance on a single pooled ICC introduces measurement error into standard errors and weights for three studies. The authors mitigate rare-event estimation issues (zero cells) by using Bayesian per-study models and, in sensitivity checks, continuity corrections; both approaches yield similar pooled results, which increases confidence in the find of approximately 25–28% lower odds of all-cause under-5 mortality. A second key limitation is heterogeneity in follow-up durations and child age distributions across trials. The analysis interprets study-specific odds as under-5 mortality odds, and it shows this introduces negligible bias from compounding in short-vs-long trials, but this choice sacrifices the precision available from time-to-event methods. The authors propose future survival analyses (age-specific hazards) for the subset with ILD; doing so would address differential exposure time and test for age-gradient heterogeneity, which is particularly relevant to neonatal and infant mortality risks. The meta-regressions explore moderators (diarrhea effects, compliance, baseline diarrhea prevalence, year, intervention type, unit of randomization), but power is very limited, as the authors demonstrate with simulation. As a result, the analysis cannot credibly characterize effect moderators or trends over time (e.g., changing vaccine coverage or ORT) beyond establishing a positive correlation with diarrhea effects. Compliance is measured very heterogeneously across trials (different thresholds, different sampling protocols, sometimes announced vs unannounced), which undermines clean comparisons and any effort to model effects as proportional to compliance. The cost-effectiveness analysis is clear, transparent about assumptions, and intentionally conservative in counting only child mortality benefits (excluding morbidity and adult benefits). The main modeling step scales the posterior-predictive effect by the ratio of program-specific effective take-up to the meta-analysis average compliance; this proportionality assumption is plausible as a first pass but is untested and could misstate benefits if dose-response is non-linear or if compliance proxies differ in construct validity across contexts. Cost inputs are program-specific (e.g., dispenser costs from a large implementer in Kenya; inline chlorination costs from two Indian states with planned scale-up; MCH coupons from randomized evaluations), appropriately documented, and varied by program type. However, cost-effectiveness inevitably depends on implementation details: compliance trajectories over time (which may decay), baseline use of chlorination

in controls, and local under-5 mortality rates. The paper partially addresses this by using context-specific U5MR and by presenting threshold analyses (cost per DALY as a function of OR), but would benefit from more quantitative uncertainty propagation (e.g., probabilistic sensitivity analysis over costs and take-up) in the main text rather than only scenario-based checks. The comparison to policy thresholds uses older WHO-CHOICE guidance (1x–3x GDP per capita per DALY), which many now consider outdated; the authors also reference DCP3’s more stringent guidance and provide results that remain robust even under a \$200 per DALY threshold. One wording issue: the paper at times calls 1x GDP per capita a “value of statistical life” proxy for net benefit calculations, but the calculation used is actually “value per DALY” (GDP per capita per DALY), not a VSL; clarifying this terminology would reduce confusion. On identification, most included trials were unblinded, but mortality ascertainment is plausibly less prone to reporting bias than caregiver-reported diarrhea. Two included studies combine water treatment with other WASH or cookstove interventions; sensitivity analyses excluding such studies yield similar or stronger effects, suggesting they do not drive the pooled mortality reduction. Control contamination motivated excluding two studies from the primary analysis; including them in sensitivity checks leaves the main result intact. The internal consistency of numbers and the breadth of robustness checks are strong points: leave-one-out analyses, alternative continuity corrections, fixed vs random effects, alternative definitions (e.g., combining two related Kenyan cohorts), and stratifications by follow-up length all show pooled ORs within roughly 0.66–0.81 and significant in most specifications. The communication is generally excellent: the rationale for focusing on mortality (objective outcome) is clear; the tension between the pooled mortality effect and a simple “diarrhea-attributable fraction $\times$ diarrhea reduction” calculation is confronted directly, and plausible mechanisms (Mills–Reincke-type cross-disease effects, prevention of severe disease, better perinatal hygiene) are offered. The code and replication materials are said to be public, which, combined with pre-registration and PRISMA adherence, supports reproducibility; one remaining practical risk is that some ILD obtained by author contact may not be redistributable, but Table S3 provides the study-level event counts used in the meta-analysis, which should suffice to reproduce the pooled results. Overall, the evidence synthesis on mortality is careful and unusually comprehensive for this topic, the main effect estimate is robust to many reasonable choices, and the cost-effectiveness framing is relevant and likely conservative. Remaining uncertainties center on (1) selection into availability of mortality data, (2) unmodeled heterogeneity (age, time, baseline risk), (3) the proportionality assumption between compliance and impact used for CEA, and (4) reliance on thresholds for valuation. Addressing these through survival modeling with ILD, moderator analyses powered by additional studies, and probabilistic CEA would further strengthen an already compelling case.

i Gemini 2.0 Flash Assessment

This paper presents a meta-analysis of the effect of water treatment interventions on child mortality in low- and middle-income countries. The authors systematically reviewed the literature, identified 18 relevant RCTs, and combined the data using frequentist and Bayesian methods to estimate the impact of water treatment on under-5 mortality. They also examined the cost-effectiveness of different water treatment approaches. The study finds a statistically significant reduction in the odds of all-cause under-5 mortality associated with water treatment interventions. The authors estimate that water treatment is a cost-effective health program.

Overall, the paper addresses a relevant and important question, uses appropriate methods, and presents the results clearly. However, there are several limitations and potential weaknesses that warrant attention.

One limitation is the relatively small number of included studies (18), which may limit the statistical power of the meta-analysis. While the authors conducted power simulations to address this issue, the power to detect heterogeneous effects might be low. Also, there is potential publication bias, as the authors acknowledge. Although they performed tests for publication bias and adjusted for it using the Andrews and Kasy method, the tests have limited power due to the small number of studies. In addition, the authors included only studies for which authors reported that mortality data were collected and remained available, which could introduce selection bias. Another limitation is the heterogeneity in the interventions, study designs, and populations across the included studies. The authors used random-effects models to account for this heterogeneity, but there remains a risk that the combined effect estimate is not representative of any specific intervention or context. Also, there's substantial heterogeneity in the follow-up periods of the included studies. The authors address this in a sensitivity analysis by excluding studies shorter than a number of weeks.

Additionally, the generalizability of the cost-effectiveness analysis is limited by the assumptions and data used. The cost and effectiveness estimates are based on specific contexts and may not be applicable to other settings. The compliance data also come from specific implementations and may vary across contexts or change over time. The analysis only considers the benefits of reduced child mortality and does not include other potential health gains or economic benefits. It makes strong linearity assumptions (see section 7, comparing meta-analysis estimates with model predictions; these assumptions seem unlikely to be correct). Additionally, in many cases the mortality data were collected as a secondary outcome, as part of internal respondent tracking systems, or for IRB reporting purposes by the authors. As such, it is potentially subject to reporting bias.

Finally, while the authors used both frequentist and Bayesian methods, they should have better justified the choice of priors in the Bayesian models, as the results could be sensitive to the prior specification. Despite the use of various sensitivity analyses, the interpretation of the results should be cautious, considering the limitations and potential biases.

In summary, the paper provides a valuable contribution to the literature by synthesizing evidence on the effectiveness of water treatment interventions on child mortality and assessing the cost-effectiveness of different approaches. However, the limitations regarding statistical power, potential for publication bias, heterogeneity, data quality, and generalizability should be considered when interpreting the results and drawing conclusions.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.36 Kubo et al. 2023

C.36.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash
Overall	71 [60–82]	65 [45–80]
Claims	67 [55–78]	70 [50–85]
Methods	65 [52–77]	75 [60–85]
Adv. Knowledge	68 [57–80]	70 [50–85]
Logic & Comms	76 [66–86]	75 [65–85]
Open Science	60 [45–75]	80 [70–90]
Global Relevance	72 [60–84]	75 [60–85]

i GPT-5 Pro Assessment

This paper investigates unintended spillovers from a February 2020 Japanese ban on commercial capture/trade of three threatened species (giant water bug, Tokyo salamander, golden venus chub) using 10 years of online auction data (Yahoo! Auction via Aucfan). The authors apply a synthetic difference-in-differences (SDID) design with a two-step residualization to control for species, year, and seasonal (quarter) fixed effects. They report significant increases in sales volumes of non-banned “substitute” species within each taxon (water bugs: +17.54 per quarter, 95% CI [14.03, 21.06]; salamanders: +10.06 [2.73, 17.39]; freshwater fish: +6.19 [0.12, 12.25]). The study provides timely evidence that trade bans can shift demand to unregulated species; it also offers multiple robustness checks (manual species-only control sets; monthly aggregation; inclusion of the announcement quarter; and placebo tests). Nonetheless, several identification and measurement issues limit causal interpretation strength and policy generality.

Most consequential for identification is the construction of “treated” (substitute) species and the composition of controls. Treated units (three per taxon) were selected using a three-step procedure: (i) same Japanese common name but legally tradable; (ii) Google Trends related queries; and (iii) phylogenetically closest species where (i)-(ii) yielded no candidates. While pragmatic, these criteria may not align with consumer substitution in this market. The paper acknowledges this and argues (Note S2) that misassignment of treated vs control species likely biases estimates downward. However, that illustrative argument assumes homogeneous treatment effects and a simple DID; under SDID weighting and heterogeneous responses, contamination could bias estimates in either direction depending on how weights load on partially affected controls and on true cross-species demand shifts. Evidence of heterogeneity in the single-species robustness (Table S4)—including a significant negative effect for *Hynobius nigrescens* in the manual dataset—underscores the need to probe substitution patterns more directly (e.g., using price responses, cross-elasticities, or search data beyond a related-query screen). A more formal pre-analysis selection protocol for substitutes (e.g., discrete choice or text-embedding similarity trained on pre-ban co-mentions and co-listing behaviors) would strengthen confidence.

Control units are defined as all other trades in the same auction category (e.g., amphibian, fish) excluding the selected substitute species. This has advantages (rich donor pool; same platform and category exposure) but risks contamination: other unlisted but plausible substitutes may sit in the control pool and also respond to the ban and/or pandemic, attenuating estimated effects. The manual control subset (species with scientific names in the same families) partially addresses this and yields qualitatively similar signs, but with notable variation and wider inter-

vals. Additional designs could help: (i) narrowing controls to close but non-candidate species with demonstrated non-substitutability (e.g., pre-ban cross-price elasticity = 0); (ii) using sub-category tags or learned text similarities to better align product types; (iii) excluding obvious non-animal items if present in categories. Presenting pre-treatment fit metrics (e.g., RMSPE) and placebo-in-time tests would further bolster validity beyond the control-contribution plots already provided.

Timing is a second major challenge. The post-period spans a single year (Feb 10, 2020–Feb 9, 2021), overlapping entirely with COVID-19. The two-step residualization with year and quarter fixed effects plus SDID weighting help control for common shocks and seasonal patterns, but this cannot rule out differential pandemic effects within categories (e.g., shifts toward home-based hobbies, logistic disruptions, or platform-specific policy changes) that correlate with the treated species. Monthly models (Table S5) help with power and dynamics but still reflect pandemic-era heterogeneity; freshwater fish effects there are not significant in some specifications. An event-study presentation of dynamic effects and pre-trends (using monthly data) and a triple-difference (e.g., comparing to similar categories not plausibly affected by the ban) would help separate ban-specific spillover from broader contemporaneous shocks.

Outcome measurement relies on counts of completed auction transactions per species. This is a reasonable proxy for demand on this major platform, but several points warrant clarification: (i) whether listings commonly bundle multiple individuals (and, if so, how volumes were normalized); (ii) how re-listed/unsold items were handled (the dataset includes final prices, implying sold listings, but explicit confirmation would help); (iii) whether supply constraints or platform moderation changed around the ban. The paper provides price series descriptively (Fig. 1–2) but does not estimate causal effects on prices for substitute species. Given standard substitution theory, parallel evidence of price increases for non-banned species would substantially strengthen the demand-shift interpretation; conversely, flat or falling prices with higher volumes would point to supply responses and/or compositional changes (e.g., shifts toward captive-bred stock). Treating no-trade quarters as price zero in the plots (rather than missing) is not ideal, though this does not affect the main SDID on volumes.

Species identification combines NLP on listing titles and site categories with manual confirmation for banned and treated species. For controls, many units in the main analysis are not identified to scientific names; the manual-only control robustness (smaller sample) is valuable but should be accompanied by precision/recall diagnostics for the NLP pipeline, dictionaries used, handling of code words, and examples of ambiguous terms. Because substitution may be mediated by colloquial naming and coded jargon, a more detailed taxonomy of name variants would aid replication and interpretation. The noted taxonomic revisions in salamanders are handled by aggregating to older taxa; this is sensible but should be explicitly linked to how titles naming conventions changed over time.

Inference and robustness are generally careful given constraints. Excluding the announcement quarter (2019Q4) is appropriate and sensitivity with/without it is reported. Placebo tests (100 rounds per setting) are near-nominal 5% rejection in most cases, though the salamander NLP dataset shows 10/100 rejections, suggesting some mild size distortion or idiosyncratic volatility; addressing this with block/permutation standard errors or alternative variance estimators would be useful. Reporting pre-treatment fit, weight distributions, and goodness-of-fit statistics would further increase transparency.

External validity and policy interpretation merit caution. The study is restricted to one large

platform in one country and three taxa; it cannot resolve whether increased online auction activity reflects net increases in total exploitation versus displacement from other channels or shifts toward captive breeding. The discussion appropriately frames risks (overharvesting, invasive species, disease spillovers), but several claims extend beyond the data (e.g., cross-border effects, feedback loops to future bans). These could be flagged more clearly as hypotheses for future work. The core contribution—evidence that a ban can stimulate trade in non-banned substitutes in the same taxon on a major online platform—is credible and important; stronger tests of mechanisms (price, seller entry/exit, geography of listings) and consumer behavior would increase policy relevance.

Open science and reproducibility are partly satisfactory. Methods are described, R packages are specified, and the SI indicates that data and code will be released on GitHub upon publication. At present (as per the PDF), replication would require access to Aucfan and the authors' parsing dictionaries and scripts; including these materials now, with detailed documentation (keyword lists, code for quarterly aggregation, species mapping tables, and checks for duplicates), would reduce replication frictions.

Overall, the paper tackles a relevant question with a sensible identification strategy and substantial data work. The main results are plausible and reasonably robust, but the short post-period, substitute/control construction, and limited mechanism testing constrain causal strength and generality. With additional analyses on prices, dynamic effects, substitute selection, and expanded transparency of the NLP pipeline, this would be a strong contribution to evidence-based wildlife trade policy evaluation.

Gemini 2.0 Flash Assessment

This paper investigates the unintended spillover effects of wildlife trade bans on non-banned, substitutable species using a synthetic difference-in-differences (SDID) approach. The study focuses on three threatened species in Japan (giant water bugs, Tokyo salamanders, and golden venus chub) and uses 10 years of online auction data to assess the impact of trade bans on related, non-banned species. The core finding is that trade bans can lead to an increase in the trade volume of non-banned species within the same taxa, suggesting that demand shifts to substitutes, which the authors term 'conservation side effects.'

The study uses a rigorous methodology, employing SDID to address potential biases and confounding factors, such as seasonality and the COVID-19 pandemic. They identify potential spillover species using multiple criteria, including shared Japanese names, Google Trends data, and phylogenetic proximity. Robustness checks, including sensitivity analyses and placebo tests, support the main findings.

Several potential issues affect the interpretation and generalizability of the results:

1. **Data limitations:** The analysis relies exclusively on online auction data from Yahoo! Auctions Japan. This may not represent the entire wildlife trade, especially if a significant portion occurs offline or through other online platforms. The generalizability of the findings is therefore limited by the scope of the data.
2. **Substitutability assessment:** The criteria for determining substitutability between banned and non-banned species (shared names, Google Trends, phylogenetic proximity) are somewhat subjective and may not perfectly capture consumer preferences or ecological relationships. The authors acknowledge this limitation, but a more direct measure

of substitutability (e.g., consumer surveys) would strengthen the analysis.

3. **Limited Post-Ban Period:** The post-ban period is only one year (February 2020 to February 2021). This short timeframe may not be sufficient to fully capture the long-term effects of the trade ban or to distinguish between short-term fluctuations and sustained trends.
4. **Control Group Composition:** While the SDID method helps address some issues, the construction of the control group raises concerns. The study includes species within the same categories on the auction website but excludes the potential spillover species. This method may not fully account for all relevant factors influencing trade volume.
5. **Scope of Spillover Effects:** The study primarily focuses on spillover effects within the same taxon. It does not explore potential spillover effects on species in different taxa or on the broader ecosystem. This narrow focus may underestimate the full consequences of the trade bans.
6. **Policy Implications:** While the paper suggests policy implications such as consumer research, monitoring, and international cooperation, it does not provide specific recommendations or implementation strategies. The discussion could be strengthened by offering more concrete guidance for policymakers.

In summary, the paper identifies a relevant problem and provides preliminary evidence of spillover effects associated with wildlife trade bans. However, the limitations related to data scope, substitutability assessment, short time frame, control group selection, and lack of broader consideration of spillover effects temper the strength of the conclusions. Further research with more comprehensive data and longer time horizons is needed to confirm and extend these findings.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.37 Kudymowa et al. 2023

C.37.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash
Overall	72 [60–82]	75 [60–85]
Claims	65 [50–78]	80 [70–90]
Methods	62 [45–75]	70 [55–80]
Adv. Knowledge	70 [55–83]	70 [55–80]
Logic & Comms	80 [70–90]	85 [75–95]
Open Science	60 [45–75]	70 [50–80]
Global Relevance	78 [65–88]	80 [70–90]

This commissioned review aims to assess and refine the construction of GiveWell’s discount rate(s) for global health and development CEAs. It is well-structured, transparent about uncertainties, and engages seriously with both theory (SRTP vs SOC, Ramsey rule) and practice (institutional surveys and parameter choices). The authors’ central recommendations—retain an SRTP framework, use an explicit wealth-effect formula adapted to percent-based consumption modeling [$r = \rho + (\gamma - 1)g$], update g to 3.3% (implying a 2.0% wealth effect), keep $\rho = 0\%$ for pure time preference, tentatively retain 1.4% for temporal uncertainty, and keep 0.9% for compounding non-monetary benefits while increasing reasoning transparency—are plausible and generally well justified, though several are explicitly tentative. They appropriately flag confidence levels across sections and identify key open questions (e.g., discounting health outcomes, internal consistency in γ). Still, there are notable issues and limitations that constrain the strength and immediate implementability of the recommendations. Most importantly, the paper highlights but leaves unresolved a core inconsistency: GiveWell’s use of log-utility ($\gamma = 1$) for modeling income benefits versus $\gamma = 1.59$ for deriving the wealth effect used in the discount rate. The authors correctly note this inconsistency risks over-discounting and recommend adopting a consistent γ ; however, they defer the choice of γ as out of scope, even though this choice potentially shifts the overall rate markedly (e.g., to $\sim 2.3\%$ if $\gamma = 1$). This leaves a principal driver of the rate unsettled. The adaptation from the standard Ramsey wealth effect (γg) to $(\gamma - 1)g$ is a useful insight given GiveWell’s percent-change modeling of consumption. The authors provide an intuitive derivation and a more formal sketch (Appendix E), including a small spreadsheet correction for the wealth effect (using $r = UC0/UC1 - 1$ rather than $1 - UC1/UC0$). However, because this involves discounting consumption growth (log consumption) rather than consumption, clarity on units is critical, and the manuscript could go further in auditing all downstream interactions with CEAs to ensure this unit choice does not inadvertently mix frameworks (e.g., how to combine this with temporal uncertainty and compounding benefits treated as components of ρ in a way that remains consistent with economic welfare theory). The decision to treat “compounding non-monetary benefits” as part of ρ (in a Ramsey lens) is unconventional; it risks conceptual double counting (e.g., growth in valued outcomes vs separate compounding channels) and is only weakly justified. The paper acknowledges the thin basis for the current 0.9% figure and keeps it mainly due to lack of strong counterarguments, explicitly calling for greater transparency. This is a sensible stopgap but not a robust result; further work should clarify whether this belongs in the discount rate, in modeled benefits, or via an SPC-like adjustment, and quantify it with empirical or structural evidence. The consumption growth recommendation ($g = 3.3\%$) relies on IMF near-term GDP per capita projections, Goldman Sachs long-term growth projections, and UN population forecasts, with a population-weighted SSA+SA composite. The method is straightforward and generally appropriate, but one finding—that the composite CAGR is lower than each component’s CAGR over the 2026–2066 window—remains unexplained; the authors openly flag this, concede uncertainty, but “stand by” the result. Given its centrality, this anomaly should be resolved or robustly stress-tested (e.g., alternative weightings, base-year normalization, decomposition explaining cross-term effects) before adoption. The proposed update cadence (a 5.2-year “period” based on an expected -0.1 pp change in the discount rate) appears over-precise relative to underlying forecast uncertainty and may encourage false precision. More defensible would be a simpler schedule (e.g., quinquennial) anchored to substantial revisions in long-run growth outlooks. The discussion of temporal uncertainty is

concise and candid about limited depth; it distinguishes GiveWell's broader definition (beyond extinction risk) from the narrower norm in public guidelines. The 1.4% figure is retained on the grounds of rough alignment with a wide range of existential/catastrophic risk estimates (0.1–2.3%), but the paper doesn't decompose risks relevant to specific intervention contexts or quantify hazard rates rigorously. As such, the recommendation is more a “do no harm for now” stance than a principled estimate. The treatment of health discounting is balanced and tracks the literature's move away from equal discounting of costs and health outcomes; the authors argue that health should likely be discounted less than consumption, yet, due to dependence on utility function shape and other modeling choices, they stick with GiveWell's current practice (only temporal uncertainty) and avoid making a numeric recommendation. While cautious, this limits practical traction; a parametric range or decision framework (e.g., under separability, cross-elasticities, or empirically plausible vH growth) would add more value. The institutional scan is helpful but clearly non-exhaustive; conclusions that other organizations' practices do not offer lessons for GiveWell reflect real context differences but also the limited sample and frequent deference in the field. The paper's strongest elements are: (i) aligning the discounting approach (SRTP) with GiveWell's welfare-maximizing, cross-intervention comparison goals and rebutting common objections (e.g., crowd-out) in this philanthropic context; (ii) clarifying unit consistency and providing a tailored wealth-effect formula; (iii) explicit error-spotting and parameterized recommendations with stated confidence; and (iv) broad relevance to donors and evaluators working in LMIC contexts. Main weaknesses include: (a) leaving the inconsistency unresolved; (b) tentative justification for compounding non-monetary benefits and broad temporal uncertainty; (c) an unexplained anomaly in the growth-rate composite and arguably over-precise update rules; and (d) limited progress on health discounting beyond high-level endorsement of differential discounting. Replicability is fair-to-moderate: the methods depend on accessible forecasts and transparent equations, but spreadsheets and code are not fully embedded in the report, and some steps (e.g., interpolation choices, composite construction) would benefit from open, auditable code. In sum, this is a careful and practically useful review that improves clarity and fixes a few issues, but some pivotal choices remain underdetermined and several recommendations are best seen as provisional pending deeper analysis and modeling. The work is likely to inform practice for GiveWell and similar evaluators, but users should be aware of the noted uncertainties and the need for further empirical or structural support on key components.

i Gemini 2.0 Flash Assessment

This paper provides a detailed review of GiveWell's current approach to calculating discount rates, with the aim of providing recommendations for improvement and reinforcing justifications for existing practices. The analysis covers theoretical approaches to discounting (SRTP and SOC), institutional approaches, and a component-by-component review of GiveWell's discount rate (consumption growth, pure time preference, temporal uncertainty, compounding non-monetary benefits). The report is well-structured and clearly written, presenting a balanced view of different perspectives and acknowledging areas of uncertainty. The authors' recommendations are generally cautious, often suggesting that GiveWell maintain its current approach but with some refinements, such as using a more explicit formula for calculating the wealth effect and increasing the consumption growth rate estimate. The study acknowledges limitations, such as the shallow engagement with the literature of pure time preference, and

it expresses a desire to see further research to support the recommendations. One notable issue is the inconsistent use of utility functions for consumption, with different values of () assumed in different parts of the calculations; the authors recommend reconsidering this inconsistency. The appendices provide supplementary information, including a history of discount rates in global health research and derivations of key equations. The report is commissioned by GiveWell but maintains an independent perspective. The overall assessment is that this is a thoughtful and well-researched review, providing valuable insights for GiveWell and other organizations concerned with social discounting, but I have some uncertainty in my assessment because I am not an expert in social discounting.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.38 Liang et al 2023

C.38.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash
Overall	86 [76–93]	65 [50–80]
Claims	78 [68–88]	75 [60–85]
Methods	80 [70–89]	65 [50–75]
Adv. Knowledge	84 [74–92]	70 [55–80]
Logic & Comms	87 [80–94]	80 [70–90]
Open Science	65 [55–75]	70 [50–80]
Global Relevance	82 [72–90]	70 [55–80]

i GPT-5 Pro Assessment

This manuscript assembles a large panel of ecological assemblage time series (BioTIME) from 1960–2015 across >15,000 U.S. sampling locations to quantify how economic production correlates with and causally affects biodiversity outcomes. It defines biodiversity via abundance, species richness, and interannual compositional stability (Jaccard; with checks using Shannon, Gini, Sørensen). The paper documents negative within-site associations between state GDP per capita and biodiversity, explores heterogeneity by taxa and distributional quantiles, and then moves to identification: (i) a shift–share style quasi-experiment leveraging heterogeneous state exposure to national military procurement buildups; (ii) an instrumental variables strategy for the pollution channel using upwind transported pollution constructed from MERRA-2 AOD, wind direction, and LASSO-based selection of upwind counties beyond 300 km; and (iii) a policy shock analysis using Clean Air Act (CAA) nonattainment designations. A decomposition suggests that 20–60% of the biodiversity impacts from production shocks are transmitted through air pollution. There is additional correlational evidence that urbanization (MODIS land cover) is negatively associated with biodiversity and that protected areas may attenuate

the GDP–biodiversity gradient, especially when less fragmented.

Key strengths and contributions: (1) A rare, macro-scale perspective linking production to biodiversity across multiple taxa with long temporal coverage; (2) transparent construction of biodiversity metrics from consistent within-study protocols; (3) multiple complementary identification strategies (military shift–share; upwind-pollution IV; regulatory shocks), with broadly consistent signs and magnitudes and extensive fixed effects; (4) careful discussion of external validity limits, nonrepresentativeness, and cross-checks across taxa (notably non-bird subsamples); (5) informative sensitivity checks (winsorization, exclusion of large deviations, vintage/duration splits, alternative biodiversity indices, spatial binning and alternative economic units, dynamic specifications and panel VAR); and (6) policy relevance via CAA results and an explicit mechanism for air pollution.

Most important issues and identification threats: - Representativeness and detection/effort biases in ecological sampling: Although BioTIME enforces fixed within-study protocols, nonrandom site selection and taxonomic coverage remain. Detection probability can vary over time with economic activity (e.g., increased anthropogenic noise in higher GDP years can depress detectability in auditory surveys such as BBS; observer composition may change). The paper partially mitigates this via location×taxon fixed effects, non-bird analyses (which often rely less on auditory detection), and sampling-vintage/duration robustness. Still, explicit checks for time-varying detectability correlated with economic activity (e.g., using site-level noise proxies, road density, traffic intensity, observer fixed effects where possible, or BBS metadata on noise/observer identity) would strengthen internal validity. The Jaccard-based turnover metric could also be sensitive to varying detection; additional robustness using occupancy or detection-adjusted estimators where available would be valuable. - Spatial scale mismatch and measurement error in economic exposure: State GDP per capita is coarse relative to site-level ecological sampling, risking attenuation and spatial confounding. The paper’s robustness with county income and ecoregion aggregates, and spatial binning, is helpful (effects attenuate but persist), yet more localized sectoral production proxies (e.g., county manufacturing output, plant openings/closures, energy generation) could refine exposure and reduce concerns that state-level shocks capture heterogeneous local conditions. - Military shift–share identification: The design relies on exogeneity of national buildups driven by geopolitics and differential state exposure captured by initial-period defense shares. This is standard, but the exclusion restriction requires that states’ initial defense intensity interacts with national buildups only through production effects relevant to biodiversity, not via concurrent unobservables that differentially trend with biodiversity. The first stage is borderline in the full sample (K-P F 7) but strong in the non-bird subsample (F 35), which suggests caution in interpreting 2SLS magnitudes for the full sample. Added robustness with state-specific linear trends (beyond Census division trends) and placebo tests (e.g., pre-trends around large buildups) would further guard against differential underlying biodiversity trajectories correlated with defense exposure. - Pollution IV construction and exclusion: The upwind IV combines MERRA-2 AOD with wind geometry and LASSO selection of contributing upwind counties beyond 300 km, yielding strong first stages. However, the exclusion restriction could be compromised if upwind regions share macro-environmental or climatic shocks that independently affect downwind biodiversity (e.g., regional drought, temperature anomalies) or if long-range transported aerosols co-move with other regional factors. The paper includes unit and year fixed effects but does not report controls for time-varying local meteorology (temperature, precipitation, humidity), which are known to affect both AOD and ecological dynamics. Including me-

meteorological controls or conditioning on reanalysis meteorology, back-trajectory diagnostics (e.g., HYSPLIT), and alternative pollutant measures (e.g., TROPOMI NO₂, ground PM_{2.5} where available) would increase credibility. Additionally, the aggregation from daily to annual county-level pollution could mask short but ecologically relevant exposures; reporting how annualization is done and checks with seasonal windows would help.

- CAA nonattainment designations: Nonattainment is determined by monitored concentrations relative to standards and is not random. The analysis treats the number of designations as a policy shock; while literature supports effects on emissions and activity, causal interpretation for biodiversity would be strengthened with event-study plots, pre-trend tests, or threshold-based RD around attainment cutoffs, as in existing environmental economics work. The alignment of estimated magnitudes with the military-shock-based elasticities and the pollution decomposition is reassuring, but a clearer identification narrative for nonattainment would reduce concerns about selection on deteriorating air quality or mean reversion.
- Mechanism decomposition: The pollution share is computed by multiplying the effect of the policy on pollution by the IV-based biodiversity–pollution elasticity. This assumes linearity and transportability of the IV LATE across contexts and pollutants; uncertainty propagation across both steps is not fully characterized. A bootstrap-based joint uncertainty interval and sensitivity analysis to alternative pollutant metrics/compositions would improve the precision and interpretation of the 20–60% range.
- Land use channel and protected areas: The urbanization results are correlational and vulnerable to confounding (e.g., sites closer to cities differ systematically; MODIS classification changes). Cross-validation with NLCD, night-lights, or parcel-level development permits, and research designs exploiting exogenous infrastructure expansions or zoning changes, would strengthen inference. Similarly, the protected-areas moderation analyses are informative but subject to targeted placement and fragmentation endogeneity; matching, border discontinuities, or staggered adoption designs could clarify causal effects.
- Inference and clustering: Most models cluster at the state level. Given county-level pollution IV and possible spatial correlation across neighboring states, exploring two-way or spatial HAC clustering, or clustering at meteorological basin/air-shed or ecoregion levels, would be informative for the pollution and biodiversity regressions.

Replication and transparency: Data sources (BioTIME, BEA, MERRA-2, EPA Greenbook, MODIS, WDPA) are public and methods are described in detail, including formulas for similarity and the upwind instrument. However, the paper does not indicate an available replication package; given the complex data construction (e.g., LASSO selection of upwind counties, spatial joins), a public code repository would substantially reduce replication barriers.

Overall, the paper provides credible, policy-relevant evidence that production growth reduces abundance, richness, and stability in ecological assemblages, with air pollution a major channel and environmental regulation mitigating impacts. The triangulation across designs, careful caveating of external validity, and breadth across taxa are major strengths. The main open issues relate to detectability/detection effort, additional meteorological controls and diagnostics for the pollution IV, stronger causal designs around CAA and land-use channels, and fuller uncertainty characterization in the mechanism quantification. Addressing these would further solidify the causal claims and practical guidance for policy.

This paper investigates the environmental impacts of economic production on biodiversity in the United States from 1960 to 2015, using a novel dataset that compiles longitudinal ecological sampling information from the BioTIME database. The authors find a negative correlation between economic production and biodiversity, measured by species abundance, richness, and similarity. They address causality by exploiting variation in local economic production driven by U.S. military buildups, finding that increased production resulting from military spending has a significant, negative effect on biodiversity outcomes. The paper also explores the channels through which economic production affects biodiversity, focusing on air pollution and land use changes. The authors use an instrumental variable strategy to isolate the causal effect of air pollution on biodiversity, finding that pollution accounts for 20-60 percent of the reduced form effect of military shocks. They also present descriptive evidence indicating that urbanization correlates with a significant decrease in biodiversity outcomes. Furthermore, the paper considers the role of environmental regulations, finding that Clean Air Act regulations had a notable, positive impact on biodiversity outcomes. Finally, the paper examines protected areas policy and finds that it may help to alleviate the negative effects of production shocks on the environment.

Strengths of the paper include the use of a novel dataset, a clear research question, a well-defined methodology, and a comprehensive analysis of the relationship between economic production and biodiversity. The authors address causality using a quasi-experimental design and explore the channels through which economic production affects biodiversity. The paper also considers the role of environmental regulations and protected areas policy.

Potential weaknesses include the reliance on observational data, which may be subject to omitted variable bias and measurement error. While the authors address causality using a quasi-experimental design, the validity of the instrumental variable may be questioned. The paper also focuses on the United States, which may limit the generalizability of the findings to other countries. The BioTIME database, while novel, is not fully representative of all species and ecosystems. The study's takeaway is more qualitative in nature, and broader applicability requires further comparative analyses with studies of similar approach and scope.

Specific issues include: * **Identification Threats:** The military buildup instrument, while commonly used, is still subject to debate regarding its exogeneity. There might be unobserved factors correlated with both military spending and biodiversity outcomes, even within a state and year fixed effects framework. * **Data Limitations:** The BioTIME dataset has limitations in terms of species representativeness and geographic coverage. The data may not be representative of all species and ecosystems in the United States, which could limit the generalizability of the findings. * **Channel Identification:** While the authors attempt to disentangle the channels through which economic production affects biodiversity, it is difficult to fully isolate the effects of air pollution, land use changes, and other factors. The instrumental variable strategy for air pollution relies on transported pollution from upwind cities, which may not fully capture the local effects of pollution. * **Generalizability:** The findings are specific to the United States and may not be generalizable to other countries with different economic and environmental conditions. * **Interpretation of Elasticities:** The authors acknowledge that the biodiversity-GDP elasticities should not be interpreted as the causal effect of a ceteris paribus increase in production, as GDP is an accounting concept and only varies because of changes in real economic activities. The estimates encompass the total effect of various determinants of biodiversity that vary with economic production. *

Omitted channels: The study focuses primarily on air pollution and land use change, but other channels such as water pollution, habitat fragmentation, and invasive species could also play a significant role. * **Functional form:** While some sensitivity checks for linearity are performed, there may still be non-linearities, interactions, or other important functional forms that are missed by this approach.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.39 Liang et al. 2025

C.39.1 Ratings Comparison

Metric	Claude Sonnet 4	GPT-5 Pro	Gemini 2.0 Flash
Overall	65 [55–75]	78 [68–87]	75 [60–90]
Claims	60 [50–70]	75 [65–85]	80 [65–90]
Methods	58 [48–68]	76 [66–86]	70 [55–85]
Adv. Knowledge	72 [62–80]	82 [72–90]	75 [60–90]
Logic & Comms	75 [68–82]	72 [62–82]	85 [70–95]
Open Science	45 [35–55]	45 [30–60]	70 [50–85]
Global Relevance	78 [70–85]	85 [75–92]	80 [65–90]

i GPT-5 Pro Assessment

This paper assembles a novel county-by-county rollout chronology for China’s Systematic Management of Children (SMC) program and links it to several large datasets to estimate long-run effects of early-life growth and development monitoring. The empirical strategy is a staggered cohort DID using the share of ages 0–6 overlapped with SMC as treatment, with county and birth-year fixed effects and province-by-birth-year fixed effects. The main finding—about a 5–6% increase in adult monthly income for full exposure—is plausible and is supported by an event-study pattern showing larger gains for earlier ages at first exposure and near-zero effects for placebo exposure beyond age six. The authors devote substantial effort to checking identification: (i) they document substantial variation in SMC timing across counties and years; (ii) show the reported timing is not predicted by a range of county characteristics (and that missingness in gazetteer timing does not obviously select on observables with province FE); (iii) present robustness to alternative exposure timing assumptions (start/mid/end-year), winsorization levels, clustering schemes, county-specific pre-trends, and inclusion of time-varying pollution controls; (iv) conduct placebo tests shifting adoption earlier; (v) use randomization inference; and (vi) implement modern staggered-DID estimators (Callaway & Sant’Anna; de Chaisemartin & d’Haultfoeuille). They also explore mechanisms using CFPS (health, mental health, cognitive tests), the Census (schooling), and CHNS (parental investments), finding improvements consistent with higher productivity rather than hours worked. Notwithstanding

these strengths, several issues merit caution and further work. 1) Treatment measurement: Exposure is assigned using hukou registration county and birth year, assuming early-life residence and service access align with hukou; in practice, temporary migration and living arrangements could create non-classical mismeasurement. The hukou rationale and restrictions in 2005 mitigate but do not eliminate concerns that some children may have lived away from the hukou county (potential attenuation) or accessed services where they resided (potential contamination); the paper would benefit from bounding exercises or sensitivity analyses (e.g., Oster, partial identification) gauging how much mismeasurement would be needed to overturn findings. 2) Sample selection and external validity: Only 873 of 2,868 counties report precise start dates. While observable differences are small with province FE, selection on unobservables remains possible. Generalizability beyond reporting counties should be discussed. 3) Concurrent policies and omitted variables: The paper addresses One-Child Policy fines and compulsory education timing (substituting province-by-cohort FE with province-specific trends to include them) and finds robustness; it also shows time-varying county economic indicators do not move with SMC. However, potential co-movement with other maternal/child health expansions (e.g., immunization, local MCH initiatives) that differentially affect young cohorts remains a residual threat. The event-study placebo for >6-year-olds and the dose-response shape mitigate this, but additional controls or narrative evidence on the timing of other pediatric programs (if available) would strengthen the case. 4) Staggered DID heterogeneity: Appendix E.10 reports a sizable discrepancy between alternative estimators: de Chaisemartin & d'Haultfoeuille (0.160) versus Callaway–Sant’Anna (0.050) and TWFE (0.058). This divergence is material and is not discussed in the main text. The authors should explain differences in estimand/weighting, reconcile magnitudes, and make clear which parameter they prefer and why; this matters for interpretation and external comparisons. 5) Binary outcomes magnitudes and model choice: Several CFPS and CHNS binary outcomes exhibit linear probability model (LPM) coefficients whose implied changes exceed baseline means (e.g., -0.159 for “upset,” baseline 0.048 ; $+0.123$ for “excellent/very good health,” baseline 0.939 ; -0.142 for “doctor visit,” baseline 0.101), yielding predicted probabilities outside $[0,1]$ for full exposure. LPM can extrapolate beyond bounds, but the magnitudes imply very large effects that are difficult to reconcile with the baseline rates. Re-estimation with logit/probit (or reporting marginal effects bounded in $[0,1]$), plus checks of distributional shifts or index robustness, would improve credibility. Relatedly, the reported mental health index effect (-0.76 SD) and several midlife chronic disease reductions (e.g., hypertension -0.162) appear exceptionally large; careful diagnostics (including balance on pre-determined health proxies, sensitivity to controls/specifications, and alternative constructions of indices) would be useful. 6) Income measurement and selection: The 2005 Census income analysis relies on ln monthly income; it is not explicit how zero/non-earners and current students are handled. Because the cohorts are age 18–29 in 2005 and SMC raises schooling and college completion, selection into continued education could dampen current income (downward bias) while selection into employment could inflate estimated wage effects (upward bias). Reporting effects on employment/earnings incidence, treatment-on-extensive margin, and robustness to excluding current students would clarify. The hours decomposition uses last-week hours to construct hourly wages, which may add measurement error and should be discussed. 7) Survey design and weighting: CFPS and CHNS employ complex sampling; it is unclear whether survey weights are used. Using appropriate weights and design-based inference (or showing robustness to weighting) would strengthen claims. 8) Program intensity/quality: The paper treats

SMC as homogeneous and “identical across counties,” but real-world implementation quality likely varied. If such variation correlates with timing or county characteristics, estimates could be biased. Where feasible, incorporating proxies for health system capacity and compliance (beyond hospital beds) or exploring heterogeneity by resource measures would help.

9) Cost-benefit framing: The back-of-the-envelope benefit of 198 RMB per adult-year versus a 270 RMB total child cost is informative, but the phrase “long-run annual return 73%” is potentially misleading. The analysis should explicitly: (i) state the horizon over which the annual gain is expected; (ii) discount future benefits; (iii) discuss persistence over the life cycle (income is observed at one cross-section, early in careers); and (iv) quantify public vs private costs, spillovers, and health care savings.

10) Transparency and reproducibility: The study relies on a substantial hand-collected dataset from gazetteers; while its construction is well described and cross-validated with literature, sharing the compiled SMC timing dataset and replication code (subject to data-use agreements for Census/CFPS/CHNS) would materially improve reproducibility, facilitate follow-up work, and enable independent validation. Overall, the core income result is plausible, well-motivated, and fairly robust across many checks, with an appealing age-at-first-exposure profile; the paper makes a meaningful contribution by providing direct evidence on adult earnings and multiple mediators. The main areas for improvement are reconciling heterogeneous DID estimators, bounding/more carefully modeling binary outcome effects, clarifying sample selection in income analyses, and enhancing transparency about data and code; the cost-benefit section would benefit from a more careful welfare accounting and lifetime perspective.

Claude Sonnet 4 Assessment

This paper examines the long-term impacts of China’s Systematic Management of Children (SMC) program, which provides routine health checkups to children aged 0-6. Using a county-by-county rollout dataset linked to the 2005 Population Census and other surveys, the authors find that full exposure to SMC increases adult income by 5%. While the paper addresses an important policy question and demonstrates substantial effort in data collection, several methodological concerns limit confidence in the results. The identification strategy relies on a cohort difference-in-differences design comparing exposed children to older never-treated cohorts within counties, and across early vs. later-treated counties. However, the parallel trends assumption is not rigorously tested with pre-treatment data, and the event study shows some concerning patterns with coefficients that are not consistently zero for placebo exposures beyond age 6. The exposure measure is crude, relying on birth year and hukou registration county without accounting for actual migration during early childhood. The paper lacks discussion of potential spillover effects within counties and families. While robustness checks address some concerns (different sample compositions, alternative specifications, accounting for historical events), the core identification threats remain. The mechanism analysis showing effects on health, education, and cognitive outcomes is interesting but relatively superficial. The cost-benefit calculation appears oversimplified, comparing only program costs to income gains without considering broader welfare effects. The writing is generally clear, though some technical details could be better explained. Overall, this represents a solid empirical effort on an important topic, but the identification strategy has meaningful limitations that prevent strong causal conclusions.

Gemini 2.0 Flash Assessment

This paper investigates the long-term impacts of the Systematic Management of Children (SMC) program in China on adult income, health, education, and cognitive skills using a cohort difference-in-differences (DID) design. The SMC program provided growth and development monitoring through routine health checkups for children aged 0-6 years. The study utilizes a novel county-level dataset constructed from over 3,000 local gazetteers, combined with data from China's 2005 One-Percent Population Census, China Family Panel Studies (CFPS), and the China Health and Nutrition Survey (CHNS). The authors find that full exposure to the SMC from birth increases adult income by approximately 5%, equivalent to about one year of schooling. They also present evidence of improved physical and mental health, better educational outcomes, increased cognitive skills, and sustained use of routine health checkups among adolescents. The paper conducts several robustness checks to address potential confounding factors, such as differential trends, endogenous sample selection, and concurrent policies. It also performs a cost-benefit analysis of the SMC program. Despite the paper's strengths, several issues need consideration. First, the reliance on local gazetteers for SMC rollout data introduces potential measurement error or selection bias, as not all counties may have recorded the information accurately or consistently. While the authors address this concern through various tests, some uncertainty remains. Second, the DID design relies on the parallel-trend assumption, which may not hold perfectly, despite efforts to test its validity. Third, the study's findings are based on observational data, making it difficult to establish causality definitively. Unobserved factors may influence both SMC exposure and adult outcomes. Fourth, the generalizability of the findings to other contexts may be limited due to the specific characteristics of the SMC program and the Chinese setting. Fifth, the measurement of some variables, such as mental health, relies on self-reported data, which may be subject to bias. Finally, while the paper presents a cost-benefit analysis, the quantification of certain benefits, such as improved health, may be challenging and subject to uncertainty. The robustness checks around staggered DID are a key strength.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.40 Naidu et al. 2023

C.40.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	89 [82–94]	80 [65–90]	75 [65–88] (n=2)
Claims	87 [80–92]	85 [70–95]	—
Methods	84 [76–90]	75 [60–85]	58 [50–70] (n=2)
Adv. Knowledge	90 [83–95]	70 [55–80]	82 [75–90] (n=2)
Logic & Comms	88 [82–93]	90 [80–95]	78 [62–82] (n=2)
Open Science	62 [45–75]	80 [70–90]	80 [75–85] (n=2)

Global Relevance	93 [87–98]	75 [60–85]	80 [72–88] (n=2)
NA	—	—	4 [4–5] (n=2)
NA	—	—	4 [4–4] (n=2)
NA	—	—	85 [80–90] (n=2)

i GPT-5 Pro Assessment

This paper reports a large, well-executed randomized controlled trial studying the causal effects of offers for temporary construction work in the UAE on Indian job seekers who passed firm screening. Random assignment occurred among screened applicants at 44 recruitment events with two major UAE construction firms; five of seven screened candidates were randomized to receive an offer. Four rounds of data collection were used (baseline in India, two phone tracking rounds, and a follow-up phone survey on average 17.5 months post-baseline), supplemented with administrative contract data from the UAE Ministry of Labor (MOL) and a friends-and-family tracking survey. The first stage is strong: the treatment increases being in the UAE at follow-up by 24–29 pp, raising the migration rate roughly from 26% to 58% and implying TOT/IV scalings of about 3–4. The use of randomization-group fixed effects, clustered standard errors, and enumerator fixed effects is appropriate; the randomization groups align with actual recruitment sessions.

Core results appear credible and are carefully characterized. Intent-to-treat effects on labor-market outcomes are large: monthly total compensation rises by about INR 4.5–5.2k (26–30% of the control mean) including in-kind food and housing; monthly cash earnings alone rise by INR 2.8–3.0k (19–21%). Unemployment declines by 4–7 pp (marginal in some specifications). Hours worked increase by 3–4 per week; expressed preferences for fewer hours do not change, and commute times are similar. These findings are robust to imputations based on MOL administrative contract values via an estimated mapping to realized earnings and to Manski-style attrition bounds. The paper transparently separates cash earnings from in-kind benefits and explicitly notes that workers may value in-kind less than market value, but still reports effects on cash-only outcomes. The measurement strategy acknowledges that UAE contract salaries are lower bounds (overtime top-ups), and links them to realized pay using survey-observed pairs; this approach is sensible, though it assumes stable mapping across those with and without survey outcomes and may not fully capture overtime heterogeneity.

The paper’s most novel contribution is quantifying costs and non-pecuniary disamenities. Agent fees average INR ~64k, with treated individuals paying INR 12–14k more by follow-up (consistent with contingent payments triggered by placement). Debt increases by ~INR 5–6k; monthly remittances rise ~INR 4k. The authors convincingly argue that about 9–12% of migration gains accrue to intermediaries and finance costs, with most of the cash surplus remitted. On non-pecuniary outcomes, a standardized well-being index (8 items) declines by 0.13–0.16 SD; components indicate more physical pain and reduced enjoyment, while loneliness and stress do not significantly worsen. A job satisfaction index shows no net effect; components suggest higher physical effort and worse climate, but lower perceived accident risk and fairer, more encouraging supervision. Baseline expectations about UAE earnings among those who migrate are close to realized values (mean log gap –6%), which undercuts concerns that deception by brokers drives migration decisions; reservation wage data further triangulate large disamenities (India-residents require ~88% earnings premium to migrate; UAE-residents would return for ~20% less), consistent with the well-being losses observed.

Internal validity is generally strong. Randomization appears balanced across most baseline covariates; a small imbalance in net assets is noted and controlled. Non-compliance is two-sided; notably, ~15% of controls end up at the same experimental firms, introducing “control contamination,” though first-stage strength and ITT focus mitigate bias risks. Attrition is material and somewhat imbalanced (higher in controls), and attriters differ along some baseline traits. The authors respond with (i) inverse-probability reweighting using only baseline predictors (leave-one-out) and (ii) multiple data sources (MOL admin, tracking, friends/family), and (iii) location-conditional Manski bounds ($\pm 25\%$ SD) leveraging admin knowledge of who entered the UAE. Labor-market results remain robust under bounds; the well-being effect turns statistically indistinguishable from zero under the most conservative upper bound but remains negative and becomes robust for smaller assumed deviations ($\pm 10\text{--}15\%$ SD). The multiplicity of outcomes is handled partially by pre-specified indices for well-being and job satisfaction, but many component analyses and additional domains (attitudes, social networks) are not explicitly adjusted for multiple testing; readers should interpret secondary component-level p-values cautiously. The clustering at 44 randomization groups is acceptable, though power for finely grained heterogeneity is limited.

The linear marginal treatment effects (MTE) extrapolation is clearly presented, framed with monotonicity, and used cautiously to characterize always-takers and never-takers. With a binary instrument, nonparametric MTE curvature is not identified; the linearity assumption is strong. The extrapolated patterns—never-takers would have slightly lower pecuniary gains and much larger well-being losses—are persuasive as suggestive evidence, especially as they align with the reservation-wage exercise, but they remain contingent on the linearity assumption. The paper could underscore more forcefully that these are model-based imputations rather than experimental estimates.

External validity is discussed thoughtfully. Applicants are negatively selected on household income relative to the broader male population but positively selected on ability/education relative to those who failed screening. Firms used are representative of the construction offer distribution in the UAE. The setting (male construction workers in a kafala-style system) generalizes credibly to GCC guest worker programs in similar sectors but not to permanent migration or high-skill flows. Follow-up is mid-run; long-run effects on assets and health are unknown. The valuation of in-kind benefits may differ across contexts. Social desirability in self-reported well-being could differ by country, though the component pattern (physical pain/heat) supports face validity.

Communication is clear, with informative figures/tables and transparent construction of indices and variables (appendix). The preregistration is noted, but the paper does not clearly document public code/data availability; some data (MOL) are sensitive. The speculative welfare/efficiency calculation combining their experimental estimates with outside elasticity evidence (from prior work) is well-motivated but rests on external parameters and strong assumptions (e.g., wage equals marginal product at doubled compensation, intermediaries not capturing increases). It is appropriately framed as an illustrative policy counterfactual rather than a causal estimate.

Overall, this is a rigorous and policy-relevant RCT that substantially advances evidence on the costs and benefits of temporary migration programs, uniquely combining pecuniary, intermediation, and non-pecuniary outcomes with multiple robustness strategies. Main threats are attrition, control contamination, and the strong assumptions behind MTE extrapolations and welfare counterfactuals; these are acknowledged and reasonably mitigated. The contribution

is high for both research and policy audiences.

i Gemini 2.0 Flash Assessment

This paper investigates the benefits and costs of temporary migration programs, focusing on the India-UAE migration corridor. It employs a randomized experiment, offering UAE construction job offers to Indian male workers and examining the effects on various labor market outcomes, well-being indicators, and social relationships. The study’s strengths lie in its experimental design, the use of administrative data to supplement survey data and deal with attrition, and its consideration of both pecuniary and non-pecuniary factors influencing migration decisions. The paper provides evidence on the significant upfront costs paid to labor intermediaries, the substantial increase in earnings for migrants, and the unexpected decline in subjective well-being driven by the physical demands and climate conditions of the jobs in the UAE. The analysis uses marginal treatment effects to extrapolate potential outcomes for different types of workers (always-takers, compliers, never-takers) to understand the heterogeneity of migration decisions, with the robustness of these findings assessed via bounding exercises. Key limitations include potential selection bias in the sample of workers, attrition despite the authors’ efforts to mitigate it, and some sensitivity of the well-being estimates to assumptions made in the attrition bounding analysis, plus the linear MTE assumption. There is also the fact that the experimental firms might not be fully representative. Additionally, the study relies on self-reported measures of well-being, which may be subject to biases. The reliance on the linear MTE assumption is potentially problematic given the possibility of non-linearities in the MTE. Another significant limitation is the focus on construction workers, which may limit the generalizability of the findings to other sectors and migration contexts. The study does however include extensive analysis of the robustness of these results to different sets of assumptions using the available data, and these are adequately discussed in the text. The use of the administrative data adds credibility and significantly strengthens the paper.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.41 Salazar and Restrepo 2024

C.41.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	68 [60–76]	65 [45–80]	70 [60–80] (n=1)
Claims	62 [50–72]	70 [50–85]	—
Methods	60 [48–70]	60 [40–75]	70 [60–80] (n=1)
Adv. Knowledge	72 [60–82]	75 [55–90]	80 [75–85] (n=1)
Logic & Comms	66 [56–76]	80 [65–90]	65 [60–70] (n=1)
Open Science	35 [20–55]	60 [40–75]	65 [60–70] (n=1)

Global Relevance	82 [70–92]	70 [50–85]	80 [68–92] (n=1)
NA	—	—	4 [4–4] (n=1)
NA	—	—	4 [4–4] (n=1)
NA	—	—	70 [65–75] (n=1)

i GPT-5 Pro Assessment

This preliminary manuscript tackles an important and under-measured question: the magnitude of deforestation leakage from spatially targeted conservation policies in Brazil. The authors develop a spatial general equilibrium model with migration and trade, where deforestation endogenously supplies agricultural land, and use it to simulate two policy regimes (Priority List municipalities with enhanced enforcement; and Protected Areas including Conservation Units and Indigenous Territories). They complement this with reduced-form checks (event study for the Priority List; RD around Protected Area borders). The overall framing is clear and the structural approach is promising; however, several identification, modeling, measurement, and robustness issues limit confidence in the headline leakage estimates. The most salient concerns are: (1) Reduced-form validity and interpretation. The Priority List event-study shows clear pre-trends by design (treated are high-deforestation “hotspots”), violating parallel trends; the paper then switches to a “synthetic differences-in-differences” approach, but offers almost no design details or diagnostics (donor pool, weights, pre-fit quality, falsification/placebo tests), and reports point estimates whose sign and magnitudes appear internally inconsistent with the descriptive event-study panels (e.g., a positive 1.09% “policy impact” on annual deforestation is hard to reconcile with the stated reductions). For the Protected Areas RD, the local discontinuity in forest cover is convincing in the raw plots; still, the operational choices (10 km bins, a 50% area threshold to classify hexagons, functional-form choices, and optimal bandwidth selection) are only partially justified. The placebo RD suggests some pre-existing discontinuity when controlling for fixed effects, which the paper attributes to long demarcation lags or indigenous stewardship; this weakens strict RD validity and motivates additional robustness (covariate continuity checks, donut RD, alternative bandwidths/parametric orders, sensitivity to the hex classification rule). Spillovers and SUTVA violations are correctly recognized, but not estimated in the reduced-form; the paper relies fully on the structural model to quantify leakage, which raises the stakes for structural assumptions. (2) Structural model design and parameterization. The model innovatively adds deforestation as an investment sector producing agricultural land, with open-access forests and decreasing returns through a β parameter. Yet several strong assumptions are only lightly defended or not stress-tested: a closed economy for Brazil (no international trade in agricultural goods), a single iceberg cost structure shared across goods, constant land shares within broad agricultural bundles (pasture, temporary, permanent), a single national forest regrowth rate δ , and identical consumer substitution elasticities across space and time. Key behavioral elasticities are borrowed or only weakly identified: $\eta = 9$ from another context; $\sigma = 2$ from external literature; α from a single cross-sectional expenditure survey without dealing with endogeneity of expenditure/prices; the migration dispersion γ estimated by PPML with income but acknowledged to be potentially endogenous to amenities. The deforestation supply elasticity ϵ is inferred from two-way FE regressions of log deforestation on model-imputed land rental rates, instrumented with lags via Anderson–Hsiao; the first-stage strength, plausibility of exclusion, and sensitivity to alternative instruments are not reported. The β parameter link-

ing deforestation productivity to remaining unprotected forest is derived from a steady-state relationship and estimated in panel regressions with state-year FE; this mixes steady-state logic with transitional data and may confound demand and supply. Overall, parameter uncertainty is substantial, but the paper reports point calibrations (e.g., sets $\alpha = 0.5$ toward the high end of its own estimated range 0.28–0.52) with little systematic sensitivity analysis. (3) Model inversion and identification of regional productivities. The inversion approach recovers region- and year-specific productivities Z^A_k , Z^{NA} , Z^D by matching observed quantities (deforestation, agricultural land shares, agricultural revenues, labor shares). This risks overfitting and makes conclusions sensitive to measurement error in revenues (especially cattle, which are proxied using municipal heads, state-level slaughter weights, and national prices) and to model misspecification. While the fit plots for land prices, incomes, and value added show reasonable correlations, they lack quantitative fit statistics and do not validate out-of-sample or with orthogonal moments. (4) Counterfactual construction and “no leakage” definition. The construction of the “no-policy” counterfactual for Protected Areas—effectively unlocking additional forest supply—is conceptually coherent. However, for the Priority List (an enforcement policy with no new spatial ban), treating “policy area” T_{PT} similarly to protected forest in the Z^D adjustment may be inappropriate; enforcement intensity should enter as an effective cost or productivity shifter rather than via area of “unlocked” forest. More importantly, the “policy, no leakage” scenario fixes prices at the no-policy levels to shut down GE channels, which is counterintuitive: partial-equilibrium comparisons are typically computed at baseline-policy prices (or by selectively turning off spillover mechanisms). The leakage calculation therefore depends on a non-standard price-fixing choice that is neither derived from the model nor stress-tested against alternatives (e.g., holding wages fixed, holding goods prices fixed separately, or using observed prices). Without a formal decomposition or sensitivity, the reported leakage shares (2–3% at one year; 10–15% over a decade) could be artifacts of this choice. (5) External validity and closure. The closed-economy assumption likely biases the magnitude of price responses and spatial leakage; international substitution and trade shocks are relevant for Brazilian agriculture. The single-good iceberg costs across sectors and time, as well as uniform migration costs over 2003–2019 based on 2005–2010 flows, are strong simplifications. (6) Internal consistency and documentation. There is at least one glaring magnitude error (stating roughly “3,500 km²” under protected status, orders of magnitude too small), plus occasional inconsistencies (e.g., leakage percentages differ between abstract and later figures; the Priority List synthetic DiD result appears sign-inconsistent with narrative). The reduced-form section mentions robustness choices (keeping only 2008 entries to address staggered adoption) but does not show parallel trends after synthetic matching or alternative donor pools. (7) Open science and replicability. The study draws on publicly available data but does not provide code or a replication package. Given the complexity of the inversion and simulation, lack of code substantially limits reproducibility and cumulative use. Despite these concerns, the conceptual contribution is meaningful: explicitly modeling deforestation as an investment that supplies land, embedding migration and trade, and attempting to quantify GE leakage from real policies is valuable. The qualitative mechanism—that intensification and sectoral reallocation attenuate leakage—aligns with both stylized facts and some reduced-form patterns (shifts away from pasture; higher cattle heads per hectare). If the authors add thorough sensitivity analysis to key elasticities ($\epsilon_{L,A}$, $\epsilon_{L,D}$, $\epsilon_{L,N}$, $\epsilon_{L,P}$), clarify the “no leakage” construction, address the Priority List counterfactual modeling, systematically document the synthetic DiD and RD robustness, correct magnitude errors, and release code, the paper could

make a solid contribution to policy debates on the effectiveness of targeted conservation in the presence of GE spillovers.

i Gemini 2.0 Flash Assessment

This paper investigates the effects of spatially targeted conservation policies on deforestation in Brazil, considering the potential for leakage, where deforestation is displaced to non-targeted areas. The authors develop a quantitative spatial equilibrium model of the Brazilian economy, incorporating deforestation as an economic sector, agricultural production with varying land demands, and costly trade and migration. They simulate the model to quantify the leakage effects of policies like the Priority Municipalities policy and the establishment of Protected Areas. The key findings suggest that targeting regions with high deforestation levels can effectively curb aggregate deforestation, but leakage increases over time. After one year, 2-3% of deforestation reductions are offset by leakage, rising to 10% after 10 years. The analysis attributes the relatively small leakage to agricultural intensification and shifts towards more productive regions. The paper presents reduced-form evidence and addresses endogeneity concerns using synthetic controls and regression discontinuity designs. However, the validity of the reduced-form estimates relies on assumptions about the ex-ante structure determining leakage and the absence of spillover mechanisms. The model incorporates various parameters, including sectoral productivities, land endowments, and amenities, estimated using publicly available data. The counterfactual analysis simulates scenarios with and without policies and leakage to assess their impact on deforestation. Overall, the study provides valuable insights into the effectiveness and limitations of spatially targeted conservation policies in mitigating deforestation in Brazil, highlighting the importance of considering leakage effects in policy design. However, there are identification challenges due to pre-trends and the difficulty of estimating structural parameters. The reliance on specific functional forms (e.g. Cobb-Douglas production functions, Constant Elasticity of Substitution utility) may influence the quantitative results. The exclusion of international trade is another limitation, although the authors discuss its potential impact. The model's complexity and computational demands may pose replication challenges. Data quality and availability, especially for historical deforestation rates and land use patterns, could also affect the accuracy of the findings. The paper does not extensively explore heterogeneity in policy impacts across different regions or sectors, which could provide further insights for policy targeting. Missing robustness checks, such as varying key parameters or considering alternative model specifications, limits confidence in the conclusions.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.42 Schoenegger et al. 2025

C.42.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash
Overall	72 [60–82]	65 [50–80]
Claims	70 [58–80]	70 [55–85]
Methods	64 [52–74]	60 [45–75]
Adv. Knowledge	68 [55–78]	75 [60–85]
Logic & Comms	78 [68–88]	80 [70–90]
Open Science	62 [45–75]	70 [50–85]
Global Relevance	66 [54–78]	70 [55–85]

i GPT-5 Pro Assessment

This paper reports a preregistered, between-subjects randomized experiment (final N=991 from Prolific) testing whether interactive access to a frontier LLM assistant improves human forecasting accuracy on six continuous, prospective questions that resolved 5–6 weeks later. Two GPT-4-Turbo assistants differed only by prompt: a “superforecasting” prompt designed to elicit calibrated, base-rate-aware reasoning and a “noisy” prompt (overconfidence, base-rate neglect). A control arm used a weaker model (text-davinci-003) explicitly prohibited from providing forecasts. Primary outcome is absolute error relative to the realized value, transformed via trimming/winsorization and standardization. Main preregistered analysis (one-way ANOVA on participant-level mean accuracy across the six questions) finds both GPT-4-Turbo conditions significantly reduce error by roughly 24–28% vs control; no significant difference between the “superforecasting” and “noisy” prompts. Exploratory analyses identify a pronounced outlier effect for one item (Bitcoin hash rate), after exclusion of which the “superforecasting” condition outperforms the “noisy” condition and both beat control by larger margins. Additional preregistered analyses find no differential benefit by (noisily measured) forecaster skill, no consistent degradation of simple aggregate (median) crowd accuracy, and no reliable moderation by perceived question difficulty; again, several inferences hinge on the Bitcoin item’s influence.

Strengths and contributions: The study’s prospective design avoids label leakage and task memorization issues common in static benchmarks, and the preregistration, OSF link, clear condition descriptions (including full prompts), and transparent reporting of a protocol deviation (extreme-value handling) are commendable. The paper squarely targets a practically important question—whether interactive LLMs can augment human forecasting—extends recent human–AI interaction evidence to a forecasting context, and examines plausible concerns (heterogeneous effects by skill, potential homogenization harming the “wisdom of crowds,” moderation by difficulty). The inclusion of a deliberately biased LLM assistant is informative: even “noisy” frontier advice appears to help relative to a weaker, non-forecasting control, pointing to interaction and quantitative engagement as potential mechanisms beyond raw model accuracy. The qualitative comparison of direct, model-only predictions between the two prompts (superforecasting superior on all six items) adds useful context.

Key threats to identification and robustness: (1) Control choice. The control condition combines two features—older/weaker engine and a prohibition on forecasts—so treatment effects conflate “frontier capability,” “willingness to supply numeric predictions,” and “prompted calibration.” A cleaner isolation of mechanisms would compare GPT-4-Turbo with/without numeric-prediction permission (or with neutral prompting) and, separately, a

human-only baseline. As designed, the main causal contrast is “frontier, numerically engaged GPT-4-Turbo assistant vs a weaker, non-forecasting LLM,” which is valuable but narrower than some claims might imply. (2) Per-protocol exclusion. Participants who did not engage with the assistant (link and at least one message) were excluded as preregistered; this ensures exposure but departs from an intent-to-treat estimand and could upwardly bias effects if non-engagers differ systemically (the authors apply the criterion across all arms, mitigating but not eliminating this concern). Dose–response (number/content of exchanges) is not analyzed, and chat logs are not reported, limiting mechanism tests (e.g., whether users copied numeric advice vs used the assistant to structure reasoning). (3) Outcome construction. The paper describes “winsorization” at 5% but then “removing” the bottom and top 5%—this is trimming, not winsorizing; a second pass winsorization at 3 SD is also applied. The deviation from the preregistration (applying 5% trimming to all groups, not only control) is justified to avoid leverage by outliers, but two-stage outlier handling plus post-hoc standardization (dividing by the control group’s SD per question) injects complexity. Using the control’s SD anchors the scale to a single arm and can complicate interpretation; alternative, more standard choices (e.g., log-absolute error where feasible, percentage or scaled error using known magnitudes, or pooled SD) and pre-specified robust estimators (e.g., Huber M-estimators) would strengthen robustness. (4) Outlier sensitivity. With only six items, a single problematic question (Bitcoin hash rate) materially shifts treatment comparisons. The authors discuss plausible mechanisms (unit confusion with BTC/USD price; the “noisy” prompt’s tendency toward higher numbers), but the dependence on item composition highlights fragility. Sensitivity to alternative outlier rules and pre-registered item-level robustness checks would be valuable; likewise, reporting item-wise treatment differences with confidence intervals and multiplicity adjustments would clarify heterogeneity. (5) Main model choice. The primary test uses participant-level aggregated error in a one-way ANOVA; given repeated measures across six questions, a mixed-effects model with random intercepts by participant and random slopes by question would more fully exploit within-participant variance and allow question-level inferences without aggregation-induced information loss. The paper does employ mixed-effects models for difficulty moderation, but not for the primary treatment contrast. (6) Skill heterogeneity test. The “skill” construct is based on a brief pre-task (three items combining binary Brier scores and intersubjective forecasts), dichotomized at the median; this is likely noisy and underpowered to detect heterogeneity. Continuous skill measures, shrinkage estimates, or out-of-sample skill proxies (if available) would be preferable. (7) Crowd-effects test. The “wisdom-of-crowds” analysis focuses on median accuracy across arms; this measures aggregate accuracy, not forecast diversity or correlation. If the concern is homogenization, variance of forecasts, pairwise correlations, or reductions in entropy/spread within arms would be more diagnostic. The observed mixed pattern (no clear harm overall; direction flips when removing the Bitcoin item) further underscores the need for diversity metrics. (8) External validity. Prolific participants may differ from professional forecasters and organizational settings; the six items are heterogeneous but few; the assistants had no retrieval (unlike many real deployments), and the session temperature (0.8) plus 25-message cap may influence behavior. These choices are defensible for ecological realism but limit generalizability claims. (9) Incentives. Paying three \$100 prizes randomly among the top-10 aims to reduce extremization incentives but does not deliver full incentive compatibility for every forecast; while applied uniformly across arms, it may still shape behavior in ways that interact with assistant use.

Internal consistency and communication: The paper is generally clear, careful about the

preregistered vs exploratory boundary, and explicit about a protocol deviation. A minor inconsistency appears in the exploratory difficulty moderation write-up: signs and interpretations of interaction coefficients (with accuracy defined so lower is better) are easy to misread; reporting marginal effects with CIs would help. The otherwise strong exposition includes full prompts and interface details; reporting randomization checks, message counts, time-on-task, and assistant transcript summaries would improve transparency and mechanism insight.

Replicability and open science: The preregistration link and prompt disclosure are strong. However, reproducibility hinges on access to anonymized data, analysis code, and (ideally) chat logs; the text does not clearly state their availability. The detailed reporting of model version, temperature, message cap, and no-retrieval constraint is helpful; still, the stochasticity of temperature 0.8 means some run-to-run variability is expected.

Bottom line: The core finding—that interactive access to a frontier LLM that will make and discuss numeric predictions helps human forecasters on prospective tasks relative to a weaker, non-forecasting assistant—is credible and practically interesting. The magnitude and ordering of benefits across prompts are sensitive to item composition and to outcome construction choices. Claims are mostly calibrated to the evidence, but stronger design (cleaner controls), richer mechanism measures, and pre-registered robustness would raise confidence and policy relevance. As it stands, this is a useful contribution to human–AI forecasting augmentation with caveats about robustness and generalizability.

i Gemini 2.0 Flash Assessment

This paper investigates the impact of LLM assistants on human forecasting accuracy. The authors conducted an experiment where participants were asked to answer six forecasting questions with the help of one of two LLM assistants: one designed to provide “superforecasting” advice, and the other designed to be overconfident and base-rate neglecting. They compared these participants to a control group that received a less advanced model without numerical predictions or explicit discussion. The study ($N = 991$) found that both LLM assistants significantly enhanced prediction accuracy compared to the control group. Exploratory analyses revealed an outlier effect in one forecasting item. Further analyses explored whether LLM augmentation disproportionately benefits less skilled forecasters, degrades the wisdom-of-the-crowd, or varies with question difficulty, but the data did not consistently support these hypotheses.

Here are some potential issues and concerns I identified in the paper:

1. **Reliance on Exploratory Analyses and Outlier Removal:** The paper relies heavily on exploratory analyses and the removal of outliers. The authors acknowledge that their results are sensitive to the inclusion/exclusion of a specific forecasting question (Question 3), which significantly influences the aggregate accuracy measure. While they provide justifications for these analyses, the robustness and generalizability of the findings are questionable.
2. **Limited Understanding of Mechanisms:** The study demonstrates that LLM assistants can improve forecasting accuracy, but it does not fully explain the mechanisms driving these improvements. The authors speculate that the continuous back-and-forth with the LLM and the induced deliberation may play a role, but they do not provide strong empirical evidence to support these claims.

3. **Potential for Bias and Lack of Generalizability:** The authors acknowledge that their online sample may be subject to systematic biases, which could limit the generalizability of their findings. Additionally, the specific questions used in the study may not be representative of all forecasting tasks, and the results may not generalize to other domains.
4. **Comparison to Human-Only Condition:** The control condition included a less advanced non-forecasting LLM, rather than a human-only condition. This makes it difficult to determine whether LLM assistants have an overall negative effect on forecasting accuracy compared to unaided human forecasters.
5. **Operationalization of Forecasting Skill:** The measure of forecasting skill is an abridged attempt at capturing accuracy and intersubjective accuracy. The authors acknowledge this results in noise and only permits large-scale relative comparisons. The validity of this measure is questionable.
6. **Lack of theoretical grounding:** The paper lacks a strong theoretical framework. While it draws on previous work on superforecasting and human-AI collaboration, it does not provide a comprehensive theoretical model to explain the observed effects.
7. **Limited Exploration of Negative Effects:** The study focuses primarily on the positive effects of LLM augmentation, with limited attention to potential negative consequences such as over-reliance on AI, deskilling, or the spread of misinformation.
8. **Potential for Data Contamination:** The authors acknowledge the risk of training data memorization in LLMs and attempt to address this by using questions that resolve in the future. However, the possibility of data contamination cannot be entirely ruled out, especially given the rapid pace of LLM development and the vast amounts of data they are trained on.
9. **Potential impact of design choices:** The paper studies a synchronous and explicit chat interface between AI and human forecasters, which will influence the way people use the model, and thus also its impact. The paper claims that the control condition does not provide any forecasting aid, because it does not provide numerical forecasts, but one could argue that simply the engagement with any system could have some impact.

In conclusion, the paper makes a valuable contribution to the literature on human-AI collaboration in forecasting, but its findings should be interpreted with caution due to the methodological limitations and the potential for bias and lack of generalizability.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.43 Schuett et al. 2023

C.43.1 Ratings Comparison

Metric	Claude Sonnet 4	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	72 [60–82]	74 [62–84]	70 [55–85]	72 [62–82] (n=2)
Claims	75 [65–83]	68 [55–80]	75 [60–85]	75 [68–82] (n=2)
Methods	68 [58–76]	64 [50–77]	70 [55–80]	70 [62–78] (n=2)
Adv. Knowledge	78 [70–85]	66 [52–80]	70 [55–80]	75 [68–82] (n=2)
Logic & Comms	82 [75–87]	86 [78–92]	80 [70–90]	82 [75–90] (n=2)
Open Science	85 [80–90]	82 [68–90]	80 [65–90]	72 [60–85] (n=2)
Global Relevance	88 [82–92]	84 [72–92]	75 [60–85]	80 [70–90] (n=2)
NA	—	—	—	4 [3–4] (n=1)
NA	—	—	—	4 [4–4] (n=1)
NA	—	—	—	80 [70–90] (n=2)

i GPT-5 Pro Assessment

This manuscript reports an expert survey on proposed best practices for AGI safety and governance, inviting 92 purposively selected experts and obtaining 51 responses (55.4%). The instrument comprised 50 statements (30 required, 20 optional) phrased as what AGI labs “should” do, answered on a 5-point Likert scale plus “I don’t know.” The headline finding—very broad support across virtually all items (mean agreement 1.39 on a -2 to 2 scale; on average 85% somewhat/strongly agree; for many items 98% agreement)—is clearly presented and could be practically useful as a starting point for standards development. The study is well written, preregistered, and accompanied by materials on OSF; the figures and tabulations are transparent and easy to follow. However, several important limitations constrain inference and policy relevance. First, sampling and external validity: the sampling frame is purposive and authored by the team, with selection based on perceived expertise and availability/willingness to participate. This risks selection bias toward respondents aligned with AGI labs and the governance community sampled; AGI lab respondents form a large share and also exhibit higher overall agreement than academia/civil society. Nonresponse bias is unaddressed: 41 invitees did not respond, and no analysis compares responders vs. nonresponders. The sample size is modest overall (N=51) and often smaller at the item level because 20 items were optional (some items have n 36–39), limiting power, especially for between-group tests. Second, measurement and construct validity: the 50 items are intentionally abstract and normative. Key constructs such as “AGI,” “powerful models,” “dangerous capabilities,” and the time horizon (“now” vs. “as we approach AGI”) are left unspecified; this vagueness plausibly inflates agreement by allowing respondents to project their preferred instantiation. One item (“API access to powerful models”) uses softer “should strongly consider” language, introducing inconsistency in framing. Many statements bundle multiple mechanisms (e.g., “restrictions on who can use the model, how they can use the model, and whether the model can access the internet”), which complicates interpretation of agreement. The predominance of “should” phrasing, social desirability around safety, and potential acquiescence bias likely push responses upward; the authors acknowledge this. High rates of “I don’t know” on some items (e.g., enterprise risk management, security standards) indicate either low familiarity or unclear wording; means exclude “I don’t know,” which can further bias central tendencies upward when ambivalence clusters in that category. Third, analysis choices: Likert means are reported (treating ordinal

data as interval), which is common but not assumption-free; medians and distributional summaries are also provided. Item-level statistical tests use chi-squared with Holm–Bonferroni correction; independence and sparse cell issues are possible given small n and optional items. Group comparisons via Mann–Whitney U on aggregated responses find AGI lab respondents more supportive overall, but repeated-measures dependence within respondents (each person rates many items) is not modeled, so standard errors may be understated. The preregistered regressions were dropped due to limited power; the rationale is reasonable, but this reduces the ability to quantify sector/gender effects. A cluster analysis is reported but is not very informative given poor separation. Fourth, interpretation: the paper’s main claim is “broad consensus among leading experts” on a wide portfolio of practices. Within-sample, the evidence supports “consensus among these respondents, on these abstract statements.” However, given the sampling and measurement issues, the claim should not be generalized to the broader, more heterogeneous set of stakeholders (including safety engineering, HCI, safety-critical industries, international regulators, and diverse geographic communities). The authors are reasonably cautious, but some policy implications read more strongly than the evidence warrants; the survey does not assess feasibility, cost, trade-offs, or marginal prioritization among practices. The lack of forced prioritization or budget-constrained trade-offs (e.g., best–worst scaling, pairwise comparisons, conjoint) means that near-universal agreement provides limited guidance about sequencing or relative importance. No evidence is provided about the effectiveness of proposed practices or how to operationalize thresholds (e.g., what triggers “red teaming” or “pausing”). Fifth, reproducibility and open science: preregistration, survey instrument, code, and data availability on OSF are strong; however, demographic data and free-text responses are withheld to protect anonymity, and the sampling frame is not fully public. This is appropriate ethically but limits recontact/replication. Aggregations appear internally consistent, with transparent tables for counts/percentages. Sixth, precision and robustness: sensitivity checks are limited. For example, results are not re-estimated including “I don’t know” as neutral, or weighting respondents by sector to mitigate overrepresentation. No reliability analysis (e.g., factor analysis to identify coherent domains, internal consistency across related items) is presented; while the items intentionally span multiple domains, latent structure could inform standards development and sharpen guidance. Several items overlap conceptually (e.g., third-party model audits, red teaming, increasing external scrutiny), and disentangling them empirically would be useful. Finally, despite these issues, the work’s practical value lies in curating a concrete, expert-derived checklist of candidate practices, with initial signals on which are least contentious vs. more uncertain (e.g., notifying other labs/state actors, avoiding capabilities jumps). To increase decision-usefulness, future iterations should: broaden and systematize sampling (including other safety-critical domains and global representation), pilot-test and precisely operationalize items (thresholds, timelines, accountability), incorporate elicitation of rationales and feasibility constraints, use designs that force prioritization under constraints, and perform robustness and sensitivity analyses (including alternative scoring of “I don’t know,” and modeling within-respondent correlation). Overall, the study is a clear, honest, and timely descriptive contribution with strong communication and decent open-science practices, but limited causal or normative authority beyond “consensus within this sample on abstract practices.”

i Claude Sonnet 4 Assessment

This paper presents a survey of 51 experts on AGI safety and governance practices, finding broad consensus across 50 statements. While the work addresses an important policy-relevant topic and achieves a strong response rate (55.4%), it faces several methodological limitations. The sampling approach, while purposive and well-justified for this specialized field, is small ($N=51$) and may suffer from selection bias toward experts already engaged with AI safety discourse. The survey design uses abstract statements that may inflate agreement levels compared to more specific implementation details. Statistical power is limited for subgroup analyses, and while Mann-Whitney U and Chi-squared tests are appropriate, multiple comparison corrections reduce the likelihood of detecting real differences. The response scale and demographic categorizations are reasonable, though the binary treatment of complex governance practices may oversimplify nuanced positions. The paper's transparency is commendable, with pre-registration and open materials, though some deviations from the pre-analysis plan occurred. The cluster analysis was unsuccessful, likely due to sample size limitations. While the findings provide valuable input for policy discussions, the high agreement levels (85.2% average agreement) may reflect either genuine consensus or methodological artifacts. The work would benefit from follow-up studies with larger, more diverse samples and more concrete implementation scenarios. The practical applications for AGI labs, regulators, and standard-setting bodies are clearly articulated, though the translation from abstract agreement to actual implementation remains challenging.

i Gemini 2.0 Flash Assessment

This paper presents the results of a survey conducted among 51 experts in the field of AGI safety and governance. The survey aimed to identify best practices for AGI labs by asking participants to rate their agreement with 50 statements about what these labs should do. The study found a broad consensus among the experts, with most agreeing that AGI labs should implement the majority of the suggested practices. The paper highlights areas of strong agreement (e.g., pre-deployment risk assessments, dangerous capabilities evaluations, third-party model audits) and areas where there was less certainty or more disagreement (e.g., inter-lab scrutiny, avoiding capabilities jumps). The authors also explore differences in opinions between experts from AGI labs, academia, and civil society, as well as between genders. Finally, the paper discusses policy implications and suggests directions for future research. Potential issues include the relatively small sample size, potential biases in sample selection, and limitations in the scope and specificity of the survey statements. The study acknowledges these limitations and suggests future research directions to address them. Internal consistency appears to be good, with results generally aligning with the claims made. However, the limited sample size and potential for selection bias could weaken the generalizability of the findings. The authors also acknowledge potential issues with the abstract nature of survey questions. Replication of the study with a larger and more diverse sample would strengthen the findings. Overall, the paper provides useful insights into expert opinions on AGI safety and governance, but its limitations should be considered when interpreting the results.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.44 Trammel and Aschenbrenner 2024

C.44.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash
Overall	86 [78–92]	65 [50–80]
Claims	76 [65–85]	70 [55–85]
Methods	89 [82–95]	65 [50–80]
Adv. Knowledge	82 [70–90]	70 [55–80]
Logic & Comms	88 [80–94]	75 [65–85]
Open Science	68 [56–80]	70 [50–80]
Global Relevance	83 [72–90]	75 [60–90]

i GPT-5 Pro Assessment

This manuscript develops a stylized, continuous-time optimal-control framework to study how exogenous technological progress interacts with existential risk and safety policy. The core environment posits a technology frontier $A(t)$, consumption $C(t)=A(t) \cdot x(t)$, and an anthropogenic hazard rate that depends either on the level of technology and policy (A,x) (“state risk”) or also on its growth $(A,\hat{A},x)$ (“transition risk”). A social planner with CRRA utility $u(C)$ and exponential discounting chooses safety/consumption tradeoffs $x(t) \in [0,1]$. The paper’s central results are: (i) under “broad” conditions (notably $\beta > 1$, a convex hazard in x with elasticity to x exceeding that to A , and positive discounting), optimal policy generates an “existential risk Kuznets curve” in which λ rises early (with $x \downarrow 1$) then falls to zero fast enough that survival probability $S_\infty > 0$; (ii) accelerations of technological progress typically reduce cumulative anthropogenic risk by both shortening time spent at risky states and inducing stronger safety effort via wealth effects (lower marginal utility, higher value of the future). The analysis is then generalized to a wider admissible class of hazard functions and non-exponential $A(t)$, and a transition-risk model is studied wherein the net effect of accelerations is ambiguous absent policy but often favorable with optimal policy (with thresholds involving β and γ). Proofs are supplied, and simple simulations illustrate dynamics.

Strengths: The paper is clearly structured and mathematically careful. The baseline model is transparent, and the authors prove existence/uniqueness of the optimal policy and provide explicit comparative dynamics. They do not rely on a single functional form: after presenting the constant-elasticity baseline (with $\beta > 0$ and $\gamma > 1$), they define an admissible class (D1–D5) and obtain generalized results (Propositions 3–6). The transitional-risk section shows nuanced threshold conditions (e.g., $\beta < 1 + \gamma / (\gamma - 1)$ for instantaneous accelerations after the regime-change A^*), clarifying when acceleration helps or is neutral/harmful. Important knife-edge cases ($\beta = 1$) are worked through: the planner tolerates enough risk that $S_\infty = 0$ even though faster growth can still lengthen expected survival time. The redundancy-based hazard and a “lower Inada” tweak demonstrate that the Kuznets pattern can persist while smoothing the counterfactual implication that $x=1$ for all early times. Overall, the logic is internally consistent, and the core mechanisms—time-compression of risk exposure and wealth-effect-induced safety—are convincingly formalized.

Main issues and limitations:

- 1) Reliance on optimal policy and strong structural assumptions. The headline claim that acceleration “typically” reduces cumulative risk leans heavily on a social planner who equates the marginal utility cost of consumption sacrifices with the expected discounted marginal benefit of risk reduction. In real settings, policy failures, delayed governance capacity, and institutional frictions may be persistent and endogenous to $A(t)$. The paper acknowledges this and gives counterexamples (Section 3.1), but its positive conclusions still depend on a strong optimality benchmark. A complementary model with persistent suboptimal policy (e.g., distorted v_t , or constraints on x adjustments) would help quantify how severe and long-lived real-world policy imperfections must be to overturn the main results.
- 2) Scope of “broad conditions.” Some central conclusions require conditions that, while plausible, are not innocuous: $\beta > 0$ and $\beta > 1$; $x > A$ and weak concavity in x (D4–D5); $\beta > 0$ for all A in the baseline state-risk model to derive “stagnation is deadly.” The generalizations do relax parts of these, but survival and acceleration benefits then hinge on limits like $R^*(C)$ and $R(k)$, or on threshold comparisons in the transition-risk case. These are elegant but underscore that “broad conditions” mask substantive structural restrictions about hazard elasticities and curvature that may vary across domains (AI, bio, nuclear). The paper is strongest when it explicitly states these thresholds; occasionally the prose suggests wider generality than the mathematics strictly supports.
- 3) Exogenous technology path and direction. $A(t)$ is exogenous, and “acceleration” is defined as moving faster along a given path. The discussion notes that direction matters—accelerating a path biased toward riskier technologies could raise risk even with optimal x —and urges interpreting accelerations as along the baseline path. Still, for policy relevance, the joint choice of rate and direction is critical. Endogenizing technology composition, or adding state-dependent governance capacity or safety tech innovation, would improve external validity.
- 4) Calibration and empirical grounding. The simulation (Figure 1) and the survival probability example (65% from “today”) are illustrative, with parameters chosen to match a 0.1% annual hazard and a particular VSL/consumption ratio. The Appendix A.1 argument that recent estimates imply $\beta = 1$ (87) is thought-provoking but rests on combining short-horizon risk-reduction claims with a particular continuous-time hazard form; it is not a structural estimate of β . No attempt is made to empirically identify β , γ , or α across domains, nor to test model predictions. That is appropriate for a theory paper, but the policy claims would benefit from clear demarcation between qualitative theory and quantitative policy inference.
- 5) Discounting and ethical sensitivity. Key results use $\beta > 1$ and $\beta > 0$ with $v_t = 1/[(1 - \beta)^{-1}]$. The paper explores $\beta = 1$ and shows survival fails, which is valuable; the ethical debate around β is acknowledged. Nonetheless, the practical impact of β on welfare valuations—and thus on thresholds comparing growth vs patience—could be more extensively mapped. Section 3.4.3’s comparison of growth level effects vs reducing β is insightful; more quantitative sensitivity here would aid policymakers.
- 6) Transitional risk coverage. The transition-risk analysis clarifies that absent policy, accelerations can be neutral ($\beta = 1$) or harmful ($\beta > 1$), and with policy, instantaneous accelerations often help if $\beta < 1 + \beta/(\beta - 1)$. Yet the economically relevant case is finite-duration

accelerations that alter ν ; the paper recognizes this but relies on instantaneous results to avoid complexity. The qualitative message is useful, but formal results for finite accelerations under policy are limited, so practical implications remain contingent.

- 7) **Communication and reproducibility.** The paper is well written; notation is heavy but consistent. Proofs and general conditions are carefully stated. Simulations are simple and the authors reference code for replication; however, full reproducibility depends on external code availability, and empirical calibration is deliberately light. Tables/figures are few but clear.

Overall, this is a thoughtful and rigorous theoretical contribution. It sharpens the “time of perils” intuition, formalizes when survival is feasible, and clarifies channels by which faster growth, in the presence of responsive policy, can reduce cumulative existential risk. Its main weaknesses are the reliance on optimal policy, the strong structural conditions required for the clean conclusions, and limited quantitative calibration. Extensions that endogenize technology direction/governance capacity, relax optimality, and empirically bound key elasticities across domains would greatly enhance applied relevance. Nonetheless, as a theory paper aimed at global-priorities questions, it is a high-quality, impactful piece of work.

i Gemini 2.0 Flash Assessment

This paper explores the relationship between technological advancement and existential risk, focusing on whether accelerating technological development increases or decreases the probability of existential catastrophe. The authors build economic models incorporating factors such as hazard rates, policy choices, and wealth effects to analyze this relationship. The central argument revolves around the idea of an “existential risk Kuznets curve,” suggesting that while the hazard rate may initially increase with technological development, it eventually falls due to factors like decreased marginal utility of consumption and increased value of the future. The paper investigates the impact of policy interventions, such as sacrificing consumption for safety, and how these interventions affect the overall risk landscape. It also considers alternative possibilities, such as transitional risks associated with the rate of technological development, and examines the role of factors like relative risk aversion and the potential for stagnation.

Here’s a breakdown of the key issues:

1. **Model Simplifications and Assumptions:** The economic models used involve simplifications, such as constant elasticities and specific functional forms. While these facilitate analysis, they may not fully capture the complexities of real-world interactions between technology, policy, and existential risk.
2. **Exogenous Technology Growth:** The models treat technological growth as largely exogenous, which means it’s not determined within the model itself. This simplifies the analysis but may overlook feedback loops where policy choices or concerns about existential risk influence the direction and pace of technological development.
3. **Calibration and Parameter Sensitivity:** The simulation results rely on specific parameter values, such as the discount rate, risk aversion coefficient, and initial hazard rate. The conclusions may be sensitive to changes in these parameters, and the robustness of the findings to alternative calibrations could be further explored.

4. **Policy Optimality and Implementation:** The analysis assumes optimal policy responses. In reality, policy decisions may be subject to various distortions, biases, or implementation challenges, which could affect the effectiveness of interventions aimed at reducing existential risk.
5. **Scope of Existential Risks:** The paper focuses primarily on anthropogenic (human-caused) existential risks. It acknowledges the existence of natural existential risks but largely ignores the possibility that technological advances may mitigate such risks. A more comprehensive analysis could consider the interplay between technological development and both anthropogenic and natural risks.
6. **Transition Risk Modeling:** The model’s treatment of transition risk might be overly simplistic, potentially impacting the assessment of acceleration’s influence on the long-term risk. The paper uses the “Russian roulette” model, in particular, as a starting point but recognizes its limitations.
7. **Limited Empirical Validation:** The paper is primarily theoretical and relies on economic modeling. Empirical validation of the model’s predictions and assumptions is limited. Connecting the model to real-world data and observations could strengthen its relevance and credibility.
8. **Communication to Policymakers:** While the paper addresses global priorities, the communication is highly technical and relies on economic jargon. Translating the findings into more accessible language for policymakers and decision-makers could enhance its practical impact.

Overall, while the paper provides a valuable framework for analyzing the complex relationship between technological advancement and existential risk, the identified issues suggest areas for further research and refinement to enhance the robustness, realism, and practical relevance of the analysis.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.45 Walker et al. 2023

C.45.1 Ratings Comparison

Metric	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	86 [78–93]	75 [60–85]	88 [78–98] (n=3)
Claims	84 [75–90]	75 [65–85]	90 [80–100] (n=1)
Methods	82 [70–90]	75 [65–85]	85 [75–95] (n=3)
Adv. Knowledge	90 [82–96]	75 [60–85]	83 [73–93] (n=3)
Logic & Comms	88 [80–94]	85 [75–90]	88 [80–97] (n=3)
Open Science	60 [45–70]	70 [60–80]	87 [77–97] (n=3)
Global Relevance	90 [83–96]	85 [75–95]	95 [90–100] (n=3)
NA	—	—	4 [4–4] (n=2)

i GPT-5 Pro Assessment

This manuscript uses a well-documented school-level randomized phase-in of mass deworming in western Kenya (PSDP, 1998–2003) combined with 23 years of longitudinal follow-up (KLPS) to estimate intergenerational effects on offspring mortality. The main claim—a 22% reduction in under-5 mortality (−17 per 1,000; LPM coefficient −0.017, s.e. 0.008) among children of treated individuals—is clearly stated, statistically supported, and robust across alternative estimators (logit/probit) and weighting schemes, with appropriate time controls (child year-of-birth fixed effects). Tracking and survey completeness are unusually strong for an LMIC setting; effective ever-surveyed rates exceed 86%, and there are no statistically significant treatment–control differences in attrition (Table A.1). The treatment is assigned at the school cluster level (75 schools), and standard errors are clustered at that level; with this many clusters, conventional cluster-robust inference is credible. Several identification threats are acknowledged and partially addressed, though some remain. First, child mortality is constructed from parent-reported birth histories following DHS conventions. While this is standard practice, underreporting of neonatal and infant deaths in surveys is a known issue; the study does not directly validate reports against administrative records. Any reporting bias would have to be differentially related to PSDP assignment to bias the ITT; given randomization and absence of differential attrition, this seems unlikely, but cannot be fully ruled out, especially because treatment increased education and healthcare contact (which might in principle shift reporting propensities). Second, deworming in this context is known to generate epidemiological spillovers; the original design faced both within- and cross-school externalities. This could attenuate ITT estimates and complicate the dose–response analysis. The authors appropriately focus on ITT and present years-of-assigned-treatment analyses as secondary; however, the “years assigned” variable embeds baseline grade (age) and exposure to cost-sharing periods, which are not randomized, so the dose–response gradient should be interpreted descriptively rather than as a second experimental estimate. Third, spousal exposure: roughly 10% of respondents report being married to someone from a PSDP treatment school. Because the analysis attributes outcomes to the respondent’s own treatment assignment, some estimated effects may partly reflect spousal treatment. The authors note this and argue the share is small; still, assortative mating induced by treatment could bias interpretation toward a household-level “treated dyad” effect rather than a pure individual-level effect. Fourth, multiple births per respondent induce within-parent correlation in outcomes; clustering at the school level is conservative for treatment assignment, but does not explicitly model respondent-level clustering. With 75 school clusters and treatment at that level, the chosen clustering is standard; still, supplemental checks with multi-way clustering or randomization inference would strengthen robustness. Fifth, the study excludes individuals treated by two later cross-cutting RCTs and reweights remaining controls to restore representativeness. This is sensible, but it adds complexity to the weighting scheme; the paper provides extensive detail and demonstrates robustness to alternative weights (Figure A.3, Table C.2), which is reassuring. Sixth, the mechanism analysis is necessarily exploratory given the original design: correlations between parental adult outcomes and child mortality are combined with treatment effects on those adult outcomes to form “implied” contributions. The authors are transparent that this is not causal mediation and emphasize timing and “bad control” concerns.

Still, the mechanism results are informative: improved parental health (including fewer miscarriages), more schooling (especially for women; +0.43 years, +7.6 pp any secondary), higher consumption (+14%), more urban residence (+4 pp), slightly delayed fertility (+0.44 years at first birth), and greater ANC/institutional delivery use all move in directions consistent with lower under-5 mortality. The infant (under-1) mortality reduction (−15%) is not statistically significant; the paper candidly reports this and discusses plausible reasons (different causal pathways for IMR vs U5MR; possible selection from reduced miscarriages). Seventh, external validity is discussed appropriately: the sample is from rural Busia with very high baseline helminth prevalence; the findings are most relevant where STH prevalence remains above WHO mass-treatment thresholds. Eighth, the cost–benefit analysis is ambitious but mixes strong and weak elements. On the positive side, it isolates benefits from intergenerational survival only, employs conservative assumptions on healthy life years, and presents both a low revealed-preference DALY value (USD PPP 67) and a much higher stated-preference DALY value (USD PPP 3,611) derived from a bespoke 2016 Busia survey. However, the stated-preference component relies on convenience sampling, single-bounded WTP questions, and health-state descriptions that some respondents found hard to interpret; it is therefore hard to treat as externally valid or policy-sufficient. The revealed-preference calculation yields an IRR 41% even before adding known adult earnings gains, which already suggests very high cost-effectiveness; the much larger IRR from stated preference (124%) should be treated as illustrative rather than definitive. Ninth, transparency and reproducibility are decent but not maximal: the registry (AEARCTR-0001191) and thorough appendices on variable construction and weights are strong points, but no replication code or data access details are provided here; given confidentiality constraints, a replication package or restricted-access pathway would materially improve reproducibility. Finally, communication is careful and balanced. The main claims are appropriately circumscribed; the non-significant IMR estimate is not overplayed; heterogeneity (by parent sex and age) and dose–response are presented as supportive but not definitive; and limitations—including mediation infeasibility and external validity—are explicitly acknowledged. Overall, the identification strategy is credible for the core U5MR result, the evidence is substantial relative to typical LMIC RCT follow-up work, and the contribution—first causal evidence on intergenerational child survival effects from an adolescent health intervention—is meaningful for policy and for the economics of health and human capital. Remaining concerns center on measurement and mediation constraints, possible treatment spillovers and spousal exposure, and the stated-preference valuation; these temper, but do not outweigh, the paper’s strengths.

i Gemini 2.0 Flash Assessment

This paper investigates the intergenerational impact of a school-based deworming program (PSDP) in Kenya on child mortality, using a 23-year longitudinal dataset from the Kenya Life Panel Survey (KLPS). The central claim is that the deworming intervention led to a 22% reduction in under-5 mortality among children of those who received the treatment. The study explores potential mechanisms through which deworming may have contributed to this reduction, including improvements in parent health, education, living standards, fertility patterns, and healthcare access. A cost-benefit analysis is also conducted to assess the economic value of increased child survival due to deworming. The research leverages the experimental design of the PSDP and the long-term panel data of the KLPS, offering a unique opportunity

to study the intergenerational transmission of health outcomes. However, several issues must be considered when evaluating the study’s findings. These include potential selection bias due to sample attrition, the challenges of disentangling the causal effects of multiple mechanisms, and the limitations of relying on self-reported data. While the authors attempt to address these concerns through robustness checks and sensitivity analyses, some uncertainty remains regarding the magnitude and interpretation of the results. A key concern is the external validity of the findings, as the KLPS sample is not nationally representative and is drawn from a specific region in Kenya. The long follow up period however allows a better understanding of longer term impacts, the cost-benefit analysis does rely on willingness to pay estimates which might be context specific. The dose response analysis provides further evidence of the interventions effects.

GPT-5 Pro reasoning trace available in the [online HTML version](#).

C.46 Xing et al. 2023

C.46.1 Ratings Comparison

Metric	Claude Sonnet 4	GPT-5 Pro	Gemini 2.0 Flash	Human
Overall	72 [60–82]	82 [74–89]	80 [65–90]	75 [66–92] (n=2)
Claims	68 [55–78]	78 [70–85]	85 [70–95]	68 [62–72] (n=2)
Methods	65 [50–75]	72 [60–82]	75 [60–85]	74 [69–78] (n=2)
Adv. Knowledge	78 [68–85]	86 [78–93]	80 [65–90]	78 [71–82] (n=2)
Logic & Comms	75 [65–82]	80 [70–88]	90 [80–95]	72 [68–75] (n=2)
Open Science	58 [45–70]	40 [25–55]	80 [70–90]	69 [84–94] (n=2)
Global Relevance	82 [75–88]	90 [82–96]	75 [60–85]	84 [75–90] (n=2)
NA	—	—	—	4 [4–5] (n=2)
NA	—	—	—	4 [4–5] (n=2)
NA	—	—	—	84 [75–90] (n=2)

i GPT-5 Pro Assessment

This is an ambitious and policy-relevant paper that assembles multiple data sources (government planting maps, remote-sensing NDVI and AOD, in-situ pollution and pollen monitors, wind, and citywide ER claims) to estimate environmental and health effects of large-scale urban afforestation in Beijing. The conceptual approach—explicitly decomposing Policy → Greenery → Environment → Health and addressing spatial spillovers through wind-transport—is a major strength. The documentation of post-2012 greening and the spatial break at MMP boundaries using NDVI is persuasive, with appropriate pre-period checks and some robustness. The use of monitor-level outcomes near population centers is well motivated. However, several identification and measurement issues merit attention and temper confidence

in some key quantitative claims. First, the wind-based “upwind NDVI shock” design risks conflating the effect of vegetation with meteorological and source-region confounding intrinsic to wind direction. Day-to-day wind direction in Beijing is systematically associated with both meteorology (temperature, humidity, mixing height, precipitation, pressure) and the location of external emission sources (e.g., industrial corridors vs mountains), which independently shift pollution and possibly pollen. Because the regressors are constructed from the cross-sectional distribution of NDVI around each monitor and wind selects the sector, the design largely exploits cross-direction variation rather than within-direction variation. Without including rich meteorological controls and/or wind-sector fixed effects (e.g., monitor-by-wind-octant FE) to absorb direction-specific air quality baselines, U may pick up non-vegetation differences between sectors (cleaner mountain air also tends to be the greener sector). The downwind-placebo test helps rule out some mechanical artifacts but does not fully exclude wind-direction confounding, since meteorology and source regions are inherently asymmetric up- vs downwind. Relatedly, event-study leads based on monthly NDVI interacted with daily wind provide some reassurance but are not definitive against direction-specific confounding. Adding explicit meteorology (temperature, humidity, precipitation, wind speed, boundary layer height if available), monitor-by-wind-sector FE, and city-date FE, and verifying results under those controls would strengthen causal interpretation. Second, in the remote sensing boundary analysis, the NDVI–AOD pair is subject to a potential measurement link: MA-IAC AOD retrieval depends on surface reflectance, and areas with sharp vegetation changes could plausibly bias inferred AOD. The authors note this and switch to ground PM_{2.5} later; nonetheless, showing the AOD gradient disappears when conditioning on retrieved retrieval quality flags, or using alternative satellite products, would further mitigate concern. Third, the MMP polygon layer is digitized from a 2022 planning document summarizing achievements over 2012–2022, and timing within the decade is not observed. While the pre/post NDVI gradient is convincing for presence, the lack of phase timing introduces uncertainty for dose and dynamics. Measurement error is acknowledged (e.g., gradients extend ~400 m beyond mapped boundaries), but a more formal sensitivity analysis to boundary and timing error—e.g., buffering boundaries and excluding small patches—would help (a partial version appears in the appendix). Fourth, there are small internal inconsistencies: the pollen monitor count is stated as 20 in the text but 12 in Table 1; and there is a unit error (“1.17 million mu (780,000 hectares)”—this overstates hectares by ~10x; elsewhere, 2.07 million mu 135,000 ha is correct). These do not overturn results but should be corrected. Fifth, the health effect estimates use OLS with district-by-month, year×month, DOW, and holiday FE. This is a reasonable starting point, but two key improvements would raise credibility: (i) jointly controlling for both PM_{2.5} and pollen (and temperature/humidity) to reduce omitted-variable bias when estimating each exposure’s effect, and (ii) instrumenting exposures where feasible (e.g., for pollen, upwind NDVI interacted with wind as an instrument, carefully validating exclusion; for PM_{2.5}, standard wind or thermal inversions IV). The paper shows low pollen–PM_{2.5} correlation for most months and provides an independent comparison against prior PM literature; still, simultaneous models and meteorology controls would better isolate each agent’s effect. Also clarify how zeros in pollen counts are handled in log models (restricting to the monitoring season likely limits zeros, but explicit treatment—e.g., small offset or Poisson/negative binomial—should be reported). Sixth, the mapping from exposure to population is plausible but underexplained for the headline “4.2%” PM_{2.5} reduction in average population exposure and “7.4%” rise in pollen exposure attributable to MMP. The calculation appears

to multiply (policy→green) × (green→environment) and average across monitors, but the weights (population vs equal, within- vs out-of-season averaging for pollen, spatial coverage) and uncertainty propagation are not detailed. Providing the exact aggregation formula, population weights, seasonal treatment, and a bootstrapped confidence interval for these citywide impacts is important. Seventh, in the ER analysis, outcomes are classified by medication categories rather than diagnoses; this is innovative and largely credible (top drug lists help), but residual misclassification may remain. Clarify coverage (the Urban Employee program reportedly covers 91% of permanent residents) and patient flow across districts; sensitivity restricting to ERs occurring in the same district as residence (if inferable) or high-urgency cases would bolster interpretation. Eighth, the cost–benefit narrative has a key inconsistency: using the paper’s own numbers, net annual health benefits from PM2.5 mortality plus healthcare (minus pollen healthcare) are roughly 5.1–5.9 billion CNY per year, while the program cost is ~75 billion CNY; this implies a nominal payback closer to 13–15 years, not “over the next decade.” That statement should be qualified or corrected, or additional benefits (heat mitigation, carbon, energy savings, property values) should be explicitly quantified to justify the 10-year horizon. Ninth, while the paper suggests tree-species recommendations, it does not link MMP species mixes to species-resolved pollen responses; connecting planting inventories to species-level pollen monitors would greatly strengthen policy guidance. Finally, openness and replicability are limited: many datasets are administrative or derived; no code or data repository is mentioned; key constructed variables (e.g., upwind/downwind cones, monitor-level aggregation, exact aggregation to citywide impacts) would be hard to reproduce without code. Providing code, derived data products (to the extent legally allowed), and a replication guide would substantially increase the work’s value. In sum, the study makes a strong, novel contribution, but the central wind-based identification would benefit from additional controls and design checks; the citywide impact aggregation needs clearer methods and uncertainty; minor internal inconsistencies should be fixed; and the payback claim should be aligned with the reported magnitudes. With these addressed, the paper would be very competitive in top field outlets and informative for practice.

i Claude Sonnet 4 Assessment

This paper examines urban forests as a policy tool for air pollution mitigation by studying a large afforestation program in Beijing that planted approximately 2 million mu of greenery over a decade. The study employs remote sensing data to document vegetation growth and uses wind direction-based quasi-experimental designs to estimate causal effects on air quality and pollen exposure, combined with medical claims data to assess health impacts. While the paper makes valuable contributions by quantifying both benefits (pollution reduction) and costs (pollen-related health impacts) of urban afforestation at scale, several methodological concerns limit confidence in the results. The identification strategy relies heavily on wind direction variation, but the paper provides limited evidence that daily wind direction changes are truly exogenous to other factors affecting air quality and health outcomes. The satellite-based vegetation measures (NDVI) may not perfectly capture the specific vegetation changes attributable to the MMP policy, and the cross-sectional analysis comparing areas near vs. far from planting sites could be confounded by systematic differences in location characteristics. The health analysis using medical claims data suffers from potential ecological fallacy issues since patient residential locations are unknown, and the assumption that district-level aggre-

gation adequately captures individual exposure is strong. The paper also acknowledges but does not fully address potential mechanical relationships between satellite-based vegetation and pollution measurements. While the wind-based identification strategy is creative and the scale of analysis is impressive, the combination of these identification concerns, measurement limitations, and aggregation issues reduces confidence in the precise causal estimates reported, particularly the specific quantitative conclusions about pollution reduction (4.2%) and health benefits.

i Gemini 2.0 Flash Assessment

This paper examines the effects of urban afforestation in Beijing, focusing on air quality and health outcomes. The authors leverage a large-scale afforestation project (Million Mu Project, MMP) as a policy intervention and use a combination of remote sensing data, ground-level monitoring data, and medical claims data to assess the impact of urban forests on air pollution, pollen exposure, and related health effects. They find that the MMP contributed to a substantial greening of Beijing, leading to improvements in air quality (reduced PM_{2.5}) but also increased pollen exposure. The analysis further explores the health implications of these changes, finding that increased pollen exposure is associated with increased emergency room visits, particularly for respiratory and sensory issues. They compare the health costs of pollen exposure with the health benefits of pollution reduction, suggesting that the benefits outweigh the costs, though pollen's health costs are also notable.

Important issues to note include:

- **SUTVA violation:** The authors acknowledge the violation of the Stable Unit Treatment Value Assumption (SUTVA) due to the spatial spillover effects of afforestation on air quality and health. They address this by estimating separate links in the causal chain (Policy -> Greenery -> Environment -> Health) and using a wind directivity design to capture spatial spillover.
- **Data limitations:** The study relies on insurance claims data, which does not contain primary diagnoses, which affects the ability to directly observe some health outcomes like patient death.
- **Generalizability of findings:** The context-specific nature of the study (Beijing) raises questions about the generalizability of the findings to other urban areas with different environmental conditions, tree species, or population characteristics.
- **Ozone effects:** The study finds ambiguous effects of afforestation on ozone levels, which could be due to the complex interactions between urban forests and ozone formation. This could be explored more deeply to see if specific tree types are driving the adverse effect.
- **Mortality:** The study does not have mortality data, which is an important outcome, but they expect the effect to be small given that most pollen-related ER visits do not lead to hospitalization.
- **Causality:** While the study employs a quasi-experimental design and various robustness checks, establishing causality remains a challenge due to potential unobserved confounders and the complex interplay of factors influencing air quality and health outcomes.
- **Interpretation of coefficients:** The authors interpret their reduced-form coefficients as having a causal interpretation. To do so requires several assumptions, which, while

discussed, may not hold in reality.

- Quantification of benefits: More work could be done to explore heterogeneity in the relationship between pollution and ER visits, including consideration of non-linear effects. This would strengthen the conclusions about the health benefits of the program. Some more details in the paper about calculation of VSL would be helpful.
- External validity: How do these results generalize to other places? The local meteorology and tree species may have a large impact on the relative amounts of pollution and pollen, but more discussion of this would be helpful.

Overall, this is a carefully done study that provides valuable insights into the environmental and health effects of urban afforestation. The authors address the challenges in their study effectively. The paper is clearly written and well-organized. The data sources are comprehensive, and the econometric methods are sound. This paper makes a valuable contribution to the literature on environmental economics and urban planning.

GPT-5 Pro reasoning trace available in the [online HTML version](#).